

HANDBOOK OF ANALYTICAL QUALITY BY DESIGN

Edited by

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Preface

Analytical methods are one of the indispensable tools for the analysis of pharmaceutical drug products and the estimation of drugs, their metabolites, degradation products, and related substances. The International Conference on Harmonization (ICH) of technical requirements for the registration of pharmaceuticals for human use and the US Food and Drug Administration (US FDA) initiated the implementation of quality-by-design (QbD) principles for producing the drug products with optimum quality and robustness coupled with efficacy and safety, as per the Q8–Q10 guidances. Although as per the mandatory requirements by USFDA, QbD is quite popular in formulation development these days, yet its application in analytical science is still in infancy. The analytical method development tends to involve a gamut of steps including the tedious processes of sample extraction and sample preparation, critical control of analytical instrumentation, and analysis of statistical data. However, development of a robust method coupled with robust and optimum performance is very difficult in the practical scenario, ostensibly owing to a high degree of variability associated during monitoring of the multiple factors. In this regard the application of QbD tools for analytical development has been greatly facilitated by the federal agencies, which not only helps in the systematic development and maintenance of the analytical methods but also provides end-to-end solutions for complete analytical life-cycle management. The idea behind designing a book of its first kind on the topic *analytical quality by design (AQbD)* will greatly help the analytical scientists and chemists working in the area of analytical development to keep them abreast with the principles and methodology of the proposed concept and its applications in developing analytical methods of the drugs, metabolites, impurities, and related substances in bulk drugs and pharmaceutical formulations.

Besides, the current news on guidance (i.e., Q14) principally focuses on “analytical procedure development,” which harmonizes the scientific approaches of analytical development, and providing the principles related with the description of analytical procedures. Implementation of this guidance in analytical development will improve the communication between the industry and regulators, thus provides more efficient, scientific, and risk-based method development with maximal regulatory flexibility for addressing the postapproval changes. Moreover, in this guidance, ICH has also

indicated that analytical scientists must use Q14 and Q2 (analytical method validation) guidances in tandem with ICH Q8–Q12 guidances for hassle-free development and build effective regulatory communication of analytical procedure–related changes. In nutshell, the proposed book will take care of every aspect of the analytical development by meeting the current regulatory demands.

The book consists of 11 chapters on various salient aspects of QbD-based method development, the federal aspects, and regulatory perspectives on a score of instrumental methods of analysis. A brief aspect of each of the chapters included in the book has been described in the following paragraphs.

Chapter 1, Introduction to Analytical Quality by Design, has discussed the key basics and fundamental aspects of the implementation of QbD principles for excellence in analytical method development. The chapter also covers key aspects of the regulatory requirements and current expectations of the regulatory agencies with respect to QbD applications for analytical methods.

Chapter 2, Analytical Quality by Design for Spectrophotometric Method Development, has discussed the implementation of QbD principles for the development of spectrophotometric methods and related applications.

Chapter 3, Analytical Quality by Design for Gas Chromatographic Method Development, has discussed the implementation of QbD principles for the development of gas chromatography methods and related applications.

Chapter 4, Analytical Quality by Design for Size-Exclusion Chromatography, has discussed the implementation of QbD principles for the development of size-exclusion chromatography methods and related applications.

Chapter 5, Analytical Quality by Design for Liquid Chromatographic Method Development, has discussed the implementation of QbD principles for the development of liquid chromatography methods and related applications.

Chapter 6, Analytical Quality by Design for High-Performance Thin-Layer Chromatography Method Development, has discussed the implementation of QbD principles for the development of high-performance thin-layer chromatography methods and related applications.

Chapter 7, Analytical Quality by Design for Capillary Electrophoresis, has discussed the implementation of QbD principles for the development of capillary electrophoresis methods and related applications.

Chapter 8, Quality by Design–Based Development of Vibrational Spectroscopy Methods, has discussed the implementation of QbD principles for the development of vibration spectroscopy methods such as Fourier transform infrared spectroscopy and other related applications.

Chapter 9, Quality by Design–Based Development of Nondestructive Analytical Techniques, has discussed the implementation of QbD principles for the development of NIR, Raman, Terahertz methods, and other related techniques.

Chapter 10, Risk Assessment and Design Space Consideration in Analytical Quality by Design, has discussed the implementation of risk

assessment techniques and the establishment of design space as vital QbD tools for the development of analytical methods for analytical excellence and regulatory compliance.

Chapter 11, Design of Experiments Application for Analytical Method Development, has discussed the vital aspects of experimental design techniques for factor screening and optimization in analytical method development.

The abovementioned chapters in the book are considered as very essential for the development of various analytical methods in a single textual repertoire for the analytical and organic chemists, pharmaceutical scientists, biochemists, and many others who are particularly involved in the application of the aforementioned analytical techniques.

Chapter 1

Introduction to analytical quality by design

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1.1 Introduction

The principal endeavor of a pharmaceutical scientist has been to develop drug products delivering the desired patient-centric attributes of quality, efficacy, and safety. Escalating concern and criticism, raised in over a decade on quality and reliability of drug products, has led to the espousal of systematic approaches in the pharmaceutical industry. Accordingly, a holistic and methodical approach of *quality by design* (QbD) has lately been in vogue during drug product development [1,2]. Emphasis during QbD has been on “building” the quality into the product and/or process, rather than merely by periodical “testing and inspection.” The ICH instituted a series of quality guidances such as Q8, Q9, Q10, Q11, and Q12 in order to harmonize the implementation of quality practices in pharmaceutical product development, all accentuating enactment of systematic tools as its 21st-century quality initiatives. Subsequent endorsement of such QbD paradigms by US-FDA, EMEA, and many other key global regulatory agencies offer unequivocal testimony to its immense significance for all the potential stakeholders, namely patients, scientists, and regulators [3,4].

QbD chiefly embarks upon envisioning and planning the product quality, based upon the predefined objectives. Verily, it is postulation and application

of rational attitude of doing things right from the first step through science- and risk-based understanding, while adopting cogent and structured approaches [5,6]. It revolves around designing a product catering to desired needs, while comprehending the process performance and allowing continuous improvement. The entire QbD exercise, therefore, aims at unraveling the scientific minutiae during systematic product development and manufacturing process(es), which would have hitherto remained as unearthed. QbD tends to ameliorate product quality, robustness, and customer satisfaction. Besides, it also reduces defects, recalls, and rejects, thus ultimately provides significant saving of the resources like time, effort, and cost [7,8].

1.2 Analytical quality by design principles and fundamentals

Analytical development is an indispensable phase not only for the characterization of drug substance, but for analysis of drug(s) in dosage forms, biological matrices, and stability samples too. Analytical methodology, accordingly, is the cardinal part of the pharmaceutical development owing to the high degree of criticality observed during method development, validation, and transfer. The traditional approach of method development is quite tedious and ineffective owing to high degree of variability involved during each stage of method development [9,10].

Efforts have, accordingly, been made by the pharmaceutical analysts to extend the documented benefits of QbD approach to analytical method development too, which is popularly termed as “Analytical QbD (AQbD).” Several conferences and workshops have lately been held, especially since 2013, providing tangible impetus on the implementation of the QbD precepts to analytical method development. The modern analytical scientists have started benefiting from integration of the QbD principles for developing far more robust, effectual, and cost-effectual analytical methods [1,9,11]. Much akin to QbD, AQbD is also a science- and risk-based paradigm for analytical method development, striving to attain the predefined objectives for enhanced method performance, robustness, ruggedness, and flexibility for continual improvement. Nevertheless, there is hardly sufficient experience or sizable exposure available among the analytical researchers today on envisioning and implementing AQbD approach in developing apt analytical methods [3].

This holistic strategy of AQbD offers an outstanding feature as the ability to accurately portray or foretell the performance of an analytical process, accurate quantification of drugs, and estimation of the possible synergisms or antagonisms among multiple method variables. Application of AQbD tends to save tangible volume of resources, as illustrated in Fig. 1.1. AQbD can also extricate and rectify a “problem,” thus facilitating analysis of drugs from drug products, efficiently and economically. By and large, AQbD offers numerous benefits over traditional analytical approach, particularly in the industrial milieu, as indicated in Table 1.1.

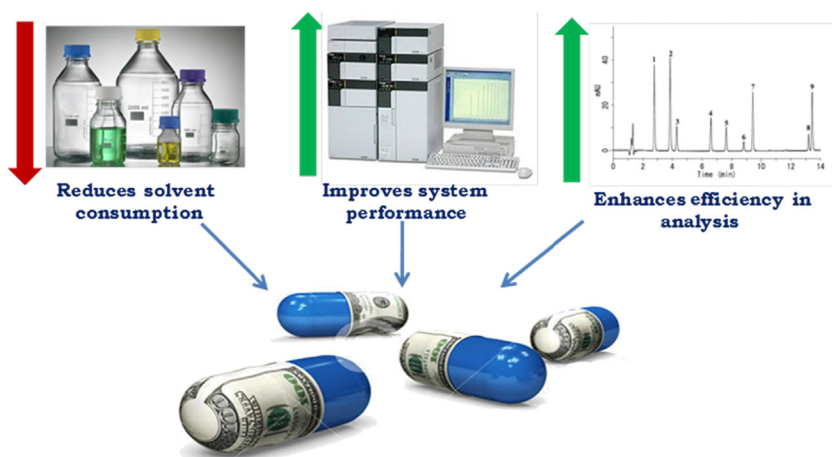


FIGURE 1.1 AQbD tends to facilitate enormous saving of resources.

1.3 Basic analytical quality by design terminology

AQbD, on the heels of QbD, endeavors for accomplishing the predefined analytical objectives. The holistic AQbD strategy revolves around five fundamental elements, namely defining the analytical target profile (ATP) or quality target method profile (QTMP), identification of critical analytical attributes (CAAs), critical method variables (CMVs) or critical method parameters (CMPs), and critical process parameters, selection of apt experimental designs for carrying out analysis based upon design of experiments (DoE), precise demarcation of analytical design space (ADS) or method operable design region (MODR), and analytical control space or normal operating region, to delineate the optimum analytical solution, and postulation of control strategy for continuous improvement. The major objective of AQbD has been to establish robust MODR or ADS within meaningful system suitability criteria and continuous life-cycle management [10–13]. Table 1.2 enumerates the key terms employed while implementing AQbD during method development.

1.4 Analytical quality by design strategic principles and implementation steps

Adoption of QbD principles to analytical methods can be rationally explained on the basis of the fact that a multitude of variables tend to influence the method outcomes. These variables encompass sample characteristics, instrument settings, method parameters, and selection of calibration models [14]. Chromatographic techniques being the commonest analytical tools in pharmaceutical quality control, the number of variables involved in

TABLE 1.1 Cardinal benefits of implementing quality by design (QbD) during analytical method development.

Stellar benefits of analytical QbD

- To understand, reduce, and control the source of variability
- To facilitate an explicit and implicit understanding of the method
- To go beyond the traditional ICH procedure of method validation
- To collect in-process information for timely control decision(s)
- To delineate real causes of variability by eliminating variability due to environmental impact
- To reduce variability in analytical attributes for improving the method robustness
- To predict real-time impact of high-risk variables on CAAs, prior to product development
- To attain flexibility in analysis of API, impurities in dosage forms, stability samples, and metabolites in biological samples
- To keep the values of analytical attributes within the pharmacopoeial monographs, and out of specification limits
- To obviate the requirements of revalidation within MODR
- To select a design with high probability of successful operation and safety
- To assess potential failure modes and their impact on system operation
- To provide the basis for plausible in-house troubleshooting procedures
- To discriminate between fault-detection and performance monitoring tools
- To smoothen the process of method transfer to the production level

API, active pharmaceutical ingredient.
Source: Adapted from Beg S, Sharma T, Saini S, Kaur R, Kaur R, Singh B. Analytical Quality by Design for robust chromatographic methods. *Cutting-Edge (Spinco Biotech)* 2020;10(2):9–17.

analytical method development phase is quite comparable to the ones involved during formulation development. Implementation of QbD provides an opportunity to achieve regulatory flexibility, but requires high degree of robustness, product quality, and understanding of process, product, and analytical method [15]. The process of QbD implementation in analytics is quite analogous to that with product QbD. Fig. 1.2 illustrates the systematic five-phase execution of analytical method development in a pharmaceutical setup, employing AQbD approach. Implementation of AQbD exercise, on the whole, can be successfully accomplished through these five key steps, which have been quite explicitly discussed below.

Step 1: Establishment of ATP or QTMP

The foundation of any analytical method developed through QbD principles is ATP, which is quite similar to *quality target product profile*, while

TABLE 1.2 Fundamental terminology employed during analytical quality by design (AQbD) paradigms.

Key terms	Explanations
Analytical target profile or quality target method Profile	Statement defining the objectives of the method employed to drive method selection, coupled with design and development activities. It constitutes the nuances of analytical methods of interest as the goal of a QbD exercise.
Potential method parameters	All the possible variables involved in an analytical method
Critical method parameters (CMPs)	Independent analytical variables influencing the method performance
Potential analytical attributes	Dependent variables related to an analyte or an analytical process characteristics that should be within an apt range to ensure desired method performance criteria
Critical analytical attributes (CAAs)	Potential analytical method attributes that are influenced by CMPs with probability to go beyond appropriate limit or range
Experimental runs or trials	Analytical experiments conducted under defined conditions, i.e., combinations of factors at varied levels for each of the CAAs to be measured
Failure mode	Specific manner by which a failure tends to occur w.r.t. function under investigation
Risk priority number	Severity of the event $\times$ probability of the event occurrence $\times$ detectability of predicting the event before it occurs
Analytical knowledge space	Defined “n” dimensional space on the basis of prior knowledge of the “n” number of CAAs and investigated CMPs
Analytical design space or method operable design region	Multidimensional part of the analytical knowledge space, enclosed by upper and lower levels of the coded variables, demonstrated to provide assurance of quality
Analytical control space or normal operating range	Part of the analytical design space, usually employed for setting in-house specifications within the working environment of the company
Control strategy	Schematic set of various controls to surmount all the possible sources of variability to meet ATP requirement during analytical method transfer

Source: Adapted from Beg S, Sharma T, Saini S, Kaur R, Kaur R, Singh B. Analytical Quality by Design for robust chromatographic methods. Cutting-Edge (Spinco Biotech) 2020;10(2):9–17.

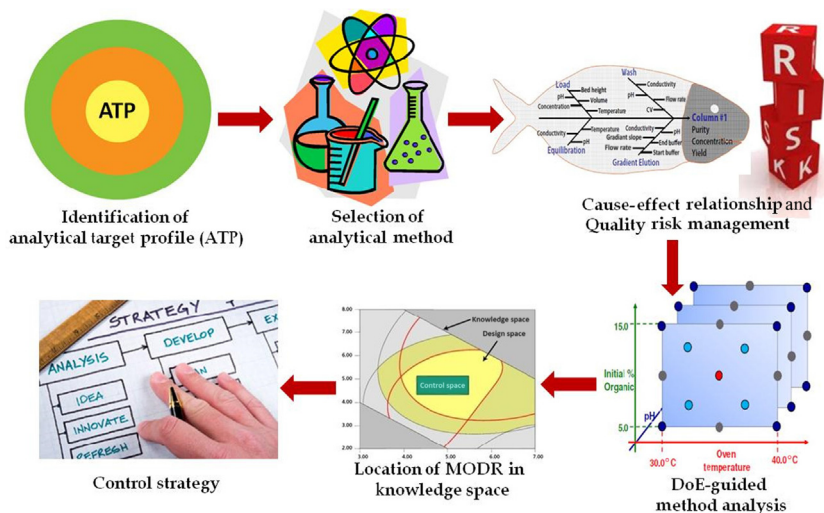


FIGURE 1.2 Five-phase analytical quality by design (AQbD)-enabled strategy for rational development of a robust analytical method.

implementing QbD for drug products. It encompasses the objective of analytical methods and quality requirements to allow predictable conclusion regarding material attributes.

Step 2: Identification of CAAs and CMVs

Risk assessment, according to ICH Q9, is performed in three steps, namely risk identification, risk analysis, and risk evaluation. Risk identification delineates all the possible method parameters, based on their potential impact on the CAAs. This can be accomplished with the help of process mapping envisioning tools, like an Ishikawa cause-and-effect fish-bone diagram, as depicted in Fig. 1.3.

Further, “vital few” CMPs critically influencing the CAAs and method performance are identified. *Quality risk management* (QRM) is the key approach widely employed to provide comprehensive understanding of the overall risks associated with various CMPs while influencing CAAs of the analytical method. CMPs are the vital input factors or independent variables related to the instrument setting parameters, column chemistry and dimension, sample preparation, etc., while CAAs include the output responses or dependent variables. In addition to the prior literature, process knowledge and experience of analytical scientist is considered pivotal in identifying the CMPs. Risk analysis is usually performed usually employing risk estimation matrix or failure mode and effect analysis (FMEA). The former identifies the variables with moderate to high degree of risk, while the latter helps in identifying the high-risk factors based on the likelihood, severity, and detectability scores. Risk analysis is accomplished on the basis of prior art coupled

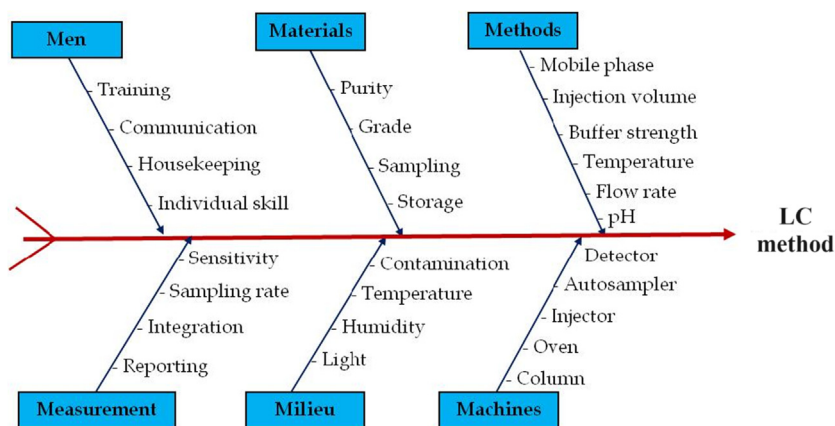


FIGURE 1.3 An Ishikawa cause-and-effect fish-bone diagram for a liquid chromatographic method development.

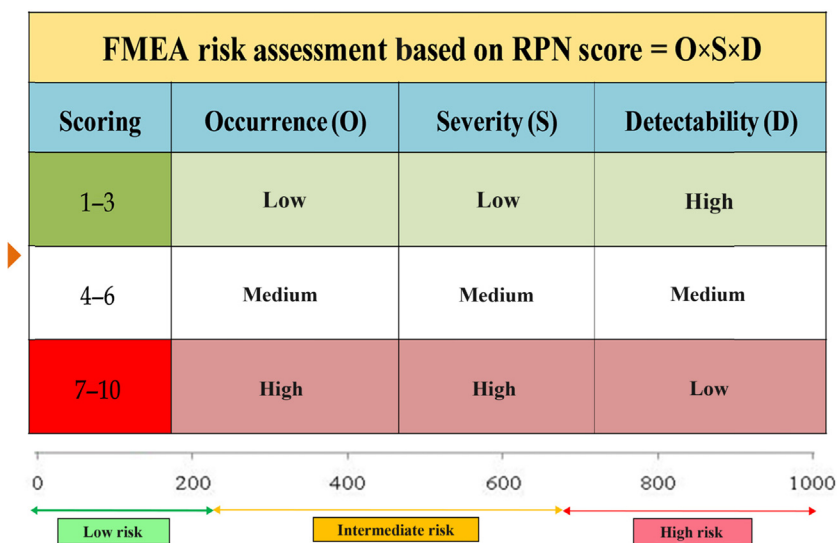


FIGURE 1.4 A typical risk assessment chart used during analytical quality by design (AQbD)-based analytical method development involving FMEA-based estimation.

with scientific prowess through intensive brainstorming among stakeholder scientists representing various sections of an industrial setup. Fig. 1.4 portrays the risk management in an industrial setup employing FMEA technique.

Besides risk assessment, identification of CMPs can also be carried out using factor-screening studies, taking into consideration the real experimental data analyzed using apposite experimental designs. Essential difference

between the risk assessment and factor-screening is the basis of discrimination of risk or factor influence; the former depends on the experiential wisdom of the scientific team, while the latter on the actual experimental analysis. Table 1.3 enlists the instances of commonly encountered CMPs and CAAs in literature during analytical method development through chromatographic approaches. A figurative account of the CAAs, particularly employed during liquid chromatographic analysis, like peak area, retention time, relative retention, resolution, tailing, asymmetry, capacity factor, and theoretical plate count.

Step 3: Design-guided method development and analysis

Application of DoE principles facilitates sound scientific understanding of the multiple method parameters and variables that tend to affect CAAs using minimal experimentation to yield the maximal information, while unraveling the prevalence of (any) interactions and reducing intricacies. For successful execution of DoE study, the knowledge of CAAs, CMPs, their ranges, and best fitting of the mathematical model(s) is obligatory. Experimental designs are indispensable during DoE-based data analysis. The experimental designs help in mapping the responses on the basis of the studied objective(s), CAAs being explored, at high (coded as +1), medium (coded as 0), or low (coded as -1) levels of CMAs. It tends to unearth the mechanistic understanding of CMPs and CAAs relationship and associated interactions among them.

First-order screening designs are useful in identifying the high-risk factors on the basis of sound statistical analysis. Instances of such screening designs include fractional factorial design, Taguchi design, and Plackett–Burman design. Response surface designs are helpful in systematic development of the analytical methods, particularly involving significant nonlinearity and/or interactions among CMPs and CAAs. Instances of such optimization designs encompass full factorial design, central composite design, Box–Behnken design, optimal design, etc. Various 3D-response surfaces and 2D graphs such as contour plots, perturbation plots, linear correlation plots, outlier plots, and Box–Cox plots constitute some of the key graphical tools demonstrating enormous analytical ability of the experimental designs.

Once the data have been collected according to the chosen design, the results can be analyzed using statistical methods such as multiple linear regression analysis, or mathematical paradigms such as artificial neural networks to draw the objective conclusions. Finally, optimum search is carried out apposite numerical (like Desirability function) and graphical optimization (like overlay plot) techniques.

Step 4: Establishment of MODR

MODR or ADS of any analytical method, also termed as “proven acceptable range (PAR),” is the multidimensional integration and interaction of input variables (i.e., CMPs during analysis) demonstrated to provide

TABLE 1.3 Commonly encountered CMPs and CAAs during various analytical method developments.

Technique(s)	CMPs	CAAs
HPLC/UPLC/ LC-MS	Mobile phase ratio, organic modifiers, pH, buffer strength, flow rate, injection volume, column type, column dimension, sonication time, column oven temperature	Peak area, retention time, theoretical plate count, asymmetry factor, capacity factor, peak resolution, relative retention time, % assay
HPTLC	Mobile phase ratio, saturation time, volume loaded, scanning speed, slit dimension, distance between tracks, migration distance, spot diameter, stationary phase, plate dimension	Peak area, retardation factor, asymmetry factor, capacity factor, peak resolution, % assay
GC	Injection volume, flow rate, carrier gas velocity, type of carrier gas, split ratio, column type, thermostat time, injector temperature, detector temperature, oven temperature, needle temperature, initial temperature, final temperature	Retention time, area, % assay, tailing factor, resolution, peak area
Technique(s)	CMPs	CAAs
HPLC/UPLC/ LC-MS	Mobile phase ratio, organic modifiers, pH, buffer strength, flow rate, injection volume, column type, column dimension, sonication time, column oven temperature	Peak area, retention time, theoretical plate count, asymmetry factor, capacity factor, peak resolution, relative retention time, % assay
HPTLC	Mobile phase ratio, saturation time, volume loaded, scanning speed, slit dimension, distance between tracks, migration distance, spot diameter, stationary phase, plate dimension	Peak area, retardation factor, asymmetry factor, capacity factor, peak resolution, % assay
GC	Injection volume, flow rate, carrier gas velocity, type of carrier gas, split ratio, column type, thermostat time, injector temperature, detector temperature, oven temperature, needle temperature, initial temperature, final temperature	Retention time, area, % assay, tailing factor, resolution, peak area

CAAs, critical analytical attributes; CMPs, critical method parameters; GC, Gas chromatography; HPLC, high performance liquid chromatography; HPTLC, high performance thin layer chromatography; LC-MS, liquid chromatography mass spectrometry; UPLC, ultra performance liquid chromatography.
Source: Adapted from Beg S, Sharma T, Saini S, Kaur R, Kaur R, Singh B. Analytical Quality by Design for robust chromatographic methods. Cutting-Edge (Spinco Biotech) 2020;10(2):9–17.

assurance of quality. Having DoE-guided method development and experimental data analysis worked out, the design space region is subsequently identified to embark upon the optimal solution. Having MODR been established with the help of experimental designs, overlay plotting, and/or numerical techniques of desirability function, it is finally validated to identify the “edge of failures,” which may circumvent any likelihood of failure of the analytical method at a later stage. Within the MODR, many a times, it is ideal to identify a region for setting in-house specifications within the firm’s working environment, also called as normal operating range (NOR) or analytical control space. The fruition of any DoE exercise depends upon several parameters, especially the experimental accuracy and measurement precision. Accordingly, the best practice before validating a MODR is to perform confirmatory validation runs to ratify the empirical model resulting from a DoE exercise by comparing the observed results with that of the predicted ones. From regulatory perspectives, working within MODR is not considered as a change, and method can be considered robust enough for use. This provides greater flexibility and leverage during method transfer process. Typically, an MODR is proposed by the applicant and approved after requisite regulatory assessment and approval.

Step 5: Control strategy and continuous improvement

A planned set of control(s) for all possible variation(s) assures that ATP requirement would be met during analytical method transfer as well as routine use. This can be attained with continuous monitoring of CAAs or system suitability parameters. Control strategy is not always a one-time exercise and it should be framed during all the critical stages of method development life-cycle for continuous improvement. Even after going through all the elements of QbD for a particular analytical method, method validation, verification, and transfer are the key exercises that ensure fitness of the method for its intended use.

1.5 Analytical quality by design in life-cycle management

AQbD is practically applicable today throughout the life-cycle of a product to facilitate the regulatory flexibility in analytical method. It implies tractability or liberty to alter the method parameters within the design space of an analytical method, usually called as MODR. Integrating all the elements of AQbD together, the analytical life-cycle management starts with defining the ATP and continues till the method applicability in the product development and beyond during quality control valuation for batch release testing. The resultant confirmation with respect to ATP is the main focus for performance qualification, for example, precision study on the site of routine use. Continual verification involves activities, which assures that the method is under control throughout its life-cycle. Once a method is established for routine use, method performance is monitored over time to ensure compliance

with the ATP framework. This can be accomplished by using control charts or other tools to track system suitability data, method related investigations, etc. Continuous monitoring allows the analyst to proactively identify and address any out-of-trend performance.

1.6 Regulatory standpoints on analytical quality by design

The fundamental precepts of QbD gravitate around thorough understanding of a process, product or objective, defined prior to initiation of a process. Earlier till 2005, US-FDA used to ask for submission of chemistry, manufacturing, and control (CMC) information from the manufacturers, as a part of new drug application (NDA). The ICH Q8, Q9, and Q10 guidance documents, on the other hand, provided stricter requirements to meet the quality standards of a drug product [16–18]. Although ICH Q8 (R2) guideline has not essentially covered the analytical method development perspective in the context of design space, yet it is extended to MODR with the aim of continuous improvement in the method robustness for improved analytical understanding.

Adopting AQbD during manufacturing process as control strategy will, therefore, warrant process performance and product quality. Implementation of AQbD is anticipated to strengthen the notion of “right analytics at right time” with momentous potential in drug product development. Strong reliance of pharmaceutical development and manufacture on robust analytical data testifies the worth of QbD-steered analytical method development as the current domain of focus and execution. The recent FDA approvals of AQbD-supplemented NDA filings tend to activate and accentuate the significance and application of QbD in analytical method development. In order to assure ongoing performance of an analytical system and associated methods, the US-FDA and USP mandatorily require system suitability testing to be established for an analytical method. However, the recent updates of USP-NF and European Pharmacopeia have permitted flexibility for analytical methods to be changed without any need for revalidation, provided AQbD approach has been implemented per se. In January 2013, a joint research program between FDA’s laboratory/review divisions and EMA was initiated. The FDA’s July 2015 Guidance for Industry: “Analytical Procedures and Method Validation for Drugs and Biologics” has covered the vital aspects of the method development, which includes critical evaluation of the method robustness with the help of DoE and multivariate testing, thus for attaining enhanced performance [19]. With the help of this guidance, it has been clear that critical attributes of the method possess definitive impact on output parameters owing to the high degree of variability.

Currently, there is no regulatory guidance dedicated to the systematic analytical method development. The upcoming ICH Q14, “Analytical Procedure Development,” along with revising the ICH Q2(R1) Guidelines on

“Validation of Analytical Procedures: Text and Methodology,” are welcomed to recommend the analytical method validation criteria meeting the stipulated ATP requirements [20]. Through method development, an understanding on what analytical method attributes impact the ATP would be required. A company’s analytical method validation standard operating procedures will commonly define the criteria for accuracy, linearity, precision, etc., assuming such criteria are consistent for all quantitative test procedures. Usage of QbD paradigms, In this regard, the use of QRM, DoE and multivariate techniques (MVT) in analytical method development provide a science- and risk-based framework for critical understanding of the root-cause variability, thus improves the method robustness and method performance.

1.7 Potential applications of analytical quality by design in analytical settings

With the growing demand of implementation of systematic approaches and quality tools into the analytical science, the popularity of AQbD has been percolated to several areas of analytical testing.

1.7.1 Analytical method development

Several literature reports have been published on the application of AQbD for developing analytical high performance liquid chromatography (HPLC), ultra performance liquid chromatography (UPLC), high performance thin layer chromatography (HPTLC), liquid chromatography mass spectrometry (LC-MS), vibrational spectroscopy, atomic absorption spectroscopy, electrophoresis methods, and many more, for the estimation of diverse drugs or during simultaneous estimation of bulk drugs and pharmaceutical dosage forms. Use of AQbD helps in identifying highly influential variables (i.e., CMPs) from diverse method parameters such as mobile phase ratio, organic modifiers, pH, buffer strength, flow rate, injection volume, column type, dimension, with prominent influence on CAAs to improve the method robustness and performance.

1.7.2 Bioanalytical method development

AQbD principles have been applied in bioanalytical method development for improving extraction recovery while chromatographic separation of analytes from biological fluids such as plasma, serum, lymph, tissue, and organ extracts. Many processing conditions, such as nature of extracting solvent, extraction time, centrifuge type, centrifugation speed, time and temperature, and sample filtration procedure, influence the recovery process, where AQbD helps in risk-based monitoring of the conditions to improve the process efficiency.

1.7.3 Identification of impurities and degradation products

Impurity profiling and identification of degradation products in bulk drugs and finished products are highly essential for maintaining product quality, safety, and efficacy. Requirement of a highly sensitive analytical method along with critical monitoring of chromatographic conditions, in this regard, facilitate estimation of impurities and degradation products, and controlling their levels within the acceptance limits for regulatory approval.

1.7.4 Nondestructive pharmaceutical analysis

Use of nondestructive tools is gaining high importance for saving time during pharmaceutical analysis of drug substances in bulk and finished products. New-age spectroscopic techniques such as near infra red, and Raman are used for nondestructive analysis to identify impurities or counterfeit drug analysis for the purpose.

1.8 Conclusion

Being relatively novel to the analytical scientists, AQbD poses tacit challenges for its implementation, especially when it is not a strict regulatory requirement. A paradigm shift is anticipated from traditional information-rich dossiers to scientifically sound and knowledge-rich documents, with no expectation of technology transfer and method validation. Moreover, harmonization of AQbD terminology and concepts at a global level is need of the hour, for example, MODR, ADS or PAR, and ATP or QTMP. Training human resources at industrial and regulatory levels is another challenge toward effective harvesting of QbD concepts in analytical arena, thus calling for definitive guidelines on documentation of knowledge generated during method development. Scanty publications, whether research or review, demonstrating the successful usage of QbD exercise in analytical research is another limitation. The current chapter is a humble endeavor from our end to furnish fundamental yet inclusive information on the emerging concept, to reinforce the existing understanding on the concept, and to provide the desired fillip toward its fruitful implementation in pharmaceutical scenario.

References

- [1] Singh B, Beg S. Attaining product development excellence and federal compliance employing quality by design (QbD) paradigms. *Pharma Rev* 2015;13(9):35–44.
- [2] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [3] Singh B, Beg S. Quality by design in product development life cycle. *Chron Pharmabiz* 2013;22:72–9.

- [4] Beg S, Hasnain MS. *Pharmaceutical Quality by Design: Principles and Applications*. New York, NY: Academic Press; 2019.
- [5] Beg S, Rahman M, Kohli K. Quality-by-design approach as a systematic tool for the development of nanopharmaceutical products. *Drug Discov Today* 2019;24(3):717–25.
- [6] Nayak AK, Ahmed SA, Beg S, Tabish M, Hasnain MS. Application of quality by design for the development of biopharmaceuticals. In: Beg S, Hasnain MS, editors. *Pharmaceutical Quality by Design*. Academic Press; 2019. p. 399–411.
- [7] Beg S, Hasnain MS, Rahman M, Swain S. Introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical Quality by Design*. Academic Press; 2019. p. 1–17.
- [8] Beg S, Rahman M, Swain S. Quality by design applications in pharmaceutical product development. *Pharma Focus Asia* 2020;1–5.
- [9] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharma Rev* 2016;14(10):61–6.
- [10] Panda SS, Beg S, Bera RAVV, Rath J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):00039.
- [11] Beg S, Sharma T, Saini S, Kaur R, Kaur R, Singh B. Analytical quality by design for robust chromatographic methods. *Cutting-Edge (Spinco Biotech)* 2020;10(2):9–17.
- [12] Molnar I, Rieger HJ, Monks KE. Aspects of the design space in high pressure liquid chromatography method. *J Chromatogr A* 2010;1217(19):3193–200.
- [13] Awotwe-Otoo D, Agarabi C, Faustino PJ, Habib MJ, Lee S, Khan MA, et al. Application of quality by design elements for the development and optimization of an analytical method for protamine sulfate. *J Pharm Biomed Anal* 2012;62:61–7.
- [14] Singh B, Raza K, Beg S. Developing “optimized” drug products employing “designed” experiments. *Chem Ind Dig* 2013;23:70–6.
- [15] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.
- [16] <https://database.ich.org/sites/default/files/Q8_R2_Guideline.pdf>.
- [17] <https://database.ich.org/sites/default/files/Q9_Guideline.pdf>.
- [18] <https://database.ich.org/sites/default/files/Q10_Presentation.pdf>.
- [19] <<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/analytical-procedures-and-methods-validation-drugs-and-biologics>>.
- [20] <https://database.ich.org/sites/default/files/Q2R2-Q14_EWG_Concept_Paper.pdf>.

Chapter 2

Analytical quality by design for spectrophotometric method development

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2.1 Introduction

Analytical methods are a fundamental part of the manufacturing process in the pharmaceutical industry. These forms the basis for the introduction of the new pharmaceuticals into the market [1]. The execution of quality-by-design (QbD) approach as suggested in the ICH guidelines [Q8 (R2)] of “Pharmaceutical Development” by the pharmaceutical industries is into focus, to pass a stringent regulation to incorporate the quality into the product [2]. QbD is being encouraged by the health authorities (HA) for quality, safe, and effective product development. QbD is said to be a mixture of product and analytical method development. Also, all the products need to undergo an analytical procedure because the analytical method provides an exact variable to measure the amount of therapeutic agent received by the end-user. Thus principles of QbD are applied to analytical methods to monitor the material method performance and the process performance involved in the manufacturing of a dosage form, to control the product quality [3]. As per ICH Q8 (R2) guideline, analytical QbD (AQbD) is a systemic approach of analysis/analytical method development with a predefined objective, that is, sustaining the accuracy and precision of the method. This, in turn, maintains the safety and efficacy of a product that is analyzed by a method. The focus is on both the process and product understanding with the knowledge

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of quality risk management. The US Food and Drug Administration, HA of the United States, has recently shown regulatory flexibility toward AQbD for a few new drug applications, which is, therefore, permitting advances in the control strategy for analytical methods. AQbD is studied because the normal documents that are submitted during filing do not have guidance on how to prove the fitness of the method for its intended use. AQbD performs this work by few steps that will be discussed below and it has added advantages like, it primarily increases the acceptance criteria for validation of the analytical procedure [4].

2.2 Why is analytical quality-by-design required?

The current research works and regulatory approval are required, because all the regulatory guidance is moving away from the conventional once-and-done validation, and these require a continuous assessment of process verification approach. This is called the “life cycle approach” [5]. This QbD approach in an analytical procedure allows the application of the previously gained knowledge, experience along with tools and by the usage of the statistical method to identify if the validation of the analytical procedure is for the intended use or not. After the application of AQbD, the obtained result is within the design space that is designed. Hence a regular assessment of the process validation is within the acceptance criteria when worked inside the design space [6]. To explain the limits that can be set to obtain a perfect design space to develop a perfect validation procedure, two main terms are needed to be described when any of the innovator company is going through the process of QbD. The first term is the out-of-trend (OOT), which refers to results that are well inside the specification limits, but the mean values of two batches of the same product differ/are outside the trend. Further, out-of-specification (OOS) refers to results that fall outside the specification limit [6]. AQbD reduces the OOT and OOS results that occur due to an error in the analytical method. Thus this makes the process robust and rugged. In OOT and OOS, the term LSL refers to the lower specification limit and the term USL refers to the upper specification limit [7].

2.3 Steps followed in analytical quality-by-design

“Quality should be tested into the product and process, but developed in,” should be followed so there must be a regulatory strategy to find out the critical process attributes (CPAs) affecting the quality, safety, and efficacy of a product. The earlier analytical methods that are stated by the HA do not provide information on how these are going to result in the method developed outside the laboratory. Also, there is a lack of knowledge of the critical parameters in an analytical method that need to be controlled. Industries must understand the variation source and must be capable to detect the existence

and the amount of variation in the product [2]. Then there should be an attempt to understand the process and product attributes and finally control them with strategies. This would help show improvement in product quality, safety, and efficacy throughout the life cycle of the methods. To bring this into the limelight, the AQbD approach is much needed. Accuracy and precision of an analytical method are the core principles that produce a robust and rugged process. The steps to be followed while implementing an AQbD approach for method development and validation are the following (Fig. 2.1) [6].

Analytical method performance characters are identified that need to be achieved to ensure the desired performance of a method, considering systemic variability, inherent variability, and system suitability. Then, the analytical target profile (ATP) for a method is defined. ATPs are then translated into critical process attributes (CPAs) of the given method. The relationship between material attributes and method parameters is established for a method. Later, the critical method material attributes (CMMAs) and critical method parameters (CMPs) are analyzed, which are said to determine the critical quality attributes (CQAs). A control strategy for CMMAs and CMPs is designed and implemented. The method performances are routinely monitored during the life cycle of the product.

ATP is said to be an equivalent of quality target product profile in QbD. It explains the goal to be achieved by a method using a definitive target.

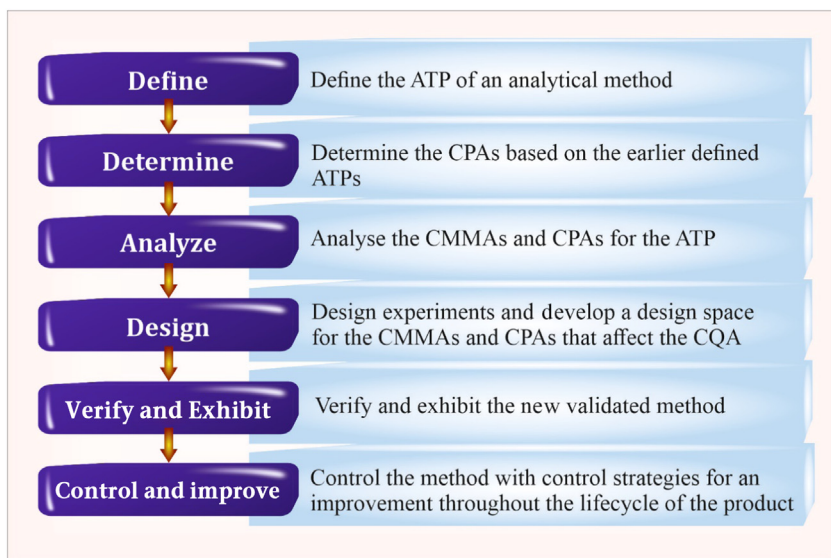


FIGURE 2.1 AQbD approach for method development and validation. *ATP*, Analytical target profile; *AQbD*, analytical QbD; *CMMAs*, critical method material attributes; *CPAs*, critical process attributes; *CQA*, critical quality attribute.

ATP is defined as “a statement that defines the performance measures to adequately define the result of a method.” This provides regulatory flexibility to the industries within the changes in the control system. It ensures that the method is “fit for use” for a particular product during its entire life cycle. After the ATP is defined, the process parameters that ultimately affect the final process of the method called critical process parameters (CPPs) are known [8]. These process parameters are, in turn, affected due to some of the materials (CMMAs) that are used and the method (CMPs) that is followed to develop the method. Hence, CPPs are, in turn, affected by the CMMAs and CMPs. After defining all the critical attributes that affect the method, the analytical control strategy must recognize the possible CQAs and their effect on the method of analysis. This process is called method designing and development. Here the target is set and the requirements are defined, and the method is designed and developed to measure the given CPA. The analytical method should satisfy the regulatory requirements as well as the need for the experiment, by satisfying the ATP. For example, if a method does not possess specificity as an ATP, but is needed as per the regulatory guidelines, then care is taken to satisfy both the process conditions and the regulatory conditions. Accuracy and precision are considered as method parameters commonly so that the method is robust and rugged [9].

2.4 Method for risk assessment

Ishikawa diagram is one of the methods to determine the risk via cause and effect (C and E) analysis. Here the risk factors are given a risk priority number or RPN, which on exceeding a certain score for a particular factor or attribute is considered critical and is further investigated. Accuracy and precision are considered as method parameters commonly so that the method is robust and rugged. To identify the robustness of a method, design of experiments (DoE) needs to be conducted to identify the C and E relationship between the performance of the method and the parameters of the method [1].

2.4.1 Design of experiments

DoE is a computational tool that collects all the variables from the method tested under AQbD. These variables are used to find the effect of different environmental and industrial conditions on the method along with a comparison to the other variables of the method [10]. Hence DoE becomes part of the AQbD method where the interaction between all the input variables of the method is reported along with the eventual effect caused by these variables [11]. In this process of AQbD, the variables are selected based on how critical they are to the method. It depends on CMMAs and CMPs and this has already been discussed before [12]. So the DoE method is used in two

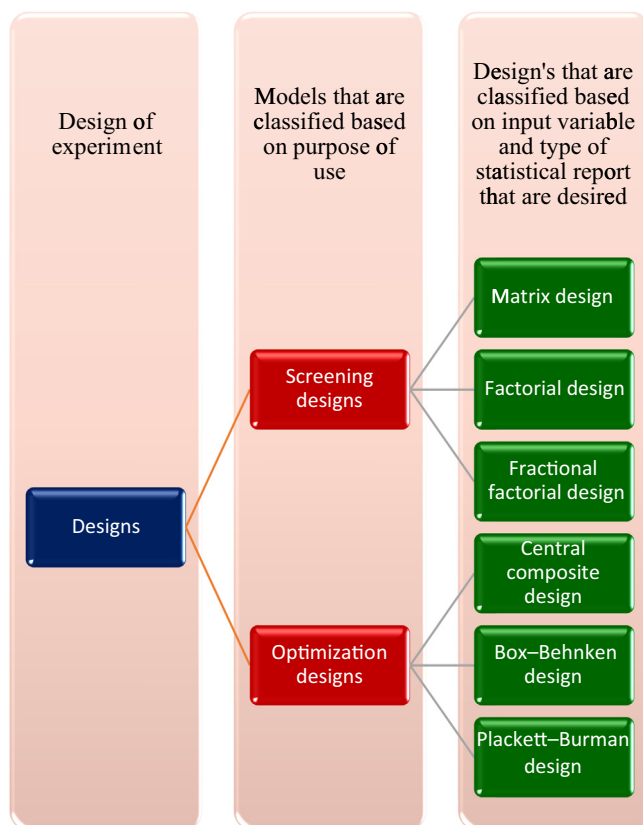


FIGURE 2.2 Different types of classification of design of experiments.

different cases, one involving screening of the variables that are used in a method and other is for the optimization of the variables that are already fixed for a particular method. Besides, the different types of design classification are shown in Fig. 2.2.

Comparative designs are applicable if there is more than one variable under investigation, but the factor of interest is only one and the question is whether this factor is significant in this method or not. Also, screening designs, called as main effects designs, possess the primary aim of finding out the critical factors from the huge number of factors that have been inserted in the DoE software. The critical factor is based on the statistical result of the method obtained when the other conditions are stable. Besides, characterization designs give a better understanding of the method that is being studied by characterizing each main effect and the interaction effects that are associated with the critical main effects. Furthermore, optimization design/response surface methodology designs are used when the critical factors are finally selected and then help to find the optimal experimental point

by predicting the response value for all possible critical factor combinations. Here, since the designs are classified based upon the selection of the critical parameters, few designs are needed that can insert the parameters and their conditions. The designs are based on the way they provide the results and are as follows (Table 2.1): (1) matrix design, (2) factorial designs, (3) fractional factorial designs, (4) central composite design, (5) Box–Behnken design, and (6) Plackett–Burman design.

Optimization is the process where the factors that are said to be critical are screened to have an optimal point in the method to give an accurate result. The use of this optimization process reduces the number of trials that happen to be more on screening the design [12]. Sometimes, even optimization designs can also be used for screening. For example, Plackett–Burman design is used for screening of factors when the factors are in multiples of 4 and factorial and fractional factorial design of 2 becomes lengthy. In the example of metformin, alogliptin, and repaglinide in binary tablets [14], the primary factors were screened for their critical parameters with the available minimum number of resources with minimum runs possible. Then these runs were optimized using face-centered composite design to get the best resolution possible. This optimization method of DoE is used rather than the traditional method because it is less time-consuming and incurred lesser cost. Also, all the experimental conditions could be varied by just changing the data in the software rather than again manually experimenting. By using this method of optimization, the plots can be studied that give information about other parameters as well. The other advantage of optimization using DoE is that multiple functions can be analyzed at the same time using any of the higher level designs. In a chromatography technique, the signal area and the response peak are optimized using a combination of response surface methodology and Plackett–Burman design [14]. The results were found in a few hours rather than days when gone through a traditional approach and were more accurate and robust than theoretical predictions. After solid dosages, when it comes to liquid dosages, it becomes tedious to obtain the best method in practice. A multiple emulsion is difficult to formulate and even more difficult to slightly vary the number of critical factors used in the formulation [15]. Here the microemulsion of more than one formulation has to be checked for its compatibility along with its robustness and accuracy to be determined in the analytical method. Hence to make the method as well as the formulation accurate and robust, the DoE software is used for optimization and reduction of the trial error trivial method. On hyphenating a small instrument with a large analytical instrument it becomes difficult to obtain the accuracy and robustness of the method. It takes more than theoretical thinking to get to what the result is, and also the trial and error needs to be run for maximum times, and there might be a shortage of resources. Hence for this DoE is used so that with minimum trial or error batches the statistical results of the critical parameters are obtained along with its optimal range

TABLE 2.1 Designs used for fitting critical parameters in the design of experiment software.

S. no.	Designs	Level	Factors	Applications
1.	Factorial design	2 or 3 or 4 or ...	$K = 1, 2, 3, 4, \dots$	It helps to identify any one factor concerning other factors when there is a change in the other factors from one level to another.
	2^k Factorial design	2 which is high and low	$K = 1, 2, 3, 4, \dots$	The factors can be screened at their higher and lower level. This is used when the levels are nearer to each other and there are fewer factors to compare. It is mostly used in pharmaceutical QbD, for tablets and capsules [13].
	3^k Factorial design	3 which is high, medium, and low	$K = 1, 2, 3, 4, \dots$	When the factors are screened at high, medium, and low levels, there is a presence of curvature due to which the prediction for each factor is of better resolution. It can be used for tablets, capsules manufacturing procedure, but most widely used in AQbD [13].

(Continued)

TABLE 2.1 (Continued)				
S. no.	Designs	Level	Factors	Applications
2.	Fractional factorial design	2 or 3 or 4 or ...	$K = 1, 2, 3, 4, \dots$	As the name suggests, it is the fraction of the factorial design. It reduces the number of runs and hence helps to find reasonable critical factors.
	2^{k-p} Fractional factorial design	2 which is high and low	2^{k-p} , for example $2^5 = 32$ runs (full factorial design) $2^{5-1} = 2^4 = 16$ runs (1/2 fraction of full factorial design) $2^{5-2} = 2^3 = 8$ runs (1/4 fraction of full factorial design) $2^{5-3} = 2^2 = 4$ runs (1/8 fraction of full factorial design)	The design runs similar to that of the full factorial design but in fractions so that only required factors are screened and also that can be run whenever there are enough time and resources to continue with the design. Theoretically, it can be used when there are more than three critical parameters.
	3^{k-p} Fractional factorial design	3 which is high, medium, and low	3^{k-p} , for example $3^5 = 243$ runs (full factorial design) $3^{5-1} = 3^4 = 81$ runs (1/3 fraction of full factorial design)	This design is not used much because most of the critical factors are found using $2K - 1$ fractional factorial design. The use of this
(Continued)				

TABLE 2.1 (Continued)

S. no.	Designs	Level	Factors	Applications
			$3^{5-2} = 3^3 = 27$ runs (1/9 fraction of full factorial design) $3^{5-3} = 3^2 = 9$ runs (1/27 fraction of full factorial design)	design is seen when there is more than one critical factor in a $2K - 1$ fractional factorial design and this needs to be brought to an optimal range to use.
3.	Central composite design	It depends on the levels used in the factorial designs as this is the extension of that design	It depends on the factors used in the factorial designs as this is the extension of that design	CCD has embedded factorial and fractional factorial design along with some center points and some star points (that gives curvature). It helps in detailed estimation and reduces the number of trials.
	Circumcised CCD			Circumcised CCD: In this design, it is possible to operate less than the lower limit and more than the upper limit of the factorial design. So, it helps to estimate more than the actual area of interest.
(Continued)				

TABLE 2.1 (Continued)				
S. no.	Designs	Level	Factors	Applications
	Face-centered CCD			Face-centered CCD: This design is used when the work cannot be performed higher than the upper limit and lower than the lower limit. To confirm that the limits are correct, this design is utilized.
	Inscribed CCD			Inscribed CCD: This design is used to know what the domains in the design are but it cannot have a value greater than the given value.
4.	Box–Behnken design	It operates at five levels (–2, –1, 0, 1, 2)	Factors are the same that are used in the fractional design	It has a primary advantage of decreasing the number of runs for the same number of factors and sufficient information is obtained to fit up to 10 coefficients. The main disadvantage is that if the factors given are extremes of the instrument, then an operation beyond that is not possible.
(Continued)				

TABLE 2.1 (Continued)

S. no.	Designs	Level	Factors	Applications
5.	Plackett–Burman design	2 which is high and low	Works in the multiples of 4 only	These are nongeometric designs because these are not expressed as cubes and hence the statistical results are difficult to interpret.

CCD, Central composite design.

using various designs. In high-performance liquid chromatography–ultraviolet (UV) spectroscopy/fluorescence to screen mostly Plackett–Burman design is preferred because it reduces the number of trials required when compared to factorial or fractional factorial. For optimization, response surface methodology along with the central composite design is used. The software creates the design and analyses the statistical reports [16]. So, optimization using DoE is much more advantageous than normal manual optimization since it reduces time, cost, and also the resources. Along with it, the results can be varied computationally, and the results may be extended until proper statistical data are obtained, which is not generally possible. Interaction effects that are not considered in the trivial method are considered here and only then a report is made. This is the advantage of having an optimization method in the DoE, but a correct understanding of the software and data is required to get the correct inference.

The other method by which risk assessment can be done is called as the process mapping. In this method, a table is created that has the columns where, in the first column, the input variables that are classified into materials (drug/reagents/chemicals/solvents/output of the previous step), plasticware (filter membranes, Eppendorf tubes) or glassware (measuring cylinders, volumetric flasks, etc.), or instruments/equipment/pipettes (weighing balances, pH meters, homogenizers, centrifuges, refrigerators, etc.) are defined. On the left-hand side of each of the steps, the outputs to the step are written. These outputs are in terms of what is obtained at the end of the step. On the

right-hand side of each output, the critical performance attribute of the analytical method that can be affected by the output is written [11].

2.5 Impact assessment of the critical method parameters on the performance

As per ICH Q9 guidelines, the impact assessment is defined as “a systematic process for the assessment, control, communication, and review of risks to the quality across the product lifecycle.” The critical method attributes are taken into consideration to identify the impact of them on the performance. Variability characters are considered here, such as the method used by the analyst, sample characteristics, preparation methods, and environmental factors. The risk factor is the product of severity concerning the CQA [17], the occurrence of failure of the product/process, and the failure detection capability of a method. Knowledge-based, C and E and failure mode and effect analysis (FMEA) are the methods through which impact analysis is done. FMEA with control noise experiments is also used nowadays.

$$\text{Risk factor} = \text{severity} \times \text{occurrence} \times \text{detectability}$$

2.6 Defining method control strategies and method validation

The control strategy is derived from the understanding of the performance of a process and the quality of the product. They are necessary to ensure the long-lasting performance of an analytical method and to deliver the predefined characteristics of a method for a given attribute. Control includes parameters related to the materials of the drug substance and drug product, equipment, and the operating conditions and specifications—in-process and finished product [18]. The basis of the control strategy is to define the parameters. A control strategy is defined as a point where techniques like end-product testing can achieve control over the method, inline measurement with the help of process analytical technology (PAT), in-process controlling, or by process parameter monitoring. Initially, the control strategy is derived based on the limited data available and the assumptions, following queries like which CQAs need a specific testing method (any of the abovementioned) and how the further process may be designed and developed based on the effect of the critical performance attribute. So, to design an AQB, some of the steps that are important to give a final perspective of the factors needed to be kept in the control state are utilized. Thus the method reaches that quality state where any modification is within the limit [19].

Initially, defining an ATP of an analytical technique gives information about the working of a process following the procedure that is set as a standard testing procedure by the industry. In this step, the factors that are

important for an analytical process to comply with that pharmaceutical process are studied so that the prepared method meets the compliance and gives the intended quality of that product. To check the quality of the product, the analytical method is used and if that analytical method that is used itself is wrong and is not evaluated, then there is no purpose of its use. Hence most of the regulatory authorities in many of the countries are now trying to work within the area that is developed by the QbD method. This ensures that any changes in this method do not affect the testing parameters and the results. Hence a group of subject matter experts (SMEs) is made to sit so that they discuss the factors that may be a reason for the risk of that process. After several rounds of this discussion, the SMEs finally arrive at those parameters that are critical in the functioning of those analytical methods, and these are called ATPs [20].

Determining the critical performance attribute from the ATP is the second step in this process. From the ATPs that are set by the SMEs, the essential factors that have to be compulsorily maintained to evaluate the method are found. This step is done because the factors considered to be critical are just through the historical knowledge and through a rough idea of the experience gained by the SMEs during their course of work in the industries [21]. To evaluate these, two parameters are generally considered. Firstly, the ATP is taken and checked that whether any change in any of the process parameters or the material parameters (either of the samples that is used in testing or the material that is involved in the process) affects the functioning of the analytical method. If it is changing the analytical technique, then it is marked to be important. Then, the next parameter is considered, which tells about the effect of the failure to meet the performance of the specification on the severity of that finished product on the end-user [22]. If the impact is severe, then this parameter is seriously considered. Further verification is done whether that parameter can be brought into the limits of the specified method. So, this second step gives an added advantage of reviewing the first step.

This is followed by determining the CMMAs and the CMPs. So, after selecting the critical factors from Step 2, then these are verified for how critical these are based upon the materials and methods that are used. This is done by using different risk assessment methods. The most widely used risk assessment methods are the Ishikawa diagram and process mapping [23]. In the Ishikawa diagram, also called the fishbone diagram, the main arrow represents the desired final result and the arrows intersecting the main arrow are the level methods and causes to be considered while developing a method. While using this, a review of all the factors left out in the previous steps is conducted, which provides a large scope of obtaining many factors for a single level [24]. The other method for this risk assessment is the process mapping. As the name, the process is mapped in a step-by-step process so that all the practical parameters are added. These include the ones that

provide an exact view of where a change needs to be made or where exactly lies the risk of going wrong. This process mapping is the one that has been used to describe the analytical methods in this chapter.

Ultimately the step of risk assessment is conducted using the FMEA diagram. In this step, all the parameters that have been collectively taken from all the three steps are considered and then a table is developed. This table helps to visualize all that happens when this step goes wrong and then the three parameters are considered to describe the risk of that process. These three parameters are probability (P), severity (S), and detectability (D). The outcome of an FMEA is RPN for each combination of failure mode severity, occurrence probability, and the likelihood of detection, which can be used to rank the risk. In the calculation of RPN, “O” is the occurrence probability or the likelihood of a failure cause to occur. The occurrence probability is ranked from 1 to 5, where highly likely to occur is assigned 5; 50:50 chance of occurring is assigned 3; unlikely to occur is assigned 1. The next parameter “S,” the severity is a measure of severity of the effect of a given failure mode. The severity of the effect of failure mode is ranked from 1 to 5, where the severe effect is assigned 5, the moderate effect is assigned 3, and no effect is assigned 1. The final parameter “D” is the detectability of a failure mode or the ease with which a failure mode can be detected. If the failure mode is easily detectable, then it presents/poses less risk to the product quality. For an easily detectable failure mode, a rank 1 is assigned; a rank 3 is assigned for moderately detectable failure modes; and for failure modes that are hard to detect, rank 5 is assigned [25]. Then, RPNs for all the failure modes are determined and the failure modes are put in a rank order based on their RPNs to know which are the ones posing maximum/highest risk toward the CQAs of drug product/process. After the FMEA, either of the risk control strategies can be adopted to reduce the RPNs of the high-risk failure modes or the information can be taken up for identifying the design space [26].

The scores of every parameter describe the risk in the following manner. First, the score for the probability of occurrence is considered.

5—Occurrence of failure mode or cause is inevitable. Based on the historical data or previous experiences, such a failure mode or cause happens every time.

4—Occurrence of failure mode or reason is high, though not inevitable. Based on the historical data or previous experience, such failure mode or cause was observed several times in a few of the batches of production.

3—Occurrence of failure mode or cause is moderate. Based on the historical data or previous experiences, such a failure mode or cause is observed occasionally.

2—Occurrence of failure mode or cause is less likely. Based on the historical data or previous experiences, such as failure mode or cause was observed very few times.

1—Occurrence of failure mode or cause is rare. Based on the historical data or previous experiences, such as failure mode or cause is observed in one of the occasions.

Now the same scheme for the next parameter severity is considered [11].

5—The failure mode severely affects the safety or efficacy of the patient taking the pharmaceutical product and therefore warrants a warning on the label. It is considered to be noncompliant with government regulation if the warning is not given on the label.

4—Patients can take the pharmaceutical product, but the efficacy and safety of the product are less than what is claimed on the label of the product, leading to customer dissatisfaction.

3—Patients can take the pharmaceutical product, and the efficacy and safety of the product are acceptable, but the customer comfort or convenience is poor.

2—Patient can take the pharmaceutical product, and the effectiveness and safety of the product are adequate, but the customer comfort or convenience is average.

1—Patient can take the pharmaceutical product, and the efficacy and safety of the product are good, but the customer with some awareness can detect the defects of the products based on the physical appearance of the product.

Now the final parameter detectability scores go in the following manner:

5—As of date, there are no PAT or detection methods available with the company to detect the change in the failure mode or cause.

4—There is less than a 25% chance to detect the change in the failure mode or cause based on the available PAT or detection methods in the company.

3—There is less than 50% chance to detect the change in the failure mode or cause based on the available PAT or detection methods in the company.

2—There is less than a 75% chance to detect the change in the failure mode or cause based on the available PAT or detection methods in the company.

1—It is almost certain to detect the change in the failure mode or cause based on the available PAT or detection methods in the company.

So, based on the previous knowledge of SMEs, the scores are given and later, the RPN scores are calculated. A threshold is set as per prior historical experience and the parameters above this threshold are considered to be risky and are monitored throughout the process and are kept under control [3].

2.7 Ultraviolet spectroscopy

UV-visible (UV-Vis) is a type of molecular absorption spectroscopy that is measured based on the electron excitation from the ground state to excited

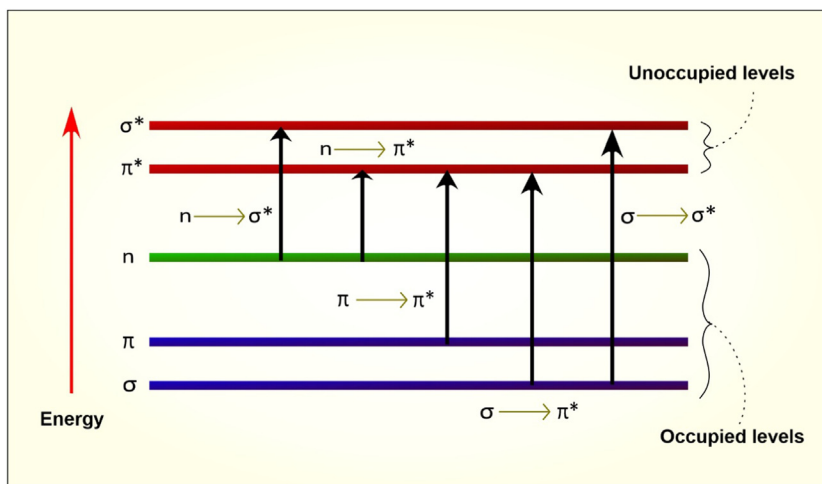


FIGURE 2.3 Transitions of electrons in UV-visible spectroscopy [4]. Where $\pi \rightarrow \pi^*$ is the pi to pi star transition, $n \rightarrow \pi^*$ is the n to pi star transition, $\sigma \rightarrow \sigma^*$ is the sigma to sigma star transition, and $n \rightarrow \sigma^*$ is the n to sigma star transition.

state or higher energy state in the UV-Vis region, that is, 200–800 nm. Their measurement is based on quantifying the absorbance from the sample. This is because the light energy being absorbed by matter rises the particles of the atoms or molecules in the sample. The total potential energy of a molecule is given as the sum of the electronic, rotational, and vibrational energies [27].

$$E_{\text{total}} = E_{\text{electronic}} + E_{\text{vibrational}} + E_{\text{rotational}}$$

where E stands for energy.

UV-Vis spectroscopy is based on electronic transitions. Quantification is possible as each molecule possesses a different amount of energy, at different levels ($E_{\text{electronic}} > E_{\text{vibrational}} > E_{\text{rotational}}$). When a particulate matter absorbs the light, there is a rise in the energy level of the atom or molecule, which results in the excitation of molecules from the ground state (higher occupied molecular orbital) to the excited state (lower unoccupied molecular orbital) [27]. Molecules with the π and the nonbonding (n) electrons absorb energy from the UV-Vis light source at the specified wavelength for UV to excite the electrons to their higher antibonding orbital (denoted with $*$). Longer the wavelength of light absorbed, more is the excitation of electrons. Later the electrons emit the absorbed energy and come back to the ground state (Fig. 2.3) [28].

UV-Vis spectroscopy follows Beer–Lambert’s law, which is the principle behind this phenomenon. The law states that “for a solution of concentration

‘ c mol/L’ having a path length of ‘ b centimeter’ and a molar absorptivity ‘ ϵ ,’ the absorbance A is the product of all the above” [27].

$$A = \epsilon bc$$

where A is the absorbance, ϵ is the molar absorptivity, b is the path length, and c is the concentration.

Molar absorptivity or molar extinction coefficient is the property of a molecule undergoing electronic transition.

Nowadays, spectrophotometers are double-beamed. Light sources such as tungsten filament or deuterium lamp are widely used, as these cover the entire UV-Vis region. The intensity of the tungsten lamp emit the radiations at 375 nm, while that of the deuterium lamp falls below 375 nm. Xenon lamps are also rarely preferred. Monochromators are composed of splits and prisms. Radiations discharged from the primary source will be dispersed by the rotating prism in the instrument. Different wavelengths from the light source are separated by the prism and selected with the help of split so that the light that entering the sample is monochromatic (light with one wavelength). Later the monochromatic light is divided into two beams by the second prism. One of the beams passes through the sample solution, while the other beam passes through the reference solution. Silica or quartz cells are usually used, as these do not absorb UV rays. The detector is a part of a spectrophotometer that converts light signals into electrical signals. Photomultiplier tubes (amplification of electric signals within the detector) or photodiodes (when light falls on a semiconductor present in this detector, electrons are emitted and thus charge is depleted) are used as detectors in UV spectrophotometer [28,29]. Instrumentation of a UV-Vis spectrophotometer is shown in Fig. 2.4.

Furthermore, the factors that contribute to the absorption of samples in a UV-Vis spectrophotometer are as follows [30]. (1) The solvent: The solvent shifts the peaks to longer or shorter wavelengths because they may interact with the sample. This is based on the chromophore present in the analyte and the polarity of the solvent influences the wavelength shift. (2) The concentration of the sample: It is directly proportional to the absorbed intensity. At a higher level, molecules interact/collide with each other causing a change in the spectra. At lower concentrations, the instrument must be able to quantify the signals. Thus the concentration must be ideal. (3) The pH of the sample: If the pH of the sample is not maintained, there may occur a shift in the equilibrium between the chemical forms of a particular analyte. Thus buffers can be used to maintain a constant value of pH. (4) Temperature: Temperature plays an important role in the expansion and contraction of solvents leading to changes in the concentration, changing the reaction rate, and varying the refractive index within the sample as a result of convection currents that affect the absorbance of solvents, particularly organic solvents. Thus the constant temperature needs to be maintained. Changes in these factors cause inaccuracy and

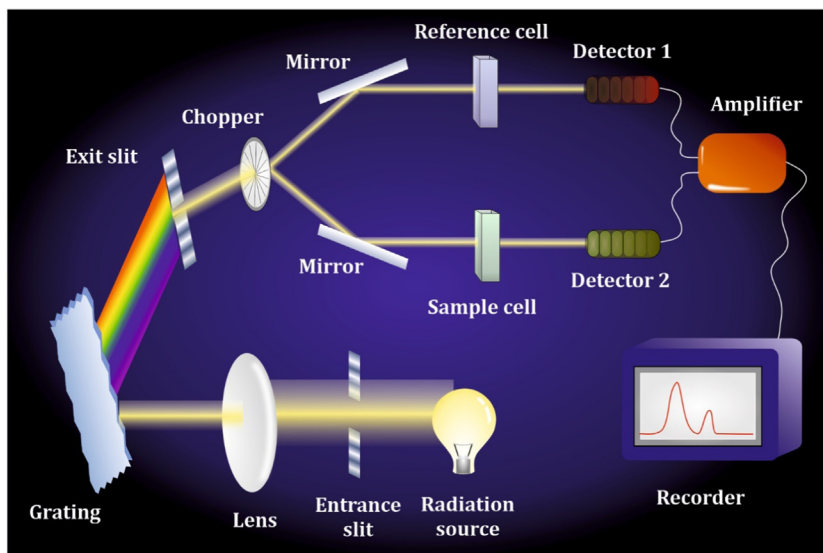


FIGURE 2.4 Instrumentation of UV-visible spectrophotometer.

precision issues by changing the wavelength ($\lambda_{\max}$) and the intensity ($\epsilon_{\max}$) of the peaks obtained as a result of absorption.

Besides, the instrumental deviations that affect the absorption are the presence of polychromatic light, and monochromators are used to avoid such incidences. The presence of stray radiations includes stray light that are of the wavelength that is outside the selected bandwidth. These are instrumental deviations as a result of reflection or scattering of light from the instrumental parts such as gratings, lenses, mirrors, and filters. In double-beam instruments, when both reference and sample solutions are simultaneously analyzed, care should be taken that the cuvettes used are of the equal path and having the same optical characters. Thus there may be a deviation in calculating the absorptivity of the sample due to mismatched cuvettes if adequate precautions are not taken [27].

Now, based on the above-discussed parameters, an AQbD may be developed for the process, which includes four main steps as follows. Defining the ATP is the first step. Table 2.2 provides the analytical validation parameters because these are the factors that are the critical for the process to be validated and changed according to AQbD. Accuracy and precision are the main factors that need to be validated because if the method complies with these two parameters, then the whole method will mostly fall in its place [31,32]. Also, an ATP may be done for the instrument. But mostly, that is not done because all the parameters of method validation ultimately result in the method by that instrument, which has all the parameters in place [33]. If the method is being newly developed, then the target of the method changes

TABLE 2.2 Analytical target profile.

ATP element		Target	Justification
Accuracy		The intended results should be close to the real value.	No or minimum systematic variability.
Specificity		The method should assess the analyte unequivocally in the presence of other components that may be expected to be present.	Specificity has to be established for achieving minimum systematic variability.
Linearity		The method should have the ability to elicit test results that are directly, or by well-defined mathematical transformation, proportional to the concentration of the analyte in the samples within a given range.	Linearity has to be established for achieving minimum systematic variability.
Precision	Repeatability (intraday precision)	The method should have a higher degree of agreement among the test results using the same operating conditions over a designated short period (typically ≤ 1 day).	To establish that the method has minimum inherent variability in the analysis of independent replicates of the analyte in the same operating conditions using the same equipment by the same analyst on the same day in the same laboratory.
	Intermediate precision (interassay precision)	The method should have a higher degree of agreement among the test results using the same operating conditions, typically within the same laboratory, over a designated period (typically ≥ 1 day). The impact of random variables on the intermediate precision of the method should be assessed. Typical variables that should be investigated include days, analysts, equipment, etc.	To establish that the method has minimum inherent variability in the analysis of independent replicates of the analyte in the same operating conditions on different equipment or when prepared by various analysts or analyzed on different days but within the same laboratory.

(Continued)

TABLE 2.2 (Continued)

ATP element		Target	Justification
	Repeatability (between laboratory precision)	The method should have a higher degree of agreement among the test results using the same operating conditions between different laboratories, typically within the same laboratory, over a designated period (typically ≥ 1 day). The impact of random variables on the intermediate precision of the method should be assessed. Typical variables that should be investigated include days, analysts, equipment, etc.	To establish that the method has minimum inherent variability in the analysis of independent replicates of the analyte when analyzed in different laboratories, where the analytical procedure needs to be transferred to another site as part of method transfer, in-sourcing, or outsourcing. Reproducibility is verified using the interlaboratory trial(s).
Quantification limit		QL: The lowest possible concentration of an analyte in the sample that can be determined with accuracy and precision.	To establish minimum inherent variability in the analysis of the lowest possible concentration of the analyte.
Range		The method should possess linearity in the concentration range between LLOQ and ULOQ with specified accuracy and precision.	The range is required to ensure that the error between actual concentration and mathematical equation predicted level (mainly unknown samples) within the range is very minimum.
Robustness		The method should remain unaffected by small but deliberate variations in procedural parameters listed in the procedure document and indicates its suitability during normal usage.	Robustness is needed to get minimum variability in the analysis of analyte due to a small deviation in the sample preparation procedures.

From the above-defined ATP, the CPAs are developed, which is the second step. *ATP*, Analytical target profile; *CPA*, critical process attributes; *LLOQ*, lower limit of quantitation; *QL*, quantification limit; *ULOQ*, upper limit of quantification.

based on the trial and error of the research and development team of industry. Here the justification for a particular parameter would be the reference value. Here in [Table 2.2](#), the target for a UV method and its justification is provided, which may be changed based upon the product/method/industry/SMEs.

Here, any change in the ATP element due to a change in the method parameters are checked for and the possibility of the failure severely affecting the process is considered. If both the scenarios happen, then the particular parameter is considered to be critical. In the second step, all the ATPs considered in the first step by the SMEs are taken. On a closer look, analytical method validation parameters are utilized as QbD parameters of the UV-Vis instrument in the first step. This is because the validation of the method is the best way to prove the highest compatibility of the method used on a particular instrument with the procedure for a specific substance. It also helps achieve compliance with the regulations to obtain the quality that is required into the product. In this step, [Table 2.3](#) illustrates the CPA that have already been explained in the previous section of this chapter. The first parameter that has been set as an ATP is accuracy because it is easily affected by a change in any formulation parameter due to an incompatibility issue that arises in the new chemical combination. The variation in the number of chemicals used may also affect the accuracy parameter. Hence any need to change the analytical technique due to any variation in the parameters ultimately affects the overall analytical method. So the validation parameter is critical on taking all these factors into account. Further specificity is directly related to accuracy and it helps to know whether the method is specific to the compound of interest. This parameter is not directly affected, but it is, in turn, related to the accuracy of the method. Thus this parameter is critical since any change affects the performance of the method. The third factor mentioned in the above table is linearity, which mainly depends upon the drug and other excipients that are used to keep the value of the compound that is tested within a specific range. This range is directly proportional to the concentration of the analyte in the sample. If there is any change in the formulation procedure or process of analysis changes, then the determination becomes wrong and hence this parameter is critical. The next parameter is precision and this is considered to be important because precision tells about the influence of the different parameters of the instrument, analyst, and the place on the results. Hence this parameter is considered to be critical most of the time. The parameters other than these are detection limit, quantification limit, and robustness. Detection limit and the quantification limit do not change much when compared to all other parameters because it mainly has a role to play with the compound of interest and not with different parameters. Hence these two parameters are said to be noncritical while robustness is considered to be critical because it is the major factor that hugely affects the performance attribute [8].

TABLE 2.3 Critical process attributes.

ATP element/ performance attribute	Does change in method material and/or method parameters affect the performance attributes of analytical method?	Does failure to meet the PA severely affect the systematic and inherent variability of the analytical method?	Is the PA critical or not critical?
Accuracy	Yes	Yes	Critical
Specificity	Yes	Yes*	Critical*, but in turn related to accuracy
Linearity	Yes	Yes*	Critical*, but in turn related to accuracy
Precision— repeatability (intraday precision)	Yes	Yes	Critical
Precision— intermediate precision (interassay precision)	Yes	Yes	Critical
Precision— repeatability (between laboratory precision)	Yes	Yes	Critical
DL and QL	Yes	No	Not critical
Robustness	Yes	Yes	Critical

*means critical, but in turn related to accuracy.
DL, detection limit; QL, quantification limit.

The third step involves determining CMMAs and CMPs. Based on the above results, the CMMAs and CMPs of the analytical method that relates to that particular part of the nine fundamental parameters affecting the analytical procedure are taken. This is done using process mapping ([Table 2.4](#)).

TABLE 2.4 Process mapping through different performance attributes.

Input	Step	Output of the step	Performance attribute
Weighing of all ingredients and sample preparation 1. Materials: a. API b. Buffers c. Solvents d. Water 2. Plasticware/glassware: a. Eppendorf tubes b. Measuring cylinder c. Volumetric flasks 3. Equipment/pipettes: a. Weighing balance b. Micropipettes c. pH meter d. Vortex mixer	Sample preparation Calibration curve standard solutions QC standards solutions Test samples	CC standards with a defined concentration range samples with a known concentration, test sample whose concentration is to be determined	Accuracy (linearity, specificity, resolution) Precision (S/N ratio—sensitivity, peak shape)
Analyzing the sample in the UV-visible spectroscopy 1. Materials: a. Sample in cuvette 2. Plasticware/glassware: a. Cuvette 3. Equipment/pipettes: a. Detector b. Type of detector c. Flow cell volume	Detection of analyte(s)	Spectral separation, good resolution between analyte and impurities, acceptable signal-to-noise ratio (S/N)	Accuracy (linearity, specificity, resolution) Precision (S/N—sensitivity, peak shape)

API, active pharmaceutical ingredient; CC, calibration curve; QCs, quality controls.

In the third step, a process mapping has been conducted for the risk assessment step instead of an Ishikawa diagram. Ishikawa can also be used, but process mapping has been used for easy representation and understanding. Here in the input of the process, the materials are firstly mentioned, where all the materials and the instruments used in that step are mentioned and the output of that step is given. The output of the first step becomes the input parameter for the second step. Finally, after the end of all the processes, the last column mentioned as a performance attribute is provided. In the first row, during weighing and preparing the sample, the performance attributes of the accuracy and precision are affected. In the second row, in which the sample is analyzed, the same two validation parameters are found to be affected. In [Table 2.5](#), the instrument parameters that are found to be affecting these validation parameters are studied. The fourth step is conducting the risk assessment using the FMEA diagram [\[11\]](#).

From the above FMEA, a threshold limit has been optimized and the factors that are above this threshold limit are considered as the critical factor that affects the process and the validation parameters. Here as explained in the earlier sections, scores are allotted to each parameter and the criticality of these parameters is checked. First, the analyte is considered, where it has been scored as 1 for probability. This is because the occurrence of this going wrong is less because the purity of the drug is known before analysis. But the severity is 5 because if the drug is potent, then this may cause an adverse effect to the end-user. Detectability is 1 because if the analyte is not the drug, then it can be easily deduced by the result of the first analysis itself. Next, the reagents and solvents are considered, which may be given the same scores because these are also made up of chemicals and their analysis results can also tell whether these are the correct reagents or not. The parameter of the cuvette was given the probability of 2 because even though the cuvette is composed of the desired material, complete information and insight regarding the grade and influence on absorbance may be missing. If these factors cannot be determined, then these may be severe in affecting the performance, and hence the value of 3 is given. It can be detected by using some in-line analytical techniques but it may not be so easy and hence the value is given as 4. The light serving as the important source is divided into polychromatic and stray light. A less RPN score is given to polychromatic light because it can be easily adjusted and its occurrence is less because the instrument at the time of calibration is checked. But the stray light cannot be detected and hence more RPN score is given to it. Lastly, the detector is the one responsible for the result and any fault in the detector gives the wrong result. This would ultimately fail the method that has been developed. Hence the QbD method is useful in rectifying these types of defects. Here this is given the highest score when compared to all others because the probability of the detector going wrong is high due to the environmental conditions and thus it also becomes severe. The in-line detectability has not been yet started

TABLE 2.5 Process and parameters of risk assessments.

Method component	Material attribute	Failure mode	(Failure mode) effect on IP/FP CQAs	P	S	D	RPN
Drug/analyte	The purity of the analyte(s)	Low	Concentration of sample—accuracy	1	5	1	5
Reagents and solvents	Purity	Low	Desire pH is not achieved—accuracy or precision	1	5	1	5
Cuvette	Make of cuvette	Low	Absorbance due to cuvette	2	3	4	24
Light source	Polychromatic light	Medium	Altered absorbance—accuracy	1	5	1	5
	Stray light	High	Altered absorbance—accuracy	2	5	3	30
Detector	Wavelength	Not optimized	Resolution/peak shape and response	4	4	2	32

IP/FP, in process/finished product.

and hence it has less detectability. So, for assigning the scores, SMEs are required and a brainstorming session is needed to create and perform the steps [9].

2.8 Spectrofluorimetry

Fluorescence is a parameter that provides an easy and successful way in the quantitative determination of certain elements that produce fluorescence and does not require separate derivatization to measure it [34]. This fluorescence property can be measured using a spectrofluorimeter and it possesses higher sensitivity, selectivity, and specificity over the other currently available techniques. Fluorescence of a molecule can be defined as an effect where a particle releases electromagnetic radiation while absorbing another form of energy, but stops to emit the radiation immediately ($<10^{-5}$ seconds) upon the termination of the input energy (or) the emission of a given wavelength by a particle that is activated by light of a different wavelength (or) the emission of radiation with a longer wavelength by a substance as a result of absorption of energy from shorter-wavelength radiation, continuing as long as the stimulus is present. Fluorescence of an element is mainly due to the fluorophore molecule and this molecule shows some excitation and de-excitation phenomenon. The excitation occurs when the electrons absorb the energy required to jump from the ground state to the excited state [35]. An excited molecule will get back to its ground state in two of these steps—fluorescence and phosphorescence that involve the emission of a photon of radiation. Fluorescence occurs when the molecule jumps from its higher excited state to the ground state and phosphorescence is observed when this molecule shows the transition from triplet state to singlet state [36]. The spectrofluorimeter mainly consists of: (1) light source: The light source that is mainly used is xenon lamps or mercury vapor lamps. Nowadays a combination of both is also used to get an exact excitation of particular molecules. (2) Monochromator: The monochromator is of two types, grating and filters. Filters were previously used but it does not intensify the results, whereas the grating monochromators are used to intensify and amplify the small radiations that are produced. (3) Detector: The detectors used for spectrofluorimetry are the same as that of UV-Vis but the most widely used detector is the photomultiplier tube. Instrumentation of spectrofluorimetry is shown in Fig. 2.5.

AQbD can be applied for spectrofluorimetry since the instrumentation is the same as that of the UV-Vis spectroscopy and most of the steps would have a common interpretation. The steps of defining the ATP and CPA of this analytical procedure are similar to the UV-Vis method. Now the step-by-step process of AQbD is reviewed.

Defining the ATP is the first step. The ATP for this analytical method is the same as that as described in Table 2.2 because most of the analytical parameters ultimately must be specific to all the parameters that have been

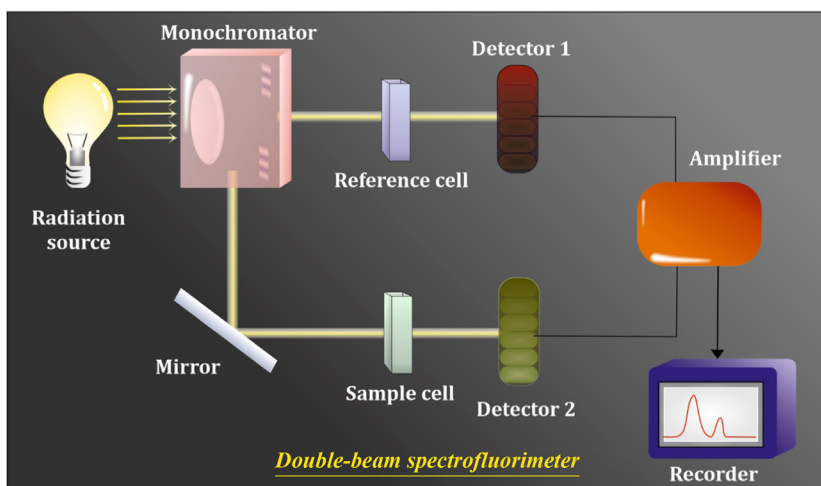


FIGURE 2.5 Instrumentation of spectrofluorimetry (double-beam spectrofluorimeter).

mentioned. The second step deals with critical process attributes. This is the same as that in Table 2.2 because the parameters and the flow as done for the UV-Vis are the same. If needed, a new component table may be designed for this analytical method separately but the inference would be the same. Determining CMMAs and CMPs is the third step. Initially, the 5M's (man, material, machine, manufacturing, and method) involved in an AQbD according to Ishikawa way of risk identification are thoroughly studied. A process mapping may be done because this is easy for a beginner as it is a step-by-step process. In process mapping, the input of the second step is the output of the first step. The interrelated nature of the process and the critical parameters are easily represented. In the last column of process mapping, the validation parameters critical to the analytical method are provided. Based on the above results the CMMAs and CMPs of the analytical method relating to the particular part of the nine fundamental parameters affecting the analytical procedure are studied. This is done using the Ishikawa diagram or process mapping [37].

Risk assessment is performed using the FMEA diagram in the fourth step. From the above risk assessment, a threshold is identified to decide the factors that are needed to be controlled to obtain an optimum design space. From the third step, the parameters that are important or critical are noted. In Step 4 the criticality of the parameters is determined, based upon which a threshold is kept. This ensures that the actual critical factors are found out and the factors that are least critical are eliminated. The first four parameters that are given in this table are the same as that of UV-Vis spectroscopy and hence the scores that are given have been already discussed. Monochromators have filters and gratings that are the objects that are

responsible for splitting a polychromatic light into a monochromatic light. There might be defects in monochromators during manufacturing or calibrations, which might affect the analysis. Since the defects may be known during calibration the occurrence of this is low and hence the value is given as 1. If this is not rectified, this might be a bit severe, which might affect the results of analysis and give the results that are above the range kept as limits and are the reference range. Hence the score for severity is given as 3. If these changes had occurred, it cannot be detected so easily because there are no tools that can detect its severity, and hence the value is given as 2. The detector has the same explanation as that in UV-Vis. The column in between the table mentions faulty mode in both the spectrophotometric techniques, which signifies the influence of this failure mode on the performance of an analytical method. The threshold limit is set by SMEs after a brainstorming session between several of them. Hence, a single individual cannot carry out an AQbD for a method and determine the critical parameters [37].

2.9 Conclusion

QbD is primarily related to the application of ICH Q8 and Q9 via significant utilization of DoE to provide a multidimensional design space. DoE helps in validation along with characterizing any changes via the design space obtained by QbD in spectrophotometric techniques. DoE will be an integral part of analytical techniques such as spectrophotometry due to the growing regulatory needs. The spectrophotometric method should ideally be robust to ensure the application for a prolonged period with fewer risks of failure. The process of implementing AQbD leads to robust analytical techniques, that are very critical in drug product development. Complete attention should be paid to develop a spectrophotometric method in analytical laboratories along with identifying and controlling failures. Thus QbD in spectrophotometric techniques would prove to be an essential association between operational laboratories and spectrophotometric development. The validated spectrophotometric method driven by QbD will rise in the coming years. This would need the harmonization of concepts, along with training of human resources in regulatory bodies and industries and necessary guidelines for documenting data generated through method development.

References

- [1] Godambe R, Disouza J, Jamkhandi DCM, Kumbhar P. Development of spectrophotometric and fluorometric methods for estimation of darunavir using QbD approach. *Int J Curr Pharm Res* 2018;10:13.
- [2] Singh B, Beg S. Quality by design in product development life cycle. *Chron Pharmabiz* 2013;22:72–9.
- [3] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.

- [4] Yadav LDS. Introduction to spectroscopy (spectrometry). Organic spectroscopy. Dordrecht: Springer; 2005.
- [5] Weitzel MLJ. The estimation and use of measurement uncertainty for a drug substance test procedure validated according to USP <1225>. *Accredit Qual Assur* 2012;17(2):139–46.
- [6] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [7] de Sousa J, Holt D, Butterworth PA. Analytical method design, development, and life-cycle management. *Pharm Qual Des* 2018;257–79.
- [8] Wenclawiak B, Hadjicostas E. Validation of analytical methods – to be fit for the purpose. In: Wenclawiak B, Koch M, Hadjicostas E, editors. *Quality assurance in analytical chemistry*. Berlin, Heidelberg: Springer; 2010.
- [9] Beg S, Hasnain MS, Rahman M, Swain S. Chapter 1 – Introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. Academic Press; 2019. p. 1–17.
- [10] Panda SS, Beg S, Kumar BVVR, Sahu J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.
- [11] Panda SS, Rath J, Ravi Kumar Bera VV. QbD driven development and validation of UV spectrophotometric method for estimation of paliperidone in extended release tablet dosage form. *Anal Chem Lett* 2018;8(4):510–18.
- [12] Adiki S, Prashanti M, Dey B, Katakam P, Assaleh F, Hwisa N, et al. Design of experiment assisted UV-visible spectrophotometric and RP-HPLC method development for ambrisentan estimation in bulk and formulations. *World J Anal Chem* 2014;2:23–30.
- [13] Mabood F, Gilani SA, Hussain J, Alshidani S, Alghawi S, Albroumi M, et al. New design of experiment combined with UV-Vis spectroscopy for extraction and estimation of polyphenols from Basil seeds, Red seeds, Sesame seeds and Ajwan seeds. *Spectrochim Acta A Mol Biomol Spectrosc* 2017;178:14–18.
- [14] Mahrouse MA, Lamie NT. Experimental design methodology for optimization and robustness determination in ion pair RP-HPLC method development: application for the simultaneous determination of metformin hydrochloride, alogliptin benzoate and repaglinide in tablets. *Microchem J* 2019;147:691–706.
- [15] Abou-Taleb NH, El-Sherbiny DT, El-Enany NM, El-Subbagh HI. Multiobjective optimization of microemulsion-thin layer chromatography with image processing as analytical platform for determination of drugs in plasma using desirability functions. *J Chromatogr A* 2020;1619:460945.
- [16] Silva JM, Azcarate FJ, Knobel G, Sosa JS, Carrizo DB, Boschetti CE. Multiple response optimization of a QuEChERS extraction and HPLC analysis of diclazuril, nicarbazin and lasalocid in chicken liver. *Food Chem* 2020;311:126014.
- [17] Cuadros-Rodríguez L, García-Campaña AM, Linares C, Ceba M. Estimation of performance characteristics of an analytical method using the data set of the calibration experiment. *Anal Lett* 1993;26:1243–58.
- [18] Darwish IA, Amer SM, Abdine HH, Al-Rayes LI. New spectrofluorimetric method with enhanced sensitivity for determination of paroxetine in dosage forms and plasma. *Anal Chem Insights* 2008;3:145–55.
- [19] Sangshetti JN, Deshpande M, Zaheer Z, Shinde DB, Arote R. Quality by design approach: regulatory need. *Arab J Chem* 2017;10:S3412–25.

- [20] Hildebrandt C, Gopireddy SR, Scherließ R, Urbanetz NA. Assessment of material and process attributes' influence on tablet quality using a QbD and DEM combined approach. *Powder Technol* 2019;345:390–404.
- [21] Jackson P, Borman P, Campa C, Chatfield M, Godfrey M, Hamilton P, et al. Using the analytical target profile to drive the analytical method lifecycle. *Anal Chem* 2019;91(4):2577–85.
- [22] Rozet E, Ziemons E, Marini RD, Boulanger B, Hubert P. Quality by design compliant analytical method validation. *Anal Chem* 2012;84(1):106–12.
- [23] Osakwe O. Chapter 8 – Pharmaceutical formulation and manufacturing development: strategies and issues. In: Osakwe O, Rizvi SAA, editors. *Social aspects of drug discovery, development and commercialization*. Boston, MA: Academic Press; 2016. p. 169–87.
- [24] Zobel-Roos S, Schmidt A, Mestmäcker F, Mouellef M, Huter M, Uhlenbrock L, et al. Accelerating biologics manufacturing by modeling or: is approval under the QbD and PAT approaches demanded by authorities acceptable without a digital-twin? *Processes* 2019;7:94.
- [25] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharma Rev* 2016;14(10):61–6.
- [26] Patil A, Shirkhedkar AA. Application of quality by design in the development of HPTLC method for estimation of anagliptin in bulk and in-house tablets. *Eurasian J Anal Chem* 2017;12:443–58.
- [27] Beilby AL. Principles of instrumental analysis (Skoog, Douglas A.). *J Chem Educ* 1972;49(6):A362.
- [28] Alexander A, Chaurasia R, Khan J, Swarna SS, Patel S. Spectrophotometric method of standard curve preparation and calculation for metronidazole. *Int J Pharma Prof Res* 2011;2.
- [29] Rapalli VK, Kaul V, Gorantla S, Waghule T, Dubey SK, Pandey MM, et al. UV Spectrophotometric method for characterization of curcumin loaded nanostructured lipid nanocarriers in simulated conditions: method development, in-vitro and ex-vivo applications in topical delivery. *Spectrochim Acta A Mol Biomol Spectrosc* 2020;224:117392.
- [30] Jayakumar S. Components, principle and applications of UV Vis-spectrophotometer, 2016.
- [31] Pradhan R, Krishna KV, Wadhwa G, Taliyan R, Khadgawat R, Kachhawa G, et al. QbD-driven development and validation of HPLC method for determination of Bisphenol A and Bis-sulphone in environmental samples. *Int J Environ Anal Chem* 2020;100(1):42–54.
- [32] Dubey SK, Duddelly S, Jangala H, Saha R. Rapid and sensitive reverse-phase high-performance liquid chromatography method for estimation of ketorolac in pharmaceuticals using weighted regression. *Indian J Pharm Sci* 2013;75:89–93.
- [33] Giram P. QbD approach for analytical method development and validation of serotonin by spectroscopic method. *Int J Pharm Pharm Res* 2017;10:98–117.
- [34] Nahata A. Spectrofluorimetry as an analytical tool. *Pharm Anal Acta* 2011;2:107e.
- [35] Prabu SL, Shahnawaz S, Kumar CD, Shirwaikar A. Spectrofluorimetric method for determination of duloxetine hydrochloride in bulk and pharmaceutical dosage forms. *Indian J Pharm Sci* 2008;70(4):502–3.
- [36] Barnett K, McGregor PL, Martin GP, Blond DJ, Weitzel J, Ermer J, et al. Analytical target profile: structure and application throughout the analytical lifecycle. *Pharmacop. Forum* 2016;42.
- [37] Beg S, Rahman M, Kohli K. Quality-by-design approach as a systematic tool for the development of nanopharmaceutical products. *Drug Discov Today* 2019;24(3):717–25.

Chapter 3

Analytical quality by design for gas chromatographic method development

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3.1 Introduction

Analysis is an integral part of the different sciences and knowledge of a society and its significance is increasing with each passing day. The scientists in their respective fields need a comprehensive understanding of the application of analysis and latest updates to continuously learn in a professional setting. In the domain of pharmaceutical research and development, a formulation would serve its intended purpose only if it is administered in a suitable amount and manner along with an absence of impurities. Impurities may develop at different phases of development, transportation, and storage, thus increasing the risk associated with a pharmaceutical. Hence the analytical exploration of the drug and excipients, intermediates, drug formulations and products, degradation products, and biological samples containing metabolites has to be undertaken. Thus this led to the development of analytical instrumentation and methods for detection and quantitation. In the analysis of pharmaceuticals, different analytical techniques and their corresponding analytical methods include chromatography, titrimetry, spectroscopy, electrochemical, kinetic, electrophoretic, flow injection, and sequential injection as well as hyphenated [1].

Chromatography is a technique that is based on the differences in polarities of various molecules in a mixture to be separated. The liquid or gaseous component acting as a mobile phase moves over a solid bed of particles known as a stationary phase. After introduction of the sample solution comprising of the

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mixture in the mobile phase, it moves through the stable phase, undergoing separation due to relative affinities for the two phases. Molecules with a greater affinity to the mobile phase travel faster than the molecules with lesser affinity [2]. The advantages of chromatography include the possibility of accurate and precise separation followed by analysis and/or purification. The chromatographic techniques require low sample volumes and can separate highly complex mixtures. The techniques are applicable to an extensive range of samples of drugs, biomatrices and tissue extracts, water and air samples, food particles, pesticides, etc. Different chromatographic methods include thin-layer chromatography (TLC), column chromatography, paper chromatography, affinity chromatography, gel permeation chromatography, ion-exchange chromatography, high-pressure liquid chromatography, and gas chromatography (GC). In GC, the mobile phase is a gas, such as helium and argon, and the stationary phase is a column in a device. This houses a stationary phase that is liquid which is adsorbed on the inert solid surface [3]. It is applied for the analysis of mixtures of volatile liquids and gases. It is a convenient, rapid, multidimensional, highly sensitive method to ensure exceptional separation using little amounts of compounds. Also the utilization of pyrolysis-gas chromatography/mass spectrometry (Py-GC/MS) provides the additional benefits of removal of sample preparation techniques of gas chromatography-mass spectrometry (GC-MS) including extraction, concentrating, dilutions, and chemical reactions [4]. Its applications involve microplastics characterization including analyzing and identifying degradation products of commercially available polymers or copolymers [5].

Furthermore, the overall objective of pharmaceutical research and development is planning of a good manufacturing process and subsequently its quality product. This is to ensure the consistent and continual deliverance of the proposed quality and performance of the final product. To achieve this, quality by design (QbD), an effective and competent approach starting with the predefined objectives that provides attention to the process and product is utilized [6]. A QbD-driven analytical method development and validation is based on validation parameters, interaction data, measurement uncertainty, control strategy, and continual improvement. Since it is derived from sound science and knowledge, it ensures the requirement of fewer resources and a decrease in probability of human error [7].

The objective of the chapter is reviewing in depth and discussing the employment of QbD and automation in GC method development and validation.

3.2 Quality by design principle

The simple notion of QbD is that quality is ideally built into a pharmaceutical product instead of testing into it [8]. The significance of QbD is aimed at meeting a predefined level of quality during the steps of researching and developing a pharmaceutical product and manufacturing processes. It

requires a thorough and proper understanding of the science and knowledge of how the formulation composition and manufacturing process variables affect the end-product quality [9]. This modern scientific approach of QbD strives to inculcate quality in a pharmaceutical product via control strategy [10]. Implementation of QbD has been undertaken in pharmaceutical industry via various initiatives including the 21st-century Food and Drug Administration (FDA) with its Current Good Manufacturing Practices (cGMP) and Process Analytical Technology (PAT). The application of QbD also involves the process validation guidance by FDA as well as the regulatory documents of International Conference on Harmonisation (ICH) Q8, Q9, and Q10 [11,12].

Q8 pharmaceutical development, Q9 pharmaceutical risk management, and Q10 pharmaceutical quality systems are the ICH guidelines that deal with QbD and related aspects with respect to quality issues [13–15]. The principles of QbD with respect to science and risk-based development of product, approach to life cycle, assessing risk, and strategy of product are described in ICH Q8, Q9, and 10. The ICH Q8 guideline is subdivided into two main portions with the first concerned with pharmaceutical development while the second is the annexure stating the QbD principles [16]. According to the guidance on pharmaceutical development, QbD is “a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.” The Q8 guideline is concerned with presentation of knowledge obtained via the application of scientific strategies and quality risk management in designing and manufacturing a pharmaceutical. Design space and PAT are two essential terms used to discuss and describe QbD as per ICH Q8. The design space can be defined as “the multi-dimensional combination and interaction of input variables (e.g. attributes of materials in formulation composition) and process parameters that have been demonstrated to provide assurance of quality.” Modeling the design space provides information regarding the process, thereby obtaining the control strategy. The analytical adaptation of the control strategy is based on the controls on input factors in a particular method. These will enable the method to meet the criteria of system suitability and various performance targets [17]. Besides, PAT is defined as “a system for designing, analysing, and controlling manufacturing through timely measurements (such as through processing) of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality.” This science of QbD in a pharmaceutical industry is primarily for the development of a product and an analytical method. The latter is also known as analytical quality by design (AQbD), for which the FDA has recently offered regulatory flexibility with respect to approval of some new drug applications (NDAs). According to ICH quality guidelines, the current trend of implementation of AQbD in analytical method development in the

pharmaceutical industry is an integral component of Q8, Q9, and Q10. Q9 is focused upon the determining the risk to quality that is derived from sound science and ultimately aimed at safeguarding the interests of the patient. The degree of work put in and paperwork of the process of quality risk management is proportional to the amount of risk. AQbD strategy in analytical method development ensures a decrease in the number of out-of-trend (OOT), out-of-control (OOC), and out-of-specification (OOS) results. This is necessary to advance product quality and decrease cost of analysis which is possible due to robustness of the developed method inside the region. AQbD also allows permission for movement inside of method operable design region (MODR), thereby assisting in the achievement of regulatory flexibility in modifying method parameters inside the MODR that is design space. Furthermore, the FDA has also established a correlation between analytical methods to risk management (ICH Q9) in 2011. The formulation quality is affected by a total of risk factors such as the lack of efficacy of a drug and an uncertainty associated with novel products and processes. An important risk factor also includes the poor detectability because of an unsuitable and inappropriate method of analysis [18,19]. Finally, Q10 aims to define an all-inclusive model to result into a competent pharmaceutical quality system that has its roots in the quality ideas of International Organization for Standardization (ISO) [15]. In the next section, the need for QbD in GC process development is introduced to highlight the importance of applying AQbD in the paradigm of analytical techniques of GC.

3.3 Need for quality by design in gas chromatography process development

Application of QbD in the pharmaceutical industry is largely focused upon the general manufacturing process of a pharmaceutical rather than its analysis. Attention is paid to the understanding of a process through applying QbD principles which may also involve developing a design space and further a control strategy specific to a process [20]. Application of QbD in the pharmaceutical industry is largely focused upon the general manufacturing process of a pharmaceutical rather than its analysis. Attention is paid to the understanding of a process through applying QbD principles which may also involve developing a design space and further a control strategy specific to a process null. Analytical methods and techniques encompass the measurement of physical and chemical characteristics of a raw material, determination of density of intermediate granules, analytical imaging of final formulation may be combined via the application of multivariate strategies. Analytical methods and techniques involve the measurement of the physical and chemical characteristics of raw materials, determination of density of intermediate granules, along with the analytical imaging of final formulations combined with the application of multivariate strategies. This may be undertaken with the aim of modeling the design space with respect

to a pharmaceutical method. In spite of generation of the model for the method, the overall AQbD strategy ultimately depends upon the quality of analytical techniques and obtained results [21].

A GC method is developed for novel products in case of absence of availability of any official methods. The steps of method development broadly include conducting literature survey to understand the physicochemical characteristics of analyte, selecting chromatographic conditions, developing an approach for analytical method, preparing sample, and optimizing the method followed by its validation [22]. An alternative method for existing product may be pursued for the ultimate aim of improving ruggedness and precision at a lower cost and time [23]. A comparison is made between the existing procedure and the alternative including the respective advantages/disadvantages. Ultimately for GC method development, column and carrier gas selection are of paramount importance. The former involves considering the stationary phase and its dimensions, column length, internal diameter, and film thickness while the latter depends upon choice of inert gas and its flow rate. Temperatures have to be carefully programmed including the initial temperature, final temperature, initial and final hold, ramp rate along with controlling injector and detector temperature [24]. Furthermore, as per the FDA guidelines, a three-stage approach has been proposed along with method validation for analytical technique such as GC. Stage 1 is method design which involves defining the requirements and needs of a method and conditions along with recognizing critical controls. Stage 2 is method qualification which helps in determining whether a method is able to meet its design. Stage 3 is life cycle management that has the objective to obtain continuing assurance that the method is within a state of control during routine utilization [25]. The importance of the individual stages is discussed in great detail in the later section as a part of the implementation of QbD in GC. The next section focuses on describing the methodological aspects associated with AQbD in GC.

3.4 Methodological aspects

Instead of utilizing GC in QbD application, the better alternative is to apply the principles of QbD to the development of the GC analytical method itself. This topic can have possible applications in a great assortment of GC methods and its hyphenated techniques as well. More robust and better GC method is the aim which will have positive implications on the QbD applied during the entire drug development process. This is because the data obtained from GC and other analytical techniques form the backbone of the application and strategy behind QbD in a process [20]. The selection of GC as a method of analysis depends upon the stability of analyte in solution. If the analyte is not stabilized by physical control or matrix deactivation, derivatization is necessary. Derivatization helps to improve the volatility required

in GC and also improve ionization for MS. If the analyte is sufficiently volatile, the detection method is further chosen. If adequate sensitivity is available, GC flame ionization (FID) is usually preferred. In case of sensitive and halogenated compounds, GC electron capture detector (ECD) is the alternative with GC-MS utilized for halogenated or nonhalogenated compounds [26]. A structured approach to GC analytical method development that is transcendental to the technique utilized is essential [27]. The method development strategy (MDS) is such an approach that considers the different available analytical methods that may have potential advantages from QbD [28,29].

The steps of MDS are defining the method goal, scouting and evaluation among available methods, choice of method for the intended goal, risk assessment via structured tools, developing control strategy for method performance and validating the method. The MDS approach is based on the QbD idea of utilizing scientific principles in each step of method development rather than end-product testing to determine quality. The first step of MDS is to know the intended use of an analytical method such as GC and considerations for an end user such as the GC instrument, laboratory, and analyst. It also includes literature survey gathering essential information regarding the physicochemical properties of the analyte and matrix. The defined method goal has to be practically and realistically obtained for which appropriate methods are evaluated to identify a specific approach. This naturally leads to selection of a particular method with a set of final method conditions. Besides, a key component of MDS, that is, risk assessment makes use of tools of design of experiments (DOE) and measurement systems analysis (MSA) to perform studies to evaluate robustness and ruggedness. The utility of these tools is to provide sufficient information about an existing method and permitting more efficient control strategies with respect to critical parameters. Thus understanding of critical method variables is elevated along with establishing the proven acceptable ranges. The penultimate step of development of control strategy for continual method performance involves performing system suitability tests for the analytical instrument, for example, GC and preparing a mitigation plan with the aim of controlling critical method variables. The final step in MDS is method validation which may be considered to be an individual and separate stage after the complete MDS process. It is done to fulfill the needs and requirements of ICH guidelines, thereby verifying and confirming built-in method performance. Sometimes the method development may not be specified in QbD approach and AQbD is a subset of the whole control strategy. This control strategy is with respect to a synthetic or formulation process, and thus a relationship between the analytical activities and complete formulation development QbD tasks is established. Various generalized approaches to QbD analytical method development have been further proposed and described as well [17,30]. A schematic pictorial for GC-based methodology is given in Fig. 3.1.

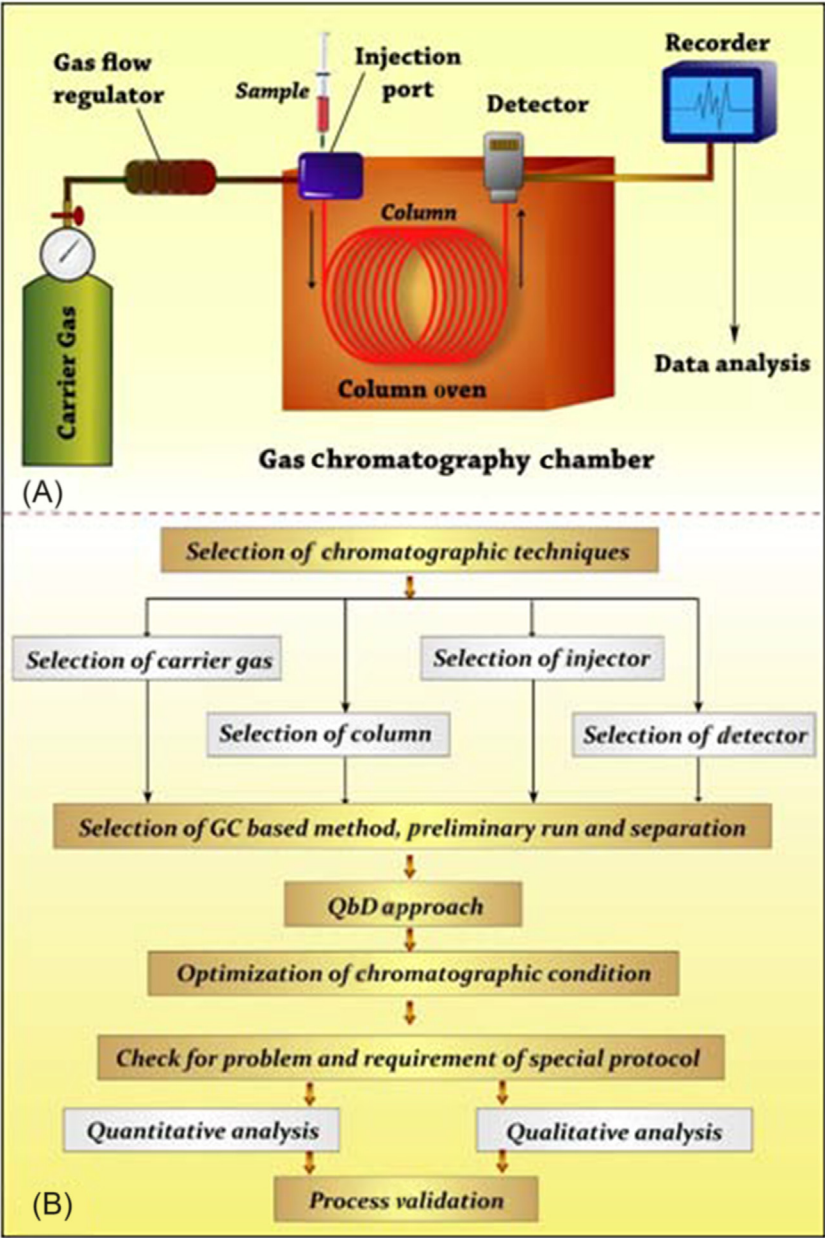


FIGURE 3.1 Gas chromatography (GC) instrumentation and the steps involved in GC method development and validation.

3.5 Implementation of quality by design in gas chromatography

GC method development is of primary concern for an analytical professional since this step decides the performance as per the required usage. Usually two main approaches are applied for the same which are one factor at a time (OFAT) and the AQbD. The former has its foundation in trial and error in which a single parameter is optimized for the response while other parameters are kept constant. This approach is highly prone to failure due to risks arising from a narrow robust behavior for instrument variables utilized during developing a method. This OFAT approach has an increased cost as well since it requires revalidation protocol especially during method transfer or when an alternative method is developed [31]. The latter is AQbD that is aimed at exploring scientific knowledge and understanding in the implementation of an analytical method for the ultimate optimization in formulation process. AQbD is a suggested and preferred approach in analytical method development to decrease OOS, OOT, and OOC. AQbD makes use of DOE that signifies the interaction among the input variables of a quantitative and qualitative type. These are then found to affect the method responses or outcomes which finally influences the results. The misconception surrounding DOE is its time-consuming nature due to more trials or difficulty in selection of design or interpretation of statistical data but DOE and its models are highly successful if utilized in an efficient manner. Literature and reference for AQbD and its application are available in European Pharmacopoeia (EP) and United States Pharmacopoeia (USP-NF). Information about flexibility for an analytical method and its change without the investment of revalidation in case of implementation of AQbD is covered but many other guidelines are not necessarily available [32]. Scientists and researchers have taken part in compilation of information with respect to AQbD in chromatographic methods such as GC [33,34]. A three-stage approach for method validation with respect to FDA has been explained further.

Stage 1 begins with the first step in method lifecycle which is selection of the analytical method suitable for the intended usage. The parameters to be considered include the analyte properties, impurities, accessibility of GC instrument, method sensitivity, and time of analysis. Since the instrument of choice is GC, cost of reagents and solvents is an important factor. For a molecule with no chromophore, consideration of different chromatographic or derivatization techniques over GC is essential. Method design in AQbD also involves defining analytical target profile (ATP), establishing critical quality attributes (CQAs), performing risk assessment, generating design space, experimental design screening, and optimization. The needs of any analytical technique, that is, GC, high-performance liquid chromatography (HPLC), chiral HPLC, and ion chromatography are defined via the ATP. The ultimate aim of an analytical technique is to fulfill the criteria with respect to validation

parameters and analytical criticalities required for regulatory requirement [9]. CQA, as defined by ICH Q8, is “a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality” [16]. The sample concentration and diluent used, flow of gas, oven program and temperature, and injection temperature are the usual factors affecting method performance in GC [35]. These may also differ among different projects and are broadly classified in three main categories which are parameters of analyte, parameters of GC instrument, and parameters of operating conditions [25]. The next step may involve the application of risk identification and risk assessment tools including the Ishikawa diagram and failure mode effect analysis (FMEA's) respectively. The risk that a GC method is vulnerable against during its operation needs to be listed out and assigned a rank. The risks that show a direct impact on the GC method need to be documented that may serve as a reference through the GC method development. Furthermore, the MODR is generated that contains a combination of all set of GC method condition variables that have had their predefined GC method objectives met. Optimization of method parameters is undertaken with the optimized method parameters lying somewhere around the MODR. Many trials of different columns, flow rates, column temperature, etc., are employed for optimization in GC. The outcomes undergo documentation and the best possible and suitable results are subjected to statistical evaluation. Ultimately experimental designs used are mainly of four main types such as factorial designs, mixture design, D-optimal design, and central composite design. The runs are performed and effect of factors on responses are evaluated and analysis of results is via software such as Design Expert, Minitab, or even Microsoft Excel [25,35]. Stage 2 is initiated only when the analytical chemist operating GC is assured that the designed and developed GC method is as per operational requirement. Method qualification process has a relation with the GC equipment qualification and is categorized in three main categories, namely installation qualification (IQ) followed by operational qualification (OQ) and then the performance qualification (PQ). These are done to ensure meeting check points on GC instrument calibration, GC method development validation parameters as per ICH Q2 (R1) chapter and GC method performance via the instrument, facilities, samples, and analyst [36]. Stage 3 considers the fact that during the product lifecycle, the GC method may undergo changes via unintentional deviations, operation of method in a new laboratory or facility, and continuous improvement tasks. Method redesign is a viable option when the modified GC operating conditions fall outside MODR. Generation of new ATP due to the change in process along with partial requalification may be undertaken for attaining reliable results. Method IQ activities such as knowledge transfer due to transfer of method to a new location are also included in stage 3 [25]. The different elements of QbD for analytical techniques such as GC and their instrumentation is shown in Fig. 3.2.

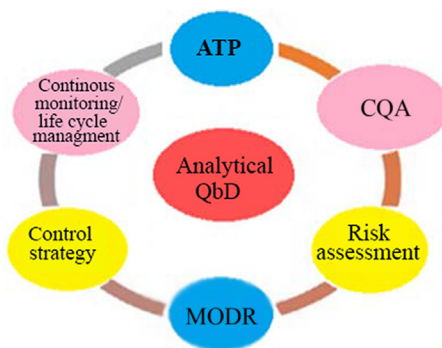


FIGURE 3.2 Elements of analytical quality by design (AQbD).

3.6 Statistical tools supporting gas chromatography-quality by design

The primary objective of the shift from the paradigm of quality by testing (QbT) to QbD is to enhance the science and understanding of products and processes. QbD is closely aligned and consolidated quality risk management that encourages the use of various designing tools to perform statistical analysis including histograms, acceptance control charts, process capability analysis, and Pareto charts. Different quality tools and methods that are a part of QbD to plan, design, and analyze an experiment include multivariate analysis, statistical quality control (QC), DOE, etc. Examples of multivariate data analysis software package also include the Monte Carlo simulation or Plackett–Burman design. Software modeling packages such as Agilent GC, Clarus GC, and TurboMass GC/MS software are used in GC instrument operation. A change in the GC column dimensions and particle size, temperatures, and flow rate can help in forecasting the movements of peaks and chromatographic separations. The analyte molecular structure as well as logP and pKa are commonly used in GC method development to determine retention and optimum separation [35]. An analytical and pharmaceutical professional must be well versed with all the literature and tools that are a fundamental part of research and development. Drug development process is accelerated due to the correct and appropriate usage of these tools and improvement in the quality of data obtained from analysis. The effort invested in obtaining a deep comprehension of these is easily compensated by their wide-ranging applications in a research laboratory [6].

DOE approach in method optimization is a tool extensively used by analytical and pharmaceutical personnel involved in research and development. The benefits include the assessment of factors and their interactions with the highest proficiency. Since the factors are studies at different levels that are

varied, the results and conclusion are applicable to an extensive range of conditions. The data obtained from main effects and interactions are maximally utilized in GC method screening and optimization. One of the most commonly used statistical designs intended to determine interactions of input variables on the response is factorial designs. They are comprising of an experimental series with the individual factors being varied in a simultaneous fashion. It is of two main design types namely full factorial and fractional factorial designs. Full factorial design is the study of effect of all factors (n) at different levels (x). This also includes the interaction in between the individual factors with x^n being the total number of experiments to be conducted. They are represented in a matrix format which is also referred to as the design matrix. A two-level factorial design helps to examine a single factor at two separate and unique levels. The formula $n = 2^x$ is utilized to calculate the number of required experiments with x being the number of factors that are studies and n being the number of experiments. Factorial designs are either symmetric with the number of factors being the same as number of levels (2^2 or 3^3) or asymmetric in which the number of factors is different from the number of levels (2^3 or 3^2). The study of multiple factors in an efficient manner is conducted via fractional factorial design. The number of runs is lesser due to the fact that there are no interactions between two or more treatments [37,38].

In addition to the use of DOE and MSA in method development, numerous statistical strategies and approaches have been utilized in the contribution to AQbD. There has been a proposal that any alterations in the possibility of moving a method outside of the MODR can be evaluated and judged. This is achieved via a validation exercise that confirms the acceptability of the criteria associated with method performance. For providing assistance to such changes, a novel and advanced statistical tool derived from two one-sided tests (TOST) was applied. It had certain benefits in intermediate precision related studies to demonstrate method equivalence and for the purpose of consistent intermethod comparisons to well-known specifications [39]. Design and the sample size of the equivalency study have been found to be linked with the setting of acceptance criteria. This signifies an acceptable level of bias between the changed method and the original method [40].

3.7 Experimental design

As per the requirement of ICH Q8 guidelines, MODR can be established in the GC method development phase with the ultimate aim of increasing the robustness while decreasing the cost. MODR, formed as result of output from DOE, is an operating range for the input variables that are critical in generating a result that continually meets the goals as set by ATP. Flexibility is provided by MODR with respect to input method parameters so as to meet

the appropriate criteria for method performance and suitable response for method without resubmitting to FDA. Suggestions by FDA include the simultaneous conductance of MODR and GC method validation. After it is defined, the required method controls are appropriately set and method validation is done with. Application of DOE in the GC method development step needs a thorough understanding of the science of selecting input variables and corresponding output response. The approach of DOE in AQbD covers the following important stages [9].

3.7.1 Screening

Screening involves the process of screening out qualitative input variables also referred to as categorical variables that are discontinuous in nature. Identification of critical method parameters is an important part of optimization experiments. The conclusion of screening is the isolation of critical method parameter (CMP) which have to be controlled. They may also be subjected to DOE in the process of MODR optimization. Different tools and selection strategies used include full factorial or fractional factorial design/Taguchi methods (optimization and/or screening), Plackett–Burman method (identification of some critical factors from many variables), pseudo-Monte Carlo sampling (pseudorandom sampling for quantitative risk analysis), Doehlert design (optimization), and Box–Behnken (three levels of each factor being -1 , 0 and $+1$).

3.7.2 Optimization

In optimization, incorporation of quantitative measures for CMP obtained from screening or from risk assessment is conducted. A rudimentary information about the science of relationship between the CMP (input variables and their quantities which are of critical risk to the method) and output responses is obtained through optimization. This ultimately provides knowledge regarding the substantial effect in performance of method and ATP.

3.7.3 Selection of design of experiment tools

In the process of optimization, different approaches are employed for the derivation of a mathematical relationship, that is, model. The total number of input variables, information regarding controlled parameters, and an understanding of the link between variable and result are considered while selecting a DOE tool. Interpretation of interaction and contribution of variables in responses is based on sound knowledge of statistics. This further enables in selection of the variables at their optimum levels. Examples include selection of the Plackett–Burman tool in case of studying large number of input

variables in absence of any interaction. Fractional factorial design or Taguchi method is employed for lower number of experimental trials or runs on comparison with full factorial designs. However resolution of the interactions confounded is of major importance as well. These tools and the rationale of selection are depicted in [Table 3.1](#).

3.7.4 Method operable design and surface plot

MODR model contour plot, that is, two-dimensional (2D) plot is obtained with two axes representing the input variables of GC under selection. At the same time, other factors which may include flow rate and configuration of GC instrument are under control. The coded level of particular variables employed in DOE are represented on both axes in numbers such as -1 , -2 , $+1$, and $+2$. A contour plot may be appropriate for the response in case of nonlinearity. The relationship between the input variable and the output response shows a pronounced curvature. Mathematical models may be employed later for selection of design space from contours. The real and actual experimental run may be used to verify the predicted value of response which is a part of validation of model. Another type of surface model may be achieved which offers a modification in response against variables that is appropriate for a linear association by usage of simulation. A systematic design space is shown in [Fig. 3.3](#).

3.7.5 Model validation and verification

Before the choice from graph or contour, validation of the predicted values of the chosen method response has to be performed by actual experimental runs. This is followed by regression analysis for statistical validation of the model. Validation is performed according to ICHQ2 (R1) guidelines with the utilization of normal operating conditions (NOC). It is the optimized condition consisting of set of variables at a particular point. In conjunction with method validation according to regulatory guidelines, method verification via joint precision and accuracy determination at numerous method factor points inside the GC separation space, that is, MODR. The greatest potential of ability of method to meet the ATP requirements is signified by the multipoint verification inside the MODR. This multipoint verification is conducted beyond the normal robustness test limits which is more than two points. This has both deviations but care is taken that the points are within the MODR. For example, the verification of column temperature is possible between 35°C and 45°C . These studies of validation and verification should possess robustness throughout the range of parameters from high to low through the target value as well [\[9\]](#).

TABLE 3.1 Choice of design of experiments (DOE) statistical tools in analytical quality by design.

Sr. no.	Name of design	Benefits of design	Limitations of design	Number of variables and/or rationale for use of design
1	Full factorial design	Identification of the main effect and interaction effect without aliasing	An increase in number of variables leads to an increase in experimental trials	2–5 variables/ optimization of variables
2	Fractional factorial design/ Taguchi methods	Requires lesser number of experimental trials	Resolution of confounding effects of interactions is a complicated issue	Screening and/ or optimization of variables
3	Box–Behnken design	The design points fall inside the design region	Two-factor design was not given	For utilization for three levels of each factor (–1, 0, +1)
4	Pseudo-Monte Carlo sampling method/ pseudorandom sampling	Preferred in the case where accurate and precise calculations are possible; convenience and speed of investigation of behavior and changes to the model	Issues in employing this sampling method for nonconvex design spaces; random numbers which are created from a random number generating algorithm	Optimization/ quantitative risk analysis
5	Plackett–Burman method	Requires less runs for a great number of variables	No insights about interaction effects are provided	Screening/or identification of critical few factors from a great number of variables
6	Doehlert design	Utilized in response surface analysis	-	Optimization of variables

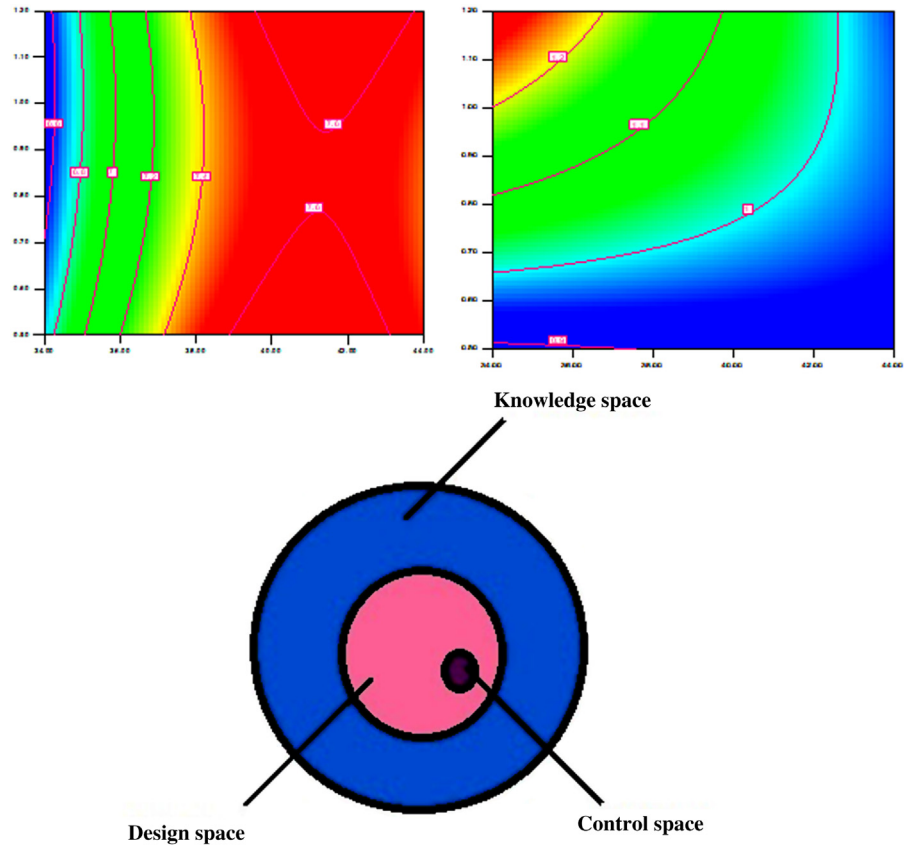


FIGURE 3.3 Contour plot for method operable design region (MODR) and a design space.

3.8 Areas explored

The applications of GC method by QbD approach have been covered in the following specific areas of interest.

3.8.1 Qualitative and quantitative analysis of phytoconstituents

Silvia Robu et al. aimed to analyze essential oils from *Salvia officinalis* and determine its chemical composition by developing and optimizing a simple, robust, economical GC coupled to mass spectrometry GC-MS method via QbD application. The characterization of the oils obtained is very important due to the varying chemical composition within one botanical species. Hence the multi-dimensional GC-MS has been the gold standard method for qualitatively and quantitatively analyzing very similar essential oil components [41]. The two phases of this research work were screening for identification of critical method variables (CMVs) that affect the critical analytical attributes (CAAs) and finally the method performance and optimizing GC conditions. Out of all factors that impact the analysis via GC method, factors such as dimensions of capillary column, type and flow rate of mobile phase, change in injected sample value or injector splitting ratio, temperatures are studied. This is done to ultimately enhance the GC separation quality and identifying along with quantifying individual components. The risk assessment was initiated with risk identification utilizing Ishikawa diagram for the same as depicted in Fig. 3.4.

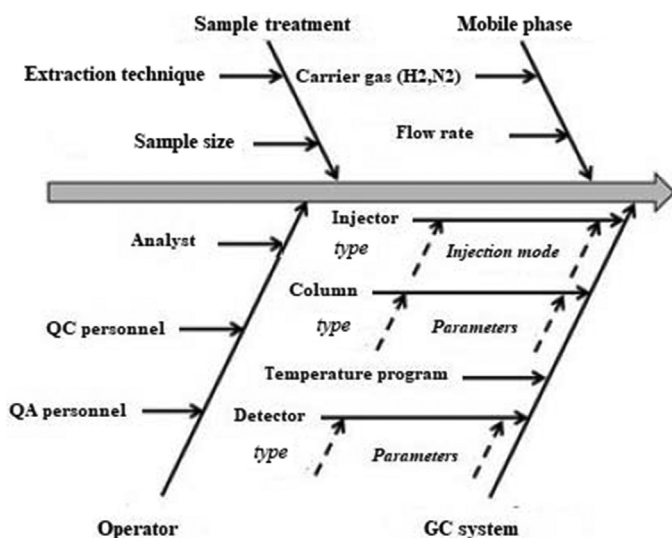


FIGURE 3.4 Ishikawa fishbone diagram depicting the variables affecting the analytical performance of a generic gas chromatography (GC) method.

The method optimization and development was undertaken using a QbD approach by studying the analysis time, separated peaks, and retention times. An experimental design as per QbD consisted of a standard set of 2 injection volumes, 3 oven temperatures, and 2 flow rates which led to a total of 24 chromatographic conditions. This was evaluated by employing the two-tiered approach with the first level containing evaluations for retention times, analysis time and number of separated peaks resulting in 3 chromatographic conditions. This was due to 3 oven temperature programs $\times$ 1 injection volume $\times$ 1 split ratio $\times$ 1 flow rate at first level. The second level had evaluations of three conditions at even more stringent criteria. These included peak symmetry, quality of identification (more than 80%), and largest peak area (affecting the method sensibility). Due to the fact that the final method is chosen against attributes of method, there is a high probability that the method is consistent and reliable and will be operational across the product lifetime. Ultimately the final conditions were 1 μ L oil being injected at temperature of 250°C in a split/splitless injector. The separation is carried out via the mobile phase being helium (1 mL/minute) and capillary column DP-5MS (30 m $\times$ 0.25 mm, 0.25 μ m film thickness) across an increment in the oven temperature between 40°C and 280°C being 10°C/minute with the separated compounds being transferred into the detector at 280°C. The analysis time was less than 25 minutes with the mass range of 15–450 AMU and source temperature and quadrupole temperature being 230°C and 150°C, respectively. The results obtained showed a chemical composition of *Salvia officinalis* essential oil being; 2.04% of naphthalene, decahydro-, 2.75% of Camphene, 2.93% of beta-pinene, 3.57% of trans-caryophyllene, 4.23% of alpha-thujone, 5.38% of alpha-humulene, 5.59% of 13-epimanool, 6.39% of alpha-pinene, 7.44% of viridiflorol, 8.68% of camphor, 10.9% of 1,8-Cineole, and 19.10% of beta-thujone [42].

3.8.2 Quantitation of residual solvents

The control of solvents throughout in process control is extremely significant for a pharmaceutical industry. Specifications of the drug product intermediates (DPI) usually contains determination of residual solvents. The application of a GC instrumentation and method is responsible for the control performed in QC laboratories [43]. Therefore, a wide variety of GC methods have to be carefully managed in QC laboratories on a day-to-day basis. This leads to consumption and wastage of significant resources which has an effect on the efficiency and productivity of the laboratory. Cátia Sousa and coworkers aimed to develop a generic GC method (headspace present), along with the application of scientific approach of QbD for quantifying solvents employed in manufacturing of DPI. For all solvents in the range between 50 ppm and ICH levels, the method was confirmed to be robust, selective, and accurate. The method gave rise to a profound understanding of the

influence of critical method attributes in the method responses. This permitted easy prediction and extrapolation of GC method behavior to various other instruments and their conditions with an aim of improving robustness. The control strategies for method life cycle were defined and described allowing continual improvement of method. The method development was divided into two unique stages with the first being determination of best GC instrumentation conditions, utilizing DOE methodology and the second including testing of matrix effect. This led to the generation of a development flow chart intended for use by all users with various DPIs. Both the stages of method development involved the application of QbD principles.

3.9 Method control strategy

The control strategy in QbD of an analytical technique such as GC is an excellent concept since it is an assurance that the performance of GC method is as intended on a routine schedule. Risk assessment helps to identify different risk factors to be considered during the implementation of control strategy. For an analytical technique such as GC, control strategy is the control of input factors required to meet both system suitability requirements in a traditional sense and other needs associated with performance [44,45]. A new appropriate GC method may also be needed in certain cases depending upon the findings of the risk assessment such as if it gives an indication that the general understanding of the performance can be elevated along with a high risk of acquiring reliable data and issues in its management. In the case that the risks are low and manageable, then the GC method control strategy may be defined. This may be comprised of correct system suitability criteria associated with management of risk and ensuring that the required attributes of the method are delivered. Thus system suitability testing of GC instrumentation might be the one control element for appropriate performance of the GC method. Such tests help in ensuring method performance along with recognizing failure modes and preventing the production of errors [46]. The system suitability tests may also be supplied with additional controls in the proper management of a risk. These may include environmental controls to keep a check on relative humidity, temperature, and light available in a research laboratory. Ultimately information obtained from system information provides vital understanding of an analytical system such as GC [47,48]. Thus the principles associated with risk assessment as mentioned in ICH Q9 guidelines help in recognizing and identifying a particular control strategy.

3.10 Validation and post method consideration

Method validation is an essential step that is conventionally carried out after completion of GC method development. It is a separate and unique exercise detached from development, occurring only once. Method development by

QbD application includes defining intent of a method, conductance of an experimental design, evaluation of experimental results, and selection of final method conditions culminating in risk assessment with evaluation of robustness and ruggedness. The ultimate goal of a GC analytical method is separation and quantification of the analyte of interest. Experimental design helps in formation of a GC database assisting in understanding a method, its optimization, and selection. If the GC column used is commercially unavailable, then the database helps in evaluation and implementation of change in GC method. The evaluation of experimental results is done via the symmetry of peak, fronting and tailing of the peak. This may be followed by more stringent criteria of system suitability parameters. The ultimate step in method development is to verify and finalize the method along with an application of ICH Q8 and Q9 such as a fishbone diagram for identification of probable risks. After a QbD-based GC method is developed and undergoes risk assessment, suitable definition and control of GC method parameters is attained. This is a sign for the initiation of formal method validation. A proper and correct method development and its assessment for risk renders validation following ICH Q2 as just a formality, as issues with this GC method are unlikely to rise during validation [35].

Selection of GC depends upon the sample volatility, solubility, composition (analyte and matrix), melting point, and aggregation state. Special care is given to GC relevant characteristics including boiling point, functional groups, polarity, reactivity, and stability on exposure to air at room temperature [49]. Careful handling may be required in case of hazardous substances, acids, bases, and thermally or chemically labile contents [50]. Derivatization is employed in the case of nonvolatile compounds or pyrolysis GC to pyrolyze a sample before entry into column [51]. Certain samples need a pretreatment before GC to remove interferences or to concentrate the compounds. This is followed by selecting a column which is dependent upon stationary phase, length, film thickness, and internal diameter [52,53]. These considerations are essential in terms of control over analysis speed and time, resolution, and efficiency of column. The order of elution of analytes is in turn influenced by solubilities in stationary phase, latent vapor pressures, affinities, and interactions in the stationary phase which all depend upon temperature [54]. A selection has to be done by the researcher from over 400 GC capillary columns available for separating 10,000 compounds. A column may have two analytes coeluting so a different column with another phase of chemistry is selected [24]. This column will reside in an oven so the GC temperature program including an initial and a final temperature along with ramp, that is, degree increase per minute is important. Resolution is improved on reducing the initial temperature or increasing the initial hold time. An excess resolution between peaks is decreased by elevating the ramp rate [55]. The next step is introducing sample in GC system via manual or auto sampler system which should ultimately be reproducible. Injector

temperature is necessary to quickly vaporize the liquid sample in the gas phase to be carried into the column for separation. The major primary techniques to vaporizing a sample and subsequently transferring it on the inlet of analytical column in capillary and micro packed GC are direct, on-column, split, splitless, and injections. Split injection is used when sample concentration is high enough to permit sufficient amount at the detector for producing a signal while some portion is discarded during injection [56]. The routinely used GC detector is FID employed for organic compounds. Temperature of the detector and relative flow rate of carrier gas into the detector are critical parameters for GC operation. Evaluation of detector parameters of sensitivity, linear and dynamic range, drift, etc., is achieved via defining a set of standards. The difference in response of detector also depends upon the type of detector, that is, concentration or mass dependent. Examples of former include thermal conductivity detector and photo ionization detector while those of the latter are flame ionization detection and flame photometric detection [57]. GC method development is ultimately governed by parameters of system suitability. These include more than 5 minutes of retention time, more than 2000 theoretical plates, more than resolution of 5 between 2 peaks, less than a tailing factor of 2 [22].

Validation results are used in judgment of quality, consistency, and reliability of analytical method thus making method validation an essential component of good analysis. As per FDA and ICH, the validation parameters are specificity, range and linearity, precision (method precision/repeatability and intermediate precision/reproducibility), accuracy (recovery), limit of detection (LOD), limit of quantification (LOQ), robustness, system suitability, and solution stability. Specificity is the ability of the GC method for accurate measurement of compound in presence of interference such as excipients, enantiomers, synthetic precursors, and degradation products that are possibly in the sample [58]. Linearity is represented as the confidence limit around the slope of the regression line and for establishing the same, minimum five concentrations are suggested as per ICH [59–64]. Accuracy is calculated by method application to samples to which known quantities of analyte were added and these should be analyzed in comparison with standard and blank to ensure absence of interference. LOD and LOQ are calculated via signal-to-noise ratio through the formula $s = H/h$, where H is height of peak of the component and h is absolute value of noise fluctuation from the chromatogram baseline of blank [60–63]. Robustness provides an idea about the reliability throughout normal utilization, and it is determined by changing a single parameter to study the effect on the GC method via analysis of samples or system suitability. System suitability is judged via peak resolution, peak tailing, theoretical plates and it is based on the notion that the GC instrument comprising of equipment, electronics, operations of analysis along with samples contribute to an integral system. Solution stability is concerned with the stability of samples and standards under normal conditions of

storage and determines the need of special conditions like refrigeration or light protection.

There is a need of a repository to store the substantial data acquired during AQbD method development and validation of GC. This repository may have constant updates by the research and development as well as QC laboratories throughout the method life cycle [27]. Due to method scouting, knowledge evaluation and risk assessment, alterations to method are possible in the repository. GC methods developed via AQbD help in setting specifications that may be improved beyond conventional specification setting. A level of comfort is possible in setting specifications when GC method does not result in OOS results, thereby leading to accommodation of requests from regulatory authorities for tightening specifications. The following section deals with the implementation of the various principles in current practice.

3.11 Implementation in current practice

AQbD needs to be introduced in the method development and validation of GC for method performance along with validation practice in the future. For implementation of AQbD in a particular drug product, the following needs to be considered. The quality target product profile (QTPP), that is, the profile depending upon specifications of product as mentioned in FDA has to be constructed. Each and every specification of the product is analyzed for criticality. The GC method development along with its suitability in providing support to criticality has to be assessed and justified. The analytical method chosen for analysis, that is, GC should be able to meet the ATP and the QTPP as well. Risk assessment has to be performed for GC method. The quantitative and qualitative variables that influence method performance and the method outcomes should be identified. An appropriate DOE has to be utilized for optimization of variables and establishing scientific knowledge and understanding as well. The MODR has to be found for assessing robustness and economic operations for the method variable or factor. The models and MODR are validated via experimental verification along numerous points so as to prove robustness. The GC method has to be validated in the operable region for the purpose of method performance which should be able to be subjected to control strategy an improvement [9].

3.12 Regulatory consideration for current and future

The concept of QbD was accepted by FDA in the year of 2004 and a detailed definition and description of the same was provided in “pharmaceutical cGMPs for 21st century – a risk based approach.” Stringent requirements and demands regarding product quality have been given in ICH Q8, Q9, and Q10. FDA also emphasizes the significance of PAT in quality of

formulations and their processes that is a framework for innovation in pharmaceutical development, manufacturing, and quality assurance. Regulatory strategies focused upon risk are necessary for science and understanding of control-related processes and methods for product quality and method performance. ICH Q8, that is, pharmaceutical development, ICH Q9, that is, quality risk management, and ICH Q10, that is, pharmaceutical quality system form the backbone of QbD and its underlying principles. QbD primarily implements Q8 and Q9 via substantial utilization of DOE which is integral in obtaining a multidimensional design space. DOE is basically utilized to determine ranges for GC instrument operating parameters so as to understand variations in sample preparation as well as method precision. DOE in GC method validation seeks to validate it for a range of amounts so that any change in the formulation concentration does not need additional validation. This is because the changes are characterized in the design space. The trend of application of DOE to analysis has risen only in recent years. DOE is utilized in method development for novel methods or those that require improvement, method validation, and quantifying the effect of analytical methods on product and process acceptance criteria. Due to new regulatory requirements, DOE will be an essential tool in analytical method development including GC which will be unavoidable in the future. Currently, regulatory documents are not available to guide in estimation of quality of design space which must be addressed in the nearest possible future [35].

3.13 Conclusion

The science of QbD is likely to continue growing in the pharmaceutical industry since its applications have provided many favorable outcomes. ICH guidelines address the major principles associated with QbD which are covered in Q8 pharmaceutical development, Q9 pharmaceutical risk management, and Q10 pharmaceutical quality systems. QbD is primarily focused upon suitable risk assessment via tools such as Ishikawa diagrams and flow-down maps along with DOE and PAT. These are dependent upon the particular intended use of a product/process and require a skilled researcher using mathematical/statistical knowledge to establish a design space. Furthermore, the strategy of QbD in analytical method (AQbD) such as GC is acceptable and encouraged due to its various variables that have a profound effect on method outcomes. AQbD should enable complete and in depth understanding from development of a GC method to its transfer. It includes the components of ATP, CQA, optimization and development of method via DOE, MODR, and control strategy, validation of method and its continual improvement. The chromatographic technique of GC is routinely used in pharmaceutical analysis and its variables include GC instrument settings, sample properties, selection of calibration models, and method parameters. The GC method should be highly robust which enables it to endure long

term utilization with a low possibility of failure. This method needs to undergo risk assessment, possible amendments, and optimization which is evaluated with appropriate tests before release in other laboratories. Ideally, there should be an overall focus on developing the GC method along with identification and control of its failures. Hence AQbD with respect to GC is an essential link between the analytical development and operational laboratories. With growing complexity in the pharmaceutical and analytical fields, the significance of a well-constructed QbD-driven analytical technique of GC will rise.

3.14 Conflicts of interest

There are no conflicts of interest.

References

- [1] Siddiqui MR, AlOthman ZA, Rahman N. Analytical techniques in pharmaceutical analysis: a review. *Arab J Chem* 2017;10:S1409–21.
- [2] Coskun O. Separation techniques: chromatography. *North Clin Istanbul* 2016;3(2):156.
- [3] McNair HM, Miller JM, Snow NH. Basic gas chromatography. John Wiley & Sons; 2019.
- [4] Peter K. Pyrolysis-gas chromatography: mass spectrometry of polymeric materials. World Scientific; 2018.
- [5] Kusch P. Application of pyrolysis-gas chromatography/mass spectrometry (Py-GC/MS). *Compr. Anal. Chem.* 2016;75:1–39.
- [6] Orlandini S, Pinzauti S, Furlanetto S. Application of quality by design to the development of analytical separation methods. *Anal Bioanal Chem* 2013;405(2–3):443–50.
- [7] Gupta K. Analytical quality by design: a mini review. *Biomed J Sci Tech Res* 2017;1(6):1–5.
- [8] Patel H, Parmar S, Patel B. A comprehensive review on quality by design (QbD) in pharmaceuticals. *Development* 2013;4:5.
- [9] Ramalingam P, Jahnavi B. QbD considerations for analytical development. *Pharmaceutical quality by design*. Academic Press; 2019. p. 77–108.
- [10] Kumar VP, Gupta NV. A review on quality by design approach (QbD) for pharmaceuticals. *Int J Drug Dev Res* 2015;7:52–60.
- [11] Guideline IH. Guidance for industry: Q8 (R2), pharmaceutical development. In: International Conference on harmonisation of technical requirements for registration of pharmaceuticals for human use, EMA/CHMP/ICH/167068/2004, European Medicines Agency; 2009.
- [12] Swarbrick J. In: Nash RA, Wachter AH, editors. *Pharmaceutical process validation*. New York, NY: Marcel Dekker; 2003.
- [13] Kakodkar S, Sharmada S. Pharmaceutical quality-by-design (QbD): basic principles. *Int J Res Methodol* 2015;1(1).
- [14] Guideline IH. Quality risk management Q9. In: International conference on harmonisation of technical requirements for registration of pharmaceuticals for human use; 2005 Nov 9.
- [15] Guideline IH. Q10: Pharmaceutical quality system. In: International conference of harmonization; 2008 Jun.
- [16] Guideline IH. Guidance for industry: Q8 (R2), pharmaceutical development. In: International conference on harmonisation of technical requirements for registration of pharmaceuticals for

- human use (ICH), US Department of Health and Human Services, Food and Drug Administration, Rockville, MD, 2009.
- [17] Borman P, Chatfield M, Nethercote P, Thompson D, Truman K. The application of quality by design to analytical methods. *Pharm Tech* 2007;31(12):142–52.
 - [18] Chatterjee S. QbD considerations for analytical methods – FDA perspective. In: US IFPAC annual meeting; 2013 Jan 25.
 - [19] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharm Rev* 2016;14(10):61–6.
 - [20] Vogt FG, Kord AS. Development of quality-by-design analytical methods. *J Pharm Sci* 2011;100(3):797–812.
 - [21] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
 - [22] Charde MS, Welankiwar AS, Kumar J. Method development by liquid chromatography with validation. *Int J Pharm Chem* 2014;4(2):57–61.
 - [23] Sood S, Bala R, Gill NS. Method development and validation using HPLC technique – a review. *J Drug Discov Ther* 2014;2(19):23–9.
 - [24] Singh PK, Pande M, Singh LK, Tripathi RB. Steps to be considered during method development and validation for analysis of residual solvents by gas chromatography. *Int Res J Pharm App Sci* 2013;3(5):74.
 - [25] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharm Times* 2014;46(8):71–5.
 - [26] Sun M, Liu DQ, Kord AS. A systematic method development strategy for determination of pharmaceutical genotoxic impurities. *Org Process Res Dev* 2010;14(4):977–85.
 - [27] Schweitzer M, Pohl M, Hanna-Brown M, Nethercote P, Borman P, Hansen G, et al. Implications and opportunities of applying QbD principles to analytical measurements. *Pharm Technol* 2010;34(2):52–9.
 - [28] Li Y, Terfloth GJ, Kord AS. HPLC-A systematic approach to RP-HPLC method development in a pharmaceutical QbD environment. *Am Pharm Rev* 2009;12(4):87.
 - [29] Zhou L, Vogt FG, Overstreet PA, Dougherty JT, Clawson JS, Kord AS. A systematic method development strategy for quantitative color measurement in drug substances, starting materials, and synthetic intermediates. *J Pharm Innov* 2011;6(4):217–31.
 - [30] Graul TW, Barnett KL, Bale SJ, Gill I, Hanna-Brown M. Quality by design for analytical methods. *Chem Eng Pharm Ind R&D Manuf* 2010;543–62.
 - [31] Molnár I, Rieger HJ, Monks KE. Aspects of the “design space” in high pressure liquid chromatography method development. *J Chromatogr A* 2010;1217(19):3193–200.
 - [32] Peraman R, Bhadrara K, Padmanabha Reddy Y. Analytical quality by design: a tool for regulatory flexibility and robust analytics. *Int J Anal Chem* 2015;2015.
 - [33] Reid GL, Cheng G, Fortin DT, Harwood JW, Morgado JE, Wang J, et al. Reversed-phase liquid chromatographic method development in an analytical quality by design framework. *J Liq Chromatogr Relat Technol* 2013;36(18):2612–38.
 - [34] Beg S, Rahman M, Kohli K. Quality-by-design approach as a systematic tool for the development of nanopharmaceutical products. *Drug Discov Today* 2019;24(3):717–25.
 - [35] Panda SS, Beg S, Kumar BVVR, Sahu J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.
 - [36] Guideline IH. Validation of analytical procedures: text and methodology Q2 (R1). In: International conference on harmonization, Geneva; 2005 Nov 10, pp. 11–12.

- [37] Box GE, Hunter JS. The $2^k - p$ fractional factorial designs. *Technometrics* 1961;3(3):311–51.
- [38] Singh B, Raza K, Beg S. Developing “optimized” drug products employing “designed” experiments. *Chem Ind Dig* 2013;23:70–6.
- [39] Borman PJ, Chatfield MJ, Damjanov I, Jackson P. Design and analysis of method equivalence studies. *Anal Chem* 2009;81(24):9849–57.
- [40] Chatfield MJ, Borman PJ. Acceptance criteria for method equivalency assessments. *Anal Chem* 2009;81(24):9841–8.
- [41] Kubeczka KH, Formáček V. Essential oils analysis by capillary gas chromatography and carbon-13 NMR spectroscopy. John Wiley & Sons Ltd; 2002.
- [42] Robu S, Romila A, Buzia OD, Spac AF, Diaconu C, Tutunaru D, et al. Contribution to the optimization of a gas chromatographic method by QbD approach used for analysis of essential oils from *salvia officinalis*. *Management* 2019;23:24.
- [43] Grodowska K, Parczewski A. Analytical methods for residual solvents determination in pharmaceutical products. *Acta Pol Pharm* 2010;67(1):13–26.
- [44] Piepel G, Pasquini B, Cooley S, Heredia-Langner A, Orlandini S, Furlanetto S. Mixture-process variable approach to optimize a microemulsion electrokinetic chromatography method for the quality control of a nutraceutical based on coenzyme Q10. *Talanta* 2012;97:73–82.
- [45] Debrus B, Lebrun P, Kindenge JM, Lecomte F, Ceccato A, Caliaro G, et al. Innovative high-performance liquid chromatography method development for the screening of 19 antimalarial drugs based on a generic approach, using design of experiments, independent component analysis and design space. *J Chromatogr A* 2011;1218(31):5205–15.
- [46] Gavin PF, Olsen BA. A quality by design approach to impurity method development for atomoxetine hydrochloride (LY139603). *J Pharm Biomed Anal* 2008;46(3):431–41.
- [47] Furlanetto S, Orlandini S, Giannini I, Beretta G, Pinzauti S. Pitfalls and success of experimental design in the development of a mixed MEKC method for the analysis of budesonide and its impurities. *Electrophoresis* 2009;30(4):633–43.
- [48] Orlandini S, Gotti R, Giannini I, Pasquini B, Furlanetto S. Development of a capillary electrophoresis method for the assay of ramipril and its impurities: an issue of cis–trans isomerization. *J Chromatogr A* 2011;1218(18):2611–17.
- [49] Bhardwaj SK, Dwivedi K, Agarwal DD. A review: GC method development and validation. *Int J Anal Bioanal Chem* 2016;6(1):1–7.
- [50] Wilde KD, Engewald W. *Practical gas chromatography: A comprehensive*. Springer; 2014.
- [51] Godswill NN, Frank NE, Edson MY, Emmanuel Y, Martin BJ, Hermine NB, et al. GC-FID method development and validation parameters for analysis of palm oil (*Elaeis guineensis* Jacq.) fatty acids composition. *Res Plant Sci* 2014;2(3):53–66.
- [52] Barry EF, Grob RL. *Columns for gas chromatography: performance and selection*. John Wiley & Sons; 2007.
- [53] Chauhan A, Goyal MK, Chauhan P. GC-MS technique and its analytical applications in science and technology. *J Anal Bioanal Tech* 2014;5(6):222.
- [54] Ong R, Marriott P, Morrison P, Haglund P. Influence of chromatographic conditions on separation in comprehensive gas chromatography. *J Chromatogr A* 2002;962(1–2):135–52.
- [55] Kupiec T. Quality-control analytical methods: gas chromatography. *Int J Pharm Compounding* 2004;8:305–9.
- [56] Klee MS, Blumberg LM. Theoretical and practical aspects of fast gas chromatography and method translation. *J Chromatogr Sci* 2002;40(5):234–47.

- [57] Barwick VJ. Sources of uncertainty in gas chromatography and high-performance liquid chromatography. *J Chromatogr A* 1999;849(1):13–33.
- [58] Shrivastava A, Gupta VB. HPLC: isocratic or gradient elution and assessment of linearity in analytical methods. *J Adv Sci Res* 2012;3(2).
- [59] Kumar V, Bharadwaj R, Gupta G, Kumar S. An overview on HPLC method development, optimization and validation process for drug analysis. *Pharm Chem J* 2015;2(2):30–40.
- [60] Dubey SK, Duddelly S, Jangala H, Saha RN. Rapid and sensitive reverse-phase high-performance liquid chromatography method for estimation of Ketorolac in pharmaceuticals using weighted regression. *Indian J Pharm Sci* 2013;75(1):89–93. Available from: <https://doi.org/10.4103/0250-474X.113535>.
- [61] Krishna KV, Saha N, Puri A. Pre-clinical compartmental pharmacokinetic modeling of 2-[1-hexyloxyethyl]-2-devinyl pyropheophorbide-a (HPPH) as a photosensitizer in rat plasma by validated HPLC method. *Photochem Photobiol Sci* 2019;18:1056–63. Available from: <https://doi.org/10.1039/c8pp00339d>.
- [62] Krishna KV, Saha N. Hydrochloride by a validated HPLC method. *RSC Adv* 2018;8:24740–9. Available from: <https://doi.org/10.1039/c8ra03379j>.
- [63] Pradhan R, Krishna KV, Wadhwa G, Taliyan R, Khadgawat R, Kachhawa G, et al. QbD-driven development and validation of HPLC method for determination of Bisphenol A and Bis-sulphone in environmental samples. *Int J Environ Anal Chem* 2020;100:42–54. Available from: <https://doi.org/10.1080/03067319.2019.1629585>.
- [64] Dubey SK, Saha RN, Jangala H, Pasha S. Rapid sensitive validated UPLC-MS method for determination of venlafaxine and its metabolite in rat plasma: application to pharmacokinetic study. *J Pharm Anal* 2013;3:466–71. Available from: <https://doi.org/10.1016/j.jpha.2013.05.002>.

Chapter 4

Analytical quality by design for size-exclusion chromatography

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4.1 Introduction

Size-exclusion chromatography (SEC) is a kind of liquid chromatography technique that is usually used for the analysis of polymers or polymeric matrix. The principle behind the separation process involved in SEC method involves partial exclusion of the molecules, based on the size, from the apertures of the immobile phase [1]. SEC is described as one of the ancient instrumental technique used for chromatography and as per the literature, firstly in 1956 the SEC chromatograms were documented by Lathe and Ruthven [2]. Studies have demonstrated that due to lenience and compatibility with almost all types of separation techniques employed in the primary dimension [3], the SEC technique is used as the most common separation method in the secondary dimension [4]. In 1959, the discovery of Sephadex (from Separation Pharmacia Dextran), comprised of dextran cross-linked with epichlorohydrin, enhanced the mechanical strength of the system [5]. Afterward, the new technique named as gel filtration chromatography (GFC) was introduced with the modified system of SEC. This modified technique helped to perform various tasks including desalting of protein solution, purification of protein matrixes, and identification of molecular weight (M_w) of aqueous polymers [6]. Furthermore, this technique was introduced into non-aqueous solutions, primarily for the determination of average M_w of organic polymers [7]. Such technique was named as gel permeation chromatography (GPC) by Moore. In general, both the techniques GFC and GPC are based on the principle of separation and have been recognized as SEC techniques. Moreover, there have been several challenges that originate during the

processing of this technique. The use of experimental designs for optimizing the various variables and processing conditions during SEC analysis could significantly improve the technique. As per the ICH Q8 guidelines [8], the quality by design (QbD) could be recognized as a systematic approach for the development, which begins with the predefined purposes and highlights the understanding of product, processes, and process controls, based on the thorough science and quality risk management [9–11]. Furthermore, the analytical quality by design (A-QbD) approaches includes the concept of QbD into the establishment of various analytical methods. The A-QbD approaches have been used for the development of pharmaceutical/biotech processes and include risk analysis assessment at initial stages for the identification of crucial process variables, followed by the establishment of design of experiment (DoE) [12,13]. Although reported literatures have demonstrated various QbD-based process developments [14–16], yet the applications of A-QbD-based approaches are recent [17,18].

This review mainly emphasizes on the application of A-QbD on SEC techniques. In this chapter, we have discussed the various types of A-QbDs, including Box–Behnken design (BBD), Central composite design (CCD), D-optimal design (DOMD), Full factorial design (FFD), and miscellaneous, and their significant importance in establishing SEC-based strategies for optimal production of pharmaceuticals or biologicals.

4.2 Application of various analytical quality by designs in size-exclusion chromatography

4.2.1 Box–Behnken design

Gulati et al. [19] isolated various strains of *Thermomyces lanuginosus* (*T. lanuginosus*), from composting soils, which when were grown over phytase screening medium (PSM) medium produced phytase (E.C.3.1.3.8, inositol hexaphosphate phosphohydrolase). The isolation and production of phytase from various strains were optimized by using BBD, in which inoculum level (X1), concentration of sodium nitrate (X2), and inoculum age (X3) were chosen as variables against the response (enzyme activity). The strains were purified in a two-step process including high-performance anion-exchange chromatography (HPAEC) and GFC. The optimized *T. lanuginosus* strain CM (strain type of *T. lanuginosus*) comprised of wheat bran and also enhanced the phytase yield comparatively more than the other strains. Further, the phytase activity of mutant (TL-7) was comparatively more (~2.29-fold) than that of the parental *T. lanuginosus* strain CM. Also, the phytase exhibited broad substrate specificity with improved activity against adenosine diphosphate, sodium phytate, and riboflavin phosphate. Awotwe-Otoo et al. [20] evaluated the effects of various variables and their interactions over the quality of IgG3 κ monoclonal antibody (mAb), by QbD approaches including BBD

and fractional factorial design (FD). Optimization results revealed that the most vital factors that affected the selected responses (glass transition temperature, mAb concentration, unfolding transition temperature, and particle size) were pH, NaCl, and Polysorbate 20. Furthermore, the results of moisture content, SEC, protein A analysis, and SDS-PAGE of the optimized formulation showed the existence of an intact protein structure with negligible aggregation. Zhong et al. [21] extracted mung polysaccharide (PS) from mung bean hulls via microwave extraction technique (MEPS) and optimized the requisite extraction parameters through BBD, including three extraction variables: microwave power (X1), time required for extraction (X2), and water-to-raw material ratio (X3). Further, two purified PS fractions, MEPS-1 (mainly composed of mannose (MAN) and GAL) and MEPS-2 [composed of rhamnose (RHAM) and GAL], were purified using diethylaminoethyl (DEAE-52) cellulose chromatography and SEC. The radical scavenging effects of MEPS and purified fractions exhibited greater radical scavenging activities, though MEPS-2 exhibited the strongest. Hence, the mung bean hulls and purified fractions of polysaccharides (PSs) might be used as functional foods with potent antioxidant activity. Ren et al. [22] examined the culture conditions that were essential for the efficient yield of PSs, isolated from the *Paecilomyces cicadae* (*P. cicadae*) Samson, over the solid-state fermentation using Plackett–Burman design (PBD) to screen out the important variables and BBD to screen out the final levels of the culture conditions. The optimized PSs were purified via chromatography column and were composed of mainly MAN, RHAM, XYL, and ARA. The results of SEC-multiangle laser light scattering (SEC-MALLS) analysis showed that the optimized PSs having average M_w of 3.756106 g/mol adopted a Gaussian coil conformation in 0.1 M sodium nitrate solution. Moreover, DPPH (1, 1-diphenyl-2-picryldrazyl), hydroxyl, and superoxide radicals assay revealed that the PSs exhibited strong antioxidant activities; thus the PSs of *P. cicadae* Samson could be used as natural antioxidants. Mune Mune and Minka [23] produced cowpea protein hydrolysate (CPH), optimized through BBD, and also examined the effect of optimized variables (time, solid-to-liquid ratio and enzyme-to-substrate ratio) over degree of hydrolysis and nitrogen solubility (NS). For NS of 75.71%, the values of hydrolysis time, solid-to-liquid ratio, and enzyme-to-substrate ratio at the optimized conditions were found to be 208.61 minutes, 1/15 (w/w), and 2.25% (w/w), respectively. Furthermore, the peptide profile and protein breakdown followed by enzymatic hydrolysis were estimated by SDS-PAGE and SEC. Also, CPH exhibited greater oil holding ability, emulsifying action, and foaming capacity as compared to the concentrate, and thus could be used for food purposes. Liu et al. [24] extracted PSs from defatted peanut (*Arachis hypogaea*) cake and optimized the parameters for isolation using BBD with three independent attributes, including temperature and time for extraction, and ratio of water-to-raw material. Moreover, the

extraction rate increased initially and then gradually decreased with the increased content of water: raw material ratio, which might be due to the increased amounts of solvent molecules causing gelatinization. The crude defatted peanut cake was purified by SEC and the purified fraction was further characterized by Fourier transform-infrared spectroscopy (FTIR), SEC-MALLS, and HPAEC. Results showed that glucose (GLU), galactose (GAL), arabinose (ARA), and xylose (XYL) were found to be the major composition of the PSs. Chen et al. [25] isolated a novel PS from the shoots of bamboo (*Chimonobambusa quadrangularis*) using ultrasonic-assisted extraction methods and further purified the same via SEC technique. Further, the parameters associated with the isolation PSs were assessed using BBD with four independent variables including ultrasonic power (X1), ratio of water-to-raw material X2, temperature for extraction (X3), and time required for extraction (X4). Results showed that the maximum yield of crude PSs (8.76%) was acquired with the optimal conditions of X1, X2, X3, and X4 as water to 240 W, 20.2 mL/g, 49°C, and 40 minutes, respectively. SEC results showed that the nature of PSs, as per its Mw distribution, was of a polydispersed hetero-polysaccharides. Also, the PSs exhibited admirable antioxidant activities; hence PSs could be used as alternative additives for applications in pharmaceuticals and functional foods. Mzoughi et al. [26] isolated PSs from *Suaeda fruticosa* (*S. fruticosa*) through ultrasonication approach and optimized the conditions using BBD. The optimization process involved isolation temperature (X1), isolation time (X2) and medium (citric acid) pH (X3) as variables, whereas total antioxidant capacity (Y_{TAP}) and PS yield (Y_{PS}) were taken as response. The purification of the PSs was performed using SEC, which showed that PS isolated from *S. fruticosa* had an average Mw of 240 kDa. Furthermore, the results of chemical composition assays revealed that the isolated PSs was a pectin (PT)-like PS and was composed of uronic acid and neutral monosaccharides including GLU, GAL, ARA, XYL, MAN, and RHAM. Moreover, the optimized PSs exhibited good antioxidant as well as antiinflammatory activity and also showed peripheral and central antinociceptive actions. Zhang et al. [27] synthesized sulfated derivatives of galactomannan isolated from fenugreek gums and optimized the reaction conditions by using BBD. A high degree of substitution and the reaction time were found to be most important parameters, although were interdependent. Analysis of compounds showed that it consisted of sulfate group. The Mw of the sulfated derivatives were identified through SEC-MALLS. It was observed that the electrostatic interface of the sulfated groups (negatively charged) and the Mw could play significant role in the biological actions of the derived compounds. Chen et al. [28] isolated polysaccharides from *Mytilus coruscus* (MCPSS) through ultrasonic-mediated enzymatic method, optimized the isolation parameters by BBD, and finally evaluated the antioxidative properties of MCPSS. Moreover, ultrasonication power, liquid-to-constituent ratio, isolation time

and temperature, enzyme gradient, and PSs extraction amount were the factors that were essential for efficient extraction of MCPSSs. The M_w of the novel PSs were estimated by high-performance GPC. Furthermore, the results of high-pressure liquid chromatography (HPLC) showed that the novel isolated MCPSSs comprised of MAN, RHAM, glucuronic acid, GLU, and GAL, and exhibited potent antioxidative properties, thus could be used as natural antioxidants. Dara et al. [29] prepared erythropoietin (EP)-conjugated solid lipid nanoparticles (EP-SLNs) and were optimized using BBD. During optimization amount of surfactant (oil phase; A), EP amount (volume of internal aqueous phase; B), and homogenization time (C) were considered as independent variables, whereas particle size, polydispersity index (PDI), and % encapsulation efficiency (%EE) of EP were selected as dependent variables. The optimized formulation exhibited PS (280 nm), PDI (0.282), and %EE (43.4%). The results circular dichroism, SEC, SDS-PAGE, and ELISA tests showed that the EP was stable in formulation. Furthermore, results of in vitro MTT ((3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide) assay showed that the optimized SLN exhibited no cytotoxicity against human foreskin fibroblast cells (Hu02, IBRC C10309). Also, in vivo studies demonstrated that EP-SLNs effectively elevated the level of RBC, hemoglobin, and hematocrit tested in Albino male Wistar rats in a dose-dependent manner. Dammak et al. [30] extracted the PSs from the quince peels (QPPSs) using ultrasonication, optimized the extraction parameters using BBD, evaluated the significant effects of pH, extraction time and temperature, and also evaluated the antioxidative and antiproliferative activities of the QPPSs. Results of the FTIR, $^1\text{H-NMR}$ (proton nuclear magnetic resonance), and SEC-MALLS showed that existence of monosaccharides, including ARA, GAL, GLU, MAN, and XYL. Moreover, the QPPSs showed efficient antioxidant activity and improved the antiproliferative activity against Caco-2 and B-16 cancerous cells and thus could be used in tumor therapy, effectively as a potent functional food.

4.2.2 Central composite design

Duarte and Duarte [31] developed a novel approach based on optimization technique for establishing a modified SEC method for assessing the M_w of the natural organic matters (NOMs). Based on the CCD and the chromatographic response attributes (CRAs), the approach was designed for efficiently determining the M_w of NOMs. Various SEC parameters, including firmness of the chromatograms, total number of distinct peaks, and analysis time, were optimized. Moreover, the CRAs were assessed, relating the significant effects of pH, organic solvent amount, and concentration of salt of the mobile phase. Singh et al. [32] extracted the enzyme chitinase (CN), which produced B-CM18 strains (named as *Lysinibacillus fusiformis* B-CM18), from the chickpea rhizosphere by using CCD with considering temperature,

NaCl concentration, starch amount, yeast, and chitin extracts as the responsible parameters for maximum CN extraction. In vitro studies showed that the extracts exhibited significant antifungal activities against a broad variety of fungal plant-originated pathogens. Furthermore, the HPAEC and GFC studies revealed that the M_w of the extracted CN was found to be 20 kDa and these studies provided a platform to estimate the potential of CN against pathogens that affects the chickpea production. Galante et al. [33] purified the CN, isolated from *Moniliophthora perniciosa*, with ammonium sulfate, filtered using sodium phosphate (buffer medium), and optimized the isolation conditions such as pH and temperature, using CCD. Four different iso-enzymes named as ChitMp I, ChitMp II, ChitMp III, and ChitMp IV were acquired and their M_w were evaluated using GFC (Sephacryl S-200). Furthermore, the in silico studies were performed and were afterward validated using Procheck 3.0 and ANOLEA. Sathiyannarayanan et al. [34] produced the biopolymer, polyhydroxybutyrates (PHBs), with the help of *Bacillus megaterium* MSBN04 (a marine sponge-mediated bacteria) and optimized the production conditions using CCD. The significant effect of the independent attributes, including tapioca industrial wastes, palm jaggeries, horse gram flour (HGF; source for nitrogen), and solution of trace element, was assessed for the production of PHBs. The HGF and solution of trace elements were found to be the vital attributes for effective synthesis of PHBs. Results of FTIR, $^1\text{H-NMR}$, and GPC showed that the resultant was a PHB monomer, having a high M_w and low PDI value and was further used to develop PHB-mediated polymeric nanoparticles. Maharjan et al. [35] optimized the codon and soluble expression of the transmembrane receptor activator of nuclear factor (NF)-B ligand (mRANKL) in *Escherichia coli* (*E. coli*) using CCD. During optimization, the effect of various variables, including cellular density before initiation, concentration of lactose, postinitiation temperature, and postinitiation time for the expression of mRANK, were observed. Furthermore, gel electrophoresis and GFC were performed to endorse the presence of trimeric form of mRANKL. Also, the tartrate-resistant acid phosphatase assay and real-time polymerase chain reaction studies revealed that the mRANKL induced the formation of osteoclast over RAW264.7 cells. Alshammari et al. [36] extracted ovotransferrin (egg protein) from egg white, taken from two different sources (domestic and poultry) and optimized the extraction condition [percent of ethanol (v/v), pH and volume (mL) of 25 mM ferric chloride/50 mL of egg white] using CCD. In the optimal conditions, a maximum production ($\sim 85\% \pm 2.5\%$) of ovotransferrin was achieved, its purity was evaluated using SEC, and its quantification was assessed through SDS-PAGE. Thus these results provided a base for efficient production of ovotransferrin through downstream process, could be cost-effective, and could be used potentially for antimicrobial applications. Abid et al. [37] employed CCD for determining the effect of extraction parameters (extraction time, pH, and temperature) over the yield and chemical configuration of PT produced from

the peels of pomegranates. In the optimized conditions the yields ranged between 6.4% and $11.0\% \pm 0.2\%$. Additionally, with the variation in extraction conditions, the degree of methylation for PTs (low methylated) also varied. Results of high-pressure SEC (HP-SEC) showed that the optimization conditions for elution of the acid-isolated PTs were associated with the low hydrodynamic volume. Li et al. [38] extracted the PSs from *Gynura medica* (GMPSs) and optimized the extraction parameters (temperature, time, and water-to-raw material ratio) using CCD. The yield of GMPSs was found to be maximum (5.66%) with the influence of optimized conditions. The optimized GMPSs (GMPS-1) were further purified using GPC. It was observed that the GMPS-1 had a M_w of 401 kDa, mainly comprised of galacturonic acid, XYL, and GLU, and was found to be an acidic PS (FTIR results). Also, the GMPSs and GMPS-1 significantly exhibited antioxidative properties, inhibited α -glucosidase, and thus could be used in medications or functional foods. Du et al. [39] isolated greater amount of PSs (dextran) from Chinese sauerkraut juice in the presence of *Leuconostoc mesenteroides* TDS2-19 strain and the specific effects of variables, including sucrose composition, concentration of sodium acetate, and pH values, were estimated and optimized using CCD. The optimization helped to enhance the production of dextran up to 71.23 ± 2.25 g/L within 48 hours of fermentation. The M_w of the isolates was found to be 8.79×10^7 Da, as estimated HP-SEC. Results of the FTIR and NMR (nuclear magnetic resonance) studies showed the existence of dextran (linear structured with no branching) as the PS that was isolated in the presence of *L. mesenteroides* TDS2-19, and thus the bacterium could be efficiently used for the large-scale production of dextran (linear). Karbasian et al. [40] investigated efficacy of poly(ethylene) glycol (PEG) and conserved bioactivity of recombinant human growth hormone (rhGH). Moreover, the experimentation was optimized with the help of CCD, taking 6.73 molar ratio of PEG: protein and pH 7.71 as main factors. Furthermore, the purity of PEG-rhGH was assessed through SEC and SDS-PAGE studies. In vitro results revealed that the Nb2-11 cells were able to proliferate up to 48 hours; however, the rate of proliferation decreased with an increased concentration of PEG-rhGH. Also, the in vivo studies showed that the half-life of the serum was prolonged for the PEG-rhGH as compared to unmodified rhGH.

4.2.3 D-optimal design

Gruendling et al. [41] optimized the factors (ionization efficacy, fragmentation amount, and development of salt adducts) essential for the establishment of synthetic polymers using DOMD and evaluated their ionization efficacy through coupled SEC-electrospray ionization-mass spectrometry (SEC-ESI-MS) technique. Results showed that the ionization efficacy could be increased without conceding the smoothness of the ionization method. Thus DOMD provided support for determining the optimal conditions for

ionization and could be efficiently used to improve the evaluation the polymeric conjugates using the SEC-ESI-MS. Yu et al. [42] developed and optimized metronidazole-loaded nanoemulsions (MNZ-NE) and gel using DOMD for improved topical targeting. Oil (Labrafil), surfactant mixture (Smix; tetraethylene glycol: Cremophor EL, 1:2 w/w), and amount of water were the three crucial variables, whereas the skin retention, cumulative drug concentration after 24 hours, and globule size were chosen as three effective responses. The results of in vitro studies revealed that after 24 hours of treatment, the drug penetrated into the deeper tissues of skin and distributed significantly in the epidermal and/or dermal layers. Furthermore, in vivo studies in the rat model showed that as compared to gel, the targetability of the drug was improved more when administered topically as MNZ-NE. Thirugnanasambandham and Sivakumar [43] extracted the PT from lime bagasse, optimized using DOMD, and evaluated the effect of different extraction parameters, including microwave intensity, time of extraction, pH, temperature, and quantity of sample over the PT yield. It was observed that all the parameters played a crucial role in the effective yield of PT and under the optimal conditions, the yield of PT was 7.8 g/100 g. Furthermore, the *M_w* of PT and its derivatives were assessed through GFC. Ramezani et al. [44] developed immunoglobulin G (IgG)-loaded formulation using spray-drying technique, optimized the crucial parameters using DOMD, and also assessed the effects of hydroxy-propyl beta-cyclodextrin (HP- β -CD), trehalose (THL), and β -CD on the particle features and stability of these inhalable formulations. The amount of IgG aggregates was determined by SEC. Results showed that the presence of HP- β -CD significantly improved the aerodynamic activity and stability of the particles more as compared to β -CD; however, THL produced inapt aerodynamic properties and also reduced IgG accumulation.

4.2.4 Full factorial design

Yusa et al. [45] developed an apt and automated technique to determine the existence of the amount of polycyclic aromatic hydrocarbons (PAHs) in aerial matters. During this method, FFD (2^4) was used to optimize conditions of the large-volume programmable temperature vaporizer injections attached with gas chromatography-mass spectroscopy. This critical process involved isolation of PM₁₀ (particles having aerodynamic diameter < 10 μ m)-bound PAHs through accelerated solvent extraction trailed by GPC. As compared to the conventional methods, the modified method showed an increase in sensitivity. Pourjavadi et al. [46] synthesized cationic salep [S-g-P(AL-co-MAPTAC)] as novel flocculants via graft copolymerization of acrylamide (AL) and [3-(methacryloylamino)propyl] trimethylammonium chloride (MAPTAC) over salep (a multicomponent PS isolated from dried tubers of few terrestrial herbs orchids). FFD was utilized to optimize the various

reaction variables, including ratio of monomers and salep, reactant concentration, ratio of AL and MAPTAC, for attaining the utmost settling rate of the cemented suspensions. Furthermore, the optimized product was evaluated for its characteristics by FTIR, GPC, thermogravimetric analysis, and ^1H -NMR. Results showed that the optimized products could be used as a potential tool for the green fabrication of cement yields. Radmanovic et al. [47] developed a physico-mathematical model to envisage the temperature of the products isolated through the freezing process, an input factor usually used for freeze-thaw modules commercially. The significant effects of the temperature (-5°C to -30°C) over the various factors responsible for effective product development were evaluated using FFD. Moreover, the product M_w and properties were assessed by SEC and PDI of products by dynamic light scattering. Results showed that the model mAb solution comprising cryoprotectants and nonionic surfactant was considerably less affected by freeze-thaw process than the mAb solution prepared using buffers only. Rahman and Gagnon [48] conducted a series of experiments and evaluated the effects of the parameters, including pH, phosphates, chlorine, and dissolved organic matters, over the features of the iron (Fe) particulates and suspensions produced synthetically through the oxidation of ferrous (Fe^{2+}) ions in the sodium bicarbonate (NaHCO_3) buffer systems. A 2^4 -FFD was used to evaluate the significance of the factors responsible for coloration and turbidity of Fe suspension systems. The phosphate concentration was found to be the most effective factor that altered the coloration and turbidity profile. The studies revealed that the concentration of chlorine and phosphate, pH of system, and dissolved organic matter (DOM) affected the zeta potential, average particle size, and solubility of suspension systems. Also, DOM increased the coloration, however, decreased the turbidity and particle size distribution of the suspension system. HP-SEC results suggested that the higher M_w fractions of DOM had significant effects over the surface charge of the Fe particulate system. Chavez et al. [49] demonstrated the importance of DoE)-FFD (2^4), for optimizing the production and storing conditions of a model antibody (IgG3). Ultraviolet-visible spectroscopy was used for assessing the stability of the IgG3 in each varying buffer formulations and SEC-HPLC was used for determining the complete solubility and aggregation of particles. Results of the preliminary studies revealed that the acetate buffer was inappropriate as a potent storage buffer as there were clear signs of precipitation. Furthermore, it was observed that arginine (Arg) enhanced the solubility of IgG3 and also the stability of the system was increased when Arg buffer was combined with histidine buffer at a pH 6.5 condition.

4.2.5 Miscellaneous

Li et al. [50] isolated the papain from the latex of *Carica papaya*, purified it through spray-drying technique with aqueous two-phase system. Finally, the

papain was restored from the PEG phase by in situ immobilization or developing a cross-linked enzyme aggregate. Furthermore, the aqueous two-phase system processing factors were optimized by PBD and CCD. A higher papain yield was achieved through in situ enzyme immobilization method. Moreover, the purity of the purified papain was assessed using ion-exchange chromatography (IEC) and the total yield was determined using SDS-PAGE. Aranganathan et al. [51] demonstrated that an intracellular azoreductase (enzyme) was isolated from the strain *Pseudomonas oleovorans* PAMD_1, which reduced the azo bridge all over the process of azo dye decolorization. The optimization of the azoreductase expression was preliminarily screened using PBD and later on the major factors such as amount of azo dye, NADH (Nicotinamide adenine dinucleotide-reduced form), GLU, and peptone were screened through CCD. Furthermore, the purification of azoreductase was increased to 16% yield, determined by HPAEC, and the M_w of the purified azoreductase was determined through SDS-PAGE technique. Maiser et al. [52] demonstrated a usual method, which was combined with the simple PEGylation reactions and high-throughput experimentation. During this process, QbD approach was used to determine the influence of the various PEGylation parameters for lysozyme (a model protein). SEC method was used to analyze the reaction kinetics and effect of PEG, pH (buffer), and reaction time. Additionally, the mono-PEG-lysozyme analysis showed the influence of varying pH (buffer) over the isoform propagation. Kamarudin et al. [53] extracted the keratin from the chicken feather and optimized the extraction conditions using multistatistical designs (fractional FD and CCD). The various factors, including temperature, medium pH, reducing agent ratio, chicken feather mass, and time of incubation, were assessed for effective keratin production. SEC and SDS-PAGE were used to determine the yield of keratin in the total isolated protein, whereas the chemical compositions were estimated using FTIR analysis. Results showed that pH played a crucial role and was found to be the most significant factor for maximum keratin yield. Yu and Xu [54] produced pectinase (enzyme) from *Bacillus* sp. ZJ1407, optimized the extraction conditions using 2^2 -FFD and CCD simultaneously, and examined its enzymatic activities. The 2^2 -FFD was used to optimize the significant effects of the various factors, whereas CCD was used to optimize the concentration of the screened factors. Results of the optimization revealed that the lactose, tryptone, and ammonium sulfate were the most significant factors that influenced the activity of enzyme. The enzyme was purified using DEAE-cellulose-IEC and Sephadex G-100 SEC, and the M_w of the purified enzyme was estimated by SDS-PAGE. Moreover, the isolated enzymes were highly stable in a pH range of 3.0–5.0 and exhibited greater thermal stability within 80°C–90°C. Du et al. [55] demonstrated that *L. mesenteroides* DRP2–19 strain increased the production of glucanase (enzymes of high M_w that are responsible for the production of exo-PSs, which usually used in food industries). Initially, PBD was performed to

optimize the amount of sucrose, Ca^{2+} , and pH for glucansucrase yield. Subsequently, the significant effects of the factors responsible for the activity of glucansucrase were examined by CCD. Furthermore, ammonium sulfate precipitation, gel/ion-exchange chromatography (GEC/IEC) were used to purify the crude enzyme and the M_w of the glucansucrase was estimated by SDS-PAGE. Gianelli et al. [56] developed a method for quantification of the Vi-PS (a homopolymer of α -1,4-*N*-acetylgalactosaminouronic acid) based on the acidic hydrolysis with the simultaneous usage of hydrochloric and trifluoroacetic acids. The hydrolysis conditions were identified with the help of DoE. The optimized method was found to be 100-fold more delicate than the earlier reported method and produced definite monomers (de-O- and de-N-acetylated) of Vi-PS. The quantification of the isolated monomers was assessed through HPAEC, and thus this method would enable as a significant tool for the characterization and specification of Vi-based vaccines.

4.3 Conclusion and future perspectives

The SEC-based approaches for the quantification of products or by-products have shown various positive aspects, including suitability, simple, vigorous, and repeatability. Moreover, studies have shown that the SEC method has been frequently applied for size fractionation during the purification of proteins. The SEC technique is utilized for the estimation of M_w of various PSs, proteins, and protein conjugates, and also acts as a robust tool for polymer characterization; however, this method faces some issues during the process. Thus to overcome the issues associated with SEC-based processes, A-QbD has been introduced simultaneously during or within the SEC method for achieving the optimal conditions. In recent years, the pharmaceutical/biotech industries and their regulatory agencies have functioned collectively to infer the strategies for A-QbD execution in SEC-based methods. In past the key perceptions of SEC have been extensively recognized; however, in future most of the manufacturers would show significant interest in incorporating A-QbD into the different work procedures during SEC-based analysis for superior results.

Conflicts of interest

The authors declare no conflicts of interest.

References

- [1] Striegel AM, Kirkland JJ, Yau WW, Bly DD. Modern size exclusion liquid chromatography. 2nd ed. New York: Wiley; 2009. p. 512.
- [2] Lathe GH, Ruthven CR. The separation of substances and estimation of their relative molecular sizes by the use of columns of starch in water. *Biochem J* 1956;62(4):665–74.

- [3] van der Horst A, Schoenmakers PJ. Comprehensive two-dimensional liquid chromatography of polymers. *J Chromatogr A* 2003;1000(1–2):693–709.
- [4] Uliyanchenko E, van der Wal S, Schoenmakers PJ. Challenges in polymer analysis by liquid chromatography. *Polym Chem* 2012;3(9):2313.
- [5] Porath J, Flodin P. Gel filtration: a method for desalting and group separation. *Nature* 1959;183(4676):1657–9.
- [6] Hjerten S. Chromatographic separation according to size of macromolecules and cell particles on columns of agarose suspensions. *Arch Biochem Biophys* 1962;99:466–75.
- [7] Moore JC. Gel permeation chromatography. I. A new method for molecular weight distribution of high polymers. *J Polym Sci A Gen Pap* 1964;2(2):835–43.
- [8] ICH Harmonised Tripartite Guideline Q8 (R1). Pharmaceutical development. In: International conference on harmonization of technical requirements for the registration of pharmaceuticals for human use. Geneva. 2008.
- [9] Nayak A, Khatua S, Hasnain M, Sen K. Development of diclofenac sodium-loaded alginate-PVP K 30 microbeads using central composite design. *DARU* 2011;19(5):356–66.
- [10] Nayak AK, Ahmed SA, Beg S, Tabish M, Hasnain MS. Application of quality by design for the development of biopharmaceuticals. 2019. p. 399–411.
- [11] Rathore AS, Winkle H. Quality by design for biopharmaceuticals. *Nat Biotechnol* 2009;27(1):26–34.
- [12] Das SS, Singh A, Kar S, Ghosh R, Pal M, Fatima M, et al. Application of QbD framework for development of self-emulsifying drug delivery systems. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design: principles and applications*. Elsevier; 2019. p. 297–350.
- [13] Hasnain MS, Javed MN, Alam MS, Rishishwar P, Rishishwar S, Ali S, et al. Purple heart plant leaves extract-mediated silver nanoparticle synthesis: optimization by Box-Behnken design. *Mater Sci Eng C Mater Biol Appl* 2019;99:1105–14.
- [14] Ali H, Singh SK. Preparation and characterization of solid lipid nanoparticles of furosemide using quality by design. *Particul Sci Technol* 2017;36(6):695–709.
- [15] Kumar S, Singh SK. In silico-in vitro-in vivo studies of experimentally designed carvedilol loaded silk fibroin-casein nanoparticles using physiological based pharmacokinetic model. *Int J Biol Macromol* 2017;96:403–20.
- [16] Verma S, Singh SK, Verma PR. Fabrication of lipidic nanocarriers of loratadine for facilitated intestinal permeation using multivariate design approach. *Drug Dev Ind Pharm* 2016;42(2):288–306.
- [17] Pathak M, Dutta D, Rathore A. Analytical QbD: development of a native gel electrophoresis method for measurement of monoclonal antibody aggregates. *Electrophoresis* 2014;35(15):2163–71.
- [18] Vogt FG, Kord AS. Development of quality-by-design analytical methods. *J Pharm Sci* 2011;100(3):797–812.
- [19] Gulati HK, Chadha BS, Saini HS. Production, purification and characterization of thermostable phytase from thermophilic fungus *Thermomyces lanuginosus* TL-7. *Acta Microbiol Immunol Hung* 2007;54(2):121–38.
- [20] Awotwe-Otoo D, Agarabi C, Wu GK, Casey E, Read E, Lute S, et al. Quality by design: impact of formulation variables and their interactions on quality attributes of a lyophilized monoclonal antibody. *Int J Pharm* 2012;438(1–2):167–75.
- [21] Zhong K, Lin W, Wang Q, Zhou S. Extraction and radicals scavenging activity of polysaccharides with microwave extraction from mung bean hulls. *Int J Biol Macromol* 2012;51(4):612–17.

- [22] Ren X, He L, Cheng J, Chang J. Optimization of the solid-state fermentation and properties of a polysaccharide from *Paecilomyces cicadae* (Miquel) Samson and its antioxidant activities in vitro. *PLoS One* 2014;9(2):e87578.
- [23] Mune Mune MA, Minka SR. Production and characterization of cowpea protein hydrolysate with optimum nitrogen solubility by enzymatic hydrolysis using pepsin. *J Sci Food Agric* 2017;97(8):2561–8.
- [24] Liu H, Jiang N, Liu L, Sheng X, Shi A, Hu H, et al. Extraction, purification and primary characterization of polysaccharides from defatted peanut (*Arachis hypogaea*) cakes. *Molecules* 2016;21(6).
- [25] Chen G, Bu F, Chen X, Li C, Wang S, Kan J. Ultrasonic extraction, structural characterization, physicochemical properties and antioxidant activities of polysaccharides from bamboo shoots (*Chimonobambusa quadrangularis*) processing by-products. *Int J Biol Macromol* 2018;112:656–66.
- [26] Mzoughi Z, Abdelhamid A, Rihouey C, Le Cerf D, Bouraoui A, Majdoub H. Optimized extraction of pectin-like polysaccharide from *Suaeda fruticosa* leaves: characterization, antioxidant, anti-inflammatory and analgesic activities. *Carbohydr Polym* 2018;185:127–37.
- [27] Zhang Z, Wang H, Chen T, Zhang H, Liang J, Kong W, et al. Synthesis and structure characterization of sulfated galactomannan from fenugreek gum. *Int J Biol Macromol* 2019;125:1184–91.
- [28] Chen W, Jia Z, Zhu J, Zou Y, Huang G, Hong Y. Optimization of ultrasonic-assisted enzymatic extraction of polysaccharides from thick-shell mussel (*Mytilus coruscus*) and their antioxidant activities. *Int J Biol Macromol* 2019;140:1116–25.
- [29] Dara T, Vatanara A, Nabi Meybodi M, Vakilinezhad MA, Malvajerd SS, Vakhshiteh F, et al. Erythropoietin-loaded solid lipid nanoparticles: preparation, optimization, and in vivo evaluation. *Colloids Surf B Biointerfaces* 2019;178:307–16.
- [30] Dammak MI, Mzoughi Z, Chakroun I, Mansour HB, Le Cerf D, Majdoub H. Optimization of polysaccharides extraction from quince peels: partial characterization, antioxidant and antiproliferative properties. *Nat Prod Res* 2020;34(10):1470–4.
- [31] Duarte RM, Duarte AC. Optimizing size-exclusion chromatographic conditions using a composite objective function and chemometric tools: application to natural organic matter profiling. *Anal Chim Acta* 2011;688(1):90–8.
- [32] Singh RK, Kumar DP, Solanki MK, Singh P, Srivastva AK, Kumar S, et al. Optimization of media components for chitinase production by chickpea rhizosphere associated *Lysinibacillus fusiformis* B-CM18. *J Basic Microbiol* 2013;53(5):451–60.
- [33] Galante RS, Taranto AG, Koblitz MG, Goes-Neto A, Pirovani CP, Cascardo JC, et al. Purification, characterization and structural determination of chitinases produced by *Moniliophthora perniciosa*. *An Acad Bras Cienc* 2012;84(2):469–86.
- [34] Sathiyarayanan G, Kiran GS, Selvin J, Saibaba G. Optimization of polyhydroxybutyrate production by marine *Bacillus megaterium* MSBN04 under solid state culture. *Int J Biol Macromol* 2013;60:253–61.
- [35] Maharjan S, Singh B, Bok JD, Kim JJ, Jiang T, Cho CS, et al. Exploring codon optimization and response surface methodology to express biologically active transmembrane RANKL in *E. coli*. *PLoS One* 2014;9(5):e96259.
- [36] Alshammari EM, Khan S, Jawed A, Adnan M, Khan M, Nabi G, et al. Optimization of extraction parameters for enhanced production of ovotransferrin from egg white for antimicrobial applications. *BioMed Res Int* 2015;2015:934512.

- [37] Abid M, Renard CM, Watrelot AA, Fendri I, Attia H, Ayadi MA. Yield and composition of pectin extracted from Tunisian pomegranate peel. *Int J Biol Macromol* 2016;93 (Pt A):186–94.
- [38] Li F, Gao J, Xue F, Yu X, Shao T. Extraction optimization, purification and physicochemical properties of polysaccharides from *Gynura medica*. *Molecules* 2016;21(4):397.
- [39] Du R, Xing H, Yang Y, Jiang H, Zhou Z, Han Y. Optimization, purification and structural characterization of a dextran produced by *L. mesenteroides* isolated from Chinese sauerkraut. *Carbohydr Polym* 2017;174:409–16.
- [40] Karbasian M, Kouchakzadeh H, Anamaghi PN, Sefidbakht Y. Design, development and evaluation of PEGylated rhGH with preserving its bioactivity at highest level after modification. *Int J Pharm* 2019;557:9–17.
- [41] Gruendling T, Guilhaus M, Barner-Kowollik C. Design of Experiment (DoE) as a tool for the optimization of source conditions in SEC-ESI-MS of functional synthetic polymers synthesized via ATRP. *Macromol Rapid Commun* 2009;30(8):589–97.
- [42] Yu M, Ma H, Lei M, Li N, Tan F. In vitro/in vivo characterization of nanoemulsion formulation of metronidazole with improved skin targeting and anti-rosacea properties. *Eur J Pharma Biopharm* 2014;88(1):92–103.
- [43] Thirugnanasambandham K, Sivakumar V. Application of D-optimal design to extract the pectin from lime bagasse using microwave green irradiation. *Int J Biol Macromol* 2015;72:1351–7.
- [44] Ramezani V, Vatanara A, Seyedabadi M, Nabi Meibodi M, Fanaei H. Application of cyclodextrins in antibody microparticles: potentials for antibody protection in spray drying. *Drug Dev Ind Pharm* 2017;43(7):1103–11.
- [45] Yusa V, Quintas G, Pardo O, Pastor A, Guardia Mde L. Determination of PAHs in airborne particles by accelerated solvent extraction and large-volume injection-gas chromatography-mass spectrometry. *Talanta* 2006;69(4):807–15.
- [46] Pourjavadi A, Fakoorpoor SM, Hosseini SH. Novel cationic-modified salep as an efficient flocculating agent for settling of cement slurries. *Carbohydr Polym* 2013;93(2):506–11.
- [47] Radmanovic N, Serno T, Joerg S, Germershaus O. Understanding the freezing of biopharmaceuticals: first-principle modeling of the process and evaluation of its effect on product quality. *J Pharm Sci* 2013;102(8):2495–507.
- [48] Rahman MS, Gagnon GA. Bench-scale evaluation of drinking water treatment parameters on iron particles and water quality. *Water Res* 2014;48:137–47.
- [49] Chavez BK, Agarabi CD, Read EK, Boyne 2nd MT, Khan MA, Brorson KA. Improved stability of a model IgG3 by DoE-based evaluation of buffer formulations. *BioMed Res Int* 2016;2016:2074149.
- [50] Li M, Su E, You P, Gong X, Sun M, Xu D, et al. Purification and in situ immobilization of papain with aqueous two-phase system. *PLoS One* 2010;5(12):e15168.
- [51] Aranganathan V, Kanimozhi AM, Palvannan T. Statistical optimization of synthetic azo dye (orange II) degradation by azoreductase from *Pseudomonas oleovorans* PAMD_1. *Prep Biochem Biotechnol* 2013;43(7):649–67.
- [52] Maiser B, Dismar F, Hubbuch J. Optimization of random PEGylation reactions by means of high throughput screening. *Biotechnol Bioeng* 2014;111(1):104–14.
- [53] Kamarudin NB, Sharma S, Gupta A, Kee CG, Chik S, Gupta R. Statistical investigation of extraction parameters of keratin from chicken feather using Design-Expert. *3 Biotech* 2017;7(2):127.
- [54] Yu P, Xu C. Production optimization, purification and characterization of a heat-tolerant acidic pectinase from *Bacillus* sp. ZJ1407. *Int J Biol Macromol* 2018;108:972–80.

- [55] Du R, Zhao F, Pan L, Han Y, Xiao H, Zhou Z. Optimization and purification of glucansucrase produced by *Leuconostoc mesenteroides* DRP2-19 isolated from Chinese Sauerkraut. *Prep Biochem Biotechnol* 2018;48(6):465–73.
- [56] Giannelli C, Raso MM, Palmieri E, De Felice A, Pippi F, Micoli F. Development of a specific and sensitive HPAEC-PAD method for quantification of Vi polysaccharide applicable to other polysaccharides containing amino uronic acids. *Anal Chem* 2020;92(9):6304–11.

Chapter 5

Analytical quality by design for liquid chromatographic method development

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5.1 Introduction

The liquid chromatography method development involves tedious operations pertaining to initial screening of the method parameters to identify the critical method parameters having influence on the critical method variables. For years, the traditional approach of method development has been practiced for liquid chromatography method development and their optimization using the traditional hit and trial methods or using the short-gun approaches such as one factor at a time or changing one separate variable at one time [1,2]. However, such approaches have proven limited worth with respect to the huge investment of time, efforts, and resources used for the purpose.

In this regard, use of systematic analytical development methodology by applying the principles of quality by design (QbD) has recently been promoted by International Conference on Harmonization (ICH), United States Food and Drug Administration, and many other regulatory agencies [3,4]. ICH guidances (Q8, Q9, and Q10) are specifically instituted for the QbD in the last one decade, while very recently in 2019, ICH has published Q14 guidance that specifically emphasizes on the analytical life cycle management wherein various chromatographic techniques are outlined. Thus on the heels of QbD, the analytical development should primarily focus on the rational method development practices involving steps such as defining the method objectives, utility and importance of the method, and measures required for improving the robustness [5–7].

The analytical QbD (also popularly referred to as AQbD) has been quite extensively employed in the field of liquid chromatographic method

development. Nevertheless, AQbD application provides immense benefits such as reducing the method development time, ease of method transfer, and more comprehensive understanding on the troubleshooting, thus improving validity of the method utility during the entire product life cycle [8,9].

5.2 Analytical quality by design overview

In comparison to the traditional method development practice, AQbD has been highly beneficial for a variety of liquid chromatographic methods. As pharmaceutical industries primarily bank on liquid chromatographic methods for quality control analysis of the drug substances, impurities, and degradation products; thus need for a highly robust and validated chromatographic method is very essential. In this context, AQbD is highly useful for analysis of drug(s) in dosage forms, biological matrices, and stability samples too [8,9].

As already covered in several literature reports, AQbD emphasizes on science and risk-based analytical method development where method target profile is defined, critical analytical attributes and critical method parameters are identified, followed by establishment of analytical design space with suitable control strategy for continuous improvement throughout the method life cycle [10,11]. A holistic AQbD exercise requires thorough understanding of the steps for attaining method development excellence and compliance. Various steps involved in AQbD-based method development are illustrated in Fig. 5.1, while AQbD applications to various liquid chromatographic methods have been portrayed in Fig. 5.2.

5.2.1 Selection of method variables and response variables

After initiation of method development activities as per the method objectives, the input variables (method factors) and output variables (method responses) are identified. It is very essential to select the actual factors that critically influence the method responses for meaningful AQbD exercise [6]. The said exercise is carried out either by prioritization on the basis of prior literature knowledge or prior experience, while risk estimation exercise also helps in identifying the potential risk factors. Some of the very popular techniques include risk estimation matrix, risk comparison matrix, failure mode and effect analysis, etc., helps in identifying the high risk factors. ICH Q9 guidance is highly useful for understanding the risk assessment techniques and their applications in rational identification of the high risk factors. Moreover, Table 5.1 illustrates a list of commonly encountered high risk factors in various liquid chromatographic methods.

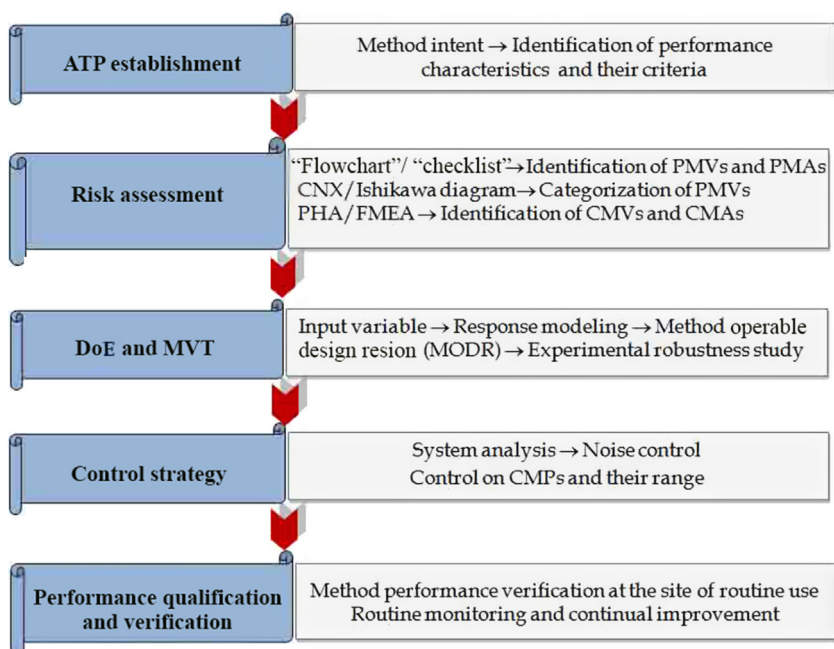


FIGURE 5.1 Salient steps involved in implementation of analytical quality by design approach. *ATP*, analytical target profile; *CMAs*, critical method attributes; *CMPs*, critical method parameters; *CMVs*, critical method variables; *CNX*, control noise experimental; *DoE*, design of experiments; *FMEA*, failure mode and effect analysis; *MVT*, multivariate techniques; *PMAs*, possible method attributes; *PMVs*, possible method variables. Adapted from Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.

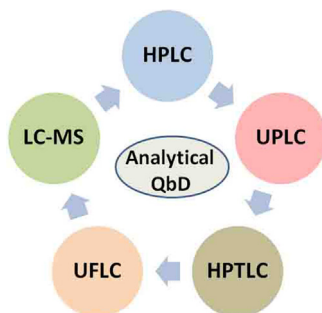


FIGURE 5.2 Applications of AQbD approach in various liquid chromatographic method developments. *AQbD*, Analytical quality by design; *HPLC*, High-performance liquid chromatography; *UPLC*, ultra-performance liquid chromatography; *QbD*, quality by design; *UFLC*, ultra-fast liquid chromatography; *HPTLC*, high-performance thin-layer chromatography.

TABLE 5.1 List of commonly encountered input and output variables in liquid chromatographic methods.

Technique(s)	Input variables	Output variables
HPLC/UPLC/ LC-MS	Mobile phase ratio, organic modifiers, pH, buffer strength, flow rate, injection volume, column type, column dimension, sonication time, column oven temperature	Peak area, retention time, theoretical plate count, asymmetry factor, capacity factor, peak resolution, relative retention time, % assay
HPTLC	Mobile phase ratio, saturation time, volume loaded, scanning speed, slit dimension, distance between tracks, migration distance, spot diameter, stationary phase, plate dimension	Peak area, retardation factor, asymmetry factor, capacity factor, peak resolution, % assay

HPLC, high-performance liquid chromatography; HPTLC, high-performance thin-layer chromatography; LC-MS, Liquid chromatography-mass spectroscopy; UPLC, ultra-performance liquid chromatography.

5.2.2 Optimization of the method factors using chemometric tools

The method factors are optimized using various chemometric tools such as multivariate statistical techniques and experimental designs. Examples of the multivariate statistical techniques include principal component analysis, partial least square analysis, cluster analysis, etc., useful in understanding the variability, while examples of experimental designs include full factorial design, central composite design, Box–Behnken design, optimal design, etc. [12]. These designs are primarily based on the multiple linear regression analysis and require mathematical modeling with the help of linear, quadratic, cubic, and quartic fitting tools for establishing factor–response relationship, which are graphically interpreted with the help of 2D contour graphs and 3D response surface maps. The nature 2D and 3D plots depend on the type of mathematical model fitting and statistical validity of the model. Besides, model diagnostic plots are also used for analyzing the impact of method variables on the method responses [13].

5.2.3 Optimum criteria demarcation in the design space

The optimum solutions obtained after experimental design–based analysis are analyzed and demarcated in the analytical design space, which provides enormous flexibility to work in a controlled environment without having any significant impact or possibility for out-of-trend results. Initially, numerical

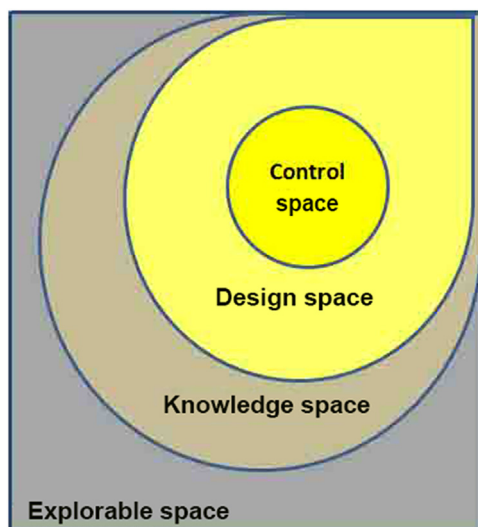


FIGURE 5.3 Pictorial depiction of various types of spaces encountered during analytical quality by design approach.

optimization and desirability function tools are used for identifying an optimum solution, which is later demarcated in the optimum design space region with boundaries defined according to the target ranges for the method factors [14]. Fig. 5.3 illustrates a typical design space used for localizing the optimal solutions after experimental designs—based product and process optimization.

5.3 Analytical quality by design application to liquid chromatographic methods

A wide variety of applications of AQbD to liquid chromatographic methods have been explored, which are now highlighted in the following subsections and also reported in Table 5.2.

5.3.1 High-performance liquid chromatography

High-performance liquid chromatography (HPLC) methods are very sought after in pharmaceutical industries for analysis of the drug products, impurities, degradation products, and active metabolites. However, such methods involve several factors and require critical monitoring of the instrumental parameters and method attributes, thus attaining good method performance. As per the AQbD-based development approach, HPLC methods are developed with the help of initial factor screening study to select the right variables and subjected to factor optimization. Several HPLC methods have been

TABLE 5.2 Analytical quality by design-based development of liquid chromatographic methods for various applications.

Drug	Technique	Designs and models	Pharmaceutical application(s)	References
Ropinirole	HPLC	Box–Behnken design	Analytical	[15]
Lapatinib	HPLC	Box–Behnken design	Analytical	[16]
Lacidipine	HPLC	Box–Behnken design	Analytical	[17]
Valsartan	HPLC	Box–Behnken design	Analytical	[18]
QuercetinSalicin	HPLC	Box–Behnken design	Analytical	[19]
	HPTLC			
Methotrexate	HPLC	Fractional factorial design	Analytical	[20]
		Central composite design		
Olmesartan medoxomil	HPLC	Box–Behnken design	Analytical	[21]
Rosuvastatin calcium	HPLC	Plackett–Burman design	Bioanalytical	[22]
		Box–Behnken design		
Sertaconazole	HPLC	Fractional factorial design	Analytical	[23]
		Box–Behnken design		
Nevirapine	HPLC	Fractional factorial design	Bioanalytical	[24]
		Box–Behnken design		

Quercetin dihydrate	HPLC	Fractional factorial design	Bioanalytical	[25]
		Box–Behnken design		
Amoxicillin	HPLC	Box–Behnken design	Analytical	[26]
Tamoxifen citrate	HPLC	Taguchi design	Analytical	[27]
		Box–Behnken design		
Ketoprofen	HPLC	Taguchi design	Analytical	[28]
		Central composite design		
Mangiferin	HPTLC	Fractional factorial design	Analytical, bioanalytical	[29]
		Central composite design		
Olmesartan	UPLC	Box–Behnken design	Bioanalytical	[30]
Valsartan	UPLC	Full fractional design	Analytical	[31]
Docetaxel	UPLC	Box–Behnken design	Bioanalytical	[32]
Lansoprazole	UFLC	Central composite design	Analytical	[33]
Telaprevir	UFLC	Box–Behnken design	Analytical	[34]
Fluoxetine	LC-MS	Box–Behnken design	Bioanalytical	[35]
AceclofenacParacetamol	LC-MS	Box–Behnken design	Bioanalytical	[36]
HPLC, high-performance liquid chromatography; HPTLC, high-performance thin-layer chromatography; UFLC, ultra-fast liquid chromatography; UPLC, ultra-performance liquid chromatography; LC-MS, Liquid chromatography-mass spectroscopy.				

reported in the literature, compiled in [Table 5.2](#), which have been principally developed using AQbD approaches.

5.3.2 Ultra-performance liquid chromatography

Ultra-performance liquid chromatography (UPLC) methods are used over the HPLC methods, as they provide leverage of time, efforts, and resources saving. However, use of UPLC methods is quite tedious as compared to the HPLC methods owing to highly critical variables involved with the same. Like HPLC, such methods are also applicable in assay, impurity profiling, degradation product characterization, and estimation of drugs and their active metabolites in biological fluids. Thus AQbD application to these analytical methods requires detailed monitoring and analysis of the critical method variables. A list of instances on the applicability of UPLC methods of the drug in the literature is provided in [Table 5.2](#).

5.3.3 Ultra-fast liquid chromatography

Ultra-fast liquid chromatography (UFLC) methods have now been popularly used in pharmaceutical analysis, owing to their high efficiency and less time required for analysis. Thus it has applicability in estimation of the drugs, impurities, and metabolites. Application of AQbD tools for such UFLC methods is become very essential for optimizing the method performance. A score of literature reports are available on the AQbD use in the development of UFLC methods for diverse pharmaceutical applications, as summarized in [Table 5.2](#).

5.3.4 Liquid chromatography-mass spectroscopy

Liquid chromatography-mass spectroscopy (LC-MS) is primarily used for bioanalytical method development for detecting very minute quantities of the drugs and their metabolites in the biological samples, where HPLC and UPLC methods provide limited sensitivity and specificity, owing to critical biological sample extraction procedures, and monitoring of the instrumental conditions and method variables. Moreover, it becomes very difficult to control and optimize the method parameters, especially the instrumental chromatographic conditions, as very limited options are available for selecting the mobile phase that can ionize the analyte to support the fragmentation process. Hence, a score of literature reports are available on AQbD application for the development and optimization of the LC-MS methods for the analytical and bioanalytical applications, which are exclusively enlisted in [Table 5.2](#).

5.3.5 High-performance thin-layer chromatography

High-performance thin-layer chromatography (HPTLC) methods are primarily used in research and development purpose, and have fewer applications in the pharmaceutical industries. Such methods are useful for estimation of constituents from the multicomponent mixtures, which are very difficult with HPLC and UPLC methods. Although HPLC methods are quite robust as compared to the HPLC and UPLC, yet they require critical monitoring of the instrumental and analytical conditions for efficient method performance. In this regard, the applicability of AQbD for HPTLC method development is enlisted in [Table 5.2](#).

5.4 Conclusion

The availability of literature instances on AQbD-based development of diverse analytical methods is testimony to the fact for real-time benefits. Hence, the federal agencies are gearing up with guidances on AQbD for establishing a framework for holistic implementation of this concept for excellence in analytical development and regulatory compliance.

References

- [1] Beg S, Rahman M, Robaian MA, Alruwaili NK, Imam SS, Panda SK. Pharmaceutical drug product development and process optimization: effective use of quality by design. New York, NY: CRC Press; 2020.
- [2] Beg S, Hasnain MS. Pharmaceutical quality by design: principles and applications. New York, NY: Academic Press; 2019.
- [3] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [4] Singh B, Beg S. Attaining product development excellence and federal compliance employing quality by design (QbD) paradigms. *Pharma Rev* 2015;13(9):35–44.
- [5] Beg S, Hasnain MS, Rahman M, Swain S. Chapter 1 – introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. Academic Press; 2019. p. 1–17.
- [6] Beg S, Rahman M, Swain S. Quality by design applications in pharmaceutical product development. *Pharma Focus Asia* 2020;1–5.
- [7] Beg S, Rahman M, Kohli K. Quality-by-design approach as a systematic tool for the development of nanopharmaceutical products. *Drug Discov Today* 2019;24(3):717–25.
- [8] Beg S, Sharma T, Saini S, Kaur R, Kaur R, Singh B. Analytical quality by design for robust chromatographic methods. *Cutting Edge* 2020;10(2):9–17.
- [9] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.
- [10] Panda SS, Beg S, Ravi Kumar SJ. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.

- [11] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharma Rev* 2016;14(10):61–6.
- [12] Singh B, Raza K, Beg S. Developing “optimized” drug products employing “designed” experiments. *Chem Ind Dig* 2013;23:70–6.
- [13] Singh B, Sharma T, Saini S, Kaur R, Jain A, Raza K, et al. Systematic development of drug nanocargos using formulation by design (FbD): an updated overview. *Crit Rev Ther Drug Carr Syst* 2020;37:229–69 [Epub ahead of print].
- [14] Nayak AK, Ahmed SA, Beg S, Tabish M, Hasnain MS. Chapter 18 — application of quality by design for the development of biopharmaceuticals. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. Academic Press; 2019. p. 399–411.
- [15] Fatima S, Beg S, Samim M, Ahmad FJ. Application of chemometric approach for development and validation of high performance liquid chromatography method for estimation of ropinirole hydrochloride. *J Sep Sci* 2019;42(21):3293–301.
- [16] Panda SS, Bera RAVV, Behera AK, Beg S. Chemometrics-assisted development of a liquid chromatography method for estimation of lapatinib in tablets: a case study on a novel quality concept. *Sep Sci Plus* 2020;3(1–2):12–21.
- [17] Panda SS, Bera RKVV, Acharjya SK, Sahoo P, Beg S. Analytical lifecycle management approach: application to development of a reliable LC method for estimation of lacidipine. *Sep Sci Plus* 2019;2(1):18–25.
- [18] Bandopadhyay S, Beg S, Katare OP, Sharma T, Singh B. Integrated analytical quality by design (AQbD) approach for the development and validation of bioanalytical liquid chromatography method for estimation of valsartan. *J Chromatogr Sci* 2020; [Epub ahead of print].
- [19] Qadir A, Aqil M, Ali A, Ahmad FJ, Ahmed S, Beg S, et al. Comparative evaluation of the liquid chromatographic methods for simultaneous analysis of quercetin and salicin in an anti-psoriasis polyherbal formulation. *Sep Sci Plus* 2020;3(4):77–85.
- [20] Jain A, Beg S, Saini S, Sharma T, Katare OP, Singh B. Application of chemometric approach for QbD-enabled development and validation of an RP-HPLC method for estimation of methotrexate. *J Liq Chromatogr Relat Technol* 2019;42(15–16):502–12.
- [21] Beg S, Sharma G, Katare OP, Lohan S, Singh B. Development and validation of a stability-indicating liquid chromatographic method for estimating olmesartan medoxomil using quality by design. *J Chromatogr Sci* 2015;53(7):1048–59.
- [22] Beg S, Panda SS, Katare OP, Singh B. Applications of monte-carlo simulation and chemometric techniques for development of bioanalytical liquid chromatography method for estimation of rosuvastatin calcium. *J Liq Chromatogr Relat Technol* 2017;40(18): 907–20.
- [23] Panda SS, Beg S, Ravi Kumar VVB, Gain A. Spectrophotometric determination of sertaconazole nitrate in pharmaceutical dosage form using quality by design (QbD) framework. *J Bioanal Biomed* 2017;9:235–9.
- [24] Beg S, Chaudhary V, Sharma G, Garg B, Panda SS, Singh B. QbD-oriented development and validation of a bioanalytical method for nevirapine with enhanced liquid-liquid extraction and chromatographic separation. *Biomed Chromatogr* 2015;30(6):818–28.
- [25] Sandhu PS, Beg S, Kumar R, Katare OP, Singh B. Analytical QbD-based systematic bioanalytical HPLC method development for estimation of quercetin dihydrate. *J Liq Chromatogr Relat Technol* 2017;40(10):506–16.
- [26] Beg S, Kohli K, Swain S, Hasnain MS. Development and validation of RP-HPLC method for quantitation of amoxicillin trihydrate in bulk and pharmaceutical formulations using box-behnken experimental design. *J Liq Chromatogr Relat Technol* 2012;35(3):393–406.

- [27] Sandhu PS, Beg S, Katare OP, Singh B. QbD-driven development and validation of a HPLC method for estimation of tamoxifen citrate with improved performance. *J Chromatogr Sci* 2016;54(8):1373–84.
- [28] Yadav NK, Raghuvanshi A, Sharma G, Beg S, Katare OP, Nanda S. QbD-based development and validation of a stability-indicating HPLC method for estimating ketoprofen in bulk drug and proniosomal vesicular system. *J Chromatogr Sci* 2015;54(3):377–89.
- [29] Khurana RK, Rao S, Beg S, Katare OP, Singh B. Systematic development and validation of a thin-layer densitometric bioanalytical method for estimation of mangiferin employing analytical quality by design (AQbD) approach. *J Chromatogr Sci* 2016;54(5):829–41.
- [30] Beg S, Jain A, Kaur R, Panda SS, Katare OP, Singh B. QbD-driven development and validation of an efficient bioanalytical UPLC method for estimation of olmesartan medoxomil. *J Liq Chromatogr Relat Technol* 2016;39(13):587–97.
- [31] Krishnaiah C, Reddy AR, Kumar R, Mukkanti K. Stability-indicating UPLC method for determination of Valsartan and their degradation products in active pharmaceutical ingredient and pharmaceutical dosage forms. *J Pharm Biomed Anal* 2010;53(3):483–9.
- [32] Khurana RK, Beg S, Lal D, Katare OP, Singh B. Analytical quality by design approach for development of a validated bioanalytical UPLC method of docetaxel trihydrate. *Curr Pharm Anal* 2015;11(3):180–92.
- [33] Panda SS, Ravi Kumar Bera VV, Beg S, Mandal O. Analytical quality by design (AQbD)-oriented RP-UFLC method for quantification of lansoprazole with superior method robustness. *J Liq Chromatogr Relat Technol* 2017;40(9):479–85.
- [34] Panda SS, Sharma K, Mohanty B, Bera RKVV, Acharjya SK, Beg S. Integrated quality by design (QbD) and design of experiments (DoE) approach for UFLC determination of telaprevir in rat serum. *J Liq Chromatogr Relat Technol* 2017;40(19):951–8.
- [35] Hasnain MS, Siddiqui S, Rao S, Mohanty P, Ara TJ, Beg S. QbD-driven development and validation of a bioanalytical LC-MS method for quantification of fluoxetine in human plasma. *J Chromatogr Sci* 2016;54(5):736–43.
- [36] Bhatt P, Hasnain MS, Nayak AK, Hassan B, Beg S. Development and validation of QbD-driven bioanalytical LC-MS/MS method for the quantification of paracetamol and diclofenac in human plasma. *Anal Chem Lett* 2018;8(5):677–91.

Chapter 6

Analytical quality by design for high-performance thin-layer chromatography method development

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6.1 Introduction

Quality by design (QbD) is a scientific and systematic strategy of development having built-in objectives and focused on understanding process and product control based on scientific principles along with quality risk management. The QbD concepts are described in the International Conference on Harmonization (ICH) guidelines Q8 (R1): pharmaceutical development, Q9: quality risk management, and Q10: pharmaceutical quality system. ICH Q10 guideline involves the qualitative study of pharmaceuticals that should be present in all steps involved in manufacturing the pharmaceuticals to maintain quality and the principle of QbD follows the same [1,2]. The United States Food and Drug Administration (USFDA) issued one guidance in 2005, called “Quality Systems Approach to Pharmaceutical CGMP Regulations” which was focused on critical attributes related to chemistry, manufacturing, and control (CMC) and their significance on the effectiveness and safety of the pharmaceutical product. After this guidance, QbD became an essential part of the pharmaceutical industry for successful implementation in product development. Also, the QbD concepts applied to analytical methods is known as analytical quality by design (AQbD). Building quality in an analytical method is the main aim of the idea of AQbD. AQbD

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approach ensures the highest reliability and quality of the developed analytical method and reduces the failure risk during the validation phase along with routine analysis [1]. Hence, a cost-effective and robust analytical method is developed using the AQbD approach. The implementation of AQbD in method development begins with defining the analytical target profile (ATP) and further critical quality attributes (CQAs). This is followed by a justification of the selection of the proposed analytical method, risk assessment study, and method development comprising screening and optimization using design of experiment (DoE). This mainly includes validation of the model and finally method control strategy. AQbD-based analytical method is also having the advantage of regulatory flexibility from the official regulators in which one parameter of the method can be changed without performing any revalidation study [1,3].

For qualitative and quantitative analysis of pharmaceuticals/drug products, chromatographic techniques are widely used due to their advantages over other nonchromatographic approaches. Among different chromatographic techniques, high-performance thin-layer chromatography (HPTLC) is one of the efficient techniques for both qualitative and quantitative analysis. HPTLC is a sophisticated instrumental technique that is the advanced/modified form of the traditional thin-layer chromatography (TLC) in which the target molecule (drug or any other molecules) is separated from the complex mixtures through the principle of adsorption [4,5]. HPTLC has some key advantages over TLC and other chromatographic techniques, such as automation, high sensitivity, attachment of suitable detection technique (UV), ease in sample preparation, use of corrosive mobile phase, precoated HPTLC plates, and low consumption of solvent with a normal analytical grade. To develop an analytical method for HPTLC, different factors are responsible, such as the coating of the plate, coating thickness, mobile phase composition and ratio, solvent purity, the dimension of the developing chamber, time for saturation, volume of sample to be spotted, solvent level in the chamber, solvent flow rate, temperature, relative humidity, and separation distance [6,7]. Hence, method development for HPTLC becomes a tough job and the main concern is that the developed analytical method should work as per the intended use. Previously for developing an analytical method, the approach used is based on taking one parameter as variable and keeping other parameters constant for getting the required response. This particular practice mostly yields a narrowly robust method for the different instrumental variables utilized in the phase of method development. Hence, this approach of method development possesses a higher risk of failure of the method and always needs a revalidation step during the transfer of the method or during alternative method development. Hence it leads to an increase in the cost of the method as well as wastage of time. To minimize and solve these hindrances of HPTLC method development, the newer and recent approach, that is, AQbD is followed to achieve quality in method [4]. Some researchers have already successfully utilized the AQbD approach for chromatographic method development. This

chapter provides an outline of the QbD applications in the development of HPTLC, the essential concepts of AQbD, and various developmental stages of the AQbD-compliant HPTLC method.

6.2 Principle of quality by design

Quality by design is having a high significance in both academics and the pharmaceutical industry after its establishment by the USFDA. ICH defined QbD as “a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.” The two things playing an important role in both the understanding and implementation of QbD are “control strategy” and “design space (DS).” ICH has defined DS as “the multidimensional combination and interaction of input variables and process parameters that have been demonstrated to provide assurance of quality.” Generally, the DS involves identifying critical attributes for the input materials and the process along with the final product. The DS mainly includes the input factor or variables (single or more) for an analytical method. It helps assure the quality of the data produced by the method, whereas control strategy deals with the final quality of the product or method [8–10]. After modeling the DS, only the control strategy is defined.

6.3 Need for quality by design in high-performance thin-layer chromatography method development

In the current scenario, chromatographic techniques are the widely used analytical methods during all stages of the product life cycle. For any analytical method development, it requires thorough experimentation with lots of trials and errors, which helps to identify the different critical process parameters (CPPs) and their significant effect on method performance, such as precision, accuracy, and robustness [11]. After the development of the method, it is validated to ensure accuracy and reliability of the method and all this process is time-consuming and complicated. In current practice, method development is based on taking one parameter as variables such as the composition of the mobile phase. This one parameter undergoes optimization for the response (R_f), while other parameters remain constant then. This practice generally leads to a less robust developed method for different instrumental variables and operating parameters. Hence, this approach of method development possesses a high risk for failure of the method and always needs a revalidation step during a transfer of the method or during the development of an alternative method. Hence it leads to an increase in the cost of the method as well as wastage of time. Hence, to minimize time, complication, and most importantly the failure of the validation, QbD is implemented for analytical

method development [4,11–13]. The advantages of QbD in HPTLC method development are as follows:

1. A systematic approach with built-in objectives, method understanding with appropriate science and risk assessment to figure out the risk factors for the HPTLC method;
2. The method performance ensures that the method is robust and performs for the life of the product;
3. The method is more precise with better system suitability and more knowledge about method parameters;
4. Consideration of the study of the DoE with systematic and multivariate strategy;
5. Using the DoE, a DS along an operating space with knowledge of all the characteristics and limitations of method performance is generated;
6. Information of CPPs of the method (stationary phase, mobile phase, plate dimension, etc.) that affect the method performance;
7. Help to determine the failure modes.

6.4 Methodological aspects

After the implementation of QbD in product development, now it is gaining popularity in analytical method development. In AQbD-based HPTLC method development, the method is verified during the initial phase of method development to ensure method performance. For any QbD-based analytical method development, method development strategy (MDS) approach is used. The MDS approach generally begins with the goal of the method with a proper and systematic literature survey, defining ATPs and CQAs, the study of risk assessment, method screening, and optimization and finally control strategy [14]. The vital element of the MDS approach is the utilization of structured risk assessment tools, including the DoE for method optimization as well as evaluation of ruggedness and robustness of the method [15,16]. In every step of method development, the approach of MDS accentuates the application of principles and scientific understanding of QbD.

6.5 Implementation of quality by design in high-performance thin-layer chromatography

Implementation of the concepts of QbD to HPTLC is justifiable, due to several factors that significantly influence the method results. Such factors may be instrumental parameters or method/process parameters. Besides, HPTLC is one of the most common and frequently used chromatographic tools for analysis in pharmaceutical quality control. The number of factors involved in the phase of developing the HPTLC method is almost equivalent to the

number of factors present in product development. Hence implementation of QbD in HPTLC is essential. Implementation of QbD provides flexibility in regulatory approval but it needs a higher degree of method robustness and method quality along with method understanding [16–18]. Implementation of QbD in HPTLC method development would be parallel to that of other analytical and product-based QbD.

6.6 Statistical tools supporting high-performance thin-layer chromatography quality by design experimental design

Some well-established statistical analysis tools such as DoE, Measurement Systems Analysis (MSA), and other such statistical strategies have been employed to aid in the QbD-based HPTLC method development.

6.6.1 Analytical target profile

The ATP is an established set of criteria defining and determining what will ideally be measured (e.g., impurity level, degradants) as well as the performance criteria to be attained through the measurement. The analyst is generally involved in defining and documenting the performance requirements or the objectives of the HPTLC method at this stage. During the construction of ATP, the characteristics of the HPTLC method are identified, which show the indicators of method performance. After identification of the characteristics, the acceptance criteria for those characteristics are determined [4,19,20]. HPTLC methods may be designed and further developed for a variety of purposes such as qualitative and quantitative analysis, stability studies of the drug product or drug itself, and impurity profiling.

6.6.2 Critical quality attribute

As per ICH Q8 guidelines, CQA is defined as “a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality.” For HPTLC, the CQAs are mostly chromatographic plates, mobile phase, sample strength, the time required for plate development, color derivatizing agent, and mode of detection [4,19–22].

6.6.3 Risk assessment

After inspecting different methods, the best method is selected for risk assessment study that comprises identifying and systematically prioritizing the risks. The step of risk assessment recognizes the critical method factors and/or variables and the different parameters having an impact on the ATP. Different risks associated with QbD-based analytical method development

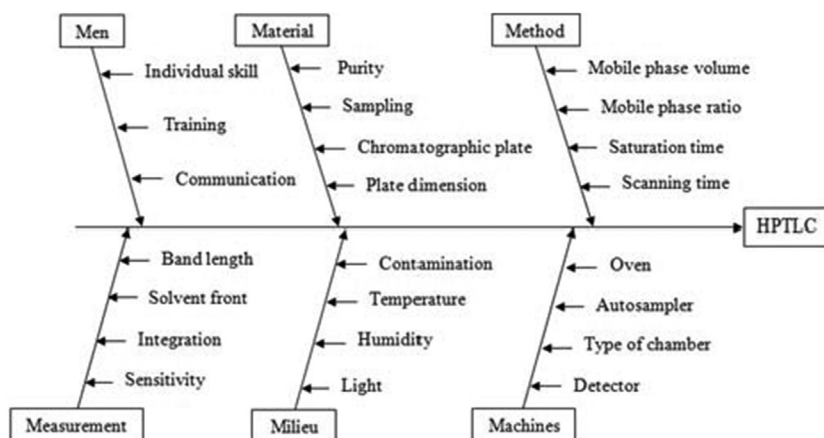


FIGURE 6.1 Typical fishbone diagram for HPTLC method development. *HPTLC*, high-performance thin-layer chromatography.

are methods of the analyst, the configuration of equipment, method parameters, properties of the sample and its preparation method, environmental conditions, and laboratories. ICH regarded risk assessment study as an essential part of QbD and provided certain guidelines (Q9) for this. For HPTLC, structured techniques and methodologies to assess risk may be utilized to meet the guidance from ICH. For a risk assessment study, different tools are used, such as failure modes and effects analysis, fishbone diagrams, and the prioritization matrix. Among these tools, the fishbone diagram, also known as Ishikawa or cause-and-effect diagram, is mostly used. A typical example of a fishbone diagram for HPTLC method development is shown in Fig. 6.1. Before performing the risk assessment study of any analytical method, a thorough literature survey is done regarding the physicochemical characteristics of the analyte. Then, the output of risk analysis can be used for generating DS or further analysis [3,4,19,20,22,23].

6.6.4 Design space

According to ICH, the DS is “the multidimensional combination and interaction of input variables and process parameters that have been demonstrated to provide assurance of quality.” The major objective of any analyst during method development is to establish the set of ranges for the factors/variables for desired efficacy. The DS may be established via interpreting the DoE results attained with a higher-order response surface design via response modeling. Graphically it may be presented as a multiresponse surface plot. In this plot, the contour plots for every response determined as a function of the factors under investigation as well as factor interactions are in an

overlapping manner. Thus the DS is an area existing inside of the limits of the domain of the experiment that is formed via a multivariate range of values of factors [3,18,24]. Care should be taken that the predefined criteria for critical material attributes (CMA) are met and the method is robust as well. For instance, in the HPTLC method, on a meticulous evaluation of all the parameters and their interactions in a method via statistical design, a design is generated utilizing the outcomes. This design is an amalgamation of the various ranges of the combination of the mobile phase, time for saturation, plate dimension rates, etc., responsible for results close to the set goals. A preliminary trial may be performed that has its basis in a thorough literature survey. Here the analyst obtains significant information for the definite method region in which any intentional alteration does not change the method performance. Hence any change inside the DS is not considered as any modification of the method, thus avoiding the need for any further regulatory approval. Thus DS is necessary to support the improved method flexibility required for regulatory approval. A DS for the particular characterization of the working performance of an analytical technique such as HPTLC is the method operable design region (MODR) [4,12,21,22].

Hence any HPTLC method operating at the set values of parameters inside the limits of MODR can give quality results with suitable probability. Furthermore, the HPTLC method is capable of being operated at any point in method parameters inside the MODR. Here all the sources of variability may be controlled utilizing an appropriate control strategy including a system suitability test. From the regulatory perspective, no revalidation of the HPTLC method is essential since any possible movement inside the MODR after the validation step is only an adjustment and not a change. Thus the MODR serves as an area representing the robustness of the HPTLC method providing flexibility to the HPTLC method while routine usage. Besides, an essential requirement of the regulatory body as per ICH Q8(R2) guideline¹³, is that the process parameters present inside the limits of DS should be able to assure quality. Thus on the creation of a DS, all the potential variations need to be accounted for that are arising from either the measurement of response or model uncertainty. This will ensure that the possibility, that the CMAs will be as per the predefined criteria, will be 99% to ensure appropriate method quality. A required level of assurance must be met that the specifications for CMA are present in the DS [3,4,12,17,19,21]. An example of DS is presented in Fig. 6.2.

6.6.5 Design of experiment

DoE is the part of QbD used for the optimization of the method. By using DoE, different factors during method development and their interactions are assessed with maximum efficiency [3,19]. The effects of these factors are

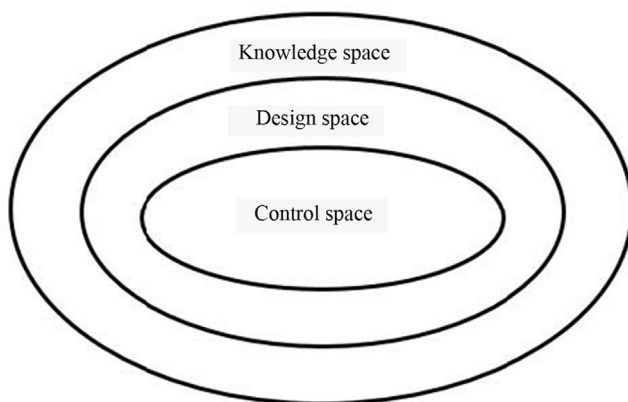


FIGURE 6.2 A schematic representation of design space.

mostly measured at varying levels and the results can be applied to a broad range of conditions. Different steps of DoE include the following:

6.6.5.1 Screening

In the first step of DoE, screening is done to screen out the different qualitative variables and to identify the different critical method parameters (CMPs) that can be used. In screening, the screening of qualitative input variables is conducted and the different CMPs for consideration, in the optimization experiments, are identified. The many tools and selection strategies and/or approaches used for screening are factorial design, central composite design, and D-optimal mixture design. These designs are used according to their uses, analyst requirements, the number of screening factors, ease of use, etc. Various CMPs are isolated as a result of screening, which should be controlled or optimized [25,26].

6.6.5.2 Optimization

Various CMPs from risk assessment and screening process are selected for optimization. During optimization, the relation between CMP and their output response is understood through scientific understanding and these responses from different CMP will show a considerable effect on the performance of the method. A mathematical relationship/model is established in the optimization stage. For optimization also, one DoE tool is selected and the selection is dependent on the number of input factors and/or variables, controlled parameters, response, etc. [3,19]. The most common experimental designs that can be used for HPTLC method development are Plackett–Burman design, Box–Behnken design, Full factorial design, Taguchi design, and Fractional factorial design. For instance, Plackett–Burman is used when a large number of variables are studied without interaction effect. Besides, Factorial design is generally used when the impact of

TABLE 6.1 A list of tools of design of experiment in analytical quality by design.

Design	Usage	Advantages	Disadvantages
Full factorial design	Optimization for 2–5 variables.	Effect of input variables on the method performance is directly determined.	Experimental run increases as per the number of variables.
Taguchi design	For optimization of more than 4 factor or factor levels.	Requires a smaller number of experimental runs.	Effects are not estimated well and resolving the confounding effects of interactions is challenging.
Plackett–Burman design	Appropriate for a substantial number of factors where even fractional factorial designs may be limited.	Requires very fewer runs for a large number of variables; hence used for preliminary experiments.	The design needs additional full factorial or fractional factorial design for optimization.
Box–Behnken design	Three levels of each factor (–1, 0, +1) are utilized.	Design points fall inside of the safe operating zone and the factors are not placed at their high levels all at the same time.	Two-factor design is not possible.

all the input variables and their interaction has to be measured. Also, the Taguchi method may be utilized with a lower number of experimental runs as opposed to factorial designs. A brief comparison of these designs is shown in [Table 6.1](#).

6.6.5.3 Surface plot

A surface plot is a 3D response plot that represents the influence of one variable on the various responses of the method. So different plots can be plotted between variables and responses. In HPTLC, the variables can be the volume of the mobile phase, the composition of the mobile phase, chromatographic plate, type of chamber, etc., and responses are band length, R_f value, etc. Variables are plotted in X-axis and the response is plotted in the Y-axis of the plot. Numbers such as –1, –2, +1, and +2 along both the axes signify

the coded level of variables employed in the DoE plan. In the DoE design, the variables are kept at different levels like -1 , -2 , 0 , $+1$, and $+2$. Besides the surface plot, a contour plot can be plotted, which is a 2D response and the plot is plotted between the same variable (X -axis) and response (Y -axis). The surface plot is generally used for linear relationships, whereas the contour plot is used for nonlinear relationships having a curvature effect [3,25–28].

6.6.5.4 *Model validation*

The predicted values of the method response obtained from the surface or contour plot have to be validated through actual experimentation. A graph is plotted between predicted values and actual values obtained through the experimental run. Subsequently, the regression analysis is done to statistically validate the method [3,25–28].

6.7 Method control strategy

The method control strategy is an essential feature and property of AQbD during method development since it ensures the performance of the method as per the intended use. From the risk assessment study, several factors are identified and considered during the implementation of the method control strategy. The method control strategy may be defined if the risks are low and thus these are easily managed. But if the risks are high and it is challenging and difficult to manage, then a more suitable and appropriate method is needed [14,18–20,22,29,30]. For example, Peter and Bernard utilized an approach based on QbD for the impurity method development (HPLC) of atomoxetine hydrochloride. During the robustness study, it was observed that the resolution of the selected impurities was following a similar trend when some method parameters such as temperature and n -propanol were varied. Therefore an early impurity pair was selected for the system suitability study and thus it became a proper method control strategy [24].

6.8 Validation and postmethod consideration

Method validation is an important process done after method development. It is used to confirm the suitability of the analytical procedure for its proposed use for specific tests. The results from the method validation may help to ensure the consistency and reliability of analytical results. After the establishment of QbD-based HPTLC method development, validation is done [3–5,21,22]. For method validation, there is certain guidance available from regulatory authorities such as USFDA guidelines and ICH Q2 guidelines [31]. Although validation is compulsory after method development but in the case of QbD-based method development, validation can be a formality because QbD-based method development follows proper and appropriate

MDS and risk assessment process. Since the method has been thoroughly developed and evaluated, issues with the method are unlikely to be appearing during the step of validation. Different validation parameters are system suitability testing, specificity, range, linearity, accuracy, precision, quantitation limit, detection limit, and robustness [32–36].

6.9 Implementation in current practice

In recent years, the use of AQbD is gaining popularity for analytical method development, mostly in HPLC [3,22,24,35,37,38]. Some researchers have already developed a QbD-based HPTLC method development [4,21]. But as QbD is a systematic and a lengthy procedure, implementation in current practice is crucial. So, a step-by-step protocol should be followed for HPTLC method development, which is as follows:

1. Construction of ATP and the HPTLC method should meet the requirement of ATP;
2. Risk assessment study using a suitable method;
3. Identification of the qualitative and quantitative variables affecting the method response and its performance;
4. Selection of suitable DoE tool to optimize the CMPs;
5. Establishing mathematical models to judge the robustness and economic operation for the method variable;
6. Validation of the models;
7. Method control strategy and improvement and finally validation of the method.

6.10 Regulatory consideration for the current and future scenario

As the principles and elements of QbD are based on ICH guidelines (Q8, Q9, Q10), it has regulatory importance [23,39,40]. In 2004 the concept of “QbD is a systematic approach to product and process design and development,” was accepted by the FDA. In 2005 USFDA launched a pilot program, that is, Office of New Drug Quality Assessment to provide the participating organizations an opportunity for the submission of CMC information representing QbD of new drug application. From January 2013 the USFDA has been accelerating the QbD drive by issuing warnings to generic manufacturers. The number of QbD containing applications has been increasing over the past 10 years [14,41,42]. In 2013 USFDA and European Medicines Agency made collaboration and the goals of that collaboration are:

1. Analytical method development (e.g., HPLC, GC, HPTLC) based on AQbD;
2. Defining the protocols for method transfer;

3. Establishing a methodology and/or procedure for the verification of the MODR on-site transfer;
4. Defining the review criteria for evaluating QbD-driven analytical methods.

6.11 Conclusion

QbD is majorly associated with the implementation of Q8 and Q9 through prominent employment of DoE to provide a multidimensional DS. The range of operating parameters for the HPTLC instrument is obtained through DoE. This goes a long way in validation along with a certain range of amounts and understanding variations in the preparation of the HPTLC sample. Thus the DS obtained by QbD in HPTLC helps to characterize any changes. Owing to the rising regulatory needs, DoE will prove to be a chief part of HPTLC method development and validation soon. Besides, the application of QbD in HPTLC is essential for comprehensive know-how of the development of the HPTLC method to the transfer. It includes the different factors displaying a significant impact on the method outcomes. These are the TLC plate, the volume of injection, selected mobile phase, the time required for the development of plate, detection method, and agent used to for the same. In conclusion, the HPTLC method should display robustness to facilitate use for a longer period along with a very low potential for failure. The HPTLC should be capable of being subjected to risk assessment, future amendments, and improvements that may be determined by suitable tests before releasing in analytical laboratories. In an ideal scenario, there has to be a complete emphasis on developing the HPTLC method and recognizing along with controlling the failures. Hence, QbD for HPTLC would serve as an indispensable association amongst HPTLC development and operational laboratories. Due to emerging complexity in the analytical and pharmaceutical domains, the importance of a well-constructed HPTLC technique applied with QbD will intensify.

Conflicts of interest

There are no conflicts of interest.

References

- [1] Ramalingam P, Jahnvi B. QbD considerations for analytical development. Elsevier; 2019. <<https://doi.org/10.1016/b978-0-12-815799-2.00005-8>>.
- [2] Beg S, Hasnain MS, Rahman M, Swain S. Introduction to quality by design (QbD): fundamentals, principles, and applications. Elsevier; 2019. <<https://doi.org/10.1016/b978-0-12-815799-2.00001-0>>.
- [3] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.

- [4] Gopani M, Patel RB, Patel MR, Solanki AB. Development of a new high-performance thin layer chromatographic method for quantitative estimation of lamivudine and zidovudine in combined tablet dosage form using quality by design approach. *J Liq Chromatogr Relat Technol* 2014;37:2420–32.
- [5] Shewiyo DH, Kaale E, Risha PG, Dejaegher B, Smeyers-Verbeke J, Vander Heyden Y. HPTLC methods to assay active ingredients in pharmaceutical formulations: a review of the method development and validation steps. *J Pharm Biomed Anal* 2012;66:11–23 <<https://doi.org/10.1016/j.jpba.2012.03.034>>.
- [6] Sonia K, Shree BB, Lakshmi KS. HPTLC method development and validation: an overview. *J Pharm Sci Res* 2017;9:652–7.
- [7] Attimarad M, Mueen Ahmed KK, Aldhubaib BE, Harsha S. High-performance thin layer chromatography: a powerful analytical technique in pharmaceutical drug discovery. *Pharm Methods* 2011;2:71–5.
- [8] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [9] Yu LX, Amidon G, Khan MA, Hoag SW, Polli J, Raju GK, et al. Understanding pharmaceutical quality by design. *AAPS J* 2014;16:771–83.
- [10] Singh B, Beg S. Attaining product development excellence and federal compliance employing quality by design (QbD) paradigms. *Pharma Rev* 2015;13(9):35–44.
- [11] Singh B, Beg S. Quality by design in product development life cycle. *Chron Pharmabiz* 2013;22:72–9.
- [12] Psimadas D, Georgoulas P, Valotassiou V, Loudos G. Molecular nanomedicine towards cancer. *J Pharm Sci* 2012;101:2271–80.
- [13] Siddhartha B, Sudheer Babu I. Analytical method development and method validation for the estimation of pantoprazole in tablet dosage form by RP-HPLC. *Der Pharma Chem* 2013;5:99–104.
- [14] Panda SS, Beg S, Kumar BVVR, Sahu J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.
- [15] Shaha M.M., Jagdale A.S., Patil P., Waghmare A. The impact of analytical quality by design (AQBD) in the method development of liquid chromatography for quality control of pharmaceuticals: a review. 2019;8:2561–74.
- [16] Sheladia S, Patel B. Implementation of QBD approach to develop and validate analytical method for simultaneous estimation of duloxetine hydrochloride and methylcobalamin in pharmaceutical dosage form by HPTLC method. *Int J Pharm Pharma Sci* 2016;8:105–13.
- [17] Khurana RK, Rao S, Beg S, Katore OP, Singh B. Systematic development and validation of a thin-layer densitometric bioanalytical method for estimation of mangiferin employing analytical quality by design (AQbD) approach. *J Chromatogr Sci* 2016;54(5):829–41.
- [18] Yuan Q., Li W., Regan L. Implementing quality by design in analytical development 2017;15:1–5.
- [19] Article R. Quality by design: a systematic approach for the analytical method. *J Drug Deliv Ther* 2019;9(3-s):1006–1012.
- [20] Gaykar D, Khadse SC. A review on analytical quality by design. *Int J Pharm Sci Rev Res* 2017;44:96–102.
- [21] Shah P, Patel J, Patel K, Gandhi T. Development and validation of an HPTLC method for the simultaneous estimation of Clonazepam and Paroxetine hydrochloride using a DOE approach. *J Taibah Univ Sci* 2017;11:121–32 <<https://doi.org/10.1016/j.jtusci.2015.11.004>>.

- [22] Gholve SB, Ajgunde RR, Bhushure OG, Thonte SS. Pelagia research library analytical method development and validation by QbD approach – a review. *Der Pharm Sin* 2015;6:18–24.
- [23] European Medicines Agency (EMA). ICH guideline Q9 on quality risk management. 2014;44:1–20.
- [24] Gavin PF, Olsen BA. A quality by design approach to impurity method development for atomoxetine hydrochloride (LY139603). *J Pharm Biomed Anal* 2008;46:431–41 <<https://doi.org/10.1016/j.jpba.2007.10.037>>.
- [25] Peraman R, Bhadrara K, Padmanabha Reddy Y. Analytical quality by design: a tool for regulatory flexibility and robust analytics. *Int J Anal Chem* 2015; <<https://doi.org/10.1155/2015/868727>>.
- [26] Das P, Maity A. Analytical quality by design (AQbD) : a new horizon for robust analytics in pharmaceutical process and automation. 2017;5:324–37.
- [27] Mustafa G, Ahuja A, Baboota S, Ali J. Box-Behnken supported validation of stability-indicating high performance thin-layer chromatography (HPTLC) method: an application in degradation kinetic profiling of ropinirole. *Saudi Pharm J* 2013;21:93–102 <<https://doi.org/10.1016/j.jsps.2011.11.006>>.
- [28] Khurana RK, Rao S, Beg S, Katara OP, Singh B. Systematic development and validation of a thin-layer densitometric bioanalytical method for estimation of mangiferin employing analytical quality by design (AQbD) approach. *J Chromatogr Sci* 2016;54:829–41 <<https://doi.org/10.1093/chromsci/bmw001>>.
- [29] Sangshetti JN, Deshpande M, Zaheer Z, Shinde DB, Arote R. Quality by design approach: regulatory need. *Arab J Chem* 2017;10:S3412–25 <<https://doi.org/10.1016/j.arabjc.2014.01.025>>.
- [30] Kovacs E, Ermer J, McGregor PL, Nethercote P, LoBrutto R, Martin GP, et al. Analytical control strategy. *Pharm Forum* 2016;42.
- [31] Guidance for industry: Analytical procedures and methods validation for drugs and biologics. USFDA; 2015.
- [32] Dubey SK, Duddelly S, Jangala H, Saha RN. Rapid and sensitive reverse-phase high-performance liquid chromatography method for estimation of Ketorolac in pharmaceuticals using weighted regression. *Indian J Pharm Sci* 2013;75(1):89–93. Available from: <https://doi.org/10.4103/0250-474X.113535>.
- [33] Krishna KV, Saha N, Puri A. Pre-clinical compartmental pharmacokinetic modeling of 2-[1-hexyloxyethyl]-2-devinyl pyropheophorbide-a (HPPH) as a photosensitizer in rat plasma by validated HPLC method. *Photochem Photobiol Sci* 2019;18:1056–63 <<https://doi.org/10.1039/c8pp00339d>>.
- [34] Krishna KV, Saha RN, Singhvi G, Dubey SK. Pre-clinical pharmacokinetic-pharmacodynamic modelling and biodistribution studies of donepezil hydrochloride by a validated HPLC method. *RSC Adv* 2018;8:24740–9. Available from: <https://doi.org/10.1039/c8ra03379j>.
- [35] Pradhan R, Krishna KV, Wadhwa G, Taliyan R, Khadgawat R, Kachhawa G, et al. QbD-driven development and validation of HPLC method for determination of Bisphenol A and Bis-sulphone in environmental samples. *Int J Environ Anal Chem* 2020;100:42–54 <<https://doi.org/10.1080/03067319.2019.1629585>>.
- [36] Dubey SK, Saha RN, Jangala H, Pasha S. Rapid sensitive validated UPLC-MS method for determination of venlafaxine and its metabolite in rat plasma: application to pharmacokinetic study. *J Pharm Anal* 2013;3:466–71 <<https://doi.org/10.1016/j.jpba.2013.05.002>>.

- [37] Bhatt DA, Rane SI. QbD approach to analytical RP-HPLC method development and its validation. *Int J Pharm Pharm Sci* 2011;3:179–87.
- [38] Puranpole A, Chavan VA, Kodrikar P, Mali G. Analytical quality by design (QbD) approach to RP-HPLC method development and validation of Meloxicam. *Asian J Pharm Technol Innov* 2016;4(20):1–10.
- [39] ICH Q8. EMEA/CHMP, 2009, ICH Topic Q 8 (R2) pharmaceutical development, Step 5: note for guidance on pharmaceutical development; 2017, 8. <http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2010/01/WC500059258.pdf> [Accessed April 24, 2017].
- [40] European Medicines Agency (EMA). ICH guideline Q10 on pharmaceutical quality system. 2014;44:1–20. EMEA/CHMP/ICH/214732/2007.
- [41] Peraman R, Bhadraya K, Padmanabha Reddy Y. Analytical quality by design: a tool for regulatory flexibility and robust analytics. *Int J Anal Chem* 2015;2015:868727 <<https://doi.org/10.1155/2015/868727>>.
- [42] Beg S, Rahman M, Kohli K. Quality-by-design approach as a systematic tool for the development of nanopharmaceutical products. *Drug Discov Today* 2019;24(3):717–25.

Chapter 7

Analytical quality by design for capillary electrophoresis

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7.1 Introduction

Quality by design (QbD) is a systemic approach to the US Food drug and administration (FDA). Now, it has gained popularity and is regarded as one of the most important models in pharmaceutical science. Analytical study with this approach has improved the standard of the product. The steps involved in systematic QbD approach of development requires defining the objectives, and applying the principles of science and risk-based optimization of the product and process parameters as per the International Conference on Harmonization (ICH) Q10 guideline applicable to analytical quality by design (AQbD). Analytical method development involving the concept of QbD is called AQbD (Fig. 7.1). AQbD differs from the classical analytical method in terms of a reduced number of out-of-trend and out-of-specification results and robustness of the method within the region [1]. AQbD is extensively adopted in pharmaceutical manufacturing and quality control. During the method development strategy (MDS) that covers QbD principles, AQbD is applied to various analytical techniques including chromatographic technique, capillary electrophoresis (CE), contamination of genotoxic substance, spectrophotometric analysis, determination of water, and tablet dissolution testing. In quality control of pharmaceuticals, CE is a well-built analytical technique that is extensively used. In the last two decades, several types of research reported CE as an efficient separation method of analysis. Developed method's validation results of parameters such as rapid analysis, simplicity, automation, ruggedness, different mechanism for selectivity, and

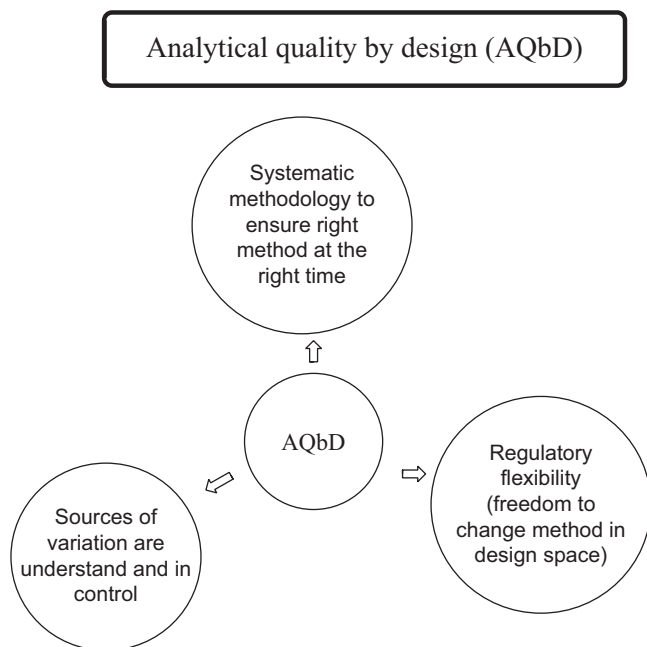


FIGURE 7.1 Analytical quality by design.

low cost required for testing of sample found more accurate in comparison to high-pressure liquid chromatography (HPLC) [2].

ICH Q8 (R2) guideline recommends the AQbD concept, in the pharmaceutical industry for improving the quality of the products [3]. The primary objective of QbD involved in the development of pharmaceutical products is to involve science and risk-based approach to develop the standard of products which is not solely limited to testing the end products. Being a relatively new approach the application of AQbD in analytical method development is not included in ICH guideline Q8(R2) [4]. The study design requires in-depth analysis of critical process parameters (CPPs) identified with the help of risk assessment and multivariate statistical techniques. The primary concept of QbD has been discussed in ICH Q8-Q11 and should be applied for good results of analytical method development. Maximum ranges of the CPPs make successful with the study of the design space (DS) which determined the multidimensional region, which leads to required values for the critical quality attributes (CQAs). Presently, chromatography is mainly a concern of AQbD [5] and a substantial lower scope of CE. The quality and standard of a pharmaceutical product are mainly controlled by our effective approach in determining the impurities. An impurity always represents a critical analytical issue. A validated method is always used in the routine analysis for the pharmaceutical dosage forms. By HPLC analysis determination of impurities in the level of 0.1% w/w with respect to active ingredient is a

comprehensive task where the instrument life exploited, here CE has been used as complementary [6]. HPLC technique has several advantages, high efficiency, and resolving power, less quantity of sample consumption, and the more chance of minutely improving the separation condition and selectivity. However, CE is a current technique that continues to grow. Researchers are continuously studying different aspects of CE related to the theory, separation modes, instrumentation, and applications. During analysis, various mode of separation is employed which is based on the nature of the analytes, for example, different types of CE like micellar CE, free solution CE, gel CE, capillary isoelectrofocusing, and capillary isotachopheresis. The principle of separation in capillary zone electrophoresis (CZE) is based on the differential migration of the analytes under the influence of a constant electric field, background medium used electrolytes [7]. In the drug development process, information's from the manufacturing process is collected to develop an analytical method for monitoring and controlling process parameters which are critical to a drug's quality [8]. Traditional analytical methods differ from the AQbD as the latter analytical approach is based on the identification, separation, and quantification, and involves QbD tools in risk assessment and design of experiments (DoE) that intensify the standard to be merged into the analytical method [9]. QbD is established by the US FDA as an essential pharmaceutical quality model for the development of pharmaceutical products. Process analytical technology (PAT) does not control the manufacturing process in the pharmaceutical industry; however, it needs to be adapted to our current drug quality system. The purpose of PAT work out on the design and development processes is to secure a predefined standard at the end of the manufacturing process. ICH Q8 had been revised in June 2009, in which an annex was added to provide the guidance of the key concept and to explain the principles of QbD [8,10–12].

In pharmaceutical development process ICH Q8 (R2) provided a direction for risk alleviation through the in-depth product and process understanding, though ICH Q9 quality risk management succeeds the principles. The pharmaceutical quality management system establishes as a model that followed the guideline ICH Q10, ICH Q8 (R2), and ICH Q9 which would be easy for innovation and continuous improvement throughout the lifetime of the product. In the drug development process, the control of impurities in drug substance and its products is of utmost importance. Several analytical methods are available; however, the selection of appropriate techniques is crucial to determine the possible impurities throughout the manufacturing process and for designing suitable control mechanism for them. The potential source of impurities in drug substance or product comes through the raw materials, intermediates, reagents, solvents, and catalysts used in the manufacturing process or drug development. Many impurities might come from the precursor materials or it may get formed during chemical synthesis. Degraded substances during product development may also be one of the sources for impurities. Interaction of the excipients with active molecules may also

produce impurities. The primary concern is to determine the source and destiny of impurities, which helps in designing the process controls, storage conditions, and specifications to ensure the impurities, must be within the limit. The screening process also helps in ruling out hypothetical impurities that are absent or insignificant [13]. Separation techniques are very powerful tools to achieve chiral analyses. Among them, liquid chromatography (LC), gas chromatography, supercritical fluid chromatography, and CE are the most employed techniques [14]. Although LC has been the most frequently used, CE has shown to have potential characteristics to carry out enantiomeric separations and it has an important impact in the last years which can be confirmed by the number of research works dealing with chiral analysis using different analytical techniques from 2015 [15].

7.2 Key aspects of analytical quality by design

QbD technique comes up as an adaptable and robust separation technique. Because of the simplicity and intrinsic flexibility, this technique fulfills the essential requirements. It is employed at all stages of drug discovery including the examination of the finished product in quality control [16]. In AQbD-based analysis, the method development and validation involve several parameters. Steps like the analytical target profile (ATP), measuring the level of specified impurity and experiment to determine the value of accuracy, precision, and range in the performance criteria are usually involved. During method development at the preliminary stage, it is important to examine the method with the purpose to achieve the highest level of selectivity with sufficient efficiency, reproducibility, and repeatability. On the basis of electrolyte composition experiments are performed. The composition of electrolytes consists of either simple buffer or it may have a pseudo stationary phase maximum of them either micelles or microemulsions. After completing the preliminary experiments, the AQbD defines the quality target product profile (QTPP) and the CQAs [17].

7.3 Quality target product profile and critical quality attributes

New regulatory concept of QbD introduced by the US FDA allows the pharmaceutical industry to focus on designing the final products without testing the products. To perform the analysis approach with QbD, the understanding of QTPP is important. QTPP is a potential compendium drug product standard characteristic that preferably will be executed to confirm the preference standard, safety parameter, and potency of the drug product. On the basis of such a definition, first, analytical methods precisely define separation. The goal of separation is to target the optimization, where selectivity will be highest with the least run time and performed within a robust region [18].

The pharmaceutical industry implies the separation of impurities from API has experimented base on regulatory requirements criteria for example validation of any new method performed according to ICH guidelines [19]. The desired pharmaceutical product standard as claimed by the ICH Harmonised Tripartite Guideline, validation of the analytical method, text, and methodology Q2 (R1)(2005) should be confirmed. During development, the various properties such as physical, chemical, biological, or microbiological are very crucial for the CQAs and it should be within acceptance criteria [20].

To attain CQAs in the development of the final product, QbD approach analysis has developed two terminologies namely critical material attributes and CPPs which explain the product-based quality. CQAs in analytical methods are the key variables on which the development of chromatogram or electropherogram will be related in the form of mathematical representation of the quality of the products [21]. CQAs are classified into two major classes “must-have” and “intend to have” on the basis of chromatogram interpretation. Instance resolution of adjacent peaks must not less than 1.5 in the case of “must-have” and 1.8 in case of “intended to” [17].

7.4 Traditional validation versus analytical quality by design

Traditional validation of analytical techniques was quite ridden with inadequacies. It also fails to assure that the analytical technique is reliable throughout its future use. Moreover, in the traditional method, the focus was on one factor at a Time. The traditional approach makes the process quite lengthy and costlier. Thus, the analytical technique was developed with narrow robustness. As a result, method failure rates during method transfer were quite high. Although all the method variables are explored but are not completely understood. After the failure of the method, the variable responsible for it was identified and then necessary changes in the process were made. This makes the whole process of validation of analytical techniques quite time-consuming to introduce any changes in the process. The need for implementation of QbD in analytical techniques was realized during late 2013 and early 2014. AQbD is a fresh approach that aims to overcome the failures faced during traditional techniques. All the variables affecting the analytical technique are thoroughly explored and all the critical performance parameters are identified and then duly prioritized. With the risk assessment approach and definite design of the experiment, the analytical procedure is designed and developed which assured the robustness and ruggedness of the analytical technique/method [22].

7.4.1 Assessment of critical method parameters by quality risk assessment

Factors that have an influence on a CQA, needs controlled methods to produce the preference quality products are called critical method parameters

(CMPs). The potential parameters in the initial list can affect CQAs that are fully extensive; however it very likely minimized and ordered by quality risk assessment (QRA). The CE is a kind of CMP parameter which considers the change in voltage, concentration of analytes, and capillary temperature. QRA approach is a scientific process that can help in the identification of CMPs; thus, removing risk and out coming is, the analytical method fulfill the QTTP under all conditions of use. The use of QRA tools helps to remove a large number of parameters, for example, cause and effect diagram, failure mode effects analysis [23]. Ishikawa fishbone diagram helps to find out the risk factors that are concern with the attribute of the CZE analysis and which will be helpful to emphasize the CMPs and it could affect the selected CQAs. Basically, five CMPs that are considered to affect CQAs are voltage, temperature, injection time, the strength of buffer, and its pH. A better result can be found between method selectivity and sensitivity from one of the CMPs parameters that are the inclusion of injection time in tuning sample loading during practical. Another approach of study follow CNX method, where

C = high-risk factor

N = potential noise factors

X = which are to experiment

The result of the experiment analyzed, if risk priority number (RPN) above a cut-off level subsequent studies required, and if factors lower, it can be eliminated from further study.

7.5 Design of experiment

After performing the QRA, continue analysis by means of DoE that give the more precise result. The data of these studies are likely to help for evaluating the effect of the higher-ranked CMPs and additionally it minimizes the number of CMPs to be studied in more detail by considering the factors which have a more effect [24]. QRA tools, basic earlier knowledge, and preliminary experiments are examined by DoE to define the knowledge space (KS) [25]. To perform preliminary experiments it necessary to selection of proper domain for the investigation, because the suitable KS will be helpful to achieve a good separation [26].

Three levels are considered for screening studies of CMPs of the given levels that is concentration, pH, voltage, temperature, etc., to covering the KS. Responses shown by the selected levels in CQAs will be monitored and it evaluates the effects of changes. Optimization of the CQAs, with the help of graphic analysis data produced by CMPs levels effects. After examining the data of the experiment it would be helpful to select the final condition. By performing the preliminary experiments DoE can be removed [27]. To design a new experiment, response surface methodology (RSM) has been

studied with aimed to obtain complete information on the effect of each CMP on the CQAs and these data will help to predict to design a map for further studies. To perform RSM, D-optimal, central composite, full factorial, Box–Behnken and Doehlert designs are used [28]. CE technique is employed for the optimization of pharmaceutical microemulsion product composition. Factors that consider for experiment are cyclodextrin concentration, voltage, buffer concentration, and temperature [29]. Piepel et al. developed an analytical method using QbD approach, and analyzed the effects of oil, surfactant and/or cosurfactant as the constituents of microemulsion [27].

7.6 Understanding design space

Experimental data are produced by CE analytical method, after that its quality can be checked by an experiment applied DS. Here, design experiment where multidimensional space considering combination of factors that has been experimented [30]. An analytical technique is used for the quantification, the method developed and validated not only for a specific condition but for a range of condition [31]. After the experiments are performed, the data analysis is carried out by mathematical and/or graphical modeling techniques to describe the relationship among the parameters. If multiple responses are involved data overlay [32]. The graphical presentation of data is possible with the help of advanced software which also provides desirability index (D). Two values ‘0’ and ‘1’ are used to explain CQAs assign desired and accepted values, ‘1’ corresponds to a highly desirable value and “0” to an unacceptable value. During new method development plan, a systemic concept to design the experiment is quite effective. Several approaches are employed during the optimization of a mathematical relationship (model). The selection of variables at optimum levels will be done by considering the statistical knowledge as basic tools to interpret the interaction and contribution. The effect of each factor that influences the separation can be calculated to find the optimum separation. Presently, the software available is for in a simulation multivariate data analysis is example Plackett–Burman design, Taguchi method, or the Monte-Carlo.

The application of Taguchi method is possible when low number of experimental runs are required as compared to factorial designs. However, if the number of input variables are high, then Plackett–Burman design is best studied.

7.7 Robustness and control strategy

The potential risk of the developed method because of a minor change of given parameters or under a variety of conditions such as different laboratories, analysts, instruments, reagents, days, etc. studied by fishbone diagram,

implemented structured methodologies for risk assessment. Performed robustness testing by considering examines all the CPPs during the screening phase. The result shown smaller changes in these factors give a deep variation of the experimental conditions, much greater than the changes usually requested when testing robustness [33]. The DS experimented to considered in a robust zone to assure the fulfillment of the requirements, but this principle did not match the demonstration that minor changes of the CPPs around the working conditions did not influence CQAs performances [34]. A control plan as key points that need to control in the robustness study as well the definition of proper system suitability criteria. During the system suitability studies, the lower and higher values of the CQAs estimated and it fixed an interval of accepted CQAs values [35]. The product developed by QbD, a control strategy has planned to ensure instant product production with the required quality. A control strategy is planned from various data collected during the method development phase and method verification process. A QbD method control strategy does not differ from the traditional method, but it must be ensured to establish a relation between purpose method and method performance. QbD approach generally used in chromatography studied [36,37]. Very few CE methods have been published consisting of MEEKC [27], solvent-modified micellar electrokinetic capillary chromatography (MEKC), and –SDS/LIF methods. CE methods are used in drug discovery and in quality control of the finished product. QbD principles to CE methods very frequently applied and implemented and the same for HPLC methods also [38].

7.8 Applications of capillary electrophoresis in the pharmaceutical, food, biomedical, or other fields

Due to its great versatility, CE enables the analysis of numerous compounds in a wide variety of samples in different fields such as the pharmaceutical, biomedical, or food analysis, among others. Here, the most relevant and recent applications of CE to the analysis of compounds in real samples are included.

7.8.1 Pharmaceuticals

Estimation of several categories of drugs such as cardiovascular diseases, osteoporosis, and leukemias are possible to be estimated through CE techniques. The analyst used the CE technique coupled with various detector modes for analysis that is in single or in a combined drug. Example of cardiovascular drug that is Metoprolol (MET) and hydrochlorothiazide (HCT) is analyzed by CE. Alnajjar et al. developed a method for the estimation of MET and HCT in their combined form [39]. Linearity of the method reported 2.5–25 $\mu\text{g/mL}$ and LOD were 0.02 (MET) and 0.01 (HCT) $\mu\text{g/mL}$

for analysis. This technique was also employed for the estimation of the cephalosporin class of antibiotics. Hancu et al. developed a simple and rapid separation of six cephalosporin antibiotics, LODs of all drugs in the range of 0.99 and 2.73 $\mu\text{g/mL}$ [40].

In the last years, new CE methods enabling the enantiomeric determination of drugs in pharmaceutical formulations were developed and validated according to the International Conference on Harmonisation guidelines (ICH guidelines). A relevant point in the pharmaceutical industry for the development of CE methodologies suitable to evaluate the enantiomeric purity of drugs marketed as pure enantiomers is the application of QbD approaches. In fact, better analytical methods can be developed by applying QbD principles. Briefly, QbD involves the definition of the ATP (e.g., the determination of the main compound and its chiral impurity at the 0.1% level with a resolution value >2.0 in an analysis time <10 min), the identification of the critical process variables (buffer concentration and pH, chiral selector concentration, voltage, temperature, etc.) affecting the CQAs (resolution and analysis time), and the establishment of the DS (that reflects the experimental conditions under which the analytical target is reached). Fractional factorial resolution designs are usually employed for the identification of the critical parameters [41] (Table 7.1).

7.8.2 Proteins, peptides, and amino acids

Analysis of such molecules with CE coupled with MS detector can accurately quantify very low concentration of the samples. The data produced by this technique will also help for peptide mapping. Whitmore and Gennaro developed two capillary electrophoresis-mass spectrometry (CE-MS) peptide analysis methods in order to map tryptic peptides: a novel sheathless interface system and a traditional sheath interface [50]. MS detector is very efficient; analysis by LC-MS methods can map 97% of the sequence of a therapeutic monoclonal antibody in a tryptic digest, but rest 3% mapped by CE-MS methods. Quantification of proteins, peptides, and amino acids from biological samples can be done by CE-MS and also other detectors. A fingerprint of amino acids performed by MS/MS provides confirmation of identities [51].

7.8.3 Carbohydrates

Different types of sweeteners are found in the food product, its separation, and detection in low-level quantity experimented by CE. A natural sweeteners glycosphingolipids are detected by this technique. Ito et al. developed a CS-ESI quadrupole ion trap and time of flight for the structural characterization of a pyridylaminated-oligosaccharide mixture derived from glycosphingolipids

TABLE 7.1 Application of AQbD in developing various types of capillary electrophoresis methods.

S. no.	Title	Outcome	References
1	Fast analysis of glibenclamide and its impurities: QbD framework in CE method development	A fast capillary zone electrophoresis method for the simultaneous analysis of glibenclamide and its impurities I(A) and I(B) in pharmaceutical dosage forms was fully developed within a QbD framework. Critical quality attributes were represented by I(A) peak efficiency, critical resolution between glibenclamide and I (B), and analysis time. A full factorial design simultaneously allowed the design space to be validated and method robustness to be tested. A control strategy was finally implemented by means of a system suitability test.	[42]
2	Chiral capillary zone electrophoresis in enantioseparation and analysis of cinacalcet impurities: use of quality by design principles in method development	A CE method for the simultaneous determination of the enantiomeric purity and of impurities of the chiral calcimimetic drug cinacalcet hydrochloride has been developed following QbD principles. A Box–Behnken design allowed the contour plots to be drawn and quadratic and interaction effects to be highlighted. Robustness testing was carried out by a Plackett–Burman matrix and finally a method control strategy was defined.	[43]
3	Development of a capillary electrophoresis method for the determination of the chiral purity of	Dextromethorphan is a centrally acting antitussive drug, while its enantiomer levomethorphan is an illicit	[44]

(Continued)

TABLE 7.1 (Continued)

S. no.	Title	Outcome	References
	dextromethorphan by a dual selector system using quality by design methodology	drug with opioid analgesic effects. As CE has been proven as an ideal technique for enantiomer analysis, the present study was conducted in order to develop a CE-based limit test for levomethorphan. The selected working conditions consisted of a 30/40.2 cm, 50 μ m id fused-silica capillary, 30 mM sodium phosphate buffer, pH 6.5, 16 mg/mL sulfated β -cyclodextrin, and 14 mg/mL methyl- α -cyclodextrin at 20°C and 20 kV. The method was validated according to ICH guideline Q2(R1) and applied to the analysis of a capsule formulation.	
4	Analytical quality by design in pharmaceutical quality assurance: development of a capillary electrophoresis method for the analysis of zolmitriptan and its impurities	A fast and selective CE method for the determination of zolmitriptan (ZOL) and its five potential impurities has been developed applying the AQbD principles. Voltage, temperature, buffer concentration, and pH were investigated as critical process parameters that can influence the critical quality attributes, represented by critical resolution values between peak pairs, analysis time, and peak efficiency of ZOL-dimer.	[45]
5	A quality by design-based approach to a capillary electrokinetic assay for the determination of dextromepromazine and levomepromazine sulfoxide	Using a QbD approach, a CE method for the simultaneous determination of dextromepromazine and the oxidation product levomepromazine sulfoxide	[46]
(Continued)			

TABLE 7.1 (Continued)

S. no.	Title	Outcome	References
	as impurities of levomepromazine	in levomepromazine was developed. The method was subsequently validated in the relative concentration range of 0.1%–1.0% of the impurities for a solution containing 0.25 mg/mL levomepromazine.	
6	Cyclodextrin- and solvent-modified micellar electrokinetic chromatography for the determination of Captopril, hydrochlorothiazide and their impurities: a quality by design approach	A fast and selective CE method has been developed for the simultaneous determination of the antihypertensive drugs captopril and hydrochlorothiazide and their related impurities in a combined dosage form. Method development was carried out implementing each step of QbD workflow, the new paradigm of quality outlined in International Conference on Harmonisation guidelines. The scouting phase was dedicated to the investigation of several operative modes in order to overcome detection and isomerization issues. The method was validated and applied to the analysis of a real sample of coformulation tablets.	[47]
7	Analytical quality by design in the development of a cyclodextrin-modified capillary electrophoresis method for the assay of metformin and its related substances	QbD is a new paradigm of quality to be applied to pharmaceutical products and processes, recently encouraged by International Conference on Harmonisation guidelines. In this paper QbD approach was applied to the	[48]
(Continued)			

TABLE 7.1 (Continued)

S. no.	Title	Outcome	References
		development of a CE method for the simultaneous assay of MET and its main impurities. Method scouting allowed CD-CZE based on the addition of carboxymethyl- β -CD to Britton–Robinson acidic buffer to be chosen as operative mode. Seven CPPs were selected, related to capillary, injection, BGE, and instrumental settings.	
8	A comprehensive strategy in the development of a cyclodextrin-modified microemulsion electrokinetic chromatographic method for the assay of diclofenac and mixture-process variable experiments and quality by design	A comprehensive strategy involving the use of mixture-process variable (MPV) approach and QbD principles has been applied in the development of a CE method for the simultaneous determination of the antiinflammatory drug diclofenac. The selected operative mode consisted of microemulsion electrokinetic chromatography with the addition of methyl- β -cyclodextrin. The critical process parameters included both the MCs of the microemulsion and the PVs. MPV experiments were used both in the screening phase and in the response surface methodology, making it possible to draw MCs and PVs contour plots and to find important interactions between MCs and PVs. The method was applied to a real sample of diclofenac gastro-resistant tablets.	[49]

AQbD, analytical quality by design; *CE*, capillary electrophoresis; *CPPs*, critical process parameters; *MET*, metformin hydrochloride; *MCs*, mixture components; *PVs*, process variables; *QbD*, quality by design.

extracted from biological materials. This method is highly sensitive and its detection limit was 25 mol/ μ L [52].

7.8.4 Bioanalysis

CE has also proven its powerfulness in the separation of analytes present in biological samples such as plasma, urine, or even single neurons. Metabolites, CE, and its clinical applications were explored by many researchers to profile body fluids for the diagnosis of diseases using biomarkers. MS was the dominant detection mode used in the development and application of these techniques [53].

7.8.5 Food analysis

Chirality of a great variety of food components makes that their enantioselective analysis has a significant role in food science and technology since it enables them to obtain information related to food quality, food processing, storage, or adulterations. Different electrokinetic chromatography (EKC) methodologies have demonstrated their potential for the quality control of food supplements [54]. The enantiomeric determination of protein amino acid in rice wine is an interesting topic to obtain information about the wine age. For this reason, Miao et al. developed an MEKC method for the determination of D-glutamic acid and D-aspartic acid in rice wine [55]. These two works demonstrate the usefulness of MEKC indirect approaches to the quality control of edible marina algae and food supplements.

7.8.6 Environmental and forensic analysis

CE-coupled detector MS is a common practice for the analysis of various environmental samples. The samples mainly examined have analytes ranged from polycyclic aromatic hydrocarbons to chloroanilines and also herbicides and water sample testing. Several researchers worked in the area of environmental area and forensic analysis and developed a new method. CE analytical technique is used to determine the triazines in herbicides that are hazardous to human health. Fang et al. reported eight triazine herbicides in vegetable and cereal samples by the on-line sweeping MEKC method. For complex matrix, the on-line sweeping technique is used to provide preconcentration without requiring purification. LOQs reported 0.1 ng/g for all eight triazines. Identification of organic acid and inorganic anions in pine bark extract by bioprocessing by using fungi [56]. Boke et al. studied a sample by this technique, identified nine organic acids within 12 min in fermented samples, and three inorganic anions detected in less than 10 min provided relevant information regarding toxicological effects or about their origin and synthesis [57]. Regarding human hair and blood samples, an

EKC-UV method using S- γ -CD as a chiral selector was proposed for the chiral separation of ketamine and norketamine which is its principal metabolite, in hair samples from 12 ketamine abusers [58].

7.8.7 Bioaffinity

Bioaffinity technique is employed for the study of analytes that have charge heterogeneity and binding affinities. Two techniques, CE-SELEX and affinity CE (ACE), are commonly employed for Heparin's high affinity for anti-thrombin III (AT-III) that gives to its anticoagulant properties. AT-III binding affinities of low molecular weight are analyzed by ACE reported by Dinges et al. A mathematical application such as linear isotherm and double reciprocal plotting is used to determine the effective binding ligand concentration that has linear isotherm data producing values between 2 and 16×10^{-8} M for the studied components [59].

References

- [1] Peraman R, Bhadrara K, Reddy YP. Analytical quality by design: a tool for regulatory flexibility and robust analytics. *Int J Anal Chem* 2015;2015:868727. Available from: <https://doi.org/10.1155/2015/868727>.
- [2] Ahuja S., Jimidar M., 2008. Capillary electrophoresis methods for pharmaceutical analysis. 1st ed., vol. 9, 2008.
- [3] Orlandini S, Pinzauti S, Furlanetto S. Application of quality by design to the development of analytical separation methods. *Anal Bioanal Chem* 2013;405:443–50.
- [4] Rozet E, Lebrun P, Debrus B, Boulanger B, Hubert P. Design spaces for analytical methods. *Trends Anal Chem* 2013;42:157–67.
- [5] Vemić A, Rakić T, Malenović A, Medenica M. Chaotropic salts in liquid chromatographic method development for the determination of pramipexole and its impurities following quality-by-design principles. *J Pharm Biomed Anal* 2015;102:314–20.
- [6] Jouyban A, Kenndler E. Impurity analysis of pharmaceuticals using capillary electromigration methods. *Electrophoresis* 2008;29:3531–51.
- [7] Chamieh J, Martin M, Cottet H. Quantitative analysis in capillary electrophoresis: transformation of raw electropherograms into continuous distributions. *Anal Chem* 2015;87:1050–7.
- [8] ICH Harmonised Tripartite Guideline. Pharmaceutical Development Q8 (R2). International Conference on Harmonisation of technical requirements for registration of pharmaceuticals for human use; 2009. <http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality/Q8_R1/Step4/Q8_R2_Guideline.pdf>. [Accessed June 14, 2012].
- [9] Orlandini S, Pinzauti S, Furlanetto S. Application of quality by design to the development of analytical separation methods. *Anal Bioanal Chem* 2012;.
- [10] Pharmaceutical CGMPs for the 21st century – A Risk-Based Approach. Final report. US Food and Drug Administration; 2004. <https://www.fda.gov/media/77391/download> [Accessed June 14, 2012].

- [11] Guidance for Industry. PAT – A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance. US Food and Drug Administration; 2004. <<http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm070305.pdf>> [Accessed June 14, 2012].
- [12] Guidance for Industry. Q8 (R2) Pharmaceutical Development. US Food and Drug Administration; 2009. <<http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm073507.pdf>> [Accessed June 14, 2012].
- [13] Olsen BA. Developing and using analytical methods to achieve quality by design and efficiency in drug development. Pharm Technol Scaling Manuf 2005;. Available from: www.pharmtech.com.
- [14] Ward TJ, Ward KD. Chiral separations: a review of current topics and trends. Anal Chem 2012;84:626–35.
- [15] Sánchez-López E, Castro-Puyana M, Marina ML. Electrophoresis. Capillary electrophoresis: chiral separations. In: 3rd edn. Worsfold P, Poole C, Townshend A, Miró M, editors. Encyclopedia of analytical science, vol. 2. Amsterdam: Elsevier; 2019. p. 334–45.
- [16] Zhu Q, Scriba GKE. Analysis of small molecule drugs, excipients and counter ions in pharmaceuticals by capillary electro migration methods—recent developments. J Pharm Biomed Anal 2018;147:425–38.
- [17] Vogt FG, Kord AS. Development of quality-by-design analytical methods. J Pharm Sci 2011;100:797–812.
- [18] Molnár I, Rieger HJ, Monks KE. Aspects of the “design space” in high pressure liquid chromatography method development. J Chromatogr 2010;A1217:3193–200.
- [19] <http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality/Q2_R1/Step4/Q2_R1_Guideline.pdf> [Accessed June 14, 2012].
- [20] <http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality/Q8_R1/Step4/Q8_R2_Guideline.pdf> [Accessed June 14, 2012].
- [21] McBrien M. Practical implications of quality by design to chromatographic method development. Chromatogr Today 2010;3:30–4.
- [22] Goswami S, Chakraverty R. A review on application of quality by design concept to analytical techniques. Int J Curr Res Health Biol Sci 2016;1(3):100–8.
- [23] <http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality/Q9/Step4/Q9_Guideline.pdf> [Accessed June 14, 2012].
- [24] Huang J, Kaul G, Cai C, Chatlapalli R, Hernandez-Abad P, Ghosh K, et al. Quality by design case study: an integrated multivariate approach to drug product and process development. Int J Pharm 2009;382:23–32.
- [25] Swartz M, Krull IS, Turpin J, Lukulay PH, Verseput R. LC GC Am 2009;27:328–39.
- [26] Debrus B, Lebrun P, Ceccato A, et al. Application of new methodologies based on design of experiments, independent component analysis and design space for robust optimization in liquid chromatography. Anal Chim Acta 2011;691(1–2):33–42.
- [27] Piepel G, Pasquini B, Cooley S, Heredia-Langner A, Orlandini S, Furlanetto S. Mixture-process variable approach to optimize a microemulsion electrokinetic chromatography method for the quality control of a nutraceutical based on coenzyme Q10. Talanta. 2012;97:73–82.
- [28] Hanrahan G, Montes R, Gomez FA. Chemometric experimental design based optimization techniques in capillary electrophoresis: a critical review of modern applications. Anal Bioanal Chem 2008;390(1):169–79.

- [29] Orlandini S, Giannini I, Navarro MV, Pinzauti S, Furlanetto S. Dual CD system-modified MEEKC method for the determination of clemastine and its impurities. *Electrophoresis*. 2010;31(19):3296–304.
- [30] Borman P, Nethercote P, Chatfield M, Thompson D, Truman K. The application of quality by design to analytical methods. *PharmTech* 2007;31:142–52.
- [31] Lionberger RA, Lee SL, Lee L, Raw A, Yu LX. Quality by design: concepts for ANDAs. *AAPS J* 2008;10(2):268–76.
- [32] Lebrun P, Govaerts B, Debrus B, Ceccato A, Caliaro G, Hubert P, et al. *Chemom Intell Lab Syst* 2008;91:4–16.
- [33] Furlanetto S, Orlandini S, Mura P, Sergent M, Pinzauti S. How experimental design can improve the validation process. *Studies in pharmaceutical analysis. Anal Bioanal Chem* 2003;377:937–44.
- [34] ICH Harmonised Tripartite Guideline, Q8(R2), Pharmaceutical development 2009.
- [35] Ermer J, Miller JH McB, editors. *Method validation in pharmaceutical analysis – a guide to best practice*. Weinheim: Wiley-VCH; 2004.
- [36] Drug product and delivery, SSCI, Crystallization Expert. <<http://www.ssciinc.com/DrugSubstance/PATandPharmaceuticalQualityByDesign/tabid/86/Default.aspx>>; 2014.
- [37] Debrus B, Guillarme D, Rudaz S. Improved quality-by-design compliant methodology for method development in reversed-phase liquid chromatography. *J Pharm Biomed Anal* 2013;84:215–23.
- [38] Orlandini S, Gotti R, Furlanetto S. Multivariate optimization of capillary electrophoresis methods: a critical review. *J Pharm Biomed Anal* 2014;87:290–307.
- [39] Alnajjar AO, Idris AM, Attimarad MV, Aldughhaish AM, Elgorashe RE. Capillary electrophoresis assay method for metoprolol and hydrochlorothiazide in their combined dosage form with multivariate optimization. *J Chromatogr Sci* 2013;51(1):92–7.
- [40] Hancu G, Kelemen H, Rusu A, Gyeresi AJ. Development of a capillary electrophoresis method for the simultaneous determination of cephalosporins. *Serb Chem Soc* 2013;78:1413–23.
- [41] Bernardo-Bermejo S, Sánchez-López E, Castro-Puyana M, Marina ML. Chiral capillary electrophoresis. *Trends Anal Chem* 2019; S0165-9936(19)30518-7.
- [42] Furlanetto S, Orlandini S, Pasquini B, Caprini C, Mura P, Pinzauti S. Fast analysis of glibenclamide and its impurities: quality by design framework in capillary electrophoresis method development. *Anal Bioanal Chem* 2015;407(25):7637–46.
- [43] Pasquini B, Orlandini S, Villar-Navarro M, Caprini C, Del Bubba M, Douša M, et al. Chiral capillary zone electrophoresis in enantioseparation and analysis of cinacalcet impurities: use of quality by design principles in method development. *J Chromatogr A* 2018;1568:205–13.
- [44] Krait S, Heuermann M, Scriba GKE. Development of a capillary electrophoresis method for the determination of the chiral purity of dextromethorphan by a dual selector system using quality by design methodology. *J Sep Sci* 2018;41(6):1405–13.
- [45] Orlandini S, Pasquini B, Caprini C, Del Bubba M, Pinzauti S, Furlanetto S. Analytical quality by design in pharmaceutical quality assurance: development of a capillary electrophoresis method for the analysis of zolmitriptan and its impurities. *Electrophoresis*. 2015;36(21–22):2642–9.
- [46] Niedermeier S, Scriba GKE. A quality by design-based approach to a capillary electrokinetic assay for the determination of dextromepromazine and levomepromazine sulfoxide as impurities of levomepromazine. *J Pharm Biomed Anal* 2017;146:402–9.

- [47] Pasquini B, Orlandini S, Caprini C, Del Bubba M, Innocenti M, Brusotti G, et al. Cyclodextrin- and solvent-modified micellar electrokinetic chromatography for the determination of captopril, hydrochlorothiazide and their impurities: a quality by design approach. *Talanta* 2016;160:332–9.
- [48] Orlandini S, Pasquini B, Gotti R, Giuffrida A, Paternostro F, Furlanetto S. Analytical quality by design in the development of a cyclodextrin-modified capillary electrophoresis method for the assay of metformin and its related substances. *Electrophoresis*. 2014;35(17):2538–45.
- [49] Orlandini S, Pasquini B, Caprini C, Del Bubba M, Squarcialupi L, Colotta V, et al. A comprehensive strategy in the development of a cyclodextrin-modified microemulsion electrokinetic chromatographic method for the assay of diclofenac and its impurities: mixture-process variable experiments and quality by design. *J Chromatogr A* 2016;1466:189–98.
- [50] Whitmore CD, Gennaro LA. Capillary electrophoresis-mass spectrometry methods for tryptic peptide mapping of therapeutic antibodies. *Electrophoresis* 2012;33(11):1550–6.
- [51] Atherton T, Croxton R, Baron M, Gonzalez-Rodriguez J, Gamiz-Gracia L, Garcia-Campana AM. Analysis of amino acids in latent fingerprint residue by capillary electrophoresis-mass spectrometry. *J Sep Sci* 2012;35:2994–9.
- [52] Ito E, Nakajima K, Waki H, et al. Structural characterization of pyridylaminated oligosaccharides derived from neutral glycosphingolipids by high-sensitivity capillary electrophoresis-mass spectrometry. *Anal Chem* 2013;85(16):7859–65.
- [53] Hyvarinen S, Mikkola JP, Murzin DY, Vaher M, Kaljurand M, Koel M. Sugars and sugar derivatives in ionic liquid media obtained from lignocellulosic biomass: comparison of capillary electrophoresis and chromatographic analysis. *Catal Today* 2014;223:18–24.
- [54] Pérez-Míguez R, Marina ML, Castro-Puyana M. Enantiomeric separation of non protein amino acids by electrokinetic chromatography. *J Chromatogr A* 2016;1467:409–16.
- [55] Miao YN, Liu Q, Wang W, Liu L, Wang L. Enantioseparation of amino acids by micellar capillary electrophoresis using binary chiral selectors and determination of D glutamic acid and D-aspartic acid in rice wine. *J Liq Chromatogr Relat Technol* 2017;40(17):783–9.
- [56] Fang R, Chen GH, Yi LX, Shao YX, Zhang L, Cai QH, et al. Determination of eight triazine herbicide residues in cereal and vegetable by micellar electrokinetic capillary chromatography with online sweeping. *Food Chem* 2014;145:41–8.
- [57] Boke N, Siren H, Petrik LF. Fungal biofermentation of pine bark producing organic acids and their quantification with capillary electrophoresis. *Ind Crop Prod* 2015;67:41–8.
- [58] Aturki Z, Rocco A, Fanali S. Forensic drugs analysis: a review of miniaturized separation techniques. *LC-GC Eur* 2015;28:18–25.
- [59] Dinges MM, Solakyildirim K, Larive CK. Affinity capillary electrophoresis for the determination of binding affinities for low molecular weight heparins and antithrombin-III. *Electrophoresis*. 2014;35:1469–77.

Chapter 8

Quality by design—based development of vibrational spectroscopy methods

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8.1 Introduction

Quality by design (QbD) for analytical methods, defined in ICH Q8 (R1), is a planned protocol for pharmaceutical development starting with predefined objectives that highlight product and process understanding as well as product and process control. The analytical method is considered as a process whose output is quality data; the concepts of QbD can be applied to develop analytical methods.

ICH Q8 (R2) discusses on pharmaceutical development where QbD elements have been described. In this regard, the application of fundamental science and risk-based methodology provides an organized approach which covers risk assessment, multifactorial optimization and design space as the tools used for systematic development.

Why QbD for analytical methods? When QbD is aligned with ICH guideline, it enhances method understanding, reducing, and controlling source of variability to facilitate continuous improvement. The traditional approach of method validation does not take an account of modern Six Sigma concepts and statistical approaches. Nevalainen et al. [1] have estimated analytical performance at the 3.85-sigma level to understand/control variability. All analytical methods used to monitor and control our manufacturing processes are directly linked to the risk assessment performed during the process control definition.

Comparison of traditional and QbD approaches to analytical methods “one-factor-at-a-time” (OFAT) approach is carried out by selecting one instrument parameter of study while holding all other parameters fixed [2].

The quality of pharmaceutical products is enhanced with a novel idea of applying QbD method which is an important part of the new modern approach to industrial quality product development; QbD is the best clarification to enhance quality in every pharmaceutical products, and nevertheless it is also a primary challenge for the industry because of its necessary to confirm procedure in time. Under this new concept of QbD right through new designing, discovery, and development of novel products, it is fundamental requirement to explain the product routine profile that helps to identify critical quality attributes (CQA), target product quality profile (TPQP), and target product profile (TPP). Based on this, we can design the product formulation and process to meet the product attributes. This leads to identifying the impact of novel raw materials, critical process parameters (CPP), critical material attributes (CMA), identification, and control sources of variability. QbD is a promising knowledge which helps pharmaceutical manufacturer along with augmented self-regulated elasticity meanwhile maintaining unyielding quality standards and actual time release of the drug dosage [3].

8.2 Quality by design tools

8.2.1 Critical quality attributes tool

CQAs may be defined as chemical, physical, microbiological, or biological characteristics that necessitate being controlled to make sure design product quality. Consequently, CQAs are applied to describe both form of new product performance and determinants of design product performance.

8.2.2 Target product profile tool

A quality target product profile (QTPP) is a new concept which is an expected extension of TPP for product quality performance. A QTPP links with the quality of drug materials or the drug molecules that is essential to convey a preferred therapeutic activity.

8.2.3 Risk assessment

A key objective of risk evaluation in pharmaceutical development is to categorize which material attributes and method parameters involve the drug product CQAs.

8.2.4 Design space

ICH Q8 explains design space, the multidimensional amalgamation, and interaction of effort variables (material attributes) and process parameters that have been demonstrated to provide reassurance of quality product.

8.2.5 Control strategy

A planned set of controls, derived from current product and process understanding that ensures process performance and product quality are the key objectives of ICH Q8 (R2) & Q10 guidances [4].

The increasing demand for new product quality development and production validation in the pharmaceutical, petrochemical, chemical, polymer, food, cosmetic pesticide, and agricultural industries has induced an important significant betterment of the vibrational spectroscopic techniques—infrared (IR), Fourier transform infrared (FT-IR) spectroscopy, and Raman spectroscopy. This tendency has partially led to the steady substitution of time being consumed in conventional analytical techniques, high-performance liquid chromatography (HPLC), gas chromatography (GC), mass spectroscopy (MS), and nuclear magnetic resonance (NMR) and nonspecific control method (pressure, pH, dosing weight, and temperature) by the extra specific and environmentally well-matched analytical technique of vibrational spectroscopy. FT-IR, Raman, and infrared (IR) spectroscopy have emerged more beneficial in the previous decade—in combination with new imaging accessories, new in-line, light-fiber optics on-line probes, and chemometric assessment procedures—as particularly powerful procedure for industrially oriented research, process monitoring, and quality control.

Polymers have become an integral ingredient of our daily life and this article will provide insight that Raman, NIR, and IR spectroscopy are enormously important depictions and control techniques for the lifetime of a polymeric pharmaceutical product [5].

NIR spectroscopy has become conventional as a key technique within the process analytical tool (PAT) toolbox and there are many examples regarding applications of conventional pharmaceutical laboratory and manufacturing functions [6]. However, much less has been published about NIR methods that utilize QbD principles throughout the development of the processes to achieve higher quality methods. The continuous manufacturing of a pharmaceutical preparation requires frequent monitoring of its CQAs, and typically this will require the removal and analysis of subsamples. In conventional manufacturing, a batch is a self-contained unit, from which subsamples statistically reflect the whole unit/sample. In a continuous process, the beginning and end of the batch are not necessarily related. Therefore, testing procedures in a continuous process are required to ensure homogeneity

throughout the manufacturing run. In the stringent financial environment that operates within most quality control laboratories, resourcing this testing is an additional challenge. As pharma continues to explore continuous manufacturing as an alternative to conventional unit operations, the ability to work within the QbD framework issued by regulators allows the application of good science and can lead to innovation and cost-effectiveness. Consequently, an automated continuous monitoring system is desirable.

Near infrared spectroscopy (NIRS) is cost-effective, flexible and has a nondestructive nature and ability to collect spectral information quickly. An NIRS method designed with QbD principles has the potential to use not only as an on-line monitoring method but as a system of control that can provide a greater understanding of the product and process and can reduce the variability of the CQAs. This enables process control in real-time through process understanding—the true essence of PAT.

The range of NIR (12,500–4000/cm), mid-infrared (MIR; 400–4000/cm), and Raman spectroscopy (Raman shift; 200–4000/cm) is under the vibrational spectroscopy. The phenomenon reflection, absorption, and emission of light in samples are based on infrared spectroscopy (NIR as well as MIR) and the scattering phenomenon are based on Raman spectroscopy. Both analytical tools provide information about the fundamental rotational and vibrational modes of the compounds. When the molecules interact with the light, there is change in the dipole moment that can be seen in the read-out device called IR spectrum bands, and due to change in the polarizability of the novel molecules given Raman spectrum. Consequently, the IR technique provides an idea about the dipole vibrations (e.g., $-\text{OH}$, $-\text{CH}$, $-\text{NH}$) while the Raman spectroscopy provides an idea about polarizable vibrations bands (e.g., $\text{C}=\text{C}$, $\text{C}\equiv\text{N}$, $\text{C}=\text{N}$) of the sample. In IR spectroscopy, the sample solution comes in contact with the IR radiation source for the development of the spectrum; the energy of light excites the molecular vibrations in the molecules. The detector of the instrument evaluates the reflected or transmitted incident radiation. In Raman spectroscopy, the spectrum is obtained when the solution of the sample is illuminated by a monochromatic laser beam source that links with the molecular vibrations. The scattered light is detected from the sample by the spectrometer [7].

Quantitation provides information that is comparable to that of the reference method and allows the method to be more easily accommodated within manufacturing, quality assurance, and regulatory functions. For this reason, it is an advantage as an alternative quantitative test method filed with the regulatory guidelines. However, it has a restricted design space, resulting in a reduced ability to compensate for the process and material changes, which may result in greater prediction errors. The development and validation of a quantitative NIR can be expensive because of its requirement to calibrate against a single parameter. This to have NIRS development samples that extend the parameter of interest beyond the range

of normal manufacturing variation, and except for reaction monitoring, are not readily available.

Use of the vibrational analytics shows rapid growth in pharmaceutical as well as in another field, the analysis of the sample performed with the QbD approach provides robust, accurate, and precise results that are more valuable compare to traditional vibrational spectroscopy. The applications of NIR are broad and used in industry for monitoring several manufacturing processes, such as granulation, drying, or mixing, in categorize to verify the end-point of these procedures. Apart from basic theoretical knowledge relating to the NIR spectral technique, these are applied for determinations of the quality and several quantities of the pharmaceutical entity. For example, measurements and control of physicochemical parameters of the end medicinal dosage, such as porosity, hardness, compression strength, thickness, disintegration time, size, and potential imitation are incorporated. NIR technique also details about plant drug analysis and biotechnology [8].

The integration of the QbD approach to NIRS methods seeks to consolidate the strengths from both approaches by expanding the quantitation design space to incorporate the relationship between the parameter of interest and the material manufacturing conditions. This will facilitate understanding of the CPPs that affect the CQAs and improve robustness.

For vibrational microspectroscopy either Raman or infrared microscopy, the readout device computer system is best for quality spectra, by interpretation of spectra finding the chemical composition at unicell or subcellular levels [9]. The best things about this method are high-resolution spectra, which is very helpful for microanalysis on minute samples [10]. This technique is applied for the diagnosis of disease also; it can differentiate the normal cells with abnormal cells or tissues because of its distinctive fingerprinting capacity and the capability to identify changes that take place during the normal cell cycle, apoptosis, or necrosis [11].

Analysis of biological compounds by Raman spectroscopy has several advantages such as the quantity of sample required is very small, water interfering capacity is a fast and changeable, noninvasive to the cells, and it allows in situ detection. Raman spectroscopy is helpful to find out the secondary structure of proteins, the link between anticancer drugs and DNA, examination of human and/or animal body fluids that is used for practice in the lab, and comparison between abnormal and normal molecules using the spectra of the sample and diagnosis of the disease [12].

8.3 Vibrational spectroscopy analysis with the quality by design approach

The design of an NIRS method by QbD principles begins with a review of the validation guidelines and specifications. These reflect the CQAs of the product, and thus, are the minimum criteria to meet. Additional CQAs which

are product specific may also exist. The sample concentration range requirements will be dictated by either existing or planned specifications or compendia limits, for example, European Pharmacopoeia (EP) assay limits 95.0%–105.0% of the label amount. Careful planning and flexibility are paramount; as the breadth of data required, it is usually obtained from a reference method, which may involve negotiations with other departments to prioritize the use of equipment and personnel who would normally have other duties and tasks to perform. A good understanding of the prospective requirements that will allow maximum efficiency and clarity to be obtained from these negotiations is therefore advisable. Every effort to leverage the extent of current knowledge from subject matter experts (SMEs) and published literature, whether located internally within an organization or externally, should be made before practical studies.

The development and validation of qualitative and quantitative NIRS methods are described in a Committee for Medicinal Products for Human Use (CHMP) guideline and is heavily influenced by the existing ICH analytical method criteria of accuracy, linearity, reproducibility, repeatability, specificity, and robustness [13,14]. There are also American Society for Testing and Materials guidance and a United States Pharmacopoeia chapter that describes the use and application of NIRS and these documents supplement existing chapters on validation [15].

More recently, QbD has been supplemented by guidelines such as ICH Q8, Q9, and Q10 which describe how integration of risk analysis, pharmaceutical quality systems, and exploratory development within the design space can lead to an increased understanding [15,16]. These guidelines were issued for the specific purpose of encouraging methods of this type and those which can be applied as part “of a system of process and controls” which could provide greater assurance to existing product quality.

As pharmaceutical companies seek to explore the benefits of continuous manufacturing using QbD and design space exploration principles, the requirement to have fast, agile, flexible, and robust control systems such as NIRS becomes increasingly important. Application of these principles to the development of NIRS ensures that provision within the method is made from the earliest design stages, resulting in a more robust measurement system that furthers the ultimate aim of feedback control. This allows changes within the design space to be made and accurately measured without significant regulatory oversight. As ultimately the measure of success of any test method may be judged by its application and acceptance by the regulatory authorities, the benefit to regulators, pharma, and the patient is obvious. With the increased attention from industry and regulators relating to PAT and QbD principles, this article will discuss the benefits of the strategic application of this philosophy when they are integrated with risk mitigation tools, design of experiments, and careful selection of equipment and methods for a NIR method on continuous processes.

Presently, the new technologies that involved a well-defined regulatory guideline and new scientific challenges carry on expanding the application and uses of Raman spectroscopy as a PAT in pharmaceutical and other industries. Advancement in Raman spectroscopy enables us for rapid tablet or capsule analysis. Off-line measurements of content uniformity primarily carried by Raman transmission and in-line Raman spectroscopy for active pharmaceutical ingredients (API) reaction monitoring or secondary pharmaceutical processes. For in-process measurements, the advance techniques like Raman spectroscopy surface-enhanced Raman spectroscopy (SERS) is used as a powerful technique, and used in bio-processing applications [17,18]. This technique is used in laboratory scale to good manufacturing practices production.

Spectroscopic techniques are used in quality control from manufacturing products to the final drug delivery system. Vibrational spectroscopy is an important tool used in the quality assurance unit in pharmaceutical and other industry and plays a very important role in the evaluation of standard material, during on-line process or off-line monitoring. The spectrums produced in the three dimensions impute well-organized and simultaneously provide spectral and spatial knowledge about the pharmaceutical samples (tablet and powder samples). Such information is important for the identification of the homogeneity in the distribution of substances in the product. Meanwhile, the additional advantage of the vibrational spectroscopy is that it gives details about the chemical bonds of the products or ingredients present in the product [19].

Vibrational spectroscopy offers several analytical tools, from absorption to reflection and dispersion techniques, extended in a large range of wave-numbers, from the visible spectrum to the microwave, thus including the NIR, MIR, and far IR (FIR) infrared regions in which the different bonds present in the sample molecules offer numerous generic and characteristic bands suitable to be employed for both qualitative and quantitative purposes. Thus it is clear that vibrational spectroscopy must be taken into consideration when evaluated specific information for the authentication of protected designation of origin foods and other food geographical indications [20].

Vibrational spectroscopy is the most powerful analytical technique that is until now to be completely developed in plant science. Formerly, such technique has been principally applied to permanent or in vitro biological samples, which do not efficiently encapsulate real-time physiological circumstances of entire organisms. Attached to multivariate analysis, this work examines the basic application of Raman or attenuated total reflectance (ATR)-FT-IR spectroscopy to establish spectral alterations analytical of healthy plant growth in leaf products of *Solanum lycopersicum*. This was achieved in the nonappearance of destructive property on leaf cells locally or on plant health systemically; Furthermore, characteristic extraction methods including PCA-LDA were used to inspect variance within a spectral

database. In vivo dimensions can successfully characterize chief key constituents of the leaf tissue and cell wall, whilst qualifying leaf growth part. The main alterations in protein and carbohydrate content of leaves were experiential, correlating with identified processes within leaf growth from cell wall expansion to leaf senescence. These conclusions demonstrate that vibrational spectroscopy is a standard technique for in vivo investigations in plant tissues [21]. FT-IR is an analytical technique that is capable to provide fingerprint-type information details about the molecule [22]. Table 8.1 reported few vibrational analyses with the QbD approach.

CU, content uniformity; *GSDs*, generalized subset designs; *FT-IR*, Fourier transform infrared; *MIR*, mid infrared; *NIR*, near infrared; *NSCLC*, nonsmall cell lung cancer; *PAT*, process analytical tool; *QbD*, quality by design; *TRS*, transmission Raman spectroscopy.

8.4 Applications of quality by design in development of vibrational approach

QbD has several applications in the analytical field that can be aligned with the previously described rational approach. These applications are divided by the analytical method and specific application and are then explained in detail to highlight the important key aspects of analytical QbD every type of method.

8.4.1 High-performance liquid chromatography for assay and impurity profile

For analysis using QbD in RP-HPLC MDS principles, first the goals of the methods and the key factors of an experiment are fixed. In a reported instance, the analysis of drug atomoxetine hydrochloride was performed by ion-pairing RP-HPLC method using QbD principles. The factors such as column temperature, mobile phase composition, and pH have been evaluated by the design of experiments approach. It has been found that this developed method is specifically sensitive to the percentage of organic modifier in the mobile phase and the column temperature. It can be concluded that this method requires controlling these factors with a planned strategy. about the method also led to a control strategy that helped to reduce the need for impurity reference standards [39].

8.4.2 Karl Fischer titration for water determination

This method determines the moisture content in the pharmaceutical product and is time consuming. Presently, it is not widely used, experimented data only used as a reference. Many spectroscopic techniques are used for the determination of moisture including vibrational spectroscopy. Recently, the

TABLE 8.1 Vibrational analyses with the quality by design (QbD) approach.

S. No.	Title	Confess	References
1.	Quality by design approach: regulatory need	In detail explanation of elements of QbD, that is, method intent, design of experiment, and risk assessment is given. Application of QbD to pharmaceutical and biopharmaceutical processes, development, and unit operation associated with it are briefly mentioned. QbD is the regulatory requirements. Pharmaceutical industry needs a regulatory compliance so as to get their product approved for marketing.	[23]
2.	Vibrational spectroscopy in analysis of pharmaceuticals	Infrared and Raman spectroscopies have shown great potentialities for drug analysis in the last decades and consequently caught the attention of the scientific world as well as of industrial developers, leading to major technological advancements. These fast, eco-friendly, and nondestructive techniques help gather essential information about the samples under examination with consistent advantages. The application of portable/ handheld NIR and Raman spectrophotometers in the analysis of pharmaceutical products for both in-process and quality control tests. Moreover, analytical methods developed by several authors are described in order to illustrate the applications explored until now.	[24]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
3.	Vibrational spectroscopy and multivariate control charts	Vibrational spectroscopy in the NIR and MIR regions coupled with multivariate control charts is a viable alternative for monitoring the stability of pharmaceuticals. The use of variable selection, combined with multivariate cumulative sum charts, improves the effectiveness of the MIR region for the identification of small changes that occur during degradation.	[25]
4.	Vibrational spectroscopic methods for quantitative analysis	Spectroscopic methods such as NIR, FT-IR, and Raman are becoming increasingly important in pharmaceutical research and manufacturing. These spectroscopic methods can be used to do rapid, nondestructive, qualitative, and quantitative analysis. Spectroscopic methods such as NIR and Raman are key PAT tools. Closely related to PAT is QbD as described in the recent ICH Q8.	[26]
5.	Active content determination of pharmaceutical tablets using near infrared spectroscopy as process analytical technology tool	The NIR method based on the transmission mode was successfully used to monitor at-line the tablet active content during the tableting process, providing better insight of the API content during the process. Hence, this improvement of control of the product quality provided by this PAT method is fully compliant with the QbD concept.	[27]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
6.	Vibrational spectroscopy: instrumentation for infrared and Raman spectroscopy	NIR spectroscopy has gained popularity. It is now an important branch of infrared spectroscopy, covering a broad range of industrial qualitative and quantitative methods of analysis. The similarities and dissimilarities can be assimilated conveniently, and for a person new to vibrational spectroscopy, it provides a more even-handed approach to these two powerful methods of analysis.	[28]
7.	Development of quality-by-design analytical methods	QbD is a systematic approach to drug development, which begins with predefined objectives, and uses science and risk management approaches to gain product and process understanding and ultimately process control. The concept of QbD can be extended to analytical methods. The current state of analytical QbD in the industry is detailed with examples of the application of analytical QbD principles to a range of analytical methods, including high-performance liquid chromatography, Karl Fischer titration for moisture content.	[29]
8.	Identification of medically relevant microorganisms by vibrational spectroscopy	Vibrational spectroscopies (infrared and Raman spectroscopy) have been developed for all sorts of analyses in microbiology. Important features of these methods are the relative ease	[30]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
		with which measurements can be performed. Finally, the use of infrared and Raman spectroscopy for the (rapid) identification of medically relevant microorganisms is discussed.	
9.	A new strategy form monitoring the quality of green coffee beans during storage	The green beans of natural coffee and pulped natural coffee were stored in three types of packaging materials in a commercial warehouse. Sensory analyses were performed, and Raman spectra were collected after 0, 3, 6, 9, 12, and 18 storage months. Raman spectra were used to construct multivariate control charts. The charts, which were constructed using principal component analysis, can only be used to identify chemical changes in the green beans from pulped natural coffee stored in different packaging materials. The use of Raman spectroscopy with Q control charts enabled the identification of chemical changes in green beans from pulped natural coffee stored in different packaging materials. The results are consistent with the results of sensory analysis.	[31]
10.	Vibrational spectroscopy as a green technology for predicting nutraceutical properties and antiradical potential of early-to-late apricot genotypes	The potential of molecular spectroscopy as a rapid tool for predicting indices of nutritional value, such as phenols, flavans, carotenoids, and antiradical potential, of early-to-late	[32]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
		apricot genotypes was investigated. Good model performances were also obtained for the prediction of antiradical potential of samples, developed herein for the first time on freeze-dried apricot samples. In general, ATR-MIR demonstrated advantages over NIR in predicting all nutraceutical parameters, whereas similar model performances were obtained for the antiradical potential tested versus ABTS + • using MIR and NIR spectral regions, both in MIR and NIR range, is adequate to predict the nutraceutical properties of apricots for screening purposes quickly and sustainably.	
11.	FT-IR spectroscopy in biopharmaceutical development	The tremendous versatility of UV-absorbance, fluorescence, and FT-IR spectroscopy creates many opportunities to leverage these techniques to support the development and manufacture of protein pharmaceuticals. FT-IR is one of the few methods capable of determining protein, secondary structure in both solid and solution. The FT-IR spectrum is sensitive to the overall secondary structural content of the protein. It is also sensitive to tertiary structure and contains contributions from the amino acid side chains. It follows that FT-IR can be a good technique to supplement a comparability study.	[33]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
12.	Vibrational spectroscopy applications in drugs analysis	Quantification can be carried out in about 10–15 minutes, including sample preparation and spectral acquisition. It is obvious that vibrational spectroscopy is capable of the analytical quantification of pharmaceutical products. With the commercial software involving chemometric approaches, the methods proposed are simple, precise, and not time consuming compared to other methods that are available in literature.	[34]
13.	Transmission Raman method for pharmaceutical tablets using quality by design (QbD) principles	To investigate the feasibility of developing a fast nondestructive at-line TRS method for core tablet potency and CU as part of a real-time release testing (RTRt) control strategy. API particle size, and concentration were studied by using a novel experimental design called GSDs. A nondestructive TRS method for core tablet potency and CU was fully validated, following ICH Q2 and EMEA NIR guidelines. The applicability of the method to process development batches was demonstrated and compared to a previously developed and validated NIR method.	[35]
14.	Monitoring strategies for quality control of agricultural products using visible and NIR spectroscopy: a review	NIR reflectance spectroscopy has become a powerful tool for the nondestructive monitoring and prediction of multiple quality and safety attributes of agro-food products. VIS-NIR	[36]

(Continued)

TABLE 8.1 (Continued)

S. No.	Title	Confess	References
		techniques, combined with chemometric methods, have shown great potential due to their fast detection speed, and the possibility of simultaneously predicting multiple quality parameters or distinguishing between products according to the objectives.	
15.	Recent advances in the vibrational spectroscopic diagnosis of nonsmall cell lung cancer	Vibrational spectroscopy has shown promising results for the detection of a variety of cancers and a limited number of studies have focused on lung cancer. vibrational spectroscopy offers an alternative or adjunct diagnostic method to be applied in bronchoscopy cytology samples. With the advent of targeted therapies, it is imperative to accurately differentiate NSCLC subtypes in order to ensure efficacy of treatment for patients.	[37]
16.	Vibrational spectroscopy as a tool for studying drug-cell interaction	As a tool in biomedical applications, use of vibrational spectroscopy for disease diagnosis, ultimately aiming toward spectral pathology. A particularly promising application is the use of vibrational spectroscopic techniques to study the interaction of drugs with cells. Many studies have demonstrated the ability to detect biochemical changes in cells in response to drug application; Vibrational spectroscopy is currently widely explored as a tool in biomedical applications. In addition to FT-IR and Raman spectroscopy.	[38]

potential of NIR spectroscopy was studied for the measurement of moisture content in tablets. Determination of water content by FT-NIR (1000–2500 nm) is used for the tablet samples prepared with different APIs and excipients, the KF method was used as the reference method. By applying the partial least square analysis model, it was found that developed KF data (water content) and NIR spectra demonstrated a strong correlation with R^2 of 0.99 and root mean square error (RMSE) of 2.55% and 2.03%, respectively. This new approach gives better and accurate results [40].

8.4.3 Quantitative color measurement

Estimation of the intensity of the color is quantitative analysis. The color of the APIs, raw material, and intermediate product of the synthesis is a very critical factor in quality control. Color variation in the product is indicative of contaminants, impurities, or degradation products. For pharmaceutical products, the color materials are specified and determined by their quality using visual appearance testing methods specified in regulatory guidelines [41].

8.4.4 Chemical identification by vibrational spectroscopy

Infrared spectroscopy is a classical method, and its advance technique is Fourier transformer infrared spectroscopy. Identification of the pharmaceutical starting materials, solvents, intermediates products, APIs, excipients, and packaging materials are experimented using vibrational spectroscopy.

8.4.5 Dissolution methods

Dissolution is very important for the pharmaceutical solid dosage form. It is necessary to know the drug release study of such formulation. The API contents of the dosage form such as tablets or capsules are released from the product; for the optimum concentration of drug to reach the desired target through the bloodstream, it first needs to get dissolved in the gastrointestinal tract followed by its absorption in the bloodstream. The drug dissolution rate fully relies on various factors such as API type, quality, and quantity of excipients, coating material used, the hardness of tablet, buffer composition, and instrument sensitivity. So, the role of quality control, to check and verify, is very crucial for the dissolution test. The old technique for performing the dissolution test is time consuming and expansive, for example, HPLC method. Therefore, an NIR spectroscopic technique in combination with multivariate statistical methods has been successfully carried out to examine the dissolution profile of drugs by several scientists [22].

Diagnostic tool in vibrational spectroscopy: in research and development of drugs the vibrational spectroscopy studies contribute to the important input for diagnosis and investigation of diseases such as cancer, diabetes, obesity, and neurological disorder. It examines other diseases also, for instance, neuro infections, epilepsy, stroke, multiple sclerosis, dementia, headache disorders, and traumatic brain injuries. Diagnosis by such a sophisticated instrument will be very helpful for the proper treatment [42].

8.4.6 API and excipients identification

Vibrational technique is employed for the identification of APIs and excipients in a drug. APIs, the active ingredient produces the desired therapeutic effects to cure the disease synthesized from the natural sources, while the excipients are the inactive or inert substances added to the finished product for a specific purpose such as appearance, test, size, or retention of quality of the formulation [43].

8.4.7 Polymorphism

The substance (here, a drug compound) which exists in more than one crystalline form having the same physicochemical properties and biological activities is said to be polymorphic. This is an important step in the development of pharmaceutical ingredients and affects different features of the manufacturing of the drug substance and the drug product. Different kinds of polymorphic forms of the active substance have different physiochemical properties in terms of melting point, bioavailability, solubility, chemical reactivity, dissolution rate, stability, and density [44].

References

- [1] Nevalainen D, Berte L, Kraft C, Leigh E, Picaso L, Morgan T, et al. Evaluating laboratory performance on quality indicators with the six sigma scale. *Arch Pathol Lab Med* 2000;124:516–19.
- [2] Schweitzer M, Pohl M. Implications and opportunities of applying QbD principles to analytical measurements. *Pharm Technol* 2010;34(2):52–9.
- [3] Pharmaceutical Quality System (2007), ICH tripartite guidelines. In: International conference on harmonization of technical requirements for registration of pharmaceuticals for human use.
- [4] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [5] Siesler HW. Vibrational spectroscopy. *Polym Sci. Compr* 2012;2:255–300.
- [6] US Food and Drug Administration, Guidance for industry: Q10 pharmaceutical quality systems, 2009.
- [7] Ozaki Y. Near-infrared spectroscopy-its versatility in analytical chemistry. *Anal Sci* 2012;28(6):545–63.

- [8] Jamrógiewicz M. Application of the near-infrared spectroscopy in the pharmaceutical technology. *J Pharm Biomed Anal* 2012;66:1–10.
- [9] Wetzel DL, LeVine SM. Biological applications of infrared microspectroscopy: infrared and Raman spectroscopy of biological materials. New York, NY: Marcel Dekker Inc; 2001.
- [10] Dumas P. Microanalysis and imaging capabilities of synchrotron infrared microscopy. *J Phys IV Fr* 2003;104:359–64.
- [11] Stuart B. Modern infrared spectroscopy. 2nd ed. Chichester: John Wiley & Sons, LTD; 1996.
- [12] Firdous S, Nawaz M, Ahmed M, Anwar S, Rehman A, Rashid R, et al. Measurement of diabetic sugar concentration in human blood using Raman spectroscopy. *Laser Phys* 2012;22:1090–4.
- [13] El-Hargasy A, D’Amico F, Drennen JK. A process analytical technology approach to near-infrared process control of pharmaceutical powder blending part I: D-optimal design for characterization of powder mixing and preliminary spectral data, evaluation. *J Pharm Sci* 2006;95(2):392–406.
- [14] Committee for Medicinal Products for Human Use – Note for Guidance on the use of Near-Infrared Spectroscopy by the Pharmaceutical Industry and the Data Requirements for New Submissions and Variations.
- [15] Validation of analytical procedures: text and methodology Q2, R1. October 1994.
- [16] Quality Risk Management Q9, International Conference on Harmonisation, November 2005.
- [17] Pharmaceutical Quality System, Q10, International Conference on Harmonisation, June 2008.
- [18] Jarvis RM, Goodacre R. Characterization and identification of bacteria using SERS. *Chem Soc Rev* 2008;37(5):931–6.
- [19] Aazam ES, Zaheer Z. Growth of Ag-nanoparticles in an aqueous solution and their antimicrobial activities against Gram-positive, Gram-negative bacterial strains, and *Candida* fungus. *Bioprocess Biosyst Eng* 2016;39(4):575–84.
- [20] Kandpal LM, Park E, Tewari J, Cho B-K. Spectroscopic techniques for nondestructive quality inspection of pharmaceutical products: a review. *J Biosyst Eng* 2015;40(4):394–408.
- [21] Butler HJ, McAnish MR, Adams S, Martin FL. Application of vibrational spectroscopy techniques to non-destructively monitor plant health and development. *Anal Methods* 2015;7(10):.
- [22] Balan V, Mihai CT, Cojocaru FD, Uritu CM, Dodi G, Botezat D, et al. Vibrational spectroscopy fingerprinting in medicine: from molecular to clinical practice. *Materials* 2019;12(18):2884.
- [23] Singh B, Beg S. Attaining product development excellence and federal compliance employing quality by design (QbD) paradigms. *Pharma Rev* 2015;13(9):35–44.
- [24] Vibrational spectroscopy in analysis of pharmaceuticals: critical review of innovative portable and handheld NIR and Raman spectrophotometers, *Trends in Analytical Chemistry*, Mafalda Cruz Sarraquc, João Almeida Lopes, 2019; 114, 251-259.
- [25] Tórras AR, Grangeiro Jr. S, Fragoso WD. Vibrational spectroscopy and multivariate control charts: a new strategy for monitoring the stability of captopril in the pharmaceutical industry. *Microchem J* 2017;133:279–85.
- [26] Long FH. Vibrational spectroscopic methods for quantitative analysis handbook of stability testing in pharmaceutical development. In: Huynh-Ba K, editor. *Handbook of Stability Testing in Pharmaceutical Development*. 2009. p. 223–40.
- [27] Chavez PF, Sacré PY, Bleye CD, Netchacovitch L, Mantanus J, Motte H, et al. Active content determination of pharmaceutical tablets using near infrared spectroscopy as process analytical technology tool. *Talanta*. 2015;144(1):1352–9.

- [28] Coates J. Vibrational spectroscopy: instrumentation for infrared and Raman spectroscopy. *Appl Spectrosc Rev* 1998;33(4):267–425.
- [29] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharm Rev* 2016;14(10):61–6.
- [30] Maquelina K, Kirschner C, Choo-Smith LP, van den Braak N, Endtz Ph H, Naumann D, et al. Identification of medically relevant microorganisms by vibrational spectroscopy. *J Microbiol Methods* 2002;51(3):255–71.
- [31] Abreu GF, Boréma FM, Oliveira LFC, Almeida MR, Alves APC. Raman spectroscopy: a new strategy for monitoring the quality of green coffee beans during storage. *Food Chem* 2019;287:241–8.
- [32] Amorielloa T, Ciccorittib R, Carboneb K. Vibrational spectroscopy as a green technology for predicting nutraceutical properties and antiradical potential of early-to-late apricot genotypes. *Postharvest Biol Technol* 2019;155:156–66.
- [33] Brader ML. UV-absorbance, fluorescence and FT-IR spectroscopy in biopharmaceutical development. *Biophysical characterization of proteins in developing biopharmaceuticals* 2020;97–121.
- [34] Bunaciu AA, Aboul-Enein HY. *Encyclopedia of spectroscopy and spectrometry*. 3rd ed. 2017. p. 575–81.
- [35] Corredor CC, Vikstrom C, Persson A, Bu X, Both D. Development and robustness verification of an At-Line transmission Raman method for pharmaceutical tablets using quality by design (QbD) principles. *J Pharm Innov* 2018;1–14.
- [36] Cortésa V, Blascob J, Aleixosc N, Cuberob S, Talensa P. Monitoring strategies for quality control of agricultural products using visible and near-infrared spectroscopy. A review. *Trends Food Sci Technol* 2019;85:138–48.
- [37] O’Deaa D, Lyngb FM, Nicholson S, O’Connell F, Maguire A, Malkin A. Recent advances in the vibrational spectroscopic diagnosis of non-small cell lung cancer. *Vib Spectrosc* 2019;104:102946.
- [38] Jamieson LE, Byrne HJ. Vibrational spectroscopy as a tool for studying drug-cell interaction: could high throughput vibrational spectroscopic screening improve drug development. 2017; 91:16–30.
- [39] Panda SS, Beg S, Kumar BVVR, Sahu J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.
- [40] Zhou L, Socha J, Vogt FG, Chen S, Kord AS. A systematic method development strategy for water determinations in drug substance using Karl Fischer titrations. *Am Pharm Rev* 2010;13:74–83.
- [41] Zhou L, Vogt FG, Overstreet PA, Dougherty JT, Clawson JS, Kord AS. A systematic method development strategy for quantitative color measurement in drug substances. *Am Pharm Rev* 2011;6:217–31.
- [42] de Oliveira Neves AC, Soares GM, de Moraes SC, Lopes da Costa FS, Porto DL, Gomes de Lima KM. Dissolution testing of isoniazid, rifampicin, pyrazinamide and ethambutol tablets using near-infrared spectroscopy (NIRS) and multivariate calibration. *J Pharm Biomed Anal* 2012;57:115–19.
- [43] Jones AW. Early drug discovery and the rise of pharmaceutical chemistry. *Drug Test Anal* 2011;3:337–44.
- [44] Thiruvengadam E, Vellaisamy G. Polymorphism in pharmaceutical ingredients: a review. *World J Pharm Sci* 2014;3:621–33.

Chapter 9

Quality by design-based development of nondestructive analytical techniques

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9.1 Introduction

There is a paradigm shifting in the pharmaceutical industry aiming to understand the processes involved that result in the improvement of product quality. To achieve this goal, the concept of quality by design (QbD) has been enacted by the regulatory body and adopted by the pharmaceutical industry [1–4]. QbD provides comprehensive understanding of the multifactor inputs that influence the product quality. Another essential element of QbD is to develop a planned strategy to evaluate the changes throughout the life cycle of a product. The extension of QbD is further applied to the analytical methods for their better understanding as well as the method performance [5–7]. The most important feature of the analytical method is to make them precise, accurate, reliable, and suitable for its intended purpose of application, which is also known as analytical target profile. Further, the analytical method is identified by a range of operating conditions that are defined by the analytical design space or method operable design region. One of the interesting aspects of defining a QbD design space is to achieve better operating robustness and flexibility in routine method operation [8]. The recent development of the integration of regulatory guidelines on analytical methods and QbD together emphasizes the significance of this approach in the field of pharmaceutical analysis [9]. In June 2018, the ICH announced to prepare a new Q14 guideline for analytical procedure development and validation, which will include a revision of the existing Q2 (R1) guideline on validation of analytical procedures Q2 (R2) and a topic of QbD concept for analytical methods, termed as analytical quality by design (AQbD) [10]. Thus the AQbD

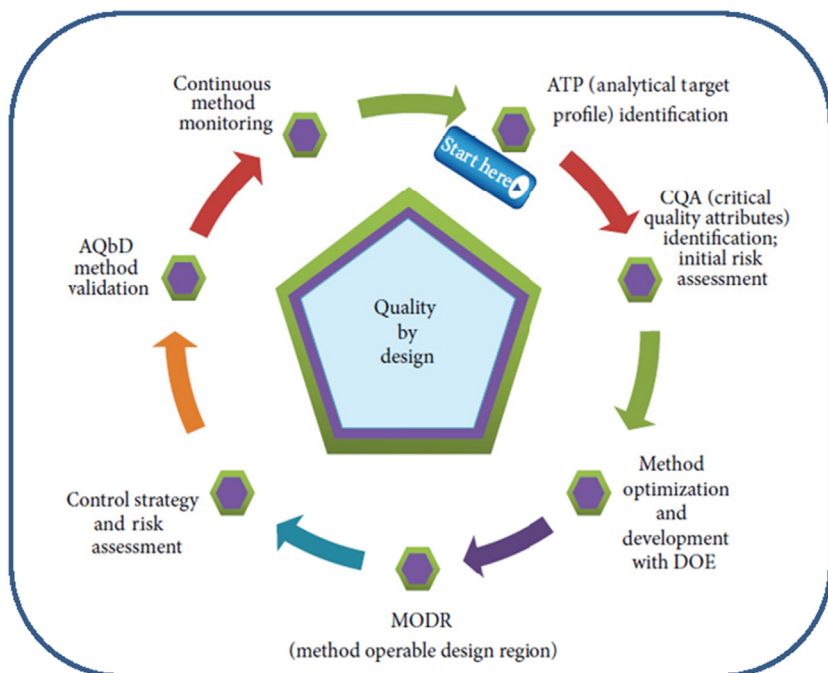


FIGURE 9.1 Elements of analytical quality by design (AQbD) tool. *Source: Adapted from International Conference on Harmonization (ICH). Guidance for industry: Q14 Analytical Process Development; June 2018.*

approach defines a proper control strategy for the analytical method to control variability and improve robustness in method performance with high quality (Fig. 9.1) [11]. Besides, another challenge faced by the pharmaceutical industry is reducing time frame, cost reduction, simplification of processes, and scale-up activities. Therefore most of the pharmaceutical companies are adopting the concept of continuous manufacturing. However, the acquisition of on/in-line analysis and real-time information is difficult on traditionally available analytical tools. Hence, to overcome these hurdles, process analytical technologies (PAT) are developed to monitor the real-time continuous processes together with the implementation of QbD approach [12]. Looking into the importance of these approaches and paradigm shift in pharmaceutical manufacturing, this review will highlight the important PAT tools and utilization of QbD in various pharmaceutical processes.

9.2 Process analytical technology

PAT is employed for analyzing real-time measurement of critical quality attributes of raw materials, and critical process parameters (CPPs) involved

in the manufacturing [13]. It interfaces the process with the analytical instrument and modifies the process conditions to ensure high quality in the final product. Several benefits associated with the implementation of PAT in the pharmaceutical industry and few are listed below:

- Ease in the continuous manufacturing
- Increasing operation efficiencies
- Increased process understanding
- Building QbD in the process
- Improvement in product quality by reducing batch variability and failure
- Reducing operation costs
- Increased regulatory compliance

Undoubtedly, pharmaceutical product-process design is quite complex and systematic understanding together with an efficient analytical instrument is very essential. The real-time monitoring tools have increasingly attracted the interests of pharmaceutical manufacturers toward the nondestructive analysis of raw materials in various conditions, namely at-line, on-line, and in-line [13]. Therefore the goals of continuous manufacturing and real-time monitoring can be accomplished using nondestructive analytical techniques such as near-infrared spectroscopy (NIRS), Raman spectroscopy, and terahertz pulsed spectroscopy (THS) [14–16].

9.3 Chemometric tools employed in quality by design

Complexity in pharmaceutical products and processes can only be effectively described by multifactorial relationships. Besides, to acquire meaningful data from the PAT framework is a tedious task. Therefore there is a need for measurement techniques for the conversion of a large amount of data and identification of multifactorial relationship through multivariate analysis [17]. Multivariate tools offer the users to develop a correlation between the raw data and processes involved. It is a valuable QbD toolkit to generate a model on the raw material or CPPs. Most commonly used multivariate tools used in the PAT framework are principal component analysis (PCA) and partial least squares (PLS) regression.

9.3.1 Principal component analysis

PCA is a mathematical procedure that transforms a large set of variables into a lower-dimensional set of new variables designated as principal components [17]. The purpose of PCA is to express the relevant information contained in the data set using a lower number of variables. Using PCA in a sample set, irrelevant variation (redundancy and noise) can be minimized and the projection of the data into a smaller number of components. This enables the use of simpler graphical representations and improving the

interpretability of the information. The application of PCA in the PAT framework is to determine the relationship between the result of process analyzer and process attributes [17].

9.3.2 Partial least squares regression

PLS is a multivariate analysis tool that combines PCA and multiple regressions [18]. This technique is used to find the correlation between two variables (X and Y) and the principal components will be selected as the directions to understand the larger amount of variance in X that is directly related to variance in Y . Because of this feature, PLS models are more robust than other multivariate techniques [18]. Besides, PLS is often applied to spectroscopy data to obtain measurements that enable real-time release testing in the pharmaceutical industry.

These multivariate tools are valuable in simplifying and interpretation of the data in real time as well as contribute to the understanding of the process knowledge. However, there are certain limitations associated with them. For instance, these models are based on data and valid only within the known space, thus requires a large amount of data for confidence on the developed model. Further, to drive a conclusion from the exploratory analysis may lead to inaccurate results.

9.4 Implementation of quality by design-based nondestructive analytical techniques in pharmaceutical unit operations

Among the PAT tools that are largely being used in the pharmaceutical industry are NIRS, Raman spectroscopy, and THS technologies. The popularity of these PAT tools in the pharmaceutical industry is because of their peculiar features, namely nondestructive nature, rapid in operation, no/less sample preparation, real-time monitoring of manufacturing processes, identification and characterization of raw materials and intermediates in various conditions such as at-line, on-line, and in-line [14]. These characteristics of PAT tools make them suitable to employ them in the monitoring of various unit operations involved in the pharmaceutical formulations, thus playing an integral role in current good manufacturing practice. Besides, the advancement in instrumentation and chemometric tools further facilitates their applications in both qualitative and quantitative purposes [17]. The paradigm shift in QbD and PAT framework has encouraged risk-based approaches in real-time testing and continuous manufacturing in pharmaceutical industries. QbD-enabled applications of PAT tools in monitoring various pharmaceutical unit operations (summarized in Table 9.1) are discussed in the following subsections.

TABLE 9.1 Applications of process analytical technology (PAT) tools in monitoring various pharmaceutical unit operations.

Unit operation	Critical process information	Mode of analyzer	Data analysis	PAT tool
Blending	Content uniformity	Off-line	PLS	NIRS
	Content uniformity	On-line	PLS	NIRS
	Content uniformity Drug distribution	On-line Chemical imaging	PC-SDA	NIRS
	Content uniformity	Off-line	PCA, PLS	THS
	Content uniformity	In-line	SNV	Raman spectroscopy
Granulation	Solid-state polymorphic conversion of API	In-line	MLR	NIRS
	Formulation homogeneity; Water-content	In-line	PCA	NIRS
	Moisture content, particle size distribution, and bulk density	On-line	PCA, PLS	NIRS
	Moisture content of granule	On-line	PCA, PLS	NIRS
	Particle size distribution	In-line	Univariate	Raman spectroscopy
Drying	Product assay Water content Drying endpoint	In-line	SNV	NIRS
	Drying endpoint	In-line	PLS	NIRS
	API phase transformation	In-line	PLS	Raman spectroscopy
	API phase transformation during drying	In-line	PLS	NIRS Raman spectroscopy
Coating	Tablet coating thickness monitoring	In-line	Univariate	NIRS
	Tablet coating thickness monitoring	In-line	PLS	NIRS

(Continued)

TABLE 9.1 (Continued)

Unit operation	Critical process information	Mode of analyzer	Data analysis	PAT tool
	Monitoring coating thickness and endpoint determination	In-line	PLS	Raman spectroscopy
	Monitoring coating thickness and determination of mean dissolution time	In-line	PLS	Raman spectroscopy TPI
	Sustained-release tablet coating	In-line	—	THS TPI
	Monitoring tablet film-coating thickness up to 1 mm	In-line	—	THS TPI
Freeze drying	Mannitol phase behavior	In-line	PCA	NIRS Raman spectroscopy
	Degree of crystallinity of glycine	Off-line	PLS	NIRS
	Degree of crystallinity of fenofibrate	In-line	PLS	Raman spectroscopy
	Degree of crystallinity of amino acids/gelatin mixtures	Off-line	PLS	THS
Miscellaneous pharmaceutical processes	Solid-state transformation of erythromycin dihydrate during pelletization process	At-line	Univariate	NIRS
	Detection of crack initiation during film coating process	In-line	—	THS
	Solid-state characterization of drug and polymer during hot-melt extrusion process	In-line	PLS	Raman spectroscopy

API, Active pharmaceutical ingredients; *MLR*, multiple linear regression; *NIRS*, near-infrared spectroscopy; *PC-SDA*, principal component scores distance analysis; *PCA*, principal component analysis; *PLS*, partial least squares; *SNV*, standard normal variate; *THS*, terahertz spectroscopy; *TPI*, terahertz pulsed imaging.

9.4.1 Blending

Quality control of pharmaceuticals has been achieved by developing a method using NIRS for assessing the quantification of paracetamol and three excipients (microcrystalline cellulose, talc, and magnesium stearate) [19]. This method was found to be effective in simultaneously quantifying both the drug and excipients in the formulation development. Similarly, a multivariate calibration approach using NIRS for uniformity determination of blend containing drug in three major excipients (crospovidone, lactose, and microcrystalline cellulose) was also investigated [20]. Multivariate PLS calibration model for on-line predictions of the active pharmaceutical ingredients (API) content during the blending process was found successful in quality control of tablet. Interestingly, high performance liquid chromatography (HPLC) of the formulated tablets confirmed the accuracy of implementing blend uniformity endpoint determination method by NIRS. In another study performed by Puchert et al. [21], the process capability of on-line NIRS was explored using advanced NIR-CI (chemical imaging) tool in terms of the spatial distribution of a drug in the tablets. Implementation of multivariate data analysis to determine a design space together with chemical imaging was found to be a powerful PAT tool for assessing blend homogeneity according to QbD principles.

The quantification of API and excipients in tablets was also monitored by THS that is also an emerging PAT tool for potential pharmaceutical applications. Theophylline concentration in the presence of lactose, magnesium stearate, starch, or avicel was investigated using spectral characteristics method, spectral superposition method, and multivariate data analysis methods (PCA and PLS) [15]. This study demonstrated that the multivariate calibration models can predict tablet concentrations quickly and reliably as compared to the other two methods. Hausman et al. [22] monitored the blend uniformity of azimilide dihydrochloride using on-line Raman spectroscopy. This PAT tool offered blend profiles rapidly in real time to collect a large number of samples during the blending process. Besides, the blend uniformity results obtained from on-line Raman spectroscopy were significantly correlated with the HPLC uniformity results.

9.4.2 Granulation

Granulation is the most widely used unit process for pharmaceutical solids. The purpose of converting a powder into uniform granules for improving the material properties and better compressibility to get finished tablet formulation. Various PAT tools are employed to monitor the water content, integrity of API, and the endpoint of granulation processes. Solid-state polymorphic conversion of a drug during the wet granulation process was monitored by in-line NIRS [23]. The formation of undesired polymorph B in the wet

granulation samples was revealed based on their narrow spectral regions unique for this solid form. Thus the undesired polymorphic conversion can be identified and quantified easily during the granulation process. Rantanen et al. [24] investigated the endpoints of the three subphases of high-shear wet granulation using in-line NIRS. Chemical information (homogeneity of the formulation and the amount of water in wet mass), as well as physical information (particle size of granules), was extracted using PCA calibration model and validated.

Fluid bed granulation is another wet granulation technique for producing granules by spraying solution on to a fluidized powder. This technique seems to be more popular in the industry because of the simplification of process and more economical. Alcala et al. [25] developed an on-line NIRS method for monitoring the fluid bed granulation process. Multivariate model (PCA) was used for monitoring different steps involved in the granulation process, while PLS model was used for predicting critical parameters such as the moisture content, particle size distribution, and bulk density. Abouzaid and coworkers [26] studied the process variables such as atomizing pressure and airflow rate and their impact on the properties of the granules using on-line NIRS. Besides, the authors proposed a PLS model to determine the moisture content of granule in the wet granulation process. Walker et al. [27] explored Raman spectroscopy to study the in situ measurement of the composition of the material within the fluidized bed in three spatial dimensions and as a function of time. This technique demonstrated accurate correlation with independent granulation experiments that provided particle size distribution analysis in a short time frame (10 seconds).

9.4.3 Drying

Online process control of the drying progress and determination of the ideal drying endpoint was measured using a continuous NIRS [28]. The calibration model was developed and critical parameters such as product assay, water content, and drying endpoint were calculated. This monitoring system was found to be automated and considerably improved product quality. Peinado et al. [29] developed a NIRS method based on QbD as well as ICH guidelines to monitor the drying endpoint of a fluidized bed process. The implementation of this in-process method allows real-time control with additional benefits of reduction in operation time and labor, sample handling, and waste generation. Kogermann et al. [30] analyzed the solid-state forms of carbamazepine dihydrate using Raman spectroscopy. This technique in combination with PLS regression enabled in-line analysis of the solid-state transformations of carbamazepine during dehydration in a fluidized bed dryer. Seminal work was also carried out by Aaltonen et al. [31] to directly monitor theophylline monohydrate granules during fluidization process using in-line NIRS and Raman spectroscopy. The transformation of monohydrate

to anhydrate was tracked quantitatively with both spectroscopic methods using the multivariate analysis (PLS). However, both NIRS and Raman spectroscopy were found to be complementary, since the former is particularly sensitive to water and the latter to crystal structure changes.

9.4.4 Coating

Among the advancement in the field of solid drug delivery system, tablets are the most important solid dosage forms for the delivery of drugs to the patients. Controlling the release kinetics, the coating is applied to the tablet either sugar or polymer coating to modify the drug release rate. Therefore the uniformity and thickness of the coating layer play an important role for several pharmaceutical dosage forms such as controlled-release tablets, film-coated tablets, and sustained-release tablets. PAT tools are being used to investigate the coating characteristics to estimate the coating endpoint in achieving the film thickness [32]. NIRS-based method has been developed to monitor the real-time coating process using univariate analysis [33]. This method was found to measure the tablet coating thickness efficiently. Gendre et al. [34] performed in-line NIRS measurement to monitor the coating process and thickness. Real-time quantitative monitoring of the coating operation was successfully performed from PLS calibration models. Besides, this method was found to be reliable when compared with the terahertz pulsed spectroscopy (TPS)-based reference method. Raman spectroscopy has also been used as an important PAT tool to monitor the coating process. Muller et al. [35,36] performed the coating analysis of diprophyl-line using in-line Raman spectroscopy. The chemometric-based method was validated according to ICH Q2 guidelines and was found to give a reliable result of coating thickness. Additionally, this method was also applicable within the restricted range by varying process parameters and measurement conditions. The same research group also developed in-line Raman spectroscopy method for monitoring of functional coating [37]. PLS model was developed to determine the mean dissolution time of sustained-release formulation. Besides, the coating thickness was measured by terahertz pulsed imaging (TPI) system. In another study, TPS was employed to analyze the sustained-release tablet coating [38]. Multidimensional coating thickness was reconstructed using TPI for the analysis of coating layer thickness, reproducibility, and uniformity. These measurements were validated with optical microscopy imaging and were found to be in good agreement with this destructive analytical technique. May et al. [39] demonstrated the terahertz in-line coating process measurement of individual pharmaceutical tablets during film coating in a pan coating unit. Terahertz sensor provides the thickness of up to 100 individual tablet coatings per minute and has the potential to determine coating thickness up to 1 mm.

9.4.5 Freeze drying

Another important process widely used in the pharmaceutical industry is freeze-drying for the conversion of solutions into solid material. However, this process is complex and time-consuming and often leads to transformations of a poor quality end product (dry cake). Therefore to improve process efficiency and to ensure the final product quality, PAT tools were found to be useful. For instance, mannitol phase behavior in the freeze-drying product was monitored using two in-line process analyzers such as NIRS and Raman spectroscopy together [40]. Both techniques were found useful for continuous control of all critical process aspects of mannitol as well as making interventions possible to avoid batch loss. Likewise, the degree of crystallinity of glycine in the freeze-dried sucrose-glycine mixtures was investigated using off-line NIRS [41]. PLS model was found to give a correlation of NIR spectral changes with the degree of crystallinity of glycine. Seminal work further was undertaken to elucidate the mechanisms for determining the size of the fenofibrate crystals during the freeze-drying process using Raman spectroscopy [13]. This study highlighted that a high freezing rate and a relatively low crystallization temperature resulted in the smallest crystals, which in turn lead to the highest dissolution rate. Darkwah et al. [42] investigated the feasibility of THS in examining the degree of crystallinity of cofreeze-dried amino-acid/gelatine mixtures in freeze-dried rapidly disintegrating tablets. THS has shown sufficient measurement sensitivity similar to other PAT tools for off-line measurements in freeze-drying process development.

9.4.6 Miscellaneous pharmaceutical processes

Solid-state transformation of a drug during the pelletization process (extrusion-spheronization and drying process) was monitored by at-line NIRS [43]. A process-induced transformation of erythromycin dihydrate to the dehydrated form was observed for the pellets dried at 60°C. The changes in the solid form were also confirmed by variable temperature-powder X-ray diffraction measurement. Another interesting study demonstrated the application of THS to detect the crack initiation in a film-coated layer on a drug tablet [44]. This technique nondestructively detects the film surface density and interface density differences between the film-coated layer and uncoated tablet, and therefore helps in the prediction of cracks in the film-coated tablet. Raman spectroscopy has also found its application to characterize the polymer-drug solid state during pharmaceutical hot-melt extrusion. Saerens et al. [45] studied the in-line quantification and solid-state characterization of metoprolol tartrate and polymer during the hot-melt extrusion process. PLS model was developed and validated to identify the partial amorphization of a drug as well as the interaction between the drug and polymer during the process.

9.5 Conclusion and future outlook

The understanding of process parameters and monitoring of these processes in real time is essentially important in developing high-quality pharmaceutical products. PAT tools that are largely nondestructive allow in-process monitoring of several unit operations in the production of pharmaceutical dosage forms. The integration of QbD and PAT framework plays a key role in studying the physical and chemical critical information for better process understanding. Besides, the growing interest of continuous manufacturing in the pharmaceutical industry requires special attention of interfacing of probes into process streams for real-time monitoring. Therefore the judicious selection of probes analyzer or complementary process analyzer is utmost important to extract the required critical process information. Another important consideration is the effective sampling volume that depends upon the appropriate location of process analyzer for the acquisition of meaningful data. Furthermore, proper data management is an important aspect of the successful implementation of the PAT framework for developing qualitative and quantitative models using suitable chemometric tools. This can be achieved by the proper training of personnel in the handling of complex data and their evaluation with the aid of chemometric analysis. The recent development in chemical imaging by integrating imaging and spectroscopy to attain both spatial and spectral information is quite popular. Therefore the potential of this emerging platform as a PAT tool has to be explored in many stages of process operation and quality control in the pharmaceutical industry. Overall, the applications of QbD-driven PAT strategy is valuable to facilitate the thorough tracking of process operations and subsequently accelerate the manufacturing of the high-quality pharmaceutical product.

References

- [1] Beg S, Hasnain MS. *Pharmaceutical quality by design: principles and applications*. New York, NY: Academic Press (Elsevier); 2019.
- [2] Beg S, Hasnain MS, Rahman M, Swain S. Introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. New York, NY: Academic Press; 2019. p. 1–17.
- [3] Beg S, Rahman M, Swain S. Quality by design applications in pharmaceutical product development. *Pharma Focus Asia* 2020;1–5.
- [4] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–84.
- [5] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharma Times* 2014;46(8):71–5.
- [6] Phil B, Phil N, Marion C, Duncan T, Truman K. The application of quality by design to analytical methods. *Pharm Tech* 2007;31(10):142147.
- [7] Swain S, Jena BR, Madugula D, Beg S. Application of quality by design paradigms for development of solid dosage forms. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. New York, NY: Academic Press; 2019. p. 109–30.

- [8] International Conference on Harmonization (ICH). Guidance for industry: Q8 (R2) pharmaceutical development, ICH harmonised tripartite guideline. Sept 4, 2009.
- [9] Vogt FG, Kord AS. Development of quality-by-design analytical methods. *J Pharm Sci* 2011;100(3):797–812. Available from: <https://doi.org/10.1002/jps.22325>.
- [10] International Conference on Harmonization (ICH). Guidance for industry: Q14 Analytical Process Development; June 2018.
- [11] Raman NVVSS, Mallu UR, Bapatu HR. Analytical quality by design approach to test method development and validation in drug substance manufacturing. *J Chem* 2015;435129. Available from: <https://doi.org/10.1155/2015/435129>.
- [12] Fonteyne M, Vercruysse J, De Leersnyder F, Van Snick B, Vervae C, Remon JP, et al. Process analytical technology for continuous manufacturing of solid-dosage forms. *TrAC Trend Anal Chem* 2015;67:159–66. Available from: <https://doi.org/10.1016/j.trac.2015.01.011>.
- [13] de Waard H, De Beer T, Hinrichs WL, Vervae C, Remon JP, Frijlink HW. Controlled crystallization of the lipophilic drug fenofibrate during freeze-drying: elucidation of the mechanism by in-line Raman spectroscopy. *AAPS J* 2010;12(4):569–75. Available from: <https://doi.org/10.1208/s12248-010-9215-z>.
- [14] De Beer T, Burggraave A, Fonteyne M, Saerens L, Remon JP, Vervae C. Near infrared and Raman spectroscopy for the in-process monitoring of pharmaceutical production processes. *Int J Pharm* 2011;417(1):32–47. Available from: <https://doi.org/10.1016/j.ijpharm.2010.12.012>.
- [15] Wu H, Heilweil EJ, Hussain AS, Khan MA. Process analytical technology (PAT): quantification approaches in terahertz spectroscopy for pharmaceutical application. *J Pharm Sci* 2008;97(2):970–84. Available from: <https://doi.org/10.1002/jps.21004>.
- [16] Gouveia FF, Rahbek JP, Mortensen AR, Pedersen MT, Felizardo PM, Bro R, et al. Using PAT to accelerate the transition to continuous API manufacturing. *Anal Bioanal Chem* 2017;409(3):821–32. Available from: <https://doi.org/10.1007/s00216-016-9834-z>.
- [17] Ferreira AP, Tobyn M. Multivariate analysis in the pharmaceutical industry: enabling process understanding and improvement in the PAT and QbD era. *Pharm Dev Technol* 2015;20(5):513–27. Available from: <https://doi.org/10.3109/10837450.2014.898656>.
- [18] Wold S, Sjostrom M, Eriksson L. PLS-regression: a basic tool of chemometrics. *Chemom Intell Lab Syst* 2001;58(2):109–30. Available from: [https://doi.org/10.1016/S0169-7439\(01\)00155-1](https://doi.org/10.1016/S0169-7439(01)00155-1).
- [19] Sarrauca MC, Lopes JA. Quality control of pharmaceuticals with NIR: from lab to process line. *Vib Spectrosc* 2009;49(2):204–10. Available from: <https://doi.org/10.1016/j.vibspec.2008.07.013>.
- [20] Sulub Y, Konigsberger M, Cheney J. Blend uniformity end-point determination using near-infrared spectroscopy and multivariate calibration. *J Pharm Biomed Anal* 2011;55(3):429–34. Available from: <https://doi.org/10.1016/j.jpba.2011.02.017>.
- [21] Puchert T, Holzhauser CV, Menezes JC, Lochmann D, Reich G. A new PAT/QbD approach for the determination of blend homogeneity: combination of on-line NIRS analysis with PC Scores Distance Analysis (PC-SDA). *Eur J Pharm Biopharm* 2011;78(1):173–82. Available from: <https://doi.org/10.1016/j.ejpb.2010.12.015>.
- [22] Hausman DS, Cambron RT, Sakr A. Application of Raman spectroscopy for on-line monitoring of low dose blend uniformity. *Int J Pharm* 2005;298(1):80–90. Available from: <https://doi.org/10.1016/j.ijpharm.2005.04.011>.
- [23] Li W, Worosila GD, Wang W, Mascaro T. Determination of polymorph conversion of an active pharmaceutical ingredient in wet granulation using NIR calibration models

- generated from the premix blends. *J Pharm Sci* 2005;94(12):2800–6. Available from: <https://doi.org/10.1002/jps.20501>.
- [24] Rantanen J, Wikstrom H, Turner R, Taylor LS. Use of in-line near-infrared spectroscopy in combination with chemometrics for improved understanding of pharmaceutical processes. *Anal Chem* 2005;77(2):556–63. Available from: <https://doi.org/10.1021/ac048842u>.
- [25] Alcala M, Blanco M, Bautista M, Gonzalez JM. On-line monitoring of a granulation process by NIR spectroscopy. *J Pharm Sci* 2010;99(1):336–45. Available from: <https://doi.org/10.1002/jps.21818>.
- [26] Abouzaid AS, Salem MY, Elzanfaly ES, El Gindy AE, Hoag S, Ibrahim A. Screening the fluid bed granulation process variables and moisture content determination of pharmaceutical granules by NIR Spectroscopy. *Eur J Chem* 2017;8(3):8. Available from: <https://doi.org/10.5155/eurjchem.8.3.265-272.1608>.
- [27] Walker GM, Bell SEJ, Greene K, Jones DS, Andrews GP. Characterisation of fluidised bed granulation processes using in-situ Raman spectroscopy. *Chem Eng Sci* 2009;64(1):91–8. Available from: <https://doi.org/10.1016/j.ces.2008.09.011>.
- [28] Mark J, Karner M, Andre M, Rueland J, Huck CW. Online process control of a pharmaceutical intermediate in a fluidized-bed drier environment using near-infrared spectroscopy. *Anal Chem* 2010;82(10):4209–15. Available from: <https://doi.org/10.1021/ac1004579>.
- [29] Peinado A, Hammond J, Scott A. Development, validation and transfer of a near infrared method to determine in-line the end point of a fluidised drying process for commercial production batches of an approved oral solid dose pharmaceutical product. *J Pharm Biomed Anal* 2011;54(1):13–20. Available from: <https://doi.org/10.1016/j.jpba.2010.07.036>.
- [30] Kogermann K, Aaltonen J, Strachan CJ, Pollanen K, Heinamaki J, Yliruusi J, et al. Establishing quantitative in-line analysis of multiple solid-state transformations during dehydration. *J Pharm Sci* 2008;97(11):4983–99. Available from: <https://doi.org/10.1002/jps.21357>.
- [31] Aaltonen J, Kogermann K, Strachan CJ, Rantanen J. In-line monitoring of solid-state transitions during fluidisation. *Chem Eng Sci* 2007;62(1):408–15. Available from: <https://doi.org/10.1016/j.ces.2006.08.061>.
- [32] Knop K, Kleinebudde P. PAT-tools for process control in pharmaceutical film coating applications. *Int J Pharm* 2013;457(2):527–36. Available from: <https://doi.org/10.1016/j.ijpharm.2013.01.062>.
- [33] Perez-Ramos JD, Findlay WP, Peck G, Morris KR. Quantitative analysis of film coating in a pan coater based on in-line sensor measurements. *AAPS PharmSciTech* 2005;6(1):E127–36. Available from: <https://doi.org/10.1208/pt060120>.
- [34] Gendre C, Genty M, Boiret M, Julien M, Meunier L, Lecoq O, et al. Development of a process analytical technology (PAT) for in-line monitoring of film thickness and mass of coating materials during a pan coating operation. *Eur J Pharm Sci* 2011;43(4):244–50. Available from: <https://doi.org/10.1016/j.ejps.2011.04.017>.
- [35] Muller J, Knop K, Thies J, Uerpmann C, Kleinebudde P. Feasibility of Raman spectroscopy as PAT tool in active coating. *Drug Dev Ind Pharm* 2010;36(2):234–43. Available from: <https://doi.org/10.3109/03639040903225109>.
- [36] Muller J, Knop K, Wirges M, Kleinebudde P. Validation of Raman spectroscopic procedures in agreement with ICH guideline Q2 with considering the transfer to real time monitoring of an active coating process. *J Pharm Biomed Anal* 2010;53(4):884–94. Available from: <https://doi.org/10.1016/j.jpba.2010.06.016>.

- [37] Muller J, Brock D, Knop K, Axel Zeitler J, Kleinebudde P. Prediction of dissolution time and coating thickness of sustained release formulations using Raman spectroscopy and terahertz pulsed imaging. *Eur J Pharm Biopharm* 2012;80(3):690–7. Available from: <https://doi.org/10.1016/j.ejpb.2011.12.003>.
- [38] Ho L, Muller R, Romer M, Gordon KC, Heinamaki J, Kleinebudde P, et al. Analysis of sustained-release tablet film coats using terahertz pulsed imaging. *J Control Release* 2007;119(3):253–61. Available from: <https://doi.org/10.1016/j.jconrel.2007.03.011>.
- [39] May RK, Evans MJ, Zhong S, Warr I, Gladden LF, Shen Y, et al. Terahertz in-line sensor for direct coating thickness measurement of individual tablets during film coating in real-time. *J Pharm Sci* 2011;100(4):1535–44. Available from: <https://doi.org/10.1002/jps.22359>.
- [40] De Beer TR, Vercruysse P, Burggraef A, Quinten T, Ouyang J, Zhang X, et al. In-line and real-time process monitoring of a freeze drying process using Raman and NIR spectroscopy as complementary process analytical technology (PAT) tools. *J Pharm Sci* 2009;98(9):3430–46. Available from: <https://doi.org/10.1002/jps.21633>.
- [41] Bai SJ, Rani M, Suryanarayanan R, Carpenter JF, Nayar R, Manning MC. Quantification of glycine crystallinity by near-infrared (NIR) spectroscopy. *J Pharm Sci* 2004;93(10):2439–47. Available from: <https://doi.org/10.1002/jps.20153>.
- [42] Darkwah J, Smith G, Ermolina I, Mueller-Holtz M. A THz spectroscopy method for quantifying the degree of crystallinity in freeze-dried gelatin/amino acid mixtures: an application for the development of rapidly disintegrating tablets. *Int J Pharm* 2013;455(1–21):357–64. Available from: <https://doi.org/10.1016/j.ijpharm.2013.06.073>.
- [43] Romer M, Heinamaki J, Miroshnyk I, Sandler N, Rantanen J, Yliruusi J. Phase transformations of erythromycin A dihydrate during pelletisation and drying. *Eur J Pharm Biopharm* 2007;67(1):246–52. Available from: <https://doi.org/10.1016/j.ejpb.2006.12.008>.
- [44] Momose W, Yoshino H, Katakawa Y, Yamashita K, Imai K, Sako K, et al. Applying terahertz technology for nondestructive detection of crack initiation in a film-coated layer on a swelling tablet. *Results Pharm Sci* 2012;2:29–37. Available from: <https://doi.org/10.1016/j.rinphs.2012.04.001>.
- [45] Saerens L, Dierickx L, Lenain B, Vervaet C, Remon JP, De Beer T. Raman spectroscopy for the in-line polymer-drug quantification and solid state characterization during a pharmaceutical hot-melt extrusion process. *Eur J Pharm Biopharm* 2011;77(1):158–63. Available from: <https://doi.org/10.1016/j.ejpb.2010.09.015>.

Risk assessment and design space consideration in analytical quality by design

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10.1 Introduction to analytical quality by design

Quality by design (QbD) is “a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management” [1]. Nowadays, QbD is a foremost prototype protocol in pharmaceutical industry to achieve quality products, besides safety and efficacy. It was mandated by US Food and Drug Administration (USFDA) in order to reduce the product recalls and six sigma (99.9999%) quality in the marketed products, it means QbD can significantly reduce the out-of-trend (OOT) results, out-of-specification (OOS) results, out-of-control, and out-of-statistical-control. In addition to USFDA, Europe Medicines Agency and International Council on Harmonization (ICH) guidelines such as ICH Q8 (R1) guideline made mandate to pharmaceutical industry through “design space.” It indicated that product and process performance characteristics must be scientifically designed as design space. In addition to ICH Q8, quality risk management (ICH Q9) and the development and manufacture [2] of drug substances (ICH Q11) have also marked the need of QbD [3].

Concept of QbD registered to analytical method development is known as analytical QbD (AQbD). AQbD assists in the evolution of a robust and cost-effective method that enables regulatory adaptability in various analytical methods. The design space concept is included as “method operable design region (MODR)” where method parameters or variables are allowed to change within a method’s design space [4]. AQbD approach results in

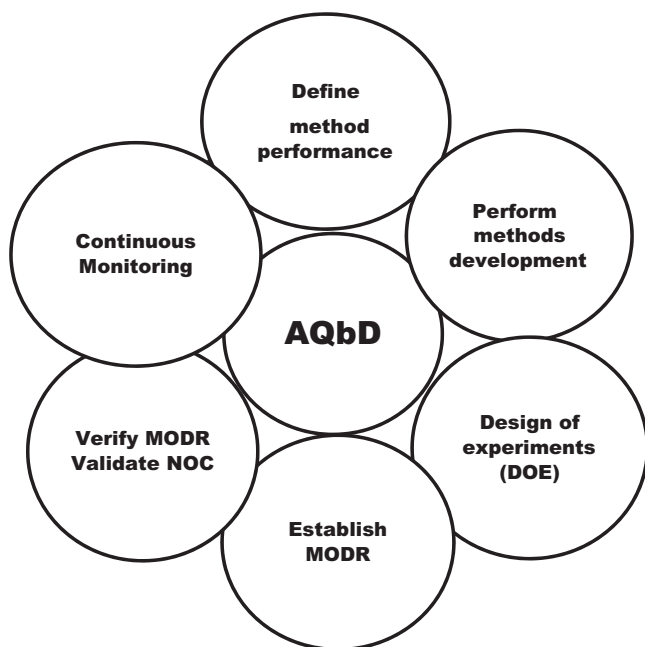


FIGURE 10.1 AQbD workflow cycle. *AQbD*, analytical quality by design; *NOC*, normal operating condition.

decrease of OOT number and OOS results because of high degree of robustness of the method employed within the region [5]. In addition this MODR also allows the analytical method to shift toward out effective or lower risk according to the analytical instrument configuration. The AQbD flow cycle is shown in Fig. 10.1.

10.2 Rewards of analytical quality-by-design approach to analytical methods

1. AQbD methods are serving as the indicators for the quality manufacturing process.
2. AQbD methods are more robust, utilize fewer resources in the investigation of OOS, and enhance the analytical confidence during the whole product testing life cycle.
3. The knowledge derived from the complete AQbD cycle would be an easy and ready reckoned during the analysis of OOS results.
4. Introduction of newer analytical methods in R&D and QC laboratory using AQbD approach will lead to a faster and higher method transfer rate vis-a-vis the traditional technology-transfer approach.

5. AQbD includes how best the method is carried out every day and “real-world” operating condition instead of the traditional “check box tick mark” approach for regulatory compliance [6].
6. In addition, the AQbD method provides scientific understanding of method variables to the method response, thereby we can control the critical method variables (CMVs).
7. In the pharmaceutical world, AQbD would be the driving force to ensure all analytical methods to work best every time where lean and six sigma techniques were established extensively.
8. AQbD greatly reduces the method development time and cost [7].

10.3 Regulatory perspective of analytical quality by design

AQbD allows the development of analytical methods, as robust and cost-effective, which can be used for testing in the life cycle of the product and provide freedom to change of method variable levels within the MODR (regulatory flexibility in analytical method). The system suitability testing of any analytical method should be as required by the United States Pharmacopeia and FDA. United States Pharmacopeia-National Formulary (USP-NF) and European Pharmacopoeia granted flexibility for an analytical method for changing the method variable within MODR without the need for revalidation, only if the method development protocols fulfilled AQbD approach. The AQbD approach is initially based on targeted product quality measurement analytical target profile (ATP), and followed by identification of CMV, selection of analytical technique, risk assessment for variables to the selected analytical techniques, optimization of variables using design of experiment (DoE), and finally design of MODR, verification, validation process for the MODR, and control strategy. Regulatory aspect of method development for AQbD and key elements of AQbD are presented in Figs. 10.2 and 10.3. In this chapter according to the title, we have focused only on risk assessment and MODR in AQbD.

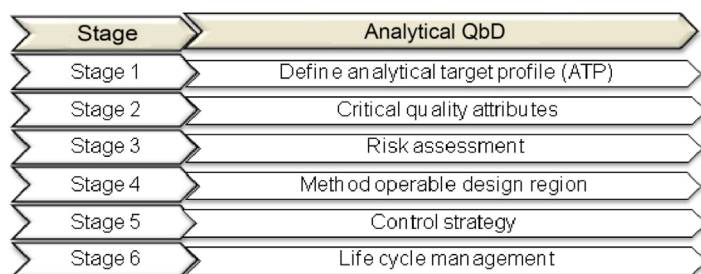


FIGURE 10.2 Regulatory perspective of AQbD. *AQbD*, analytical quality by design.

Element	Objectives
Analytical quality profile	Defines what to quantify and how to quantify.
Critical quality attributes (CQA)	Separation, identification, accuracy, precision, robustness, ruggedness requirements
Critical method attributes (CMA)	Identifying method parameters which have impact on the performance of the method, such as buffer, pH, column temperature, injection volume, organic concentration, etc.
Critical material attributes (CMA)	Evaluating the reagents, reagent grades and concentrations used in the analysis
Design of experiments (DOE)	Nested or factorial design identifies a centre method and creates a method design space
Method validation	Establishing practical proof that the method is reliably bringing quality results
Control strategy	Ensuring the method performance with accepted results

FIGURE 10.3 Key elements of AQbD. *AQbD*, analytical quality by design.

10.4 Risk assessment in analytical method

Quality risk management (ICH Q9) is “an efficient process for the assessment, control, communication and review of risks to the quality across the lifecycle” [8]. Risk analysis is the fundamental part of the AQbD approach. In AQbD, risk assessment focuses on the identification and ranking of method parameters that impact performance of method and conformance to the ATP. Risk assessments are often iterative all round the lifetime of a method, and are typically performed in the method development. In addition, risk assessment also focused on potential differences such as laboratory practices, environment, testing cycle times, and reagent sources. In the risk assessment the major differences such as equipment availability should be acknowledged and factored at the technique selection and method development stages. Risk question or problem description is the prime most steps in quality risk assessment of QbD approach. If the risk question is effectively defined, then it is feasible to assess the suitable risk management tool and the various classes of information required to tackle the risk. As per ICH Q9, risk assessment will be accomplished in three steps as shown in Fig. 10.4.

1. *Risk identification*: This is a systematic use of information to detect hazards linked with the risk question. The information includes historical data, theoretical analysis, informed opinions, and reviews of stakeholders. The risk identification regulates “What might go wrong?” that includes identifying the possible results. This provides basis for the advancement in risk management process.
2. *Risk analysis*: It is an assessment of risk allied with the recognized hazards. It is either qualitative or quantitative method associated with the possibility of development and seriousness of problems.

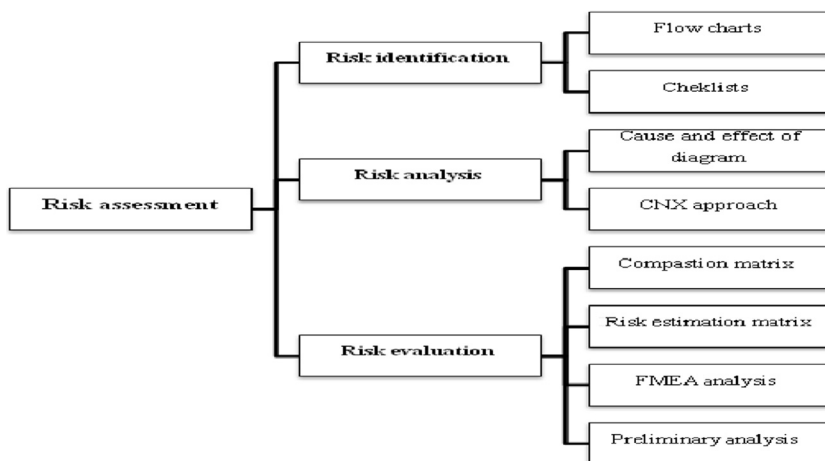


FIGURE 10.4 A sequence of steps and tools involved in risk assessment as mentioned in ICH Q9 guidelines. *ICH*, International Council on Harmonization.

3. **Risk evaluation:** Risk evaluation differentiates the recognized and examined risk counter to the specified risk criteria. Risk evaluation considers the firmness of proof for all fundamental questions. The quality of product is determined based on performing efficacious risk assessment and robustness of data. The end product of risk assessment is either qualitative or quantitative estimation of risk. Numerical probability will come into picture if risk is quantitatively measured, whereas qualitative descriptors, such as “high,” “medium,” or “low,” are used when risk is measured qualitatively (ICH Q9).

The most familiar way to execute risk assessment is fishbone diagram [ICH Q8 (R2)], commonly known as Ishikawa shown in Fig. 10.5. Appropriately risk factors are arranged as follows:

1. **High-risk factors:** This should be predetermined before the method development process, for example, sample preparation process.
2. **Noise factors:** These are exposed to multivariate statistical analysis (MSA) study. It can be prepared by staggered cross-nested study design, ANOVA, and variability charts. Noise factors were further exposed to robustness testing.
3. **Experimental factors:** For example, instrumentation and operation methods [9].

For example, the variability in separation by chromatography may be because instrumental configuration, columns, injection volumes, and flow rate are kept in a controlled manner during the experiment, whereas variables such as column temperature, pH, and % ration of mobile phase are assigned to test the robustness for the development of MODR. DoE approach is

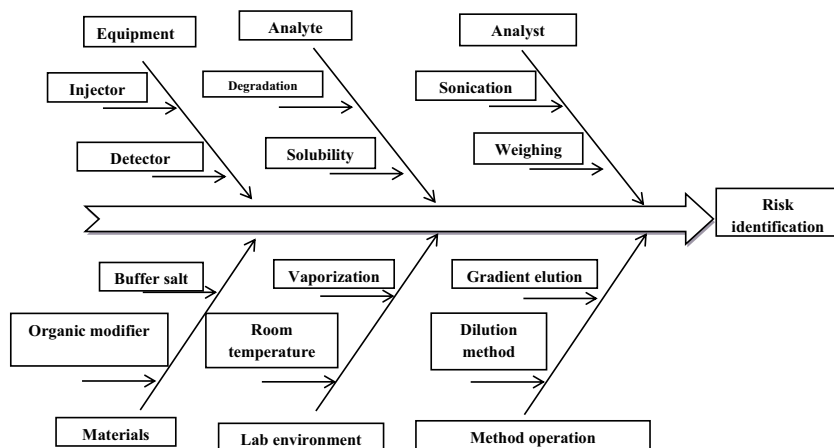


FIGURE 10.5 Fishbone diagram for risk identification.

adopted to define the MODR. Analytical method failure and risk factor are coordinated in the following manner [10]:

$$\text{Risk factor} = \text{Severity} \times \text{occurrence} \times \text{detectability}$$

Severity effects the patient is related to safety of efficacy [critical quality attributes (CQAs)]; occurrence is the chance of failure related to process knowledge, control, and product; detectability is the ability to detect a failure capability of analytical method and sampling.

Risk management approach and tools allow us to focus on factors that are critical to quality so that our efforts and resources are targeted to areas where it is really needed. As the resources are not unlimited, risk approach will lead to more effective systems and controls, and eventually reduces cost while achieving quality. Choice of analytical methods to CQA of product is shown in Table 10.1.

10.5 Risk assessment in HPLC method development

In the risk assessment for AQbD methods, analyte and column chemistry is very important. There are two types of variables in assessing CMVs; they are qualitative and quantitative.

10.5.1 Qualitative variables

Column (c8, c18, CN, or phenyl), detector (ultra violet (UV), diode array detector (DAD), etc.), elution (isocratic or gradient), type of organic phase (methanol or acetonitrile, tetrahydro furan (THF)), choice of buffer (organic

TABLE 10.1 Choice of analytical methods to critical quality attribute of product.

S. no	CQA	Method performance as per ICH	Acceptable analytical method
1	10.1 Assay	● Specificity: no interference with impurities ● Linearity: 80%–120% of the true value ● Accuracy: 98%–102% ● Precision: $\pm 2.0\%$ (RSD)	HPLC UPLC LC-MS HPTLC
10.2 2	10.3 Impurity	● Specificity (no interference from each other) ● Linearity: 50%–150% of the true value ● Accuracy: 98%–102% ● Precision: ● imp. $\leq 0.15\%$: $\pm 20\%$ of the true value, 80% probability ● imp. $> 0.15\%$: $\pm 15\%$ of the true value, 90% probability	HPLC UPLC LC-MS HPTLC

CQA, critical quality attributes; HPLC, high performance liquid chromatography; HPTLC, High performance thin layer chromatography; ICH, International Council on Harmonization; LC-MS, liquid chromatography-mass spectrometry; RSD, relative standard deviation; UPLC, ultra performance liquid chromatography.

or inorganic, then among inorganic), pH (low or high), choice of reagent (from different vendors) can be considered as the qualitative variables, which are highly critical and influential on the method performance.

10.5.2 Quantitative variables

%Organic phase (for the organic phase selected from the screening), concentration of buffer, pH of buffer, column temperature, flow rate, etc. are considered as the moderately influential factors. So it can be screened and kept controlled (provided that plates are more than 3000, it is not necessary that it should be 4000, 5000, 4500, etc.). The risk assessment of high performance liquid chromatography (HPLC) method variables is shown in Table 10.2.

Note: Depending on the nature of the project/problem, we need to select the input and output factors and then link up the expected relationship between input and output factors. Once this relationship is built, we can try to identify the degree of risk in each response (high or low).

TABLE 10.2 Risk assessment of HPLC method variable.

Method variables (X)	HPLC method responses (Y)						
	10.4 Resolution	10.5 Retention time	Theoretical plates	Tailing factor	Response factor	Precision	Peak area
pH	10.6 H	10.7 H	10.8 H	10.9 H	10.10 H	10.11 H	10.12 H
10.13 Organic phase	10.14 M	10.15 H	10.16 M	10.17 M	10.18 H	10.19 M	10.20 H
10.21 Buffer type	10.22 H	10.23 H	10.24 H	10.25 H	10.26 H	10.27 H	10.28 H
10.29 Buffer strength	10.30 M	10.31 M	10.32 M	10.33 M	10.34 H	10.35 H	10.36 H
10.37 % aqueous	10.38 H	10.39 H	10.40 H	10.41 M	10.42 H	10.43 H	10.44 H
10.45 Column type	10.46 M	10.47 H	10.48 H	10.49 H	10.50 H	10.51 M	10.52 H
10.53 Particle size	10.54 L	10.55 M	10.56 M	10.57 M	10.58 H	10.59 M	10.60 H
10.61 Column type	10.62 L	10.63 M	10.64 M	10.65 M	10.66 H	10.67 M	10.68 H
10.69 Column size	10.70 L	10.71 M	10.72 M	10.73 M	10.74 H	10.75 M	10.76 H
10.77 Flow rate	10.78 L	10.79 M	10.80 M	10.81 M	10.82 M	10.83 M	10.84 M
10.85 Sample size	10.86 L	10.87 L	10.88 L	10.89 L	10.90 L	10.91 L	10.92 L
10.93 Sample volume	10.94 L	10.95 L	10.96 L	10.97 L	10.98 L	10.99 L	10.100 L
10.101 Column temp.	10.102 M	10.103 M	10.104 M	10.105 H	10.106 H	10.107 M	10.108 H
10.109 Detector	10.110 –	10.111 –	10.112 –	10.113 –	10.114 M	10.115 L	10.116 M
10.117 Elution mode	10.118 H	10.119 H	10.120 H	10.121 H	10.122 H	10.123 H	10.124 H

H, High; *M*, moderate; *L*, low.

Where high risk is involved what countermeasures can be taken. Then remaining factors will be identified and will be used as input factors for DoE optimization.

- Above all factors, retention, resolution, theoretical plates, and tailing factor are responsible for method performance such as accuracy, precision, specificity, theoretical plates, and linearity range.
- Among all, resolution is highly responsible for specificity, selectivity, accuracy, and precision.
- Retention time also contributes to specificity, accuracy, and precision.
- Theoretical plates contribute high efficacy elution (indirectly increase the resolution), so that moderately affect the specificity, accuracy, and precision. It also affects the parameters such as limit of detection and limit of quantification, and determines the linearity range.
- Tailing factor affects specificity, accuracy, and precision.

10.6 Example: risk assessment in analytical quality by design—based HPLC method for etofenamate

10.6.1 Required information on chemical structure

- Etofenamate (ETO) is a nonsteroidal antiinflammatory agent and chemically known as 2-[[3-(trifluoromethyl) phenyl] amino] benzoic acid 2-(2-hydroxyethoxy) ethyl ester. The structure is shown in Fig. 10.6.
- It is a pale yellow viscous liquid.
- Freely soluble in methanol and practically insoluble in water.
- The pKas of ETO are 6.0 and 7.0.

Chemistry-based risk assessment for HPLC method and overall risk assessment to HPLC method development are shown in Tables 10.3 and 10.4.

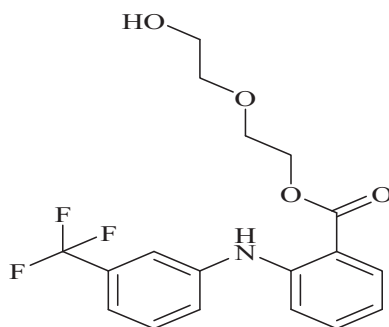


FIGURE 10.6 Structure of etofenamate.

TABLE 10.3 Chemistry-based risk assessment for HPLC method.

Molecule chemistry	Critical HPLC method variable	Risk
Freely soluble in methanol and practically insoluble in water	10.125 Most of methods depend on combination of water or aqueous buffer in mobile phase.	10.126 Small change in compositions affects the method SST and then method performance. High water content may precipitate the drug.
10.127 The pKas 6.0 and 7.0	10.128 Mobile phase pH will be in between 3 and 7.	10.129 Small change in pH affects % ionization and then affect efficiency of the method. Low pH may hydrolyze ester.
10.130 Viscous liquid	10.131 Concentration-dependent flow velocity.	10.132 Concentration-dependent flow may have effect on efficiency of the method.
SST, system suitability testing.		

10.7 Design space

It is a scientific concept employed in pharmaceutical/biopharmaceutical industry to support and assure product quality. The culmination of knowledge and information gathered during product development provides the foundation for the design space shown in Fig. 10.7. According to ICH Q8 R2, design space is defined as the multidimensional combination and compatibility of input variables (material attributes) and operational parameters that are exhibited to furnish quality assurance [11]. Design space is proposed by the applicant, working outside the design space requires a regulatory postapproval change process, and the proposed design space is subjected to regulatory evaluation and approval [12].

For the researcher, design space is a Y (quality attributes) = F (process parameters, material attributes)—a function or a relationship between (basic) process parameters and (basic) quality attributes.

Example:

Design space for a drying operation is dependent upon temperature and/or pressure over a time. End point for moisture (humidity) content is 1%–2%, so operating above the upper limit of the design space may lead to excessive

TABLE 10.4 Overall risk assessment to HPLC method development.

S. no.	Input (Xs) CMV	Output (Ys)		
		10.133 Retention (Y_1)	10.134 Theoretical plates (Y_2)	Tailing factor (Y_3)
1	10.135 Flow rate	10.136 High	10.137 High	10.138 Moderate
10.139 2	10.140 pH	10.141 High	10.142 High	10.143 High
10.144 3	10.145 % aqueous	10.146 High	10.147 High	10.148 Moderate
10.149 4	10.150 Buffer %	10.151 Moderate	10.152 Moderate	10.153 High
10.154 5	10.155 Column temp.	10.156 High	10.157 High	10.158 Moderate
10.159 6	10.160 Type of organic phase	10.161 High	10.162 High	10.163 Moderate
10.164 7	10.165 Column type	10.166 High	10.167 Moderate	10.168 High
10.169 8	10.170 Buffer type	10.171 Moderate	10.172 High	10.173 High
10.174 9	10.175 Detector	10.176 Analyte-dependent		
10.177 10	10.178 Column length	10.179 High	10.180 High	10.181 Moderate
10.182 11	10.183 Column particle size	10.184 High	10.185 High	10.186 Moderate

CMV, critical method variable.

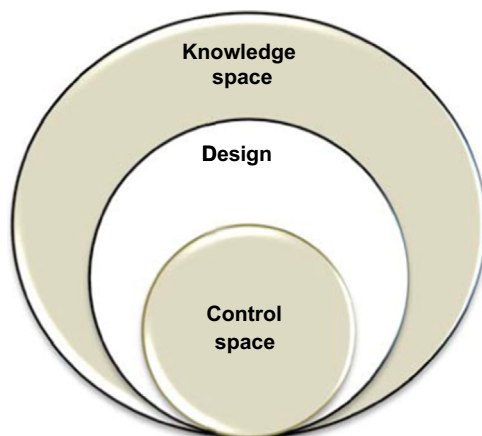


FIGURE 10.7 Design space in AQbD. *AQbD*, analytical quality by design.

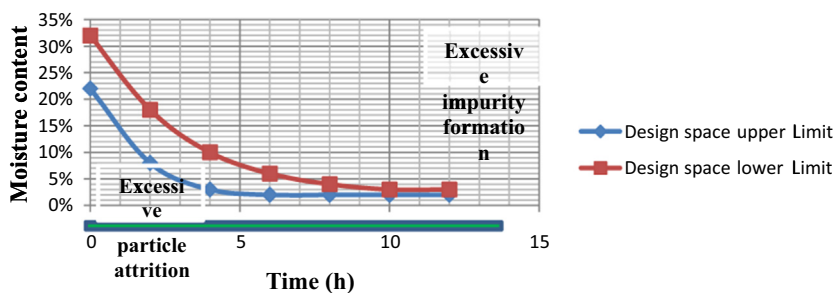


FIGURE 10.8 Design space for drying operation.

impurities, and below the lower limit of the design space may result in excessive particle attrition [ICH Q8 (R2)] as shown in Fig. 10.8.

Before establishing design space, the approach should include target product CQAs, prior scientific knowledge, and risk assessment [13]. CQAs are the properties that should be within an appropriate limit, range, or distribution to ensure the desired product quality. Implementing quality risk management [14] and QbD [15] methodologies as early as possible will help to facilitate and define the design space.

10.8 Method operational design region

It is established to design an operational area for analytical method, such as analysis time, limits, and procedure. According to ICH Q8, MODR may be accepted in method establishment phase, and such a method can be treated as cost-effective and robust method. By understanding method performance

regions, required operational conditions can be established. Analyte sensitivity and critical method parameters must be evaluated. It is the operating limit for the method input variables that produce results consistently to fulfill the objectives in ATP [16]. MODR allows adaptability in varied input method parameters to ensure the expected method performance and response without resubmission to USFDA. MODR is based on a multivariate approach, risk, and science to evaluate the effect of different factors on method performance [17].

10.9 Steps engaged in design space

Step 1: Establishment of a knowledge management system

One of the initial steps to establishing design space is to determine a process for capturing data and scientific knowledge. At present, knowledge management is the systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components [18]. Once a knowledge management process is created, existing data, prior knowledge, and new information can be added to the database. One of the simplest tools for knowledge management is the spreadsheet, which can be utilized to gather information for the database.

Risk assessment should be conducted as a major aspect of the evaluation to decide whether the new information is critical and affects product quality. Using a QbD approach during the product development will help to identify and determine the relationship between the materials and manufacturing process that impacts the product quality attributes.

Step 2: Evaluation of active and inactive components

This is the principal key area to assess concerns the materials/components used in the drug product. Suppliers should provide certificates of analysis that include the acceptance criteria or specifications, test strategies, and test results for the materials. This information should be added to the knowledge management database and evaluated to determine whether any of the material characteristics has an impact on product performance. For active pharmaceutical ingredient (API), the physicochemical and biological properties should be identified and the inactive components evaluated, their concentration and characteristics can also influence drug product performance. Information on excipients performance can be used to justify the choice and quality attributes of the material and to support the drug product specification. Implementing formal experimental design will make it easier to determine the critical material attributes that impact product quality.

Step 3: Assessment of manufacturing process parameters

This is the second key region to evaluate the manufacturing process. An initial risk assessment of the manufacturing process steps should be performed prior to the baseline characterization work, in order to determine potential critical process steps. Critical process parameters (CPPs) are those whose variability has an impact on a CQA—therefore these should be monitored or controlled to

ensure the process or produces of the desired quality. This assessment, in combination with the prior scientific knowledge, can be used for determining the criticality of process parameters under the following conditions:

- When a basic process that has comparative properties and process is applied to another commercial product (e.g., new strength or new dosage form);
- When there is a significant body of distributed information on the process;
- When test studies and commercial information are accessible, for instance, when the process validation life cycle is applied to a legacy product to validate the underlying assessment [19].

By overlaying the component CQAs with the process parameters, we can determine the CPPs that impact product quality. Experimental design should be implemented to identify the relationship between materials and process parameters that impact product quality.

Step 4: Reviewing of attributes for criticality

After primer material and process attributes are recognized, they can be reviewed to determine criticality and the impact on the quality target product profile (QTPP). Tolerances around the critical attributes and CPPs can help define the product specification limits [20].

10.10 Design space tools and design of experiments

DoE is defined as “a sorted technique utilized for determining the connection between factors influencing a process and the yield of that process” [21]. DoE can be used for comparative experiments, screening experiments, response surface modeling, and regression modeling [22].

- *Comparative experiments*: These types of experiments are used for selecting a better one between two alternatives. Here the choice relies on the comparison of average results generated from a sample of data from each alternative. For example, choosing a vendor for API from two or more vendors can be a comparative experiment.
- *Screening experiments*: These are involved in the selection of key factors affecting a response. In this type of experiments, we select somewhat small number of components that have critical effects on the preferred response. These experiments can be used as fundamental tools for creating response surface models.
- *Response surface modeling*: This strategy is used after identifying the critical factors affecting a response. Response surface model can be used for hitting an objective, augmenting or limiting a response, reducing variation, making a process robust, and looking for multiple goals.
- *Regression modeling*: This model can be used to decide an exact model that evaluates the dependence of reaction variable(s) on process inputs.

10.11 Common experimental designs

- *Completely randomized designs*: Here, the reaction variable is resolved from various degrees of a primary factor without contemplating some other factor [23].
- *Randomized block designs*: In this case, one essential factor will be there and blocking is used to expel the impact of immaterial elements that can be controlled.
- *Full factorial designs*: In this design, each degree of each factor appears with each degree of each other factor. There will be high or low levels for the factors. For “ n ” number of independent factors with two levels, the number of experimental runs required is equal to 2^n .
- *Fractional factorial designs*: Here, a deliberately chosen fraction of the runs is selected instead of carrying out all the runs as in a full factorial design. When the number of factors increases, full factorial designs will become exceptionally enormous.
- *Plackett–Burman designs*: it works, where just the fundamental impacts of the variables are significant, these designs are said to be extremely proficient. They have an experimental run number as a multiple of four. An extremely enormous number of factors can be evaluated with a minimum number of runs.
- *Three-level full factorial designs*: Here, three levels are considered for each factor. The number of experimental runs is given by 3^n , where n is the number of independent factors in consideration. Here, the levels are coded as -1 , 0 , and $+1$. A third level compared to the two-level designs helps to investigate the quadratic relationship between the response and each of the factors.
- *Response surface designs*: A quadratic or cubic model can give a total depiction of process behavior. The two classical quadratic designs include Box–Wilson central composite designs (CCDs) and Box–Behnken designs.

A CCD is utilized to construct a subsequent request (quadratic) model for the response variable. Linear regression is used in the design to acquire the outcomes. The factor levels are usually coded for the design [24]. They are two-level full factorial or fractional factorial designs. The designs are expanded by a number of center points and other chosen runs.

Box–Behnken designs are used for three-level factor experiments and are generally used to fit second-order models with the responses. The design is a combination of two-level factorial designs with incomplete block designs. These designs are practically rotatable. Box–Behnken structures are having the advantage that they require just three levels. Likewise, it has advantages that there are no runs, where all factors are at either the -1 or $+1$ levels and that there are no runs at the corner points.

10.12 Process model for design of experiment

1. It can be seen from the graph (shown in Fig. 10.9) that for the values of X_2 = aqueous phase % from 10% to 15%, at X_3 = flow rate of mobile

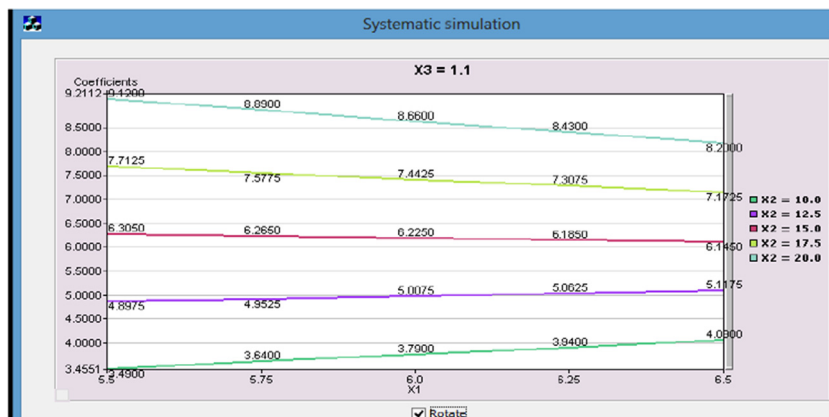


FIGURE 10.9 Process model for DoE. DoE, design of experiment.

phase 1.1 $\mu\text{mL/minutes}$, and for $X_1 = \text{pH}$ can range from 5.5 to 6.5, the retention time is between 3.5 and 6.3 minutes.

2. This itself is a robust process and also a design space.
3. Since it is inferred that the relationship between Y (retention time) and X_s variables is nonlinear, it is tried to find out by stretching the levels of variables whether the robust process is possible with wider range of variables.

10.13 Two-dimensional model for design space: contour plots

There could be different combinations that may give a number of feasible solutions for robust process. X_1 versus X_2 with X_3 as a constant, X_2 versus X_3 with X_1 as constant, X_1 versus X_3 with X_2 as constant. Out of these combinations whichever is the most desirable from the point of retention time, that can be selected as a robust process, which can be described as design space. Even the constant value can be changed depending on the sign of the variable and see where a maximum design space can be obtained [25].

For example,

Combination 1:

X_1 versus X_2 at constant level of $X_3 = 1.2$

$X_1 = -2$ to 2 level of pH and $X_2 = -1$ to 0 level of aqueous phase

The design space for RT = 4.5–6. See graph in Fig. 10.10.

Combination 2:

X_1 versus X_3 at constant level of $X_2 = 15\%$

$X_1 = -2$ to 2 and $X_3 = 0$ –2

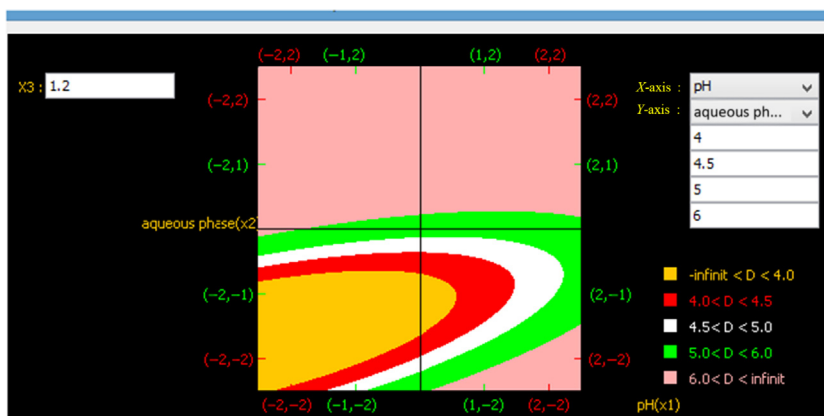
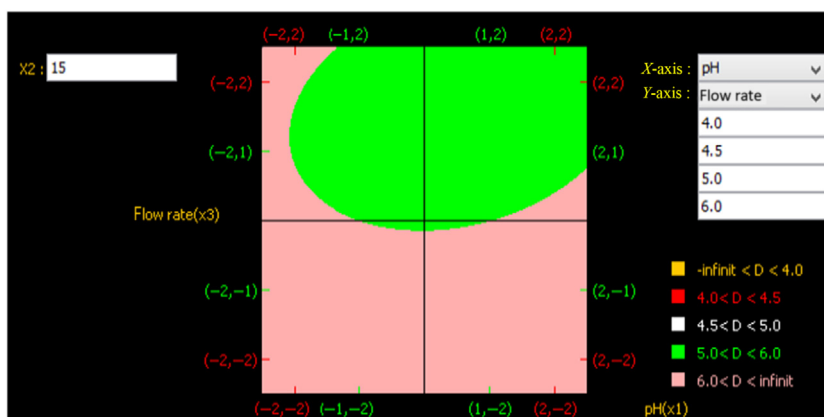
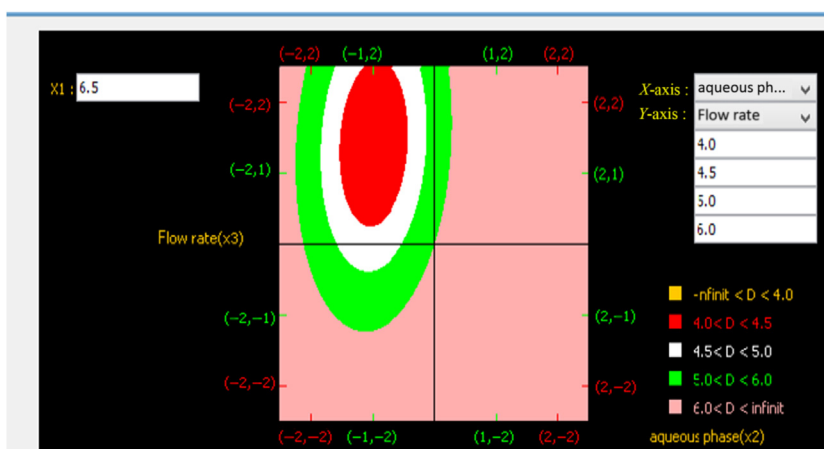
The design space for RT = 5–6. See graph in Fig. 10.11.

Combination 3:

X_2 versus X_3 at constant level of $X_1 = 6.5$ pH

$X_2 = -2$ to 2 level and $X_3 = -1.2$ to 2 level

Design space for RT = 4–6.0. See graph in Fig. 10.12.


 FIGURE 10.10 2D model contour plots of X_1 versus X_2 with X_3 as constant.

 FIGURE 10.11 2D model contour plots of X_2 versus X_3 with X_1 as constant.

 FIGURE 10.12 2D model contour plots of X_1 versus X_3 with X_2 as constant.

10.14 Three-dimensional models of design space: response surface methodology

A quadratic or cubic model that gives a complete description of process behavior. The curvature in the response surface is caused by the presence of quadratic and possibly cubic terms in the response. Quadratic models are always sufficient to the pharmaceutical industry. While a two-level factorial design cannot fit a quadratic equation, three-level factorial designs can fit but require a large number of runs when more than four factors are involved.

10.14.1 Example: in vitro drug release of Losartan potassium

- *Drug release at 2 hours (Y_1) = $29.31 + 3.03X_1 - 2.00X_2 + 5.70X_1X_2$*

The major factors affecting the drug release at 2 hours were factor X_1 , X_2 , and X_1X_2 . The first one had positive effect and latter one had negative effect. By increasing the concentration of hydroxypropyl methylcellulose (HPMC) K100 drug release increased and increasing the concentration of ethylcellulose the drug release decreased, and F_1 , F_7 had shown burst release, whereas interaction between the factors increased the drug release. Contour plots and 3D response plots have shown that in order to maintain burst release, that is, not more than (NMT) 30% in 2 hours. HPMC should be maintained at high levels and ethyl cellulose (EC) should be maintained at low levels. From Figs. 10.13 and 10.14 it is clearly shown that by maintaining the concentrations of HPMC and EC at drug release at 63.5–92.5 mg and 61.5–80 mg, respectively, the controlled drug release (88%, 90%) can be achieved to the desired target level (drug release at 2 hours NMT 30%). Contour plots (Fig. 10.13) and 3D surface plots (Fig. 10.14) showed the effect of factor X_1 and X_2 on response Y_1 (drug release at 12 hours). It was observed that in order to control the drug release, that is, 30%, the concentrations of factor X_1 and factor X_2 were kept at high level and low level, respectively.

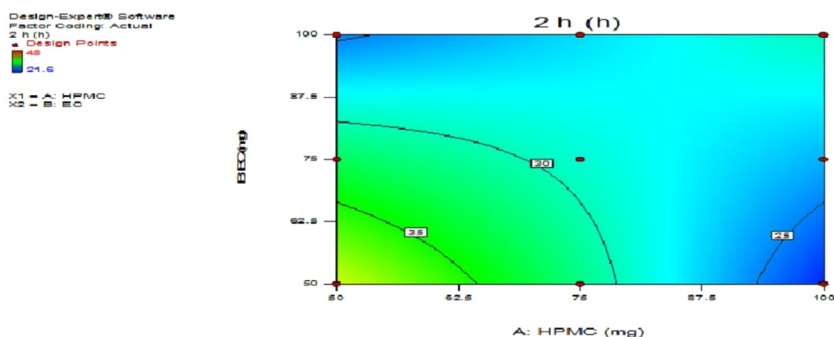


FIGURE 10.13 Contour plot showing the effect of X_1 and X_2 on drug release at 2 h.

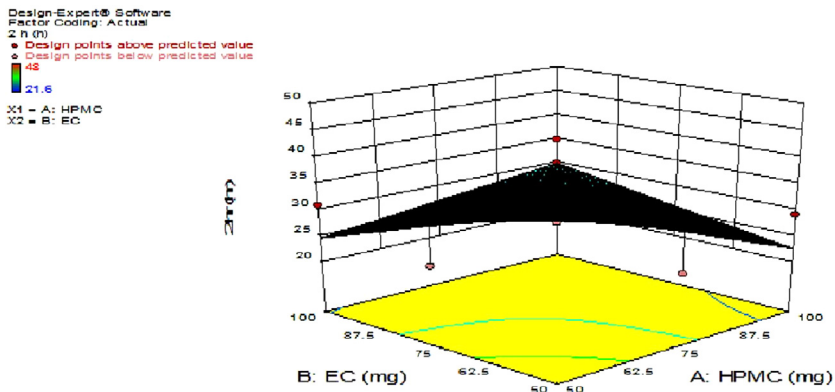


FIGURE 10.14 3D response plot showing the effect of X_1 and X_2 on drug release at 2 h.

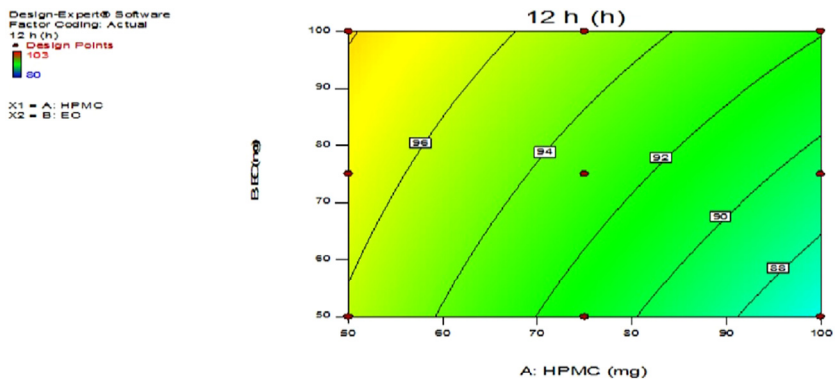


FIGURE 10.15 Contour plot showing the effect of X_1 and X_2 on drug release at 12 h.

- Drug release at 12 hours (Y_2) = $93.07 + 3.84X_1 + 2.04X_2 + 0.84X_1X_2$

The major factors affecting the drug release at 12 hours were factor X_1 , X_2 , and X_1X_2 . All the factors showed positive effect. By increasing the concentration of HPMC K100 and ethyl cellulose drug release increased and all the formulations showed sustained effect, whereas interaction between the factors increased the drug release. Contour plots and 3D response plots have shown that in order to maintain sustained release, that is, not less than (NLT) 80% in 12 hours. HPMC should be maintained at high levels and EC should be maintained at low levels. From Figs. 10.15 and 10.16 it is clearly shown that by maintaining the concentrations of HPMC and EC at 60–95 and 55–80 mg, respectively, the controlled drug release (88%, 90%) can be achieved to the desired target level (drug release at 12 hours NLT 80%). Contour plots (Fig. 10.15) and 3D surface plots (Fig. 10.16)

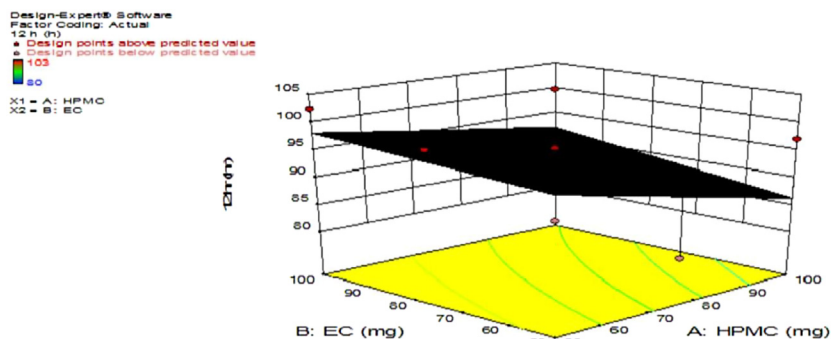


FIGURE 10.16 3D response plot showing the effect of X_1 and X_2 on drug release at 12 h.

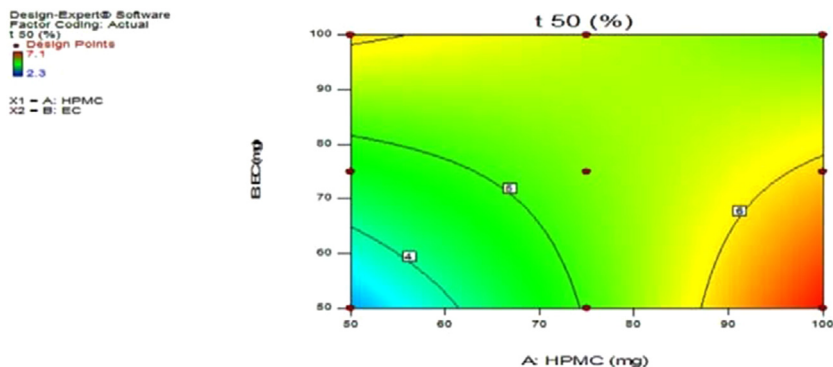


FIGURE 10.17 Contour plot showing the effect of X_1 and X_2 on t_{50} h.

showed the effect of factor X_1 and X_2 on response Y_2 (drug release at 12 hours) the required criterion is that the drug release from the dosage form is not less than 80%. In this scenario design space is selected with more drug release, and by maintaining factor X_1 at high level and factor X_2 at low level all criteria will be met [26].

- Time taken to release 50% drug (Y_3) = $5.36 + 0.75X_1 + 0.30X_2 - 1.20X_1X_2$

The major factors affecting the $t_{50\%}$ were factor X_1 , X_2 , and X_1X_2 . Contour plots (Fig. 10.17) and 3D surface plots (Fig. 10.18) showed the effect of factor X_1 and X_2 on response Y_3 ($t_{50\%}$). The 2D-contour plots and 3D-response plots are shown indicate the relationship between the input factors for their response on $t_{50\%}$. From Figs. 10.17 and 10.18 it is clearly shown that by maintaining the concentrations of HPMC and EC at 62–88 and 65–80 mg, respectively, the controlled drug release (5 hours, 6 hours) can be achieved to the desired target level ($t_{50\%}$ in NLT 3 hours) [27].

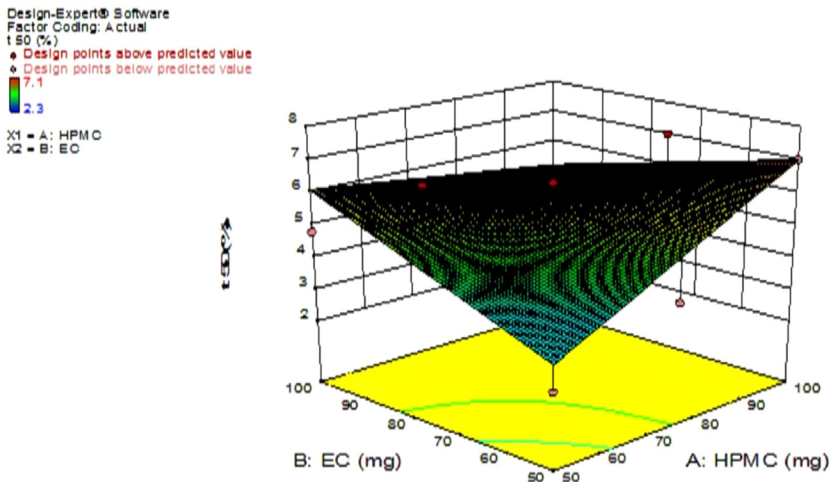


FIGURE 10.18 3D response plot showing the effect of X_1 and X_2 on t_{50} h.

10.15 Advantages of design space approach to analytical methods

- Design space approach to analytical methods will give more robust and rugged, bringing about less assets spent investigating OOS results and greater confidence in analysis testing process durations.
- Resources presently invested in performing traditional technology transfer and method validation activities will be redirected to ensure truly robust and rugged methods.
- The introduction of new analytical techniques—from innovative work research and development to quality control laboratories—using a QbD approach will prompt a higher exchange achievement rate than the traditional technology-transfer approaches.
- A predetermined procedure will help the deliberate and effective execution of the QbD philosophy and encourages a group approach.
- A genuine persistent learning process is built up using a focal corporate information repository that can be applied over all methods.

10.16 Limitations of design space approach to analytical methods

- Same facility is required at both development and QC laboratory.
- Acceptance must be gained for registration of the strategy execution criteria instead of the method conditions.
- Experts/analysts must learn new devices and skills [28].

References

- [1] Woodcock J. The concept of pharmaceutical quality. *Am Pharm Rev* 2004;7(6):10–15.
- [2] International Conference on Harmonization [ICH Q8 (R2)] Pharmaceutical Development, August 2009. <<https://database.ich.org/sites/default/files/Q8%28R2%29%20Guideline.pdf>>.
- [3] International Conference on Harmonization [ICH Q11] Development and manufacture of drug substances (chemical entities and biotechnological/biological entities), 1 May 2012. <https://database.ich.org/sites/default/files/Q11_Guideline.pdf>.
- [4] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22(1):58–64.
- [5] Peraman R, Bhadraya K, Reddy YP. Analytical quality by design: a tool for regulatory flexibility and robust analytics. *Int J Anal Chem* 2015;2015:1–9.
- [6] Singh B, Khurana RK, Kaur R, Beg S. Quality by design (QbD) paradigms for robust analytical method development. *Pharm Rev* 2016;14(10):61–6.
- [7] Panda SS, Beg S, Ravi Kumar BVV, Sahu J. Implementation of quality by design approach for developing chromatographic methods with enhanced performance: a mini review. *J Anal Pharm Res* 2016;2(6):39–43.
- [8] Singh B, Beg S. Product development excellence and federal compliance via QbD. *Chron PharmaBiz* 2014;15(10):30–5.
- [9] Bhutani H, Kurmi M, Singh S, Beg S, Singh B. Quality by design (QbD) in analytical sciences: an overview. *Pharm Times*. 2014;46(8):71–5.
- [10] International Conference on Harmonization [ICH] and USFDA Guidance for Industry Q8 (R2) Pharmaceutical Development, November 2009. <<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q8r2-pharmaceutical-development>>.
- [11] Deidda R, Orlandini S, Hubert P, Hubert C. Risk based approach for method development in pharmaceutical quality control context: a critical review. *J Pharm Biomed Anal* 2018;161:110–21.
- [12] Peterson JJ. Approach to the ICH Q8 definition of design space. *J Biopharm Stat* 2008;18:959–75.
- [13] Peraman R, Kalva B, Shanka S, Padmanabha Reddy Y. Analytical quality by design (AQbD) approach to liquid chromatographic method for quantification of acyclovir and hydrocortisone in dosage forms. *Anal Chem Lett* 2014;4(5–6):329–42.
- [14] International Conference on Harmonization [ICH Q9] and USFDA Guidance for Industry, Quality Risk Management, June 2006. <<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q9-quality-risk-management>>.
- [15] International Conference on Harmonization [ICH Q8 R2] and USFDA Guidance for Industry, Pharmaceutical Development, November 2009. <<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q8r2-pharmaceutical-development>>.
- [16] Rozet E, Lebrun P, Debrus B, Boulanger B, Hubert P. Design spaces for analytical methods. *Trends Anal Chem* 2013;42:157–67.
- [17] International Conference on Harmonization [ICH (Q9)] Quality Risk Management, 9 November 2005. <<https://database.ich.org/sites/default/files/Q9%20Guideline.pdf>>.
- [18] International Conference on Harmonization [ICH Q10] and FDA Guidance for Industry, Pharmaceutical Quality System, April 2009. <<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q10-pharmaceutical-quality-system>>.
- [19] Mitchell M. Determining criticality-process parameters and quality attributes part I: criticality as a continuum. *Bio Pharm Int* 2013;26(12).

- [20] USFDA guidance for industry, process validation: general principles and practices, January 2011. <<https://www.fda.gov/media/71021/download>>.
- [21] Beg S, Hasnain MS, Swain S, Rahman M. Chapter – 1: introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design: principles and application*. Academic Press (Elsevier); 2019. p. 1–17.
- [22] NIST/SEMATECH e-Handbook of statistical methods. [Last updated 30.10.2013, Last cited on 2009 Aug 4]. <<https://www.itl.nist.gov/div898/handbook/>>.
- [23] Hernandez M, Forthofer RN, Lee ES. *Biostatistics: A guide to design, analysis and discovery*. 2nd ed. Academic Press; 2007. p. 323–48.
- [24] Central composite design. [Last cited on 2009 Aug 5]. <<https://www.itl.nist.gov/div898/handbook/pri/section3/pri3361.htm>>.
- [25] Peraman R, Bhadraya K, Padmanabha Reddy Y, Surayaprakash Reddy C, Lokesh T. Analytical quality by design approach in RP-HPLC method development for the assay of Etofenamate in dosage forms. *Indian J Pharm Sci* 2015;77(6):751–7.
- [26] Surya prakash Reddy C, Padmanabha Reddy Y, Devanna N. Formulation and optimisation of the extended release tablets of Dalfampridine by 23 factorial design. *J Pharm Sci Innov* 2016;5(1):27–37.
- [27] Mounika CH, Ravindra Reddy J, Bhargav E, Harish E, Lakshmi Chowdary M. Formulation and optimization of Losartan Potassium sustained release tablets by statistical experimental design. *Indo Am J Pharm Sci* 2018;5(2):1068–80.
- [28] Murugan R. The importance of design space to analytical methods. *Spinco Biotech - Cutting Edge* 2014;7–11.

Chapter 11

Design of experiments application for analytical method development

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11.1 Introduction

Design of experiments (DoE) is a tool used for systematic development and optimization of the products and processes. Based on the sound principles of statistics, DoE works very well on the multifactorial experiments where one-factor-at-a-time approaches largely fail [1]. Scientific experiments are usually associated with high degree of variability due to involvement of multiple factors that are particularly responsible as the major sources of variation; thus DoE rationally helps in addressing such challenges to provide solutions with better quality consistency [2,3]. In other words, DoE helps in establishing a cause-and-effect relationship between the factors and response(s).

The merits of DoE have been extensively discussed in several literature reports. Some of the very useful advantages of such a technique include resource saving, time, and efforts saving. Besides, it reduces wastage and increases the process efficiency [4]. Thus, ICH guidance especially Q8 on pharmaceutical development that emphasizes on quality by design for product and process development also describes the merits of DoE [5–7]. For the past few decades, the applications of DoE in drug product development have been increasingly practiced. Due to enormous merits of DoE, it has lately been applied for the analytical development practice in various chromatographic methods and other related techniques [8].

11.2 Fundamental of applying design of experiments

For application of DoE, a meaningful understanding of the strategic and fundamental principles is highly essential. DoE primarily considers any

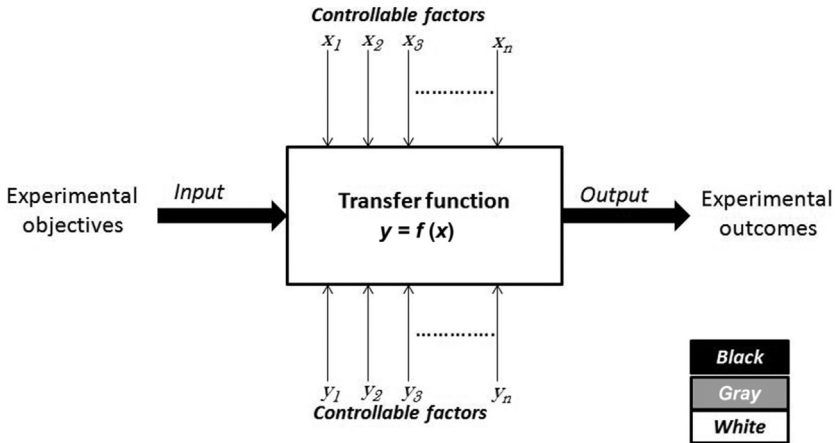


FIGURE 11.1 Fundamental concept of DoE, where x and y variables are interlinked with the help of a transfer function (black box—0% predictability, gray box—50% predictability, white box—100% predictability). DoE — design of experiment.

experimental system as an entity controlled by a group of independent and dependent variables that primarily influences the quality. Besides, the system is also influenced by group of uncontrollable variables also called as nuisance variables that are beyond the control of the system [1,2]. DoE helps in establishing a link between the input and output variables in a system and acts like a transfer function to interpret the signal by reducing the noise due to variability. The concept of DoE can be graphically better interpreted from Fig. 11.1, where “ x ” variables ($x_1, x_2 \dots x_n$) are referred to as controllable input factors, while “ y ” variables ($y_1, y_2 \dots y_n$) are referred to as controllable output variables. The quantitative relationship between these two sets of variables is determined with the help of mathematical models, which deciphers the magnitude of impact between the variables [3].

11.3 Key principles of design of experiments

DoE primarily works on three key principles such as replication, randomization, and blocking (or error control), which have been described in detail in the following subsections.

11.3.1 Replication and randomization

It is considered as an approach of repeating the experiments in a systematic way or by utilizing the random distribution to reduce the bias or error during experimentation. Ideally, minimum replicates of three or more can be used

for experimental optimization. Moreover, the replicates should be performed in a randomized order for avoiding experimental designs [3]. Also, the principles of randomization and replication are applied simultaneously during execution of experimental designs. By randomizing the experiment, one can average out the effects of extraneous factors on the experiments.

11.3.2 Blocking or error control

It is the approach of distribution of the similar types of experimental conditions in groups or blocks so that it may reduce the variability due the nuisance factors, thus avoiding its impact on the experimental outcomes. Thus block is considered as a set of homogeneous experimental conditions by nullifying the influence of the nuisance factors [3].

11.4 Steps in performing design of experiments

The application of DoE requires thorough understanding of the salient steps involved in it, which have been provided below. Fig. 11.2 also illustrates a schematic layout of steps involved in the application of DoE principles.

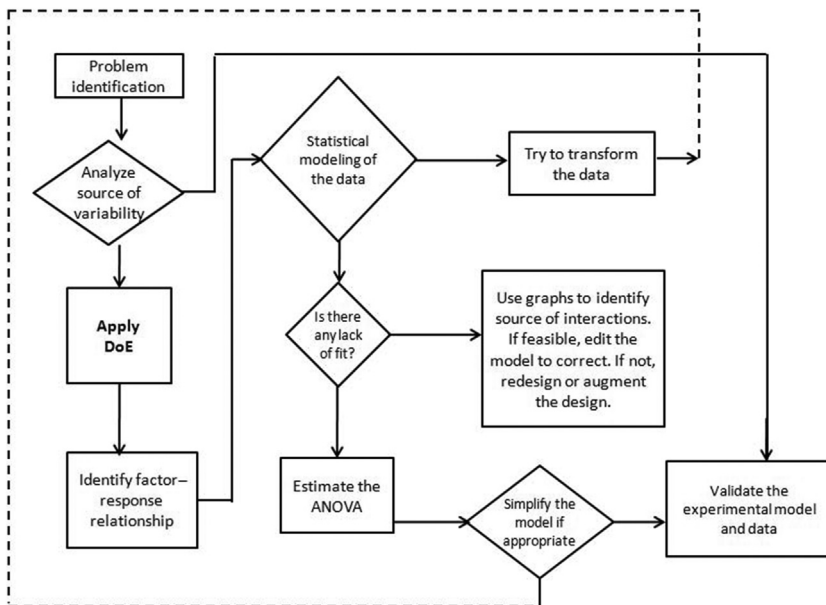


FIGURE 11.2 Flow chart indicating various steps involved in the application of DoE approach. DoE, design of experiment.

11.4.1 Problem conceptualization

For a meaningful DoE analysis, it is highly important to identify the research problems. Based on the study objectives, the experimental designs are selected for detailed optimization study. Moreover, it is quite helpful for researchers to prepare list of specific problems on the basis of preliminary studies.

11.4.2 Screening of the factors

It is highly important to identify the key influential factors before performing factor optimization studies. In this regard, factor screening studies are particularly helpful, where screening experimental designs can be applied for identifying the factors with high risk on the responses. Among the possible so many factors, only a vital few factors are identified. Some of the commonly employed screening designs include fractional factorial design (FFD), Taguchi design, Plackett–Burman design, etc. are of potential interest. Fig. 11.3A–C portrays the schematic diagrams of the screening experimental designs.

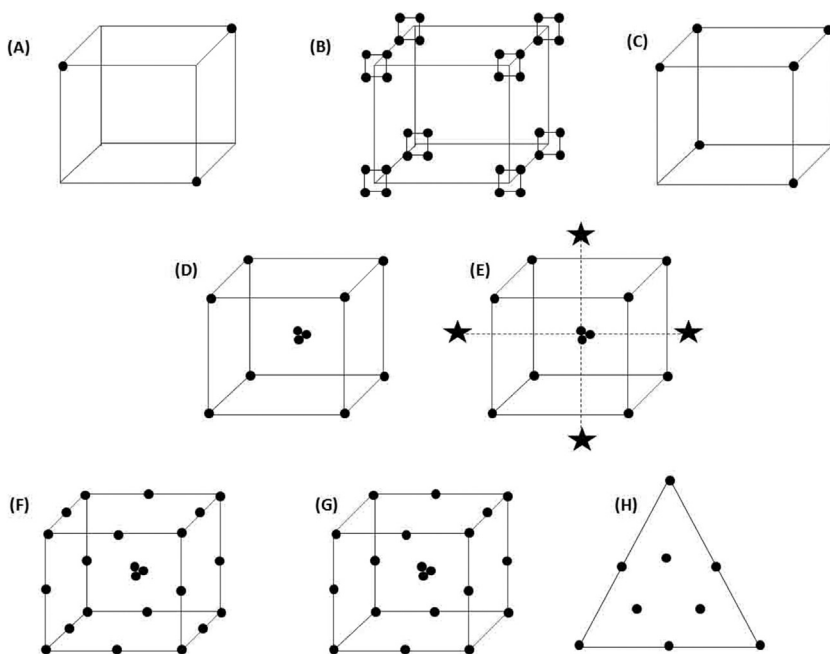


FIGURE 11.3 Graphical depiction of various experimental designs: (A) fractional factorial design, (B) Taguchi design, (C) Plackett–Burman design, (D) full factorial design, (E) central composite design, (F) Box–Behnken design, (G) optimal design, and (H) mixture design.

11.4.2.1 Fractional factorial design

The FFD is very frequently used for factor screening study, which is primarily considered as the low-resolution screening design. Such designs are expressed as in the form of X^{k-p} , where X denotes the levels of factor, k denotes the number of factors, and p denotes the fractionation used. FFD is used for screening more than three and maximum of seven factors in an experimental design. However, it uses two levels (-1 and $+1$) to establish a linear relationship between the points.

11.4.2.2 Taguchi design

Taguchi design is quite sought after for factor screening, sensitivity testing, and robustness analysis. Such design primarily applies orthogonal arrays to create the objective functions for evaluating the factor influence on the responses. Like FFD, Taguchi design is suitable for screening more than three and maximum of seven factors in an experimental setup. However, it only uses two levels (-1 and $+1$) to establish a linear relationship between the points.

11.4.2.3 Plackett–Burman design

This design is exclusively recommended in experiments containing very high number of factors ranging between 9 and 11. As a low-resolution design, it requires two levels of each of the factors, while it produces a limited number of runs for identifying the highly influential factors.

11.4.3 Optimization of the factors

After the system has been characterized and we are reasonably certain that the important factors have been identified, the next objective is usually optimization. An optimization experiment is usually a follow-up after a screening experiment. Use of customized optimization designs (also called as response surface designs) is very helpful to produce the optimal settings of the factors. Fig. 11.3D–H portrays a schematic approach used for performing factor screening study.

11.4.3.1 Full factorial design

The full factorial design (FD) is used for optimization of the factors identified from the screening designs. Such design is suitable for optimization of a minimum of three factors and requires three levels (-1 , 0 , $+1$) for the purpose. Often, FD also requires center points for more robust estimation of the outcomes and to reduce the bias due to error associated during the experimental trials. This design also helps in studying the effect of each factor on the response variables.

11.4.3.2 *Central composite design*

The central composite design (CCD) is considered as very effective in factor optimization by considering the interaction among the factors. Such design is operated at a minimum of three levels or maximum of five levels, and thus highly symmetric in nature due to high degree of rotatability. The design helps in establishing the quadratic relationship among the factor levels. CCD is also referred to as an augmented form of FD and very effective to produce the productive outcomes and is very useful for an experiment having a minimum of two or more factors.

11.4.3.3 *Box–Behnken design*

The Box–Behnken design (BBD) is very popularly used for factor optimization and operates at three levels of each factor. Like CCD, it also employs quadratic mathematical models for establishing cause-and-effect relationship among the factors with minimal number of experimental runs. However, the design shows minimum flexibility for rotatability and is very useful for an experiment giving a minimum of three or more factors.

11.4.3.4 *Optimal designs*

The optimal designs are used over the CCD and BBD, when a high number of factors are required to be optimized with minimal number of runs. Optimal designs are of various types such as D-optimal, IV-optimal, and A-optimal, which require three levels of the factors and quadratic modeling approach for producing the optimal solutions.

11.4.3.5 *Mixture designs*

The mixture designs are useful in experiments containing factor proportions sum total of 100%, where other designs are difficult to produce the symmetric solutions. Such designs are of different types such as simplex-lattice designs, simplex-centroid designs, and optimal designs. Among these, optimal designs have been extensively explored, which are again of three types such as D-optimal, IV-optimal, and A-optimal. These designs are capable of linear, quadratic, and cubic model fitting, and thus produces the best solutions.

11.5 **Application of design of experiments in analytical development**

The applications of DoE are galore in diverse segments of science and technology-driven sectors. Of course, the pharmaceutical industry has observed a paradigm shift in the past two decades to adopt the systematic approaches over the traditional approaches. Similarly, application of such tools into analytical development has also observed a paradigm shift for attaining

excellence in analytical testing and regulatory compliance [9,10]. A vast score of literature reports are available where various experimental designs have been used for analytical method development and optimization, troubleshooting, and other variegated applications, thus testifying the success of using such an approach into practice.

11.6 Conclusion

DoE has now become a versatile tool for optimizing the performance of analytical methods by reducing the method variability and multifold improvement in method performance. Although no official regulatory guidance on analytical development (Q2 and Q14) suggests for the mandatory usage of DoE tools, yet the application of the same provides greater leverage for systematic analytical testing with hassle-free monitoring of the method performance. Beyond DoE, multivariate chemometric tools are used nowadays for unearthing meaningful outcomes.

References

- [1] Singh B, Raza K, Beg S. Developing “optimized” drug products employing “designed” experiments. *Chem Ind Dig* 2013;23:70–6.
- [2] Singh B, Kumar R, Ahuja N. Optimizing drug delivery systems using systematic “design of experiments.” Part I: fundamental aspects. *Crit Rev Ther Drug Carr Syst* 2005;22:27–105.
- [3] Beg S, Swain S, Rahman M, et al. Application of design of experiments (DoE) in pharmaceutical product and process optimization. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. New York, NY: Academic Press; 2019. p. 43–64.
- [4] Beg S, Rahman M, Panda SS. Pharmaceutical QbD: omnipresence in the product development lifecycle. *Eur Pharm Rev* 2017;22:58–64.
- [5] Beg S, Hasnain MS. *Pharmaceutical quality by design: principles and applications*. New York, NY: Academic Press; 2019.
- [6] Beg S, Rahman M, Robaian MA, et al. *Pharmaceutical drug product development and process optimization: effective use of quality by design*. New York, NY: CRC Press; 2020.
- [7] Singh B, Beg S. Quality by design in product development life cycle. *Chron Pharmabiz* 2013;22:72–9.
- [8] Beg S, Hasnain MS, Rahman M, et al. Introduction to quality by design (QbD): fundamentals, principles, and applications. In: Beg S, Hasnain MS, editors. *Pharmaceutical quality by design*. Academic Press; 2019. p. 1–17.
- [9] Beg S, Sharma T, Saini S, et al. Analytical quality by design for robust chromatographic methods. *Cutting-Edge (Spinco Biotech)* 2020;10:9–17.
- [10] Beg S, Rahman M, Swain S. Quality by design applications in pharmaceutical product development. *Pharm Focus Asia* 2020;1–5.

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HANDBOOK OF ANALYTICAL QUALITY BY DESIGN

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Handbook of Analytical Quality by Design addresses current regulatory thinking on systematic analytical testing and application of science and risk-based concept of Quality by Design (QbD) for analytical method development and validation. Recently ICH has highlighted the importance of such rational method practices for an efficient and effective analytical development. The adoption of systematic approaches ensures quality can be built into the analytical practice for avoiding quality crises in the later stages. The analytical QbD approach significantly enhances method performance and robustness which are crucial during interlaboratory studies and improves the reproducibility of the analytical lifecycle of the developed method. It provides a robust and flexible operable range for an analytical method with greater excellence and regulatory compliance. The book covers sample preparation problems and usefulness of the QbD concept involving Quality Risk Management (QRM), Design of Experiments (DoE), and Multivariate (MVT) Statistical Approaches to solve them by optimization of the developed method, along with validation for different techniques like HPLC, UPLC, UFLC, LC-MS, GC-MS, NIR, electrophoresis, etc. techniques. The use of spectroscopic or spectrometry data, some of which require multivariate statistical analyses, is also covered. *Handbook of Analytical Quality by Design* will be useful for graduate students and professionals working in the pharmaceutical industry, analytical chemistry, regulatory agencies, and those in related fields.

Key Features:

- Concise language for easy understanding of the novel and holistic concepts for efficient analytical development
- Covers key aspects of systematic analytical development and validation practices
- Provides a robust and flexible operable range for an analytical method with greater excellence and regulatory compliance

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